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Lecture Notes on Classical Mechanics

Class notes for CE1010, and ME3030

Supplementary .ipynb interactive Notebook:
https://github.com/mugalan/classical-mechanics-from-a-geometric-point-of-view/blob/main/mechanics/class-notes/Mugas_Classical_Mechanics.ipynb

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Preface

This is a compilation of notes that originated as class notes for ME2204 Engineering Mechanics at the department of Mechanical and Manufacturing Engineering, Faculty of Engineering, University of Ruhuna during the period of 2006 to 2008. Subsequently they have served as class material for ME211, ME320, and ME518 at the department of Mechanical Engineering, Faculty of Engineering, University of Peradeniya during the period of 2009 to 2026. I am currently using it as class notes for CE1010 (Chapter - 1) and ME3030 (Chapter - 2 and Chapter - 3). They are far from complete and I will be frequently updating them as time permits. I am deeply indebted to all the students who have suffered through them and have tried out the exercises and have provided me with valuable input. I am sure there are many errata and will greatly appreciate if you can please bring them to my notice by sending an e-mail to mugalan at gmail.com.

The GitHub repository: classical-mechanics-from-a-geometric-point-of-view, provides a collection of interactive IPython notebooks and supplementary material, it includes lightweight Python utilities ('sims') for simulations, 3D Plotly animations, and tools for rigid body system modeling, estimation and control.

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Chapter 1
Galilean Mechanics



Fig. 1.1 Galileo Galilei — 15 February 1564 to 8 January 1642: Wikipedia

Mechanics deals with the scientific description of the world as we perceive it. The human infatuation with the subject pre-dates written history and has given rise to the well accepted customary approach of searching for scientific laws, in the form of mathematical expressions, to describe and generalize naturally observed phenomena. The sole test of the 'truth' of such laws was (and still is) experiment. Typically one would imagine, conceptualize and arrive at mathematical expressions to describe and generalize observed phenomena and then devise experiments to verify their validity. They are considered to be 'true' insofar as their predictions agree with experimental observations within the uncertainty of the measurements. Discrepancies may become apparent when more precise and accurate measurements become possible. Up until the 20th century, Galilean mechanics (or what is popularly known as Newtonian mechanics[1]) and Maxwell's electromagnetism were found to accurately describe a wide range of observed phenomena within the uncertainty of the measurements available at the time. As

[1] Galileo Galilei's contribution to the formulation of the laws as stated by Newton is monumental.

technology developed it became possible, around the start of the 20th century, to conduct experiments and observe nature with increasingly high precision. Scientists started observing phenomena that were not accurately described by classical physics. Discrepancies arose when Galilean principles were used to describe objects moving close to the speed of light, and when classical mechanics and electromagnetism were used to describe atomic and subatomic phenomena. The effort to reconcile mechanics with Maxwell's electromagnetism and accurately describe motion at speeds close to that of light led to Einstein's *special theory of relativity*, while the effort to accurately describe microscopic phenomena led to *quantum mechanics*. Both theories have been extensively tested, with predictions agreeing with observations within experimental uncertainty in their tested domains. Einstein extended his theory of relativity to incorporate gravitational interaction between objects, developing the theory of *general relativity*. Special relativity and quantum mechanics are brought together in the framework of *quantum field theory*, with electromagnetic interactions described by *quantum electrodynamics*. The unification of Einstein's general relativity and quantum mechanics remains an open challenge in the search for a single framework that could describe all fundamental interactions.

The objects of motion that we encounter in most of our engineering practice are much larger than microscopic objects and move at speeds much slower than the speed of light. The motion of such objects is sufficiently accurately described by Galilean mechanics. The Galilean picture adopted here is rooted in our ordinary sensory experience of material objects and adopts the idealization that material objects are made of *impenetrable interacting particles of matter*, with no two particles occupying the same place at the same instant [1]. Motivated by this ordinary sensory experience, the Galilean picture further assumes that *time* is absolute, independent of space, one-dimensional, continuous, isotropic[2] and homogeneous, and that *space* is Euclidean, three-dimensional, continuous, isotropic, and homogeneous. All observers agree on simultaneity and elapsed time, when expressed in the same units.

The notion of time is related to memory. Memory allows one to recognize a sequence of events. The duration between events is measured by time. Time is given operational meaning through physical change. In mechanics, we measure elapsed time by comparing motions, and in this operational sense motion is essential to our conception of time. In classical mechanics, space and time can be viewed as tools that allow us to describe the relations between objects and hence their motion.

The aim of mechanics is to describe the relation between objects using the language of mathematics. Transforming the observations into a mathematical expression involving numbers requires measurements. Measurements depend on the measuring system or in other words the observer. Although descriptions of motion depend on the observer, the laws governing motion should admit an observer-independent formulation. *Thus mechanics can be considered to be the search and study of the scientific laws of motion in the form of mathematical expressions that are observer independent*. This means that half of the problem lies in figuring out the capabilities each observer has and the measurements and observations that different observers can agree upon. Here, an observer means a measurement system that is mathematically represented by a coordinate system. Although different observers may assign different numer-

[2] Here, temporal isotropy means that the geometry of time has no preferred orientation: reversing the time coordinate preserves durations. It does not imply time-reversal invariance of physical processes.

ical coordinates to the same physical quantity, its geometric meaning is independent of these representations. Properties and relations that are independent of the chosen observer in this sense are called *observer invariant*, or *intrinsic*. For instance we observe that, as resistance and other disturbing influences are reduced, moving objects tend to maintain their velocity for longer. This motivates the idealization that, in the absence of interactions, a particle remains at rest or continues in uniform straight-line motion. An important question that will be addressed below is: for what class of observers is this character of free motion invariant?

Below we will explore these notions that will eventually help us mathematically describe the motion of everyday observed objects. This study is what is called Galilean mechanics. The application of the Galilean laws of mechanics is generally divided into three branches depending on the type of objects under consideration; *rigid-body mechanics, deformable-body mechanics* and *fluid mechanics*. In these notes we will concentrate on learning the basics of rigid-body mechanics. The study of rigid-body mechanics begins with describing the motion of a single hypothetical *particle of matter*. A general rigid body is considered to consist of a large number of such particles where the distance between every pair of constituent particles remains fixed. The geometric description of the motion of these objects is what is generally known as *Kinematics* while the study of the cause of motion is referred to as *Kinetics*. We will explore these ideas a bit more closely in the following sections.

1.1 Fundamental Laws of Galilean Mechanics

Describing the motion of a particle depends on the measurements that an observer makes (quantitative observations). As introduced above, an observer is a measuring system mathematically represented by a coordinate system. In general, different measurement systems will yield different descriptions of the motion. For example, a particle that appears to be fixed with respect to one observer will appear to be moving for another observer relative to whom its position changes. Thus, in general, the description of the position of a given particle depends on the observer. A natural question that arises is the following: do there exist notions and properties of nature whose geometric meaning does not depend on the observer? If so, what are they?

1.1.1 Inertial Observers

Our general experience suggests that motion is continuous and that, apart from the influence of local physical conditions such as gravity, there is no preferred spatial direction or location. Likewise, our measurement of elapsed time suggests a uniform notion of duration with no preferred time origin. These observations motivate modelling space and time as three-dimensional and one-dimensional Euclidean affine spaces[3], respectively. We adopt these as idealizations of the underlying space-time structure. Specifically, we assume that:

Axiom 1.1 Fundamental Assumptions of Galilean Space-Time: *There exists a nonempty class of observers called inertial observers who, using common units of measurement, agree that*

(i) time is independent of space, and simultaneity is the same for all such observers;
(ii) time is a one-dimensional Euclidean affine space, and the elapsed time between any two given events is the same for all such observers;
(iii) space at each instant is a three-dimensional Euclidean affine space, and the distance between any two given simultaneous events is the same for all such observers;
(iv) any particle trajectory with constant velocity relative to one such observer also has constant velocity relative to every other such observer, for any initial position and velocity.

Galilean mechanics rests upon these assumptions and is applicable where these assumptions and its laws of motion provide an adequate description of the phenomena under consideration. These fundamental assumptions about classical space-time are also assumptions

[3] An affine space is a space with no preferred origin that, once an origin is chosen, can be identified with a vector space; changing the origin changes this identification by a translation. A Euclidean affine space additionally carries an inner product on the associated vector space, defining distances and angles independently of the chosen origin.

about the measuring systems the observers have. The first two assumptions imply that inertial observers agree on simultaneity and elapsed time when using common units, or in other words that a *universal time* can be assigned, represented by an idealized *universal clock*. The third assumption states that the distance between simultaneous events is inertial-observer independent and that space at each instant is Euclidean[4].

The fourth assumption connects the spatial measurements made at different instants. Together with the first three assumptions, it implies that transformations between inertial observers are homogeneous in the sense that transformed coordinate differences depend only on the original coordinate differences, and not on the common spatial or temporal location of the events. This will be discussed further below.

In summary, the above axiom supposes that there exist observers, referred to as inertial observers, who agree on simultaneity, elapsed time, and Euclidean distances at each instant, and who agree on whether a particle trajectory has constant velocity.

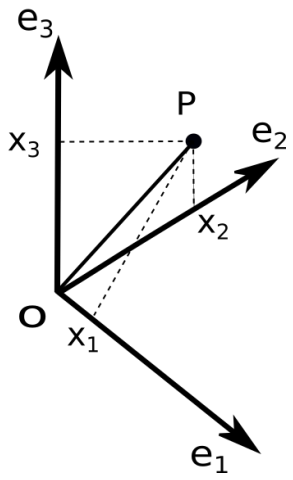


Fig. 1.2 The representation of point in space using an orthonormal frame **e**.

Below we will explore further the consequences of these assumptions. The first two assumptions allow an observer to use an idealized clock to measure universal time differences. Choosing a time origin, a unit of duration, and a positive temporal orientation then identifies each instant with a unique real number $t \in \mathbb{R}$.

The third assumption allows us to construct a Cartesian coordinate system for space at each instant, as follows. At a fixed instant, let **e** denote an inertial observer. The observer **e** can describe a point P in the 3D space that we live in by picking a point O in space and setting three mutually perpendicular axes with a common unit of length at O[5]. The assumption that

[4] A space where Euclidean geometry holds.

[5] Show that this can be done since **e** sees space to be Euclidean.

space is isotropic and homogeneous allows us to choose any origin and any orientation of these axes. The origin, together with an ordered set of unit directions along these axes, defines an *orthonormal frame of reference*. Labelling these unit directions $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ to give a right-handed orientation, we can symbolically represent the frame directions as a row matrix $\mathbf{e} = [\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3]$, where the symbols $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ represent geometric unit axis rather than numerical coordinate columns. In these notes we will always assume that the orthonormal frames are right-handed. With an abuse of notation, we use $\mathbf{e}$ to denote both the observer and its frame, suppressing the origin O when it is understood.

Using such a frame, any point P in 3D-Euclidean space can be uniquely described using only three measurements (numbers) x_1, x_2, x_3. These numbers are the signed components of the displacement from O to P along the $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ directions, respectively, as shown in figure 1.2. Conversely, the assumed three-dimensional Euclidean structure ensures that every ordered triple of real numbers $(x_1, x_2, x_3) \in \mathbb{R}^3$ corresponds to a unique point relative to the chosen origin and axes. Here, Euclidean space is understood to mean the full Euclidean space, rather than a proper region of it. Symbolically, we describe this identification as

$$OP = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + x_3\mathbf{e}_3 = \underbrace{\begin{bmatrix}\mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3\end{bmatrix}}_{\mathbf{e}} \underbrace{\begin{bmatrix}x_1 \\ x_2 \\ x_3\end{bmatrix}}_{x} = \mathbf{e}x.$$

In these notes the column matrix

$$x = \begin{bmatrix}x_1 \\ x_2 \\ x_3\end{bmatrix}$$

will be referred to as the Euclidean *representation* matrix, or simply the representation, of the position of the point P with respect to the frame $\mathbf{e}$. The components $(x_1, x_2, x_3) \in \mathbb{R}^3$ will be referred to as the position components of P, or the *Euclidean coordinates* of P, with respect to $\mathbf{e}$. With a further abuse of notation, we also denote this ordered triple by the same symbol x as its column-matrix counterpart. We will use the terms measurement system, frame, coordinate system, and observer interchangeably in this sense.

Consider two points P and Q at the same instant, with Euclidean coordinates $x = (x_1, x_2, x_3) \in \mathbb{R}^3$ and $y = (y_1, y_2, y_3) \in \mathbb{R}^3$, respectively, with respect to a frame $\mathbf{e}$. Recall that the standard *inner product* in $\mathbb{R}^3$ is defined by

$$\langle\langle x, y \rangle\rangle \triangleq x_1y_1 + x_2y_2 + x_3y_3.$$

Then, by the Pythagorean theorem for 3D-Euclidean space, the measured distance between P and Q is

$$d(P,Q) = \|x - y\| = \sqrt{\langle x - y, x - y \rangle} = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2}. \tag{1.1}$$

For $P \neq O$ and $Q \neq O$, the angle $\theta \in [0, \pi]$ between the rays from O through P and Q is given by

$$\theta = \cos^{-1}\left(\frac{\langle\langle x, y \rangle\rangle}{\|x\| \cdot \|y\|}\right).$$

The above demonstrates that the standard Euclidean inner product on $\mathbb{R}^3$

$$\langle\langle \cdot, \cdot \rangle\rangle : \mathbb{R}^3 \times \mathbb{R}^3 \mapsto \mathbb{R}$$

fills in the missing part of distance and angle measurements in space and completes the picture of identifying space with $\mathbb{R}^3$.

Thus, the fundamental assumptions of classical space-time imply that, at each instant, an inertial observer can construct an orthonormal frame and use it as a measurement system for space. This gives a globally defined Cartesian coordinate system in which the distance between any two simultaneous points with coordinates $x, y \in \mathbb{R}^3$ is given by (1.1). Although their coordinates may differ, all inertial observers obtain the same value for this distance.

Together with a clock measuring universal time differences, a chosen time origin, and common units of measurement, this construction allows an observer **e** to assign unique coordinates

$$\boxed{(t,x) \in \mathbb{R} \times \mathbb{R}^3}$$

to each space-time event, where t specifies when the event occurs and x specifies its spatial position in the observer's frame at that instant.

Let us first show that the coordinate transformations between inertial observers must be affine. Let (t,x) and (τ,ξ) denote the coordinates of the same space-time event according to two inertial observers. The first two assumptions, with common units and the same positive temporal orientation, imply that

$$\boxed{\tau = t + a},$$

where $a \in \mathbb{R}$ is constant. Write the spatial coordinate transformation as $\xi = \xi(t,x)$ and assume that it is twice continuously differentiable.

The fourth assumption requires every constant-velocity trajectory

$$x(t) = x_0 + ut$$

to have constant velocity in the other observer's coordinates. Since $d\tau = dt$, applying the chain rule to each component ξ_k, for $k = 1,2,3$, gives

$$0 = \frac{d^2\xi_k}{dt^2} = \frac{\partial^2 \xi_k}{\partial t^2} + 2\sum_{i=1}^{3} u_i \frac{\partial^2 \xi_k}{\partial t\, \partial x_i} + \sum_{i,j=1}^{3} u_i u_j \frac{\partial^2 \xi_k}{\partial x_i\, \partial x_j}.$$

At any fixed (t,x), this identity holds for every $u \in \mathbb{R}^3$. Thus all second partial derivatives of every component ξ_k must necessarily vanish[6]. Its first partial derivatives are therefore constant, so

[6] You are invited to fill in the details of this argument in exercise-1.1

$$\xi_k(t,x) = \alpha_k + \beta_k t + \sum_{i=1}^{3} R_{ki} x_i.$$

In matrix notation,

$$\boxed{\xi = \alpha + \beta t + Rx,}$$

$\alpha, \beta \in \mathbb{R}^3$, and $R \in \mathbb{R}^{3\times 3}$ are constant. Hence the full space-time coordinate transformation

$$(t,x) \mapsto (\tau,\xi) = \big(t + a, \alpha + \beta t + Rx\big)$$

is affine. These global coordinate systems, related by affine transformations, equip space-time with an affine structure.

Finally, the third assumption requires preservation of simultaneous spatial distances:

$$\|\xi(t,x) - \xi(t,y)\|^2 = \|x - y\|^2.$$

Substituting the affine form gives

$$(x-y)^T R^T R (x-y) = (x-y)^T (x-y)$$

for all $x, y \in \mathbb{R}^3$. Consequently,

$$R^T R = I_{3\times 3}.$$

Thus R is orthogonal[7]. With our convention of right-handed spatial frames, it also satisfies $\det(R) = 1$. Such matrices are called special orthogonal matrices. The space of all special orthogonal matrices are usually denoted by $SO(3)$.

Without loss of generality, we may also choose the clocks to be synchronized and the spatial origins to coincide at $t = 0$, giving $a = 0$ and $\alpha = 0$. With these conventions, the representations of the same space-time event by two inertial observers are related by

$$(\tau,\xi) = (t, \beta t + Rx).$$

For simultaneous displacements, this transformation gives $\Delta\xi = R\Delta x$. Orthogonality of R ensures that lengths and angles are preserved, consistently with the use of orthonormal frames by both observers.

Let O' be the origin of the frame used by $\mathbf{e}'$, and let $o(t)$ denote its position coordinates according to $\mathbf{e}$. The event at O' at time t has coordinates $(t, o(t))$ according to $\mathbf{e}$ and $(t, 0)$ according to $\mathbf{e}'$. Therefore,

$$0 = \beta t + Ro(t).$$

Since R and β are constant,

$$v \triangleq \dot{o}(t) = -R^T\beta, \qquad \beta = -Rv.$$

Thus the origin of $\mathbf{e}'$ moves with constant velocity relative to $\mathbf{e}$. Moreover, since R is constant, their frames do not rotate relative to each other.

[7] Notice that from the uniqueness of the inverse of invertible matrices $R^{-1} = R^T$. Thus we also have $RR^T = I_{3\times 3}$

Consequently, with synchronized clocks and origins coinciding at $t = 0$, the coordinate transformation takes the form

$$(\tau, \xi) = \big(t, R(x - vt)\big), \tag{1.2}$$

where $v \in \mathbb{R}^3$ and $R \in SO(3)$. An observer rotating or accelerating relative to an inertial observer does not, in general, describe free particles as moving with constant velocity, and therefore does not satisfy the fourth assumption. Such an observer may nevertheless use Euclidean spatial coordinates and agree on simultaneous distances.

Finally, because R is constant, we may choose the axes of $\mathbf{e}'$ to be parallel and aligned with those of $\mathbf{e}$, so that $R = I_{3\times 3}$. In these aligned inertial frames $(\tau, \xi) = \big(t, (x - vt)\big)$. More generally, inertial frames may have any constant relative orientation; they translate at constant relative velocity without relative rotation.

In summary, Axiom 1.1 postulates a class of inertial observers who agree on simultaneity and elapsed time, and who use Euclidean spatial coordinates in which the distance between any two simultaneous points is given by (1.1), independently of the observer. Their agreement on constant velocity motion implies that their frames move with constant relative velocity and without relative rotation.

1.1.2 Description of Motion

Consider a particle moving in 3D-Euclidean space. We have seen that the position $P(t)$ of the particle at a given time t can be expressed by the Euclidean representation matrix $x(t)$ with respect to some orthonormal frame $\mathbf{e}$. Since the position of the particle is changing with time, $x(t)$ is a function of time and describes a curve in $\mathbb{R}^3$ that is parameterized by time t. The velocity of a point P is always defined with respect to an orthonormal reference frame. Specifically, relative to the $\mathbf{e}$ frame, velocity describes the infinitesimal change of position per unit time, defined by the limit

$$\dot{x}(t) \triangleq \lim_{\delta t \to 0} \frac{x(t+\delta t) - x(t)}{\delta t} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix}.$$

Observe that by definition $\dot{x}(t)$ gives the tangent to the curve at $P(t)$ (Refer to figure 1.3). The components

$$\dot{x}_i(t) = \lim_{\delta t \to 0} \frac{x_i(t+\delta t) - x_i(t)}{\delta t} \quad i = 1, 2, 3$$

represent the infinitesimal change of position in the $\mathbf{e}_i$ direction. The magnitude of the velocity (the *speed*) in the $\mathbf{e}$ frame is defined to be

$$v \triangleq ||\dot{x}|| = \sqrt{\dot{x}_1^2 + \dot{x}_2^2 + \dot{x}_3^2}.$$

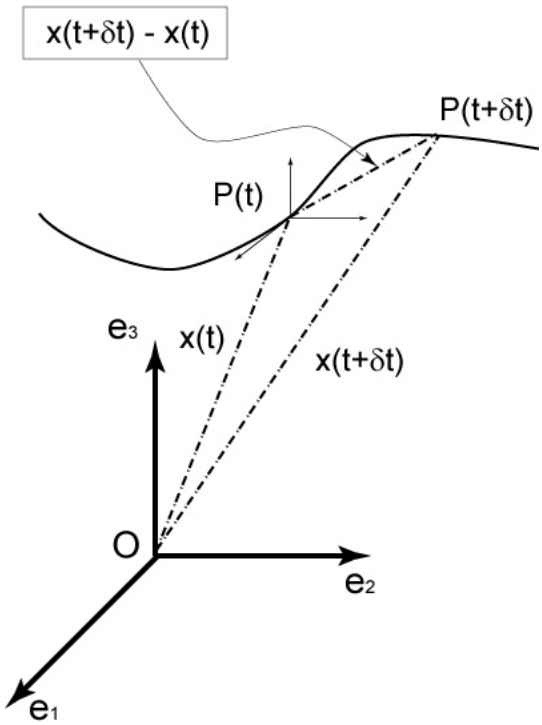


Fig. 1.3 The Velocity of a Particle described with respect to a frame **e**.

Thus we visualize $\dot{x}(t)$ as the components, relative to **e**, of a directed arrow based at $P(t)$, whose direction indicates the direction of motion and whose length represents the speed.

The acceleration of a particle is also defined in terms of orthonormal frames. Relative to the **e** frame, acceleration describes the infinitesimal change of velocity per unit time, defined by the limit

$$\ddot{x}(t) \triangleq \lim_{\delta t \to 0} \frac{\dot{x}(t+\delta t) - \dot{x}(t)}{\delta t} = \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{bmatrix}.$$

Animals have inbuilt sensors that allow them to measure accelerations. For example, the human ear contains accelerometers that allow them to sense the direction of the gravitational accelerations. It is interesting to note that animals can not sense absolute velocities and thus can not distinguish between being at rest or being in constant velocity motion.

Consider a certain space-time event A and let **e** and $\mathbf{e}'$ be two inertial observers. Let $(t,x) \in \mathbb{R}^4$ and $(\tau,\xi) \in \mathbb{R}^4$ be the representation of the space-time event A made by **e** and $\mathbf{e}'$ respectively. Assume that both observers have synchronized clocks and that the spatial origins coincide at $t = 0$. Then from the Galilean space-time axioms, we established previously that the two representations are related by $\tau = t$ and $\xi = R(x - vt)$ for some constant $v \in \mathbb{R}^3$ and $R \in SO(3)$. Thus the representation of the velocity of a particle in the two frames are related by $\dot{\xi} = R(\dot{x} - v)$ and the acceleration of a particle in the two frames are related by $\ddot{\xi} = R\ddot{x}$. Immediately we see that since $\|\ddot{\xi}\| = \|\ddot{x}\|$ all inertial observers agree on the magni-

tude of accelerations. Furthermore if the two observers are parallel and the axes are aligned then $R = I_{3\times 3}$ and hence $\ddot{\xi} = \ddot{x}$. Therefore we see that if a particular observer, **e**, sees that an object is moving at a certain acceleration then all *parallel inertial observers* with respect to **e** will also observe that the object is moving at that same acceleration. This allows us to conclude that even though the position and velocity of a particle may not be the same for all inertial observers with parallel frames the acceleration of a particle is observed to be the same for all inertial observers with parallel frames. Hence we can conclude the following:

The Galilean space-time axioms imply that all inertial observers agree on the magnitude of a particle's acceleration. Furthermore, in inertial frames with parallel and aligned axes, the acceleration components are identical, independently of the constant relative velocity of the frames.

1.1.3 The Apparent Cause of Change of Motion

Kinetics deals with the apparent causes of changes in motion. Galileo Galilei in the 17th century made the observation that a person doing experiments, with moving objects, below the deck in a ship traveling at constant velocity, without rocking, on a smooth sea; would not be able to tell whether the ship was moving or was stationary[8]. Thus he concluded that laws of nature that describe the motion of objects must be the same in all inertial frames.

Based on experiments, involving colliding rolling balls on a smooth horizontal surface, it was observed that objects generally tend to move at constant velocities or remain at rest unless brought into interaction with other objects. Galileo's work laid a crucial foundation for the *principle of inertia*. In the form adopted here, subsequently embodied in Newton's first law, this principle states that a particle subject to no interactions remains at rest or moves with constant velocity relative to any inertial observer. Thus for an isolated matter particle we would have that $\ddot{x} = 0$ in any and every inertial frame and thus that it is an invariant property observed by all inertial observers. This also means that all inertial observers will agree that an isolated matter particle is either at rest or is moving in a straight line. It was also observed that there was a certain resistance to change in this steady motion and that this resistance depended on the amount of matter present in the object. Bigger objects would tend to change its motion slower than smaller objects that were made of the same material. The measure of the degree of this resistance to change in motion is referred to as the *inertia* of the object. Careful experiments with two colliding balls on a smooth surface indicated that a number, can be ascribed to each ball such that if for each ball you multiply this number with the velocity of the ball, and added the results together, this sum remains approximately constant under these experimental conditions. The constant that multiplied the velocity was called the *mass* of the object. If it is assumed that all inertial observers agree on the mass of an object as a unique

[8] Without external reference cues, bodily sensations alone do not distinguish rest from uniform straight-line motion. Thus, in the absence of a visual change of reference, one could not differentiate between being in a ship moving at constant velocity and being in a ship at rest.

property of the object, then all inertial observers will also agree on the statement that the sum of the mass times velocity of two colliding balls remain a constant even though the constant they obtain may be different for each observer. We adopt the hypothesis that this mass is an intrinsic property of the particle, independent of the inertial observer. This is one of the crucial assumptions of Galilean mechanics:

Axiom 1.2 The Principle of Conservation of Linear Momentum: *Each particle has a definite positive, time-independent mass, whose value is the same for all inertial observers using common units. Define the linear momentum of a particle relative to an inertial frame as its mass times its velocity in that frame. For any finite isolated system of particles, meaning a system with no interactions with its surroundings, the sum of the linear momenta of its particles remains constant in every inertial frame.*

The assumptions on classical space-time stated in Axiom 1.1, together with the above principle of conservation of linear momentum, provide the foundation for our development of Galilean mechanics.

Let us investigate what, additional information, the principle of conservation of linear momentum gives us about a system of interacting but isolated set of particles $P_1, P_2, \cdots, P_n$. That is, a set of particles that interact with themselves but do not interact with any other particles of the universe. This idealizes situations in which momentum exchange with the surroundings is negligible over the time interval considered. Let m_i be the mass of P_i and $\dot{x}_i$ be the velocity of P_i in some inertial frame $\mathbf{e}$. Then the conservation of linear momentum in $\mathbf{e}$ gives us that $\sum_{i=1}^{n} m_i \dot{x}_i =$ constant. The principle of conservation of linear momentum says that in a different inertial frame $\mathbf{e}'$ this quantity still remains constant as well. However in general the two constants will not be the same. That is, although two observers in two different inertial frames will agree that the total linear momentum of the system of particles is conserved, in general they will measure two different values for this constant. Thus though conservation of linear momentum is inertial frame invariant, the total linear momentum is not.

Differentiating the expression $\sum_{i=1}^{n} m_i \dot{x}_i =$ constant one sees that in the $\mathbf{e}$-frame

$$\sum_{i=1}^{n} m_i \ddot{x}_i = 0. \tag{1.3}$$

Observe that now all inertial observers will agree that this sum is equal to zero. The expression (1.3) implies that an isolated particle ($n = 1$) will move at constant velocity in any inertial frame (*Galileo's principle of inertia or what is commonly known as Newton's* 1st *law of motion*). The above sum can be re-arranged to give

$$m_j \ddot{x}_j = -\sum_{\substack{i=1 \\ i \neq j}}^{n} m_i \ddot{x}_i.$$

Thus the rate of change of momentum of particle j balances the combined rate of change of momentum of the remaining particles. We define the *net force* acting on particle j through this

momentum exchange:

$$f_j \triangleq -\sum_{\substack{i=1 \\ i\neq j}}^{n} m_i \ddot{x}_i.$$

Then

$$m_j \ddot{x}_j = f_j.$$

With this definition of net force, the momentum balance takes the familiar form of *Newton's second law of motion*. Specific interaction laws are needed to determine f_j and hence predict the motion. Under a change of inertial frame, force components transform as $f_j' = Rf_j$; they are identical when the axes are aligned.

For an isolated system of two interacting particles, (1.3) shows that their mutual forces are equal and opposite. To extend this description to many particles, we assume that each pair retains its two-particle interaction law when other particles are present, and that these contributions add. Here, retaining the law means retaining its dependence on the pair's instantaneous state, not retaining a fixed force value.

Axiom 1.3 *Pairwise interaction laws remain unchanged in the presence of other particles, and their force contributions add to give the net force on each particle.*

Writing f_{ji} for the force contribution on particle j due to particle i, this assumption and the isolated-pair momentum balance give

$$f_j \triangleq \sum_{\substack{i=1 \\ i\neq j}}^{n} f_{ji}, \qquad f_{ji} = -f_{ij}.$$

Thus the equal-and-opposite character of each isolated-pair interaction persists when the pair forms part of a larger system.

Equal-and-opposite forces need not act along the line joining the particles. This additional requirement, corresponding to the strong form of Newton's third law, is separately postulated:

Axiom 1.4 *Each pairwise force acts along the line joining the two particles.*

We will see later that this assumption is sufficient to ensure the experimentally observed fact of conservation of angular momentum for an isolated system of particles.

Thus Newton's first law follows from conservation of linear momentum for an isolated particle, while the second law takes its familiar form upon defining net force through momentum exchange. For an isolated pair of particles, momentum conservation implies equal and opposite mutual forces. The additional pairwise-interaction axiom extends this property to each interacting pair. Together with the central-force axiom, it gives the strong form of Newton's third law.

We observe changes in motion or deformation and attribute them to interactions between particles. The notion of *force* is introduced to model these interactions through their observable effects. Thus, what we observe is the effect attributed to an interaction; force is the conceptual quantity used to describe it. In our formulation, force quantifies the rate of momentum exchange, while specific interaction laws determine its magnitude and direction.

The equal-and-opposite central pairwise forces assumed here ensure conservation of linear and angular momentum for an isolated system. Although fundamental interactions need not admit this description, the model is adequate for the mechanical systems considered in these notes, where relativistic and quantum effects are negligible.

In summary, the fundamental assumptions of the Galilean particle model adopted here are:

i. there exists a nonempty class of inertial observers who agree on simultaneity, elapsed time, Euclidean distances at each instant, and whether a particle trajectory has constant velocity. Time is a one-dimensional Euclidean affine space, independent of space, and space at each instant is a three-dimensional Euclidean affine space;
ii. material objects are modelled as impenetrable particles, each with a positive, time-independent mass that is the same for all inertial observers using common units;
iii. the total linear momentum of any finite isolated system of particles is conserved in every inertial frame;
iv. pairwise interaction laws remain unchanged in the presence of other particles, and their force contributions add to give the net force on each particle;
v. each pairwise force acts along the line joining the two particles.

Defining net force as the rate of momentum exchange gives

$$m_i \ddot{x}_i = f_i, \tag{1.4}$$

in any inertial frame. Conservation of linear momentum for an isolated pair, together with persistence of its interaction law, implies equal-and-opposite pairwise forces. Specific interaction laws determine the forces and hence the motion.

1.1.4 Predicting Motion

Knowledge of the interactions allows us to formulate force laws, and equation (1.4) can then be used to predict the motion of a particle. This is what is monumental about this law. From a mathematical perspective, it describes a second-order differential equation, generally of the form

$$m\ddot{x} = F(t, x, \dot{x}).$$

Given a force law satisfying suitable existence and uniqueness conditions, the initial position $x(t_0) = x_0$ and initial velocity $\dot{x}(t_0) = v_0$ determine the motion on the interval where the solution exists.

Example 1.1. Consider a horizontal spring with one end fixed to a support and the other attached to an object of mass m moving on a smooth horizontal table. We approximate the object as a point particle, assume that the spring's mass is negligible, and restrict the motion to the spring's axis. Damping and other resistances are assumed negligible.

Let $x(t)$ be the signed displacement along the spring's axis from the position at which the spring is unstretched. For small deformations, we assume that the force exerted on the object through its interaction with the particles making up the spring can be modeled by $f = -kx$, for $k > 0$. This linear relation is known as *Hooke's law*, where the spring constant k depends on the material and geometry of the spring. The vertical forces due to gravity and the table balance and do not contribute to the motion along the spring's axis.

Newton's second law therefore gives $m\ddot{x}(t) = -kx(t)$, which can be re-arranged as

$$\ddot{x}(t) + \omega_n^2 x(t) = 0, \qquad \omega_n = \sqrt{\frac{k}{m}}.$$

This ordinary differential equation has the solution

$$x(t) = \chi \cos(\omega_n t + \phi),$$

where the amplitude χ and phase ϕ are determined by the initial displacement and velocity. The resulting simple harmonic response is shown in Figure 1.4 for $k = 10N/m$, $m = 1\,kg$, $x(0) = 1\,m$ and $\dot{x}(0) = 0$.

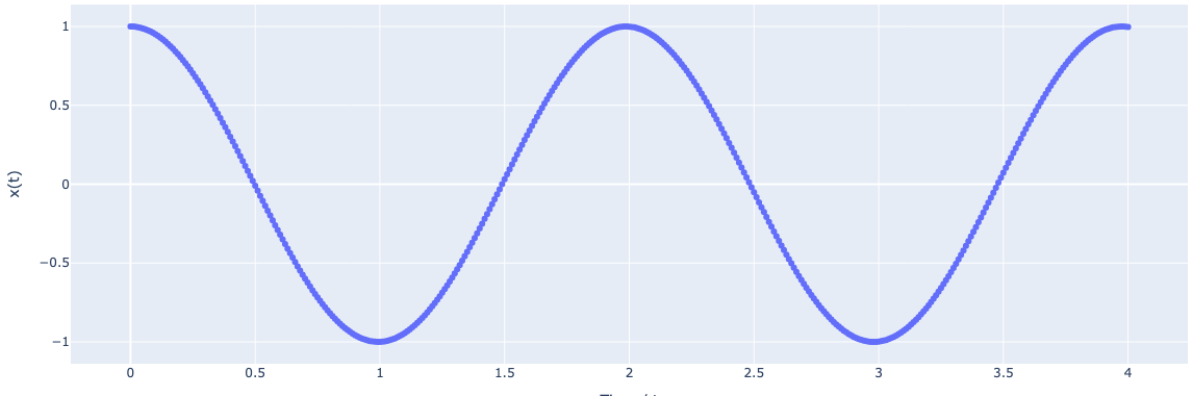


Fig. 1.4 Simple harmonic motion of a mass attached to a spring obeying Hooke's law, with angular frequency $\omega = \sqrt{k/m}$.

For another example of modeling the small amplitude vibrational motion of a multistory building you may refer to the Exercise 1.5. A few other similar examples can also be found in the following .ipynb notebook: sample_answer_to_modeling_a_simple_coupled_system.ipynb

1.1.5 Kinetic Energy

Another fundamental property of motion is the *energy* associated with it. Consider a particle of constant mass m moving under the influence of a resultant force f, arising from its interactions with the rest of the universe. The *kinetic energy* of the particle, measured with respect to an inertial frame $\mathbf{e}$, is defined by

$$\mathrm{KE}(t) \triangleq \frac{1}{2} m \|\dot{x}(t)\|^2 = \frac{1}{2m} \|p(t)\|^2,$$

where $x(t)$ is its position and $p(t) = m\dot{x}(t)$ is its linear momentum in this frame. Kinetic energy is nonnegative and vanishes precisely when the particle is at rest relative to the chosen frame.

Since velocity depends on the inertial frame, kinetic energy also depends on the frame and is not invariant under a change of inertial observer. Moreover, even for an isolated system of particles, the total kinetic energy need not remain constant: interactions between the particles can convert kinetic energy into other forms of energy.

Differentiating along the motion and applying Newton's second law gives

$$\frac{d}{dt}\mathrm{KE} = m\langle\langle \ddot{x}(t), \dot{x}(t)\rangle\rangle = \langle\langle f(t), \dot{x}(t)\rangle\rangle.$$

Thus only the component of force along the velocity changes the kinetic energy. A force component in the direction of motion increases it, while an opposing component decreases it. A force perpendicular to the velocity does not change the kinetic energy.

The quantity

$$P(t) \triangleq \langle\langle f(t), \dot{x}(t)\rangle\rangle$$

is called the *power* of the force, and its integral

$$W(t_1, t_2) \triangleq \int_{t_1}^{t_2} \langle\langle f(t), \dot{x}(t)\rangle\rangle \, dt$$

is the *work done by the force* during the interval $[t_1, t_2]$. Integrating the kinetic-energy identity therefore yields the *work–energy theorem*.

The rate of change of a particle's kinetic energy equals the power of the resultant force acting on it. Consequently, the change in kinetic energy equals the work done by that force:

$$\mathrm{KE}(t_2) - \mathrm{KE}(t_1) = W(t_1, t_2).$$

This identity follows from Newton's second law. When the work can be expressed as the decrease of a potential energy, it implies conservation of the sum of kinetic and potential energy.

Example 1.2 (The spring–mass system). Consider the horizontal spring–mass system discussed earlier, with displacement x measured from the unstretched position and spring force $f = -kx$. Its motion is

$$x(t) = x_0 \cos(\omega t + \phi), \qquad \omega = \sqrt{\frac{k}{m}},$$

where x_0 is the amplitude. Hence

$$\dot{x}(t) = -x_0 \omega \sin(\omega t + \phi),$$

and the kinetic energy is

$$\mathrm{KE}(t) = \frac{1}{2} m x_0^2 \omega^2 \sin^2(\omega t + \phi) = \frac{1}{2} k x_0^2 \sin^2(\omega t + \phi).$$

It vanishes at the turning points and is greatest when the mass passes through the unstretched position.

The work done by the spring force between times t_1 and t_2 is

$$\begin{aligned} W(t_1, t_2) &= \int_{t_1}^{t_2} -kx(t)\dot{x}(t)\, dt \\ &= \frac{1}{2} k x(t_1)^2 - \frac{1}{2} k x(t_2)^2. \end{aligned}$$

Thus the work depends only on the initial and final displacements. Defining the *elastic potential energy* of the spring by

$$\mathrm{PE}(x) \triangleq \frac{1}{2} k x^2,$$

we obtain

$$W(t_1, t_2) = \mathrm{PE}(x(t_1)) - \mathrm{PE}(x(t_2)).$$

The work done by the spring is therefore the *decrease* in its potential energy. Equivalently, the increase in potential energy is the work required to deform the spring quasistatically against its restoring force. We have chosen the potential energy to be zero at $x = 0$; adding a constant would not change any energy differences.

Along the motion,

$$\mathrm{PE}(t) = \frac{1}{2} k x_0^2 \cos^2(\omega t + \phi),$$

so the total mechanical energy satisfies

$$E = \mathrm{KE}(t) + \mathrm{PE}(t) = \frac{1}{2} k x_0^2.$$

Kinetic and potential energy are continuously exchanged: as the mass moves towards $x = 0$, the spring does positive work and potential energy becomes kinetic energy; as it moves away, the spring does negative work and kinetic energy becomes potential energy. Their sum remains constant under the assumed absence of damping and other resistances.

1.1.6 The Motion of a Set of Interacting Particles

In this section we investigate further consequences of the preceding axioms for a fixed collection of particles $P_1, P_2, \cdots, P_n$. Such a collection may model a deformable body, including a fluid, or a rigid body, in which the distance between every pair of constituent particles remains fixed.

Let x_i and $\dot{x}_i$ denote the position and velocity of particle P_i, of mass m_i, in an inertial frame $\mathbf{e}$ with origin O. Its linear momentum is

$$p_i \triangleq m_i \dot{x}_i.$$

Since m_i is constant, (1.4) gives

$$\dot{p}_i = m_i \ddot{x}_i = f_i,$$

where f_i is the net force due to interactions with particles both within and outside the collection. With our definition of net force through momentum exchange, this is the momentum form of Newton's second law.

For an isolated collection, conservation of linear momentum gives

$$p \triangleq \sum_{i=1}^{n} p_i = \sum_{i=1}^{n} m_i \dot{x}_i = \text{constant}.$$

In the presence of external interactions, this total need not remain constant. To determine its rate of change, write

$$f_i = f_i^e + \sum_{\substack{j=1 \\ j \neq i}}^{n} f_{ij},$$

where f_i^e is the resultant external force on particle i, and f_{ij} is the force on i due to particle j within the collection. By the pairwise-interaction axiom and isolated-pair momentum balance, $f_{ij} = -f_{ji}$. Hence

$$\sum_{i=1}^{n} \sum_{\substack{j=1 \\ j \neq i}}^{n} f_{ij} = \sum_{i<j} (f_{ij} + f_{ji}) = 0.$$

Defining the total external force by

$$f^e \triangleq \sum_{i=1}^{n} f_i^e,$$

we obtain

$$\dot{p} = f^e.$$

Let O_c be the *center of mass*, with position

$$\bar{x} \triangleq \frac{1}{M}\sum_{i=1}^{n} m_i x_i, \qquad M \triangleq \sum_{i=1}^{n} m_i.$$

Since the masses are constant, differentiation gives

$$M\dot{\bar{x}} = p, \qquad M\ddot{\bar{x}} = f^e.$$

The following video taken in outer space demonstrates this principle very nicely: Richard Garriott Space Video Blog: https://youtu.be/4IYDb6K5UF8.

This describes the center-of-mass motion, but does not determine the motion relative to the center of mass, such as the rotation of a rigid body[9]. To study the rotational aspect of motion, we introduce angular momentum.

Let O' be a point, fixed or moving relative to $\mathbf{e}$, with position $o(t)$, so that $OO' = \mathbf{e}\,o$.

The *angular momentum* of particle P_i in the $\mathbf{e}$ frame about O' is defined by

$$\pi_i \triangleq (x_i - o) \times m_i \dot{x}_i.$$

Here the velocity is measured in $\mathbf{e}$, even when O' is moving.

Differentiating, and using $\dot{x}_i \times m_i \dot{x}_i = 0$, gives

$$\begin{aligned}\dot{\pi}_i &= (\dot{x}_i - \dot{o}) \times m_i \dot{x}_i + (x_i - o) \times m_i \ddot{x}_i \\ &= -\dot{o} \times m_i \dot{x}_i + (x_i - o) \times f_i.\end{aligned}$$

The *moment of the force* f_i about O' is defined by

$$\tau_i \triangleq (x_i - o) \times f_i. \tag{1.5}$$

Consequently,

$$\dot{\pi}_i = -\dot{o} \times m_i \dot{x}_i + \tau_i. \tag{1.6}$$

For each interacting pair, the equal-and-opposite and central-force properties give

$$(x_i - o) \times f_{ij} + (x_j - o) \times f_{ji} = (x_i - x_j) \times f_{ij} = 0.$$

Thus the internal force moments cancel when summed over the collection:

$$\sum_{i=1}^{n} \tau_i = \sum_{i=1}^{n} (x_i - o) \times f_i^e \triangleq \tau^e,$$

where τ^e is the *total moment of the external forces about* O'.

[9] See for instance the amazing looking videos taken in outer space: https://youtu.be/bKk_7NIKY3Y, https://youtu.be/1n-HMSCDYtM.

The total angular momentum about O' is $\pi \triangleq \sum_{i=1}^{n} \pi_i$. Therefore,

$$\begin{aligned}\dot{\pi} &= -\dot{o} \times \sum_{i=1}^{n} m_i \dot{x}_i + \tau^e \\ &= -\dot{o} \times p + \tau^e = -M\dot{o} \times \dot{\bar{x}} + \tau^e.\end{aligned}$$

In particular, $\dot{\pi} = \tau^e$ whenever $\dot{o} \times \dot{\bar{x}} = 0$. This includes the cases $\dot{o} = 0$, $\dot{\bar{x}} = 0$, and $O' = O_c$.

The conditions $\dot{o} = 0$ and $\dot{\bar{x}} = 0$ need not hold for every inertial observer. However, choosing $O' = O_c$ gives $\dot{o} = \dot{\bar{x}}$ in every inertial frame, so the extra term vanishes for every such observer.

The rate of change of total angular momentum about the center of mass equals the total external force moment about the center of mass. Consequently, for an isolated collection of particles, this angular momentum is conserved in every inertial frame.

The central-force assumption, together with equal-and-opposite pairwise forces, was essential to the cancellation of internal force moments in this derivation. These assumptions are therefore sufficient to ensure conservation of angular momentum for an isolated collection.

In summary, consider a fixed collection of particles $P_1, P_2, \cdots, P_n$ interacting with one another and with particles outside the collection. Under our momentum-conservation, pairwise-additivity, and central-force assumptions, the following hold in any inertial frame:

(a) The rate of change of total linear momentum equals the resultant external force:

$$\dot{p} = f^e. \tag{1.7}$$

(b) The center of mass moves as a particle of mass $M = \sum_{i=1}^{n} m_i$ subject to the resultant external force $f^e = \sum_{i=1}^{n} f_i^e$:

$$M\ddot{\bar{x}} = f^e. \tag{1.8}$$

(c) The rate of change of total angular momentum about the center of mass equals the total moment of the external forces about that same point:

$$\dot{\pi} = \tau^e, \qquad \tau^e = \sum_{i=1}^{n} (x_i - \bar{x}) \times f_i^e. \tag{1.9}$$

Centrality of the pairwise forces is needed for the angular momentum balance; the first two results require only cancellation of equal-and-opposite internal forces.

These conclusions apply to any fixed collection of particles satisfying our assumptions, whether modelling a deformable body or a rigid body. For an isolated collection, $f^e = 0$ and $\tau^e = 0$. Consequently, its center of mass moves with constant velocity, and its total angular momentum about the center of mass is conserved in every inertial frame.

You may refer to Exercise-1.6 for a concrete example application of the ideas developed in this section.

1.1.7 Particle Collisions and Thermalization

Following the heuristic discussion of Feynman, Leighton, and Sands [10, Vol. I, Sec. 39–4], we explore how elastic collisions redistribute kinetic energy and motivate thermal equilibrium. For this section, we idealize collisions as instantaneous events between point particles, replacing the brief interaction by a jump in velocity that conserves total momentum and kinetic energy. The particles move freely between collisions, and their positions are treated as coincident at the collision instant. This is a limiting idealization of a short-range interaction, rather than a literal assertion that impenetrable particles overlap. Our discussion of repeated collisions uses statistical assumptions to develop physical intuition, rather than providing a rigorous proof of thermalization.

Let us first consider what happens when only two particles collide with each other. Let the two particles have a mass of m_1 and m_2 respectively. Let $\mathbf{e}$ be an inertial frame and $\mathbf{b}$ be a frame that moves parallel to $\mathbf{e}$ with origin coinciding with the center of mass of the two particles. We will call the frame $\mathbf{b}$ the center of mass frame. Let x_1, x_2 be the Euclidean representation of the two particles in $\mathbf{e}$ and let X_1, X_2 be the representation of the two particles in $\mathbf{b}$. Let o be the representation of the center of mass of the particles in the $\mathbf{e}$ frame. Then $x_i = o + X_i$. Since the origin of $\mathbf{b}$ is at the center of mass of the two particles we have

$$m_1 X_1 + m_2 X_2 = 0. \tag{1.10}$$

Differentiating this we have

$$m_1 \dot{X}_1 + m_2 \dot{X}_2 = 0. \tag{1.11}$$

Thus the total linear momentum of the particles in the $\mathbf{e}$ frame can be expressed as

$$m_1 \dot{x}_1 + m_2 \dot{x}_2 = (m_1 + m_2)\dot{o} \tag{1.12}$$

and the total kinetic energy of the particles can be expressed as

$$\text{KE} = \frac{m_1}{2}||\dot{x}_1||^2 + \frac{m_2}{2}||\dot{x}_2||^2, \tag{1.13}$$

$$= \frac{(m_1 + m_2)}{2}||\dot{o}||^2 + \frac{m_1}{2}||\dot{X}_1||^2 + \frac{m_2}{2}||\dot{X}_2||^2. \tag{1.14}$$

Let us consider what happens in a collision. Denote by a superscript $'$ the variables that describe the motion of the particle after collision. Principle of conservation of momentum implies that

$$(m_1 + m_2)\dot{o} = (m_1 + m_2)\dot{o}'.$$

This shows that the *center of mass velocity is unaltered by the collision*[10]. Let us assume that the *kinetic energy is not lost in the collision*[11]. Thus KE = KE′. This gives us

$$\frac{(m_1+m_2)}{2}||\dot{o}||^2+\frac{m_1}{2}||\dot{X}_1||^2+\frac{m_2}{2}||\dot{X}_2||^2=\frac{(m_1+m_2)}{2}||\dot{o}'||^2+\frac{m_1}{2}||\dot{X}_1'||^2+\frac{m_2}{2}||\dot{X}_2'||^2.$$

Since $\dot{o}=\dot{o}'$ we have

$$\frac{m_1}{2}||\dot{X}_1||^2+\frac{m_2}{2}||\dot{X}_2||^2=\frac{m_1}{2}||\dot{X}_1'||^2+\frac{m_2}{2}||\dot{X}_2'||^2.$$

From (1.11) we have

$$\dot{X}_2=-\frac{m_1}{m_2}\dot{X}_1,\quad \dot{X}_2'=-\frac{m_1}{m_2}\dot{X}_1',$$

and hence

$$||\dot{X}_1||^2=||\dot{X}_1'||^2,\quad ||\dot{X}_2||^2=||\dot{X}_2'||^2.$$

Since the particles interact only at collision, their incoming and outgoing trajectories are straight lines. At collision, the two point particles coincide with their center of mass. Thus, in the center of mass frame, the particles move along a common straight line through the origin before the collision and along a possibly different such line after the collision. In an elastic collision, each particle's speed in this frame is unchanged; only its direction of motion can change. The cosine of the angle between this line and the velocity of the center of mass can be found from $\dot{o}\cdot(\dot{X}_2-\dot{X}_1)$

$$\begin{aligned}\dot{o}\cdot(\dot{X}_2-\dot{X}_1)=\dot{o}\cdot(\dot{x}_2-\dot{x}_1)&=\left(\frac{m_1\dot{x}_1+m_2\dot{x}_2}{(m_1+m_2)}\right)\cdot(\dot{x}_2-\dot{x}_1)\\&=\frac{m_2||\dot{x}_2||^2-m_1||\dot{x}_1||^2+(m_1-m_2)\dot{x}_1\cdot\dot{x}_2}{m_1+m_2}.\end{aligned}$$

Let us now consider two particles in a gas confined within a stationary container, where particles have been colliding elastically with one another and with the walls for a long time. These collisions repeatedly change their directions of motion.

To develop an intuitive picture, suppose that repeated collisions sufficiently randomize the relative motion that it has no persistent tendency to align with or oppose the center of mass velocity. Positive and negative contributions to their scalar product would then cancel on average, suggesting

$$\langle\dot{o}\cdot(\dot{x}_2-\dot{x}_1)\rangle=0.$$

Here angle brackets denote long-time averages in the rest frame of the container. This is a statistical assumption about the relative directions of these two velocities, rather than merely the absence of a preferred direction in space.

[10] Note that as shown by (1.8) this conclusion holds in general as well.
[11] Such collisions are called elastic collisions.

Similarly, collisions with other particles and the walls may be expected to erase correlations between the velocities of the two chosen particles. If we regard these velocities as approximately statistically independent, and assume no persistent mean motion in the container, then

$$\langle \dot{x}_1 \cdot \dot{x}_2 \rangle \approx \langle \dot{x}_1 \rangle \cdot \langle \dot{x}_2 \rangle = 0.$$

We adopt both zero-average relations as idealizations. They provide a heuristic description of the randomization produced by many collisions, rather than conclusions established by the two-particle collision laws alone. Under these assumptions, the above identity gives

$$\left\langle \frac{m_1 ||\dot{x}_1||^2}{2} \right\rangle = \left\langle \frac{m_2 ||\dot{x}_2||^2}{2} \right\rangle \tag{1.15}$$

This suggests that, after sufficiently many collisions, particles of different masses may have the same average translational kinetic energy. Extending this heuristic argument to pairs of particles in a gas motivates the *equipartition of translational kinetic energy* at thermal equilibrium. This does not mean that individual particles have equal kinetic energies at every instant: their energies fluctuate as collisions exchange energy between them.

For a classical monatomic ideal gas in thermal equilibrium, this common mean kinetic energy is related to the *absolute temperature* T by

$$\left\langle \frac{1}{2} m_i \|\dot{x}_i\|^2 \right\rangle = \frac{3}{2} k_{\mathrm{B}} T,$$

where velocities are measured in the rest frame of the gas and k_{B} is Boltzmann's constant. This relation provides a kinetic interpretation of temperature: temperature measures the average energy of random translational motion, rather than the average particle speed. Thus, at the same temperature, lighter particles have greater mean-square speeds than heavier particles.

(a) In the absence of a resultant external force, the center of mass velocity of a system of particles remains constant. Collisions with container walls can change the momentum of the particles.
(b) Elastic particle collisions conserve total kinetic energy while redistributing it among the particles. Perfectly elastic collisions with stationary walls also preserve kinetic energy. This conservation does not, by itself, imply thermal equilibrium.
(c) Under the statistical assumptions described above, repeated collisions motivate equal mean translational kinetic energies at equilibrium. For a classical monatomic ideal gas, this mean energy is $\frac{3}{2} k_{\mathrm{B}} T$ per particle.

1.2 Description of Motion in Moving Frames

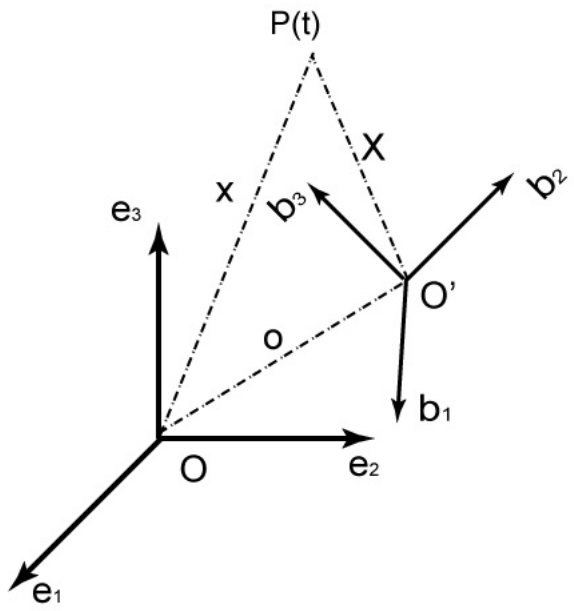


Fig. 1.5 The description of motion of a particle $P(t)$ in two different orthonormal frames $\mathbf{e}$ and $\mathbf{b}$ representing two different observers.

We can easily see that the construction of orthonormal frames is not unique and that, in general, different observers can have different orthonormal frames. For instance consider figure-1.6 where two observers have defined two right hand oriented orthonormal frames $\mathbf{e} = [\mathbf{e}_1 \ \ \mathbf{e}_2 \ \ \mathbf{e}_3]$ and $\mathbf{b} = [\mathbf{b}_1 \ \ \mathbf{b}_2 \ \ \mathbf{b}_3]$ respectively. The position of the particle $P(t)$ at a particular instant of time t is described by the two observers using the Euclidean representation matrices $x = [x_1 \ x_2 \ x_3]^T$ and $X = [X_1 \ X_2 \ X_3]^T$ respectively. Let o be the Euclidean representation of the point O' in the orthonormal frame $\mathbf{e}$. That is let $OP = \mathbf{e}x$, $OO' = \mathbf{e}o$ and $O'P = \mathbf{b}X$. We are interested in determining the relationship between the observed motion in the two different frames [12].

Let us begin by finding the relationship between the two representations x and X. Introduce another frame $\mathbf{e}' = [\mathbf{e}'_1 \ \ \mathbf{e}'_2 \ \ \mathbf{e}'_3]$, as shown in figure-1.6, with origin coinciding with O' such that its axis are of unit length and are also parallel and aligned with those of $\mathbf{e}$. Let the representation of the point P in the $\mathbf{e}'$ frame be x'. That is let $O'P = \mathbf{e}'x'$. From the Euclidean assumption of space it follows that

$$OP = OO' + O'P = \mathbf{e}o + \mathbf{e}'x' = \mathbf{e}x$$

[12] Unlike the preceding discussion of inertial observers, we now allow the frames to translate and rotate relative to one another.

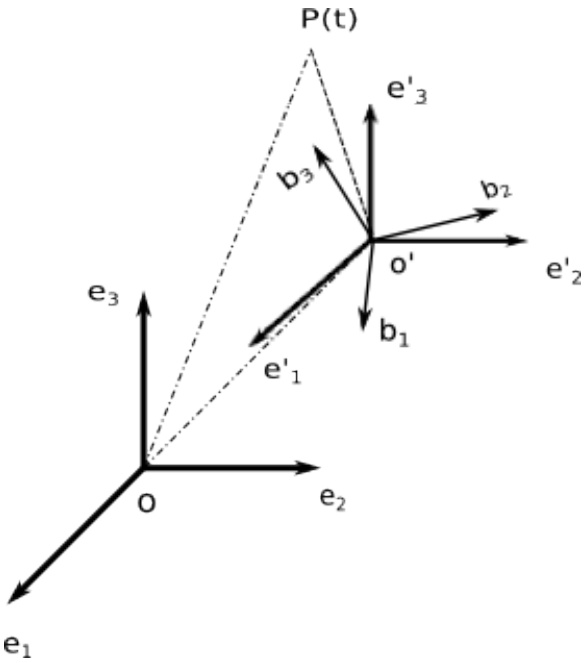


Fig. 1.6 Description of the point $P(t)$ in three orthonormal frames $\mathbf{e}$, $\mathbf{e}'$, and $\mathbf{b}$ representing two different observers.

Since $\mathbf{e}$ and $\mathbf{e}'$ are parallel to each other we have that the two representations x and x' are related by

$$x = o + x'. \tag{1.16}$$

However, what we are really interested in is the relationship between x and X.

Observe that one can represent each of the axis $\mathbf{b}_i$ using the $\mathbf{e}'$ frame as follows.

$$\begin{aligned}
\mathbf{b}_1 &= r_{11}\mathbf{e}'_1 + r_{21}\mathbf{e}'_2 + r_{31}\mathbf{e}'_3,\\
\mathbf{b}_2 &= r_{12}\mathbf{e}'_1 + r_{22}\mathbf{e}'_2 + r_{32}\mathbf{e}'_3,\\
\mathbf{b}_3 &= r_{13}\mathbf{e}'_1 + r_{23}\mathbf{e}'_2 + r_{33}\mathbf{e}'_3.
\end{aligned}$$

This can be expressed in the matrix form

$$\underbrace{[\mathbf{b}_1 \ \mathbf{b}_2 \ \mathbf{b}_3]}_{\mathbf{b}} = \underbrace{[\mathbf{e}'_1 \ \mathbf{e}'_2 \ \mathbf{e}'_3]}_{\mathbf{e}'} \underbrace{\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}}_{R}.$$

It can be shown that the assumption of Euclidean 3D-space implies that $\mathbf{b} = \mathbf{e}'R$ where R is a 3×3 special orthogonal matrix[13]. Since $O'P = \mathbf{e}'x' = \mathbf{b}X = \mathbf{e}'RX$ we have that

[13] In exercise-1.7 you are asked to prove this. The space of all 3×3 special orthogonal matrices is denoted by $\mathrm{SO}(3)$. Recall that a special orthogonal matrix R satisfies $RR^T = R^TR = I_{3\times 3}$ and $\det(R) = 1$.

$$x' = RX. \tag{1.17}$$

The columns of R are the components of the axes of $\mathbf{b}$ in the $\mathbf{e}'$ frame. Thus R converts components from $\mathbf{b}$ to $\mathbf{e}'$, while R^T performs the inverse conversion.

From the expression (1.17) we have that the two representations of the point P in the two frames $\mathbf{e}$ and $\mathbf{b}$ as depicted in figure-1.5 are related by

$$x = o + RX. \tag{1.18}$$

The above expression can also be expressed in matrix form as:

$$\begin{bmatrix} x \\ 1 \end{bmatrix} = \begin{bmatrix} R & o \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X \\ 1 \end{bmatrix}, \tag{1.19}$$

where $\mathbf{0} = 0_{1\times 3}$. This expression will be very useful when one deals with kinematic chains such as robot arms.

Notice that the preceding discussion shows that the relationship between two right hand oriented orthonormal frames $\mathbf{e}$ and $\mathbf{b}$ is uniquely given by a $(o,R) \in \mathbb{R}^3 \times SO(3)$ and that conversely any $(o,R) \in \mathbb{R}^3 \times SO(3)$ defines the relationship between two right hand oriented orthonormal frames $\mathbf{e}$ and $\mathbf{b}$ in a unique fashion.

1.2.1 Kinematics in Moving Frames

When frames translate or rotate relative to one another, the velocities and accelerations measured relative to them generally differ. These differences arise both from the motion of the frames and from the different axes used to express components. We therefore distinguish $\dot{x}$, the velocity relative to $\mathbf{e}$ expressed in $\mathbf{e}$, from $\dot{X}$, the velocity relative to $\mathbf{b}$ expressed in $\mathbf{b}$.

Thus it is important to always specify the orthonormal frame with which they are expressed. Consider the problem of describing the motion of a point $P(t)$ that is moving with respect to both frames $\mathbf{e}$ and $\mathbf{b}$ and let $\mathbf{b}$ be translating and rotating with respect to $\mathbf{e}$. Thus we have that all matrices $x(t), X(t)$ and $o(t), R(t)$ are changing with respect to time. Differentiating the expression (1.18) relating the two position representations we find that the two velocities $\dot{x}$ and $\dot{X}$ measured in the two frames are related by

$$\dot{x} = \dot{o} + \dot{R}X + R\dot{X}.$$

Similarly the two accelerations measured in the two frames $\ddot{x}$ and $\ddot{X}$ are related by

$$\ddot{x} = \ddot{o} + \ddot{R}X + 2\dot{R}\dot{X} + R\ddot{X}.$$

In the following we proceed to find if $\dot{R}$ and $\ddot{R}$ can expressed a little more conveniently. In exercise-1.7 we have seen that $R^T(t)R(t) = I$. Thus it follows that $\dot{R}^T R + R^T \dot{R} = 0$, and hence that

$$R^T \dot{R} = -(R^T \dot{R})^T = \widehat{\Omega},$$

where $\widehat{\Omega}$ is a skew symmetric matrix. The space of all 3×3 skew-symmetric matrices is denoted by $\mathrm{so}(3)$. In Exercise 1.9 you are asked to show that $\mathrm{so}(3)$ is a 3-dimensional vector space.

Thus we have that if $R(t) \in \mathrm{SO}(3)$ then

$$\dot{R} = R\widehat{\Omega}, \tag{1.20}$$

where $\widehat{\Omega}(t) \in \mathrm{so}(3)$.

Differentiating it once we have $\ddot{R} = R(\widehat{\Omega}^2 + \dot{\widehat{\Omega}})$. Substituting these expressions for $\dot{R}$ and $\ddot{R}$ in the above expressions relating velocities and accelerations in the two frames **e** and **b** we have,

$$\dot{x} = \dot{o} + R\left(\widehat{\Omega}X + \dot{X}\right), \tag{1.21}$$

$$\ddot{x} = \ddot{o} + R\left(\widehat{\Omega}^2(t)X + 2\widehat{\Omega}\dot{X} + \dot{\widehat{\Omega}}X + \ddot{X}\right). \tag{1.22}$$

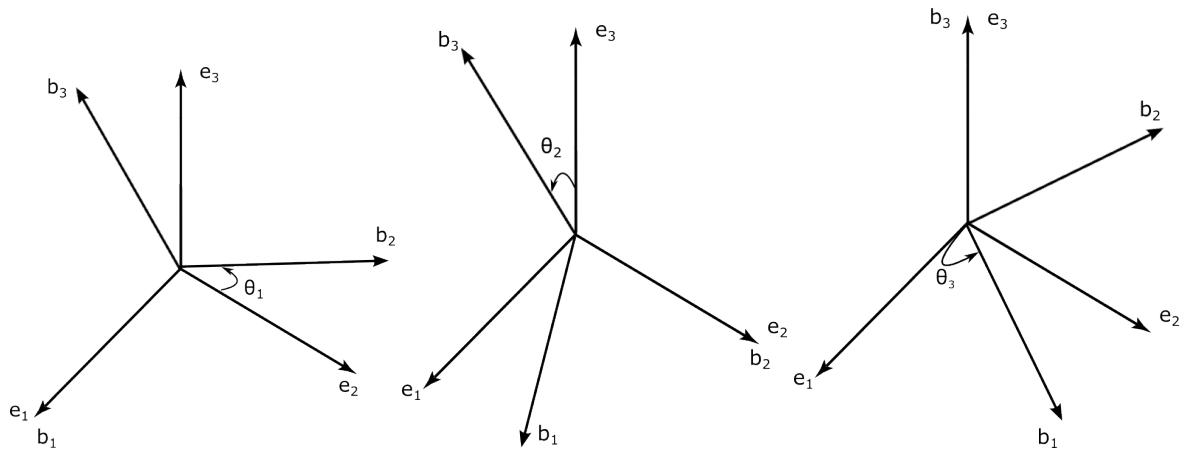


Fig. 1.7 Rotated Frames

Thus we see that in order to find these relationships one needs to compute the matrices $\widehat{\Omega}, \widehat{\Omega}^2$ and $\dot{\widehat{\Omega}}$. As an illustration let us consider, easy to visualize, three special frame rotations. We will also see that these three special type of rotating frames will become useful

when representing the motion of complicated systems as well. Consider the three rotating orthonormal frames $\mathbf{a},\mathbf{b},\mathbf{c}$ that are related to a fixed frame $\mathbf{e}$ as shown in figure 1.7. Each of the frames $\mathbf{a},\mathbf{b},\mathbf{c}$ correspond to a simple a rotation through an angle $\theta_i(t)$ about the positive $\mathbf{e}_i$ axis, with positive angles defined by the right-hand rule. Let $\mathbf{a} = \mathbf{e}R_1(\theta_1)$, $\mathbf{b} = \mathbf{e}R_2(\theta_2)$, and $\mathbf{c} = \mathbf{e}R_3(\theta_3)$. In exercise-1.8 you are asked to show using direct calculations that the following expressions hold.

$$R_1(\theta_1) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_1 & -\sin\theta_1 \\ 0 & \sin\theta_1 & \cos\theta_1 \end{bmatrix}, \quad R_2(\theta_2) = \begin{bmatrix} \cos\theta_2 & 0 & \sin\theta_2 \\ 0 & 1 & 0 \\ -\sin\theta_2 & 0 & \cos\theta_2 \end{bmatrix}, \quad R_3(\theta_3) = \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 & 0 \\ \sin\theta_3 & \cos\theta_3 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \tag{1.23}$$

and

$$R_1^T\dot{R}_1 = \widehat{\Omega}_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\dot{\theta}_1 \\ 0 & \dot{\theta}_1 & 0 \end{bmatrix}, \quad R_2^T\dot{R}_2 = \widehat{\Omega}_2 = \begin{bmatrix} 0 & 0 & \dot{\theta}_2 \\ 0 & 0 & 0 \\ -\dot{\theta}_2 & 0 & 0 \end{bmatrix}, \quad R_3^T\dot{R}_3 = \widehat{\Omega}_3 = \begin{bmatrix} 0 & -\dot{\theta}_3 & 0 \\ \dot{\theta}_3 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

and

$$\widehat{\Omega}_1^2 = -\dot{\theta}_1^2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \widehat{\Omega}_2^2 = -\dot{\theta}_2^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \widehat{\Omega}_3^2 = -\dot{\theta}_3^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Having seen how to calculate $\widehat{\Omega}$ and $\widehat{\Omega}^2$ and noticing that they have a pattern we may ask what general properties the 3×3 skew-symmetric matrices have. In Section-1.2.2 we will investigate in detail several properties of 3×3 special orthogonal matrices and 3×3 skew-symmetric matrices in order to facilitate these computations on one hand and on the other hand to get a deeper understanding of the physical meaning of $R \in \mathrm{SO}(3)$ and $\widehat{\Omega} \in \mathrm{so}(3)$.

1.2.2 Infinitesimal Rotations and Angular Velocity

In this section we will take a closer look at the physical meaning of the skew symmetric matrix $\widehat{\Omega} = R^T\dot{R}$. To do so we will have to first obtain a better understanding of special orthogonal matrices. A given special orthogonal matrix $R \in \mathrm{SO}(3)$ can be viewed in at least three different ways.

We first explain how R describes the relative orientation of two frames, a change of coordinates, and a rigid rotation of space. We then use the properties of skew-symmetric matrices to interpret $\widehat{\Omega} = R^T\dot{R}$ in terms of angular velocity, and distinguish the representations of this angular velocity in the reference and rotating frames.

1.2.2.1 Coordinate Transformations and Rigid Rotations

We have seen before that the relationship between two right hand oriented orthonormal frames with coinciding origin is uniquely determined by a special orthogonal matrix and that con-

versely every special orthogonal matrix uniquely defines a relationship between two such frames.

First to see how it represents a coordinate transformation consider the expression (1.17) a bit more closely. What this says is that R can be thought of as a coordinate transformation that relates the $\mathbf{e}'$-frame coordinates of the point P, given by the matrix x' to the $\mathbf{b}$-frame coordinates of the point P, given by the matrix X. This idea can be extended to any intrinsic property[14] of the particle such as velocity, momentum, or force, that can be considered as a *arrow in space with the foot coinciding with* O'[15] in the following manner.

Let $\mathbf{e}$ and $\mathbf{b}$ be two right-handed orthonormal frames with coinciding origins, related by $\mathbf{b} = \mathbf{e}R$, where $R \in SO(3)$. Consider a physical quantity that, once its reference observer is specified where necessary, can be represented geometrically by a directed line segment. The same directed line segment has representations γ and Γ in the frames $\mathbf{e}$ and $\mathbf{b}$, respectively, related by

$$\gamma = R\Gamma. \tag{1.24}$$

Here we change only the representation of the quantity. For example, the velocity relative to a specified observer may be expressed in either frame; this does not mean that the velocities measured relative to the two observers are related by this formula.

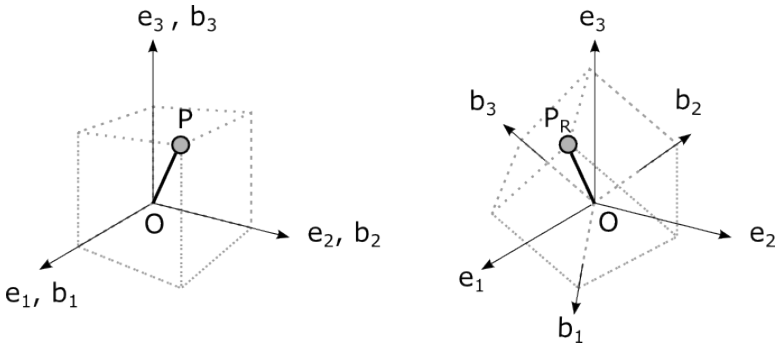


Fig. 1.8 The $\mathbf{b}$-frame is fixed on the box and the $\mathbf{e}$-frame is some 'fixed' reference frame.

On the other hand a given $R \in \mathrm{SO}(3)$ can be viewed as a *map that acts* on a point P in space to give a new point P_R in the following manner. Let $\mathbf{e}$ be some 'fixed' frame and let x be the representation of P in the $\mathbf{e}$ frame. Let P_R be the point in space that has the representation Rx in the $\mathbf{e}$-frame. That is let $OP = \mathbf{e}x$ and $OP_R = \mathbf{e}(Rx)$. This allows one to consider $R \in \mathrm{SO}(3)$ as a transformation that takes P to a new point P_R in space by mapping $x \to Rx$ and

[14] A property that does not depend on the choice of coordinates used to represent it is referred to as an intrinsic property.

[15] Also referred to as a directed line segment. This is what you would have traditionally learnt as vectors at the secondary school level.

identifying P_R with the point in space that has the representation Rx in the **e**-frame. Let Q be another point in space that has the representation y with respect to the **e**-frame. Then since $R^TR = RR^T = I_{3\times 3}$ we see that $||Rx-Ry|| = ||x-y||$ and $\langle\langle Rx, Ry\rangle\rangle = \langle\langle x, y\rangle\rangle$ and hence that this map preserves lengths and angles in space. Such maps that transform points in space to other points in space in such a way that it preserves distances between points and angles between lines are called *isometries*. Let us apply this map to all points in space and see how they transform by considering an illustration. Consider the set of points defined by a cube in space as shown in the left hand side of the figure-1.8. The cube is chosen such that the point P coincides with the vertex of this cube that is diagonally opposite the vertex at the centre of the frame O as shown in the left hand side of figure-1.8. Since R preserves lengths and angles and satisfies $\det R = 1$, it also preserves handedness. Thus when the points defining the cube are transformed by R to the new points, using the map $x \to Rx$, the new transformed points will also correspond to a cube that is identical to the initial cube with the exception of it now being 'rotated' about the vertex O. This situation is shown in the right hand side of figure-1.8. Let **b** be a frame such that it is fixed with respect to the cube such that initially both **b** and **e** coincide. It is now easy to see that the new orientation of the frame **b** fixed to the cube is related to the frame **e** by the relationship $\mathbf{b} = \mathbf{e}R$.

Thus we see that a given $R \in \mathrm{SO}(3)$ can be uniquely identified with a 'rigid rotation' of space and conversely that every rigid rotation of space about a fixed point can be identified with an $R \in \mathrm{SO}(3)$. This also shows that the configuration of a *rigid body* moving such that one of its points remains fixed in space can be uniquely identified with an $R \in \mathrm{SO}(3)$.

In the coordinate interpretation, the geometric point remains unchanged and its components change between frames. In the rotation interpretation, the reference frame remains fixed and the geometric point itself is moved.

1.2.2.2 Angular Velocity

Let us now revert our attention to the 3×3 skew symmetric matrix $\widehat{\Omega} = R^T\dot{R}$. We will see that it can be interpreted as the angular velocity of the frame **b** about **e**. To do so we will first need to be familiar with several properties of the space of 3×3 skew symmetric matrices, $\mathrm{so}(3)$.

It is straightforward to see that $\mathrm{so}(3)$ is a three dimensional real vector space under matrix addition and scalar multiplication[16]. Thus it is isomorphic[17] to $\mathbb{R}^3$. That is, there is a one-to-one and onto correspondence between elements of $\mathbb{R}^3$ and elements of $\mathrm{so}(3)$. The isomorphism $\widehat{\ }: \mathbb{R}^3 \to \mathrm{so}(3)$ that is explicitly defined by,

[16] See exercise 1.9.

[17] A vector-space isomorphism is a bijective linear map; it preserves vector addition and scalar multiplication.

$$\widehat{\Omega} = \begin{bmatrix} 0 & -\Omega_3 & \Omega_2 \\ \Omega_3 & 0 & -\Omega_1 \\ -\Omega_2 & \Omega_1 & 0 \end{bmatrix}, \tag{1.25}$$

for $\Omega = (\Omega_1, \Omega_2, \Omega_3) \in \mathbb{R}^3$ gives one such identification. It is easy to verify that $\widehat{\ }: \mathbb{R}^3 \to so(3)$ is linear. That is $\widehat{X+Y} = \widehat{X} + \widehat{Y}$ and $\widehat{\alpha X} = \alpha \widehat{X}$ for any $X, Y \in \mathbb{R}^3$ and $\alpha \in \mathbb{R}$. It is also easy to directly verify that this isomorphism satisfies

$$\widehat{\Omega} X = \Omega \times X, \tag{1.26}$$

$$\langle\langle X, Y \rangle\rangle = -\frac{1}{2} \text{trace}(\widehat{X}\widehat{Y}), \tag{1.27}$$

for $X, Y, \Omega \in \mathbb{R}^3$.

In exercises 1.9 – 1.13 you are asked to show the following very useful and interesting properties of 3×3 skew-symmetric matrices:

$$\widehat{RX} = R\widehat{X}R^T, \tag{1.28}$$

$$\widehat{X}^2 = XX^T - ||X||^2 I_{3\times 3}. \tag{1.29}$$

for any $X \in \mathbb{R}^3$ and $R \in \text{SO}(3)$.

For the following discussion, let $\mathbf{e}$ be a fixed reference frame with fixed origin O. Let us now consider smooth rotations that are parameterised by a parameter t that we may consider to be time. Let $R(t)$ be a smooth curve in the space $\text{SO}(3)$ such that $R(0) = I_{3\times 3}$. The initial alignment $R(0) = I$ simplifies the illustration; the angular-velocity relations derived below hold for any initial orientation. Then from the above discussion we see that $R(t)$ represents a smooth rigid rotation of space for all t. Let $\mathbf{b}(t) = \mathbf{e}R(t)$ be the corresponding rotating frame the origin coinciding with the origin O of $\mathbf{e}$. Since $R(0) = I_{3\times 3}$ we see that $\mathbf{b}(0) = \mathbf{e}$. Let $P(t)$ be a point in space that corresponds to $P(0)$ being 'rotated' by $R(t)$. That is if X is the representation of the point $P(0)$ in the $\mathbf{e} = \mathbf{b}(0)$ frame then $R(t)X$ is the representation of the point $P(t)$ in the $\mathbf{e}$-frame. Thus since rotations by R preserve angles and lengths in space, $P(t)$ will appear to be fixed as viewed in the $\mathbf{b}(t)$ frame and will be equal to X. That is, the representation X of the point $P(t)$ in the $\mathbf{b}(t)$ frame will not depend on t. Let $x(t)$ be the representation of the point $P(t)$ in the $\mathbf{e}$-frame. Then $x(t) = R(t)X$. The velocity of the point $P(t)$ in the $\mathbf{e}$-frame is thus given by $\dot{x}(t) = \dot{R}(t)X$. Previously we have seen that $R^T(t)\dot{R}(t) = \widehat{\Omega}(t)$ is always a skew-symmetric matrix. Thus we have that the velocity of the point $P(t)$ as expressed in the $\mathbf{e}$-frame has the representation $\dot{x}(t) = R(t)\widehat{\Omega}(t)X$. Therefore from (1.26) we see that the velocity of the point $P(t)$ can be expressed in the $\mathbf{e}$-frame as

$$\dot{x} = \dot{R}X = R\widehat{\Omega}X = R(\Omega \times X) = (R\Omega) \times (RX) = (R\Omega) \times x = \omega \times x, \tag{1.30}$$

where we have used the property $R(X \times Y) = RX \times RY$ and have set $\omega \triangleq R\Omega$ in the last equality. Notice that ω is the $\mathbf{e}$-frame representation of the quantity that has the representation Ω in the $\mathbf{b}$-frame. Also notice that $||\omega|| = ||R\Omega|| = ||\Omega||$.

At any instant for which $\omega(t) \neq 0$, points whose current position is parallel to $\omega(t)$ have zero instantaneous velocity (since $\omega(t) \times \omega(t) = 0$). Thus the line through O in the direction $\omega(t)$ is the instantaneous axis of rotation. This axis may change with time.

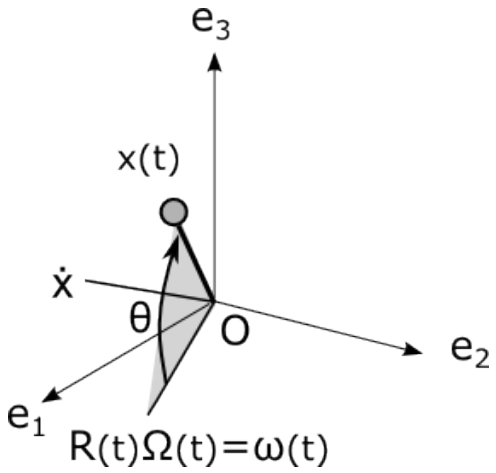


Fig. 1.9 The meaning of angular velocity.

On the other hand by the definition of the cross product in $\mathbb{R}^3$ and the last equality of the expression (1.30) we have, for $\omega \neq 0$,

$$\dot{x} = ||\omega||\,||x||\,\sin\theta\,\mathbf{n},$$

where $\mathbf{n} = (\omega \times x)/\|\omega \times x\|$ is the unit direction perpendicular to both ω and x θ is the angle between ω and x in the **e**-frame as shown in figure-1.9. The perpendicular distance from the point to the axis is $r = \|x\|\sin\theta$, so its speed is $\|\dot{x}\| = r\|\omega\|$. Thus $\|\omega\|$ is the instantaneous angular speed about this axis. Points on the axis have zero instantaneous velocity. If $\omega = 0$, all points have zero instantaneous velocity and there is no unique instantaneous axis. Thus we see that $P(t)$ is instantaneously rotating about $\omega(t)$ as viewed in the **e**-frame. Since X was arbitrary we see that this is true for every point in space. Which shows that under the 'rotation' by $R(t)$ every point in space is instantaneously rotating about ω with an angular rate of rotation equal to $||\omega|| = ||\Omega||$ as viewed in the frame **e**. Thus we have the following result.

The relation $\dot{x} = \omega \times x$ shows that ω gives the instantaneous axis and angular rate of rotation of **b** relative to **e**, expressed in **e**. We therefore call ω the spatial representation of this angular velocity. The same angular velocity has the representation

$$\Omega = R^T \omega$$

in **b**, called its body representation. Thus ω and Ω describe the same rotation of **b** relative to **e**, expressed using different frames, and $\|\omega\| = \|\Omega\|$ is its angular speed.

1.2.3 Angular Momentum in Moving Frames

Let **e** be an inertial frame and let **b** translate and rotate relative to it. We consider the angular momentum of particle P_i about the moving origin O', defined using its velocity relative to **e**. Its **e**-frame representation is

$$\begin{aligned}\pi_i &= m_i(x_i - o) \times \dot{x}_i = m_i R\left(X_i \times (\Omega \times X_i + \dot{X}_i + R^T \dot{o})\right), \\ &= R\left(-m_i \widehat{X}_i^2 \Omega + m_i X_i \times (R^T \dot{o} + \dot{X}_i)\right),\end{aligned}$$

where the last equality follows from

$$X_i \times (\Omega \times X_i) = -X_i \times (X_i \times \Omega) = -\widehat{X}_i^2 \Omega.$$

The quantity

$$\mathbb{I}_i \triangleq -m_i \widehat{X}_i^2 = m_i \left(||X_i||^2 I_{3\times 3} - X_i X_i^T\right), \tag{1.31}$$

is defined as the *moment of inertia* of the particle P about the point O' and expressed in the frame $\mathbf{b}(t)$. Using this we can now express the angular momentum of P about O' as

$$\pi_i = R\left(\mathbb{I}_i \Omega + m_i X_i \times (R^T \dot{o} + \dot{X}_i)\right). \tag{1.32}$$

The above expression shows that

$$\Pi_i \triangleq \left(\mathbb{I}_i \Omega + m_i X_i \times (\dot{X}_i + R^T \dot{o})\right), \tag{1.33}$$

is the moving **b**-frame representation of the angular momentum of P_i about O'.

Thus $\Pi_i = R^T \pi_i$ represents the same angular momentum using the axes of **b**; it is important to note that it is not defined using the particle's velocity relative to **b**.

1.2.4 Kinetic Energy in Moving Frames

Recall that the kinetic energy KE_i of a particle of mass m_i is defined with respect to an *inertial frame* **e** and is given by the relationship

$$\mathrm{KE}_i \triangleq \frac{1}{2} m_i ||\dot{x}_i(t)||^2,$$

where $||\cdot||$ is the Euclidian norm in $\mathbb{R}^3$.

Since $||RX|| = ||X||$ we also have that the kinetic energy of the particle can also be expressed as

$$\mathrm{KE}_i = \frac{1}{2} m_i ||\dot{x}_i||^2 = \frac{1}{2} m_i ||R^T \dot{x}_i||^2 = \frac{1}{2} m_i ||R^T \dot{o} + \widehat{\Omega} X_i + \dot{X}_i||^2,$$

$$= \frac{1}{2} m_i \left(||\dot{o}||^2 + 2\dot{o}^T R(\widehat{\Omega} X_i + \dot{X}_i) + ||\widehat{\Omega} X_i||^2 + 2\dot{X}_i^T \widehat{\Omega} X_i + ||\dot{X}_i||^2 \right).$$

Note that $||\widehat{\Omega} X_i||^2 = ||\widehat{X}_i \Omega||^2 = -\Omega^T \widehat{X}_i^2 \Omega$. Using the substitution $\mathbb{I}_i = -m_i \widehat{X}_i^2$ the kinetic energy of the particle can be expressed as

$$\mathrm{KE}_i = \frac{1}{2} \left(m_i ||\dot{o}||^2 + 2m_i \dot{o}^T R(\widehat{\Omega} X_i + \dot{X}_i) + \Omega^T \mathbb{I}_i \Omega + 2m_i \dot{X}_i^T \widehat{\Omega} X_i + m_i ||\dot{X}_i||^2 \right).$$

If the particle P is fixed with respect to the moving frame **b** then $X_i =$ constant and hence

$$\mathrm{KE}_i = \frac{1}{2} \left(m_i ||\dot{o}||^2 + 2m_i \dot{o}^T R(\Omega \times X_i) + \Omega^T \mathbb{I}_i \Omega \right). \tag{1.34}$$

The first and last terms describe the contributions from translation of the origin and rotation about it. The middle term arises because these two velocity contributions need not be perpendicular. Thus the kinetic energy is generally not simply the sum of separate translational and rotational energies.

The total kinetic energy of a collection of N particles is the sum of their individual kinetic energies. Define

$$M = \sum_i m_i, \qquad \bar{X} = \frac{1}{M} \sum_i m_i X_i, \qquad \mathbb{I} = \sum_i \mathbb{I}_i.$$

Summing the expressions obtained above gives the following.

The total kinetic energy with respect to the inertial frame **e**, expressed using moving-frame quantities, is

$$\mathrm{KE} = \frac{1}{2} M \|\dot{o}\|^2 + M\dot{o}^T R\left(\widehat{\Omega}\bar{X} + \dot{\bar{X}}\right) + \frac{1}{2}\Omega^T \mathbb{I}\Omega \quad + \sum_i m_i \dot{X}_i^T \widehat{\Omega} X_i + \frac{1}{2} \sum_i m_i \|\dot{X}_i\|^2.$$

If every particle is fixed relative to **b** (such an object is called a rigid body), then $\dot{X}_i = 0$ and this reduces to

$$\mathrm{KE} = \frac{1}{2} M \|\dot{o}\|^2 + M\dot{o}^T R(\widehat{\Omega}\bar{X}) + \frac{1}{2}\Omega^T \mathbb{I}\Omega.$$

If, in addition, O' is the center of mass, then $\bar{X} = 0$ and $o = \bar{x}$, giving

$$\mathrm{KE} = \frac{1}{2} M \|\dot{\bar{x}}\|^2 + \frac{1}{2}\Omega^T \mathbb{I}\Omega.$$

In this case the total kinetic energy is the sum of the kinetic energy of translation of the center of mass and that of rotation about the center of mass.

1.2.5 Newton's Law in Moving Frames

Recall that Galilean laws of mechanics states that the total linear momentum of a set of interacting but otherwise isolated set of particles is conserved when observed in any inertial frame. Thus Newton's second law hold only in inertial frames. Let **e** be an inertial frame and let the representation of the position of a particle of mass m_i in the **e**-frame be x_i. Let the force acting on the particle due to the interaction it has with the other particles of the Universe have the representation f_i in the **e**-frame. Then Newton's second law gives $f_i = m_i\ddot{x}_i$. If an observer makes measurements with respect to a different frame **b** that is translating and rotating with respect to **e** then it is natural to ask what Newton's second law looks like with respect to the measurements made with respect to the **b**-frame. Denote the representation of the origin of the **b**-frame with respect to the **e**-frame be o and let $R \in \mathrm{SO}(3)$ be the rotation matrix that relates the **b** frame to the **e** by the relationship $\mathbf{b} = \mathbf{e}R$. Furthermore let $X_i = R^T(x_i - o)$ be the representation of the position of the particle and $F_i = R^T f_i$ be the representation of the interaction force acting on the particle in the **b** frame.

We see that the acceleration of the particle in the **e**-frame is related to the **b**-frame quantities by (1.22). Substituting this in $f_i = m_i\ddot{x}_i$ we have the following :

$$F_i = m_i R^T \ddot{o} + m_i \widehat{\Omega}^2 X_i + 2m_i \widehat{\Omega}\dot{X}_i + m_i \dot{\widehat{\Omega}} X_i + m_i \ddot{X}_i.$$

Notice that this equation expresses Newton's law using the force representation F_i, the relative position X_i and its derivatives, the angular velocity matrix $\widehat{\Omega}$ and its derivative, and the acceleration of the frame's origin expressed in **b**, $R^T\ddot{o}$. Thus this expression, if you may, can be considered to be the 'appropriate version' of the Newton's equations in the rotating frame $\mathbf{b}(t)$.

If the observer mistakenly treats the moving frame as inertial[18], she will identify $m_i\ddot{X}_i$ with the resultant interaction force. However, rearranging equation (1.35) gives

$$m_i\ddot{X}_i = F_i - \left(m_i R^T \ddot{o} + m_i \widehat{\Omega}^2 X_i + 2m_i \widehat{\Omega}\dot{X}_i + m_i \dot{\widehat{\Omega}} X_i\right). \tag{1.35}$$

The discrepancy between this quantity and the actual interaction force $F_i = R^T f_i$ will then appear to require additional forces. These apparent forces represent the unaccounted motion of the reference frame; they do not arise from additional physical interactions.

Thus an observer moving with the rotating frame will, in addition to the fundamental interacting forces, feel that there exists another resultant '*apparent force*':

$$F_{app} \triangleq \underbrace{-m_i\, R^T\ddot{o}}_{Einstein}\ \underbrace{-m_i\, \widehat{\Omega}^2 X_i}_{Centrifugal}\ \underbrace{-2m_i\, \widehat{\Omega}\dot{X}_i}_{Coriolis}\ \underbrace{-m_i\, \dot{\widehat{\Omega}} X_i}_{Euler}, \tag{1.36}$$

simply due to its ignorance of the motion of its frame. The first term $-m_i\, R^T\ddot{o}(t)$ is known as the *Einstein force*, the second term $-m_i\, \widehat{\Omega}^2 X_i$ is known as the *Centrifugal* force, the

[18] Like for instance when we think of an earth fixed frame to be inertial.

third term $-2m_i\,\widehat{\Omega}\dot{X}_i$ is known as the *Coriolis* force and the last term $-m_i\,\dot{\widehat{\Omega}}X_i$ is known as the *Euler* force. Observe that all these apparent forces have mass as a multiplicative factor. Note that the Einstein apparent force is observed due to the *translational ignorance* of the one's reference frame while the Centrifugal, Coriolis, and Euler forces are observed due to the *rotational ignorance* of the reference frame.

Using these expression we can explain many physical effects. In Exercise 1.15, you are asked to explain why a person standing on a scale inside an elevator obtains twice the usual scale reading when the elevator accelerates upwards at g, and a zero reading when it accelerates downwards at g, where g is the magnitude of the gravitational acceleration. The scale measures the supporting force exerted on the person, often called the *apparent weight*, rather than the gravitational force itself.

If gravity were absent but the elevator accelerated upwards at g, the scale would again read mg, just as it would for a stationary elevator in gravity. Thus the scale reading alone cannot distinguish between:

a.) A stationary elevator in a gravitational field of magnitude g.
b.) An elevator accelerating upwards at g in the absence of gravity.

More generally, when spatial variations of the gravitational field can be neglected, experiments within the elevator cannot distinguish these two situations. This observation motivates Einstein's equivalence principle, a foundation of general relativity. In a freely falling frame, gravitational acceleration can be removed locally, much like an apparent force arising from the choice of frame.

For a collection of N particles, write

$$F_i = F_i^{\text{ext}} + \sum_{j\neq i} F_{ij},$$

where F_i^{ext} represents interactions with particles outside the collection and F_{ij} is the force exerted by particle j on particle i, all expressed in $\mathbf{b}$. Since the internal forces are equal and opposite, $F_{ij} = -F_{ji}$, they cancel upon summation. Summing (1.35) therefore gives the following.

For a system of N particles with total mass, center of mass coordinates, and total external force given by

$$M = \sum_i m_i, \qquad \bar{X} = \frac{1}{M}\sum_i m_i X_i, \qquad F^{\text{ext}} = \sum_i F_i^{\text{ext}},$$

the center of mass motion in $\mathbf{b}$ satisfies

$$M\ddot{\bar{X}} = F^{\text{ext}} - MR^T\ddot{o} - M\widehat{\Omega}^2\bar{X} - 2M\widehat{\Omega}\dot{\bar{X}} - M\dot{\widehat{\Omega}}\bar{X}. \tag{1.37}$$

Thus the center of mass obeys the same equation as a single particle of mass M subject to the total external interaction force. The remaining terms account for the translation and rotation of the reference frame.

In the following sections we will demonstrate the value of equation (1.37) in writing down the equations of motion. In particular in section-1.2.5.1 while in section-1.2.5.2 we show an example of how it can be used to describe the motion of a bead constrained to move on a rotating hoop. We also invite you to try this approach in describing similarly constrained motion that is described in exercises 1.18-1.22. Finally in the section 1.2.5.3 we use this approach to investigate the effects of the motion of Earth in describing particle motion.

1.2.5.1 Description of Particle Motion in a Plane using Polar Coordinates

For a particle restricted to move in 2-dimensions, it is sometimes convenient to write down the motion in polar coordinates (r,θ). This amounts to observing the motion in a moving frame $\mathbf{b}(t) = [\mathbf{e}_r(t) \;\; \mathbf{e}_\theta(t) \;\; \mathbf{e}_z(t)]$ (refer to figure 1.10) where $\mathbf{e}_r$ aligns along the particle P at all times. We treat $\mathbf{e}$ as inertial and take both frames to have the same fixed origin, so $o = 0$. We consider $r > 0$, where the polar directions are defined. The components F_r, F_θ, F_z describe the resultant interaction force, including any constraint reactions. Consider the orthonormal

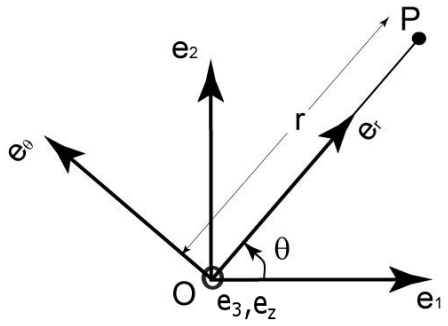


Fig. 1.10

frame $\mathbf{b}(t) = [\mathbf{e}_r \;\; \mathbf{e}_\theta \;\; \mathbf{e}_z]$. Let $\mathbf{e} = [\mathbf{e}_1 \;\; \mathbf{e}_2 \;\; \mathbf{e}_3]$ be an Earth fixed frame. Then $\mathbf{b}(t) = \mathbf{e}R(t)$ where

$$R(t) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Thus we have

$$\widehat{\Omega} = \begin{bmatrix} 0 & -\dot{\theta} & 0 \\ \dot{\theta} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \dot{\widehat{\Omega}} = \begin{bmatrix} 0 & -\ddot{\theta} & 0 \\ \ddot{\theta} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \widehat{\Omega}^2 = -\dot{\theta}^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The representation of P in this frame is

$$X = \begin{bmatrix} r \\ 0 \\ 0 \end{bmatrix}$$

and hence we see that

$$\dot{X} = \begin{bmatrix} \dot{r} \\ 0 \\ 0 \end{bmatrix} \quad \ddot{X} = \begin{bmatrix} \ddot{r} \\ 0 \\ 0 \end{bmatrix}.$$

From Newton's equations in the **b**-frame given by (1.35) we have

$$m \begin{bmatrix} \ddot{r} \\ 0 \\ 0 \end{bmatrix} = - \begin{bmatrix} -mr\dot{\theta}^2 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 2m\dot{r}\dot{\theta} \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ mr\ddot{\theta} \\ 0 \end{bmatrix} + \begin{bmatrix} F_r \\ F_\theta \\ F_z \end{bmatrix}.$$

Observe that the apparent force known as the Centrifugal force is $mr\dot{\theta}^2$ and is in the $\mathbf{e}_r$ direction, the Coriolis force is $-2m\dot{r}\dot{\theta}$ in the $\mathbf{e}_\theta$ direction and the Euler force is $-mr\ddot{\theta}$ in the $\mathbf{e}_\theta$ direction and we recover what we have learnt in our junior level physics classes. Simplifying the above equations we have that

$$\begin{aligned} m\ddot{r} - mr\dot{\theta}^2 &= F_r, \\ mr\ddot{\theta} + 2m\dot{r}\dot{\theta} &= F_\theta \\ F_z &= 0. \end{aligned}$$

If the particle is constrained to a circle centered at the origin, then r is constant and the equations reduce to

$$F_r = -mr\dot{\theta}^2, \qquad F_\theta = mr\ddot{\theta}.$$

The radial resultant provides the inward acceleration required for circular motion. The tangential resultant determines the change in angular speed; it is not required by the circular constraint itself. For example, a smooth circular guide supplies a radial reaction, while any tangential force must arise from other interactions. In uniform circular motion, $F_\theta = 0$.

In this rotating frame, the radial and tangential resultants balance the centrifugal and Euler apparent forces, respectively. Compare these results with those obtained in Exercise 1.17.

1.2.5.2 Example: Bead on a Rotating Hoop

As an illustration of the use of (1.35) that represents Newton's equations in a moving frame we will consider the problem of analyzing the motion of a bead on a rotating hoop[19]. A schematic of the system is shown in figure-1.11. Consider a bead of mass m sliding without friction on a rigid circular hoop of radius r. The hoop's center O is fixed, and its rotation about the vertical diameter is prescribed by $\theta(t)$. All three frames introduced below share the origin O, and $\mathbf{e}_3$ points vertically upwards. We neglect air resistance.

Let us choose frames as shown in Figure-1.12 and Figure-1.13 and assume that the frame **e** is an inertial frame. We will denote by P the position of the bead and by O the origin of the

[19] This section was typed and illustrated by Mr. K. G. B. Gamagedara E/09/078.

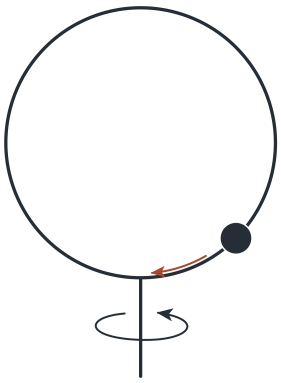


Fig. 1.11 Bead on a Rotating Hoop.

e-frame. Let **c** be another orthonormal frame such that it moves with respect to **e** in such a way that $\mathbf{c}_3 \equiv \mathbf{e}_3$ and $\mathbf{c}_1$ is always orthogonal to the plane of the hoop as shown in figure-1.12. Then the two frames **c** and **e** are related by $\mathbf{c} = \mathbf{e}R_3(\theta)$ where we use the customary notation $R_i(\theta)$ to denote a rotation about the ith axis by an angle equal to θ. Specifically in this case,

$$R_3(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

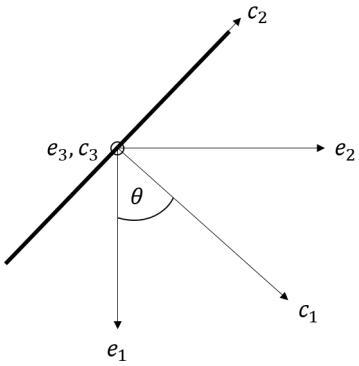


Fig. 1.12 The view of the system from the top.

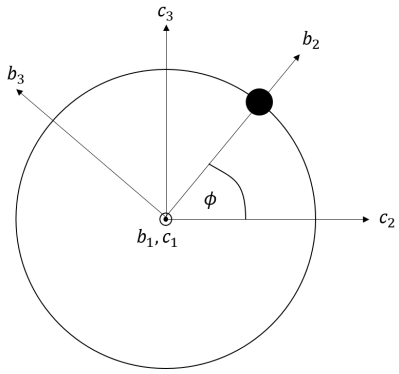


Fig. 1.13 View of the system from a direction perpendicular to the plane of the hoop.

Let **b** be another orthonormal frame such that it moves with respect to **c** in such a way that $\mathbf{b}_1 \equiv \mathbf{c}_1$ and $\mathbf{b}_2$ is always along OP as shown in figure-1.13. Then the two frames **b** and **c** are related by $\mathbf{b} = \mathbf{c}R_1(\phi)$ where

$$R_1(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}.$$

Thus we have that, the two frames $\mathbf{b}$ and $\mathbf{e}$ are related by $\mathbf{b} = \mathbf{e}R_3(\theta)R_1(\phi) = \mathbf{e}R$ which gives that $R = R_3(\theta)R_1(\phi)$.

The representation of the center of mass position of P, in the $\mathbf{b}$ frame is seen to be,

$$X = \begin{bmatrix} 0 \\ r \\ 0 \end{bmatrix},$$

and is independent of time and hence we have that $\dot{X} = \ddot{X} = 0$. In fact it is this very convenient reason why we chose the $\mathbf{b}$-frame to move with the particle. The Newton's equation (1.37) in the $\mathbf{b}$-frame then becomes, $F = m(\widehat{\Omega}^2 + \dot{\widehat{\Omega}})X$, where F is the representation, in $\mathbf{b}$, of the fundamental forces acting on the bead due to its interaction with the rest of the Universe.

Notice that it is convenient to express the constraint interactions in the frame $\mathbf{b}$ while it is convenient to express the gravitational interaction in the $\mathbf{c}$-frame. Thus if f is the representation of the force in the $\mathbf{e}$-frame and F is the representation of the force in the $\mathbf{b}$-frame. The directions $\mathbf{b}_1$ and $\mathbf{b}_2$ are normal to the hoop's tangent $\mathbf{b}_3$. Since the contact is frictionless, the constraint reaction has only the signed components F_{N1} and F_{N2} along these two directions. Thus we have that

$$\mathbf{e}f = \mathbf{b}F = \mathbf{b}\begin{bmatrix} F_{N1} \\ F_{N2} \\ 0 \end{bmatrix} + \mathbf{c}\begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} = \mathbf{b}\begin{bmatrix} F_{N1} \\ F_{N2} \\ 0 \end{bmatrix} + \mathbf{b}R_1^T(\phi)\begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix}$$

Hence we have

$$F = \begin{bmatrix} F_{N1} \\ F_{N2} \\ 0 \end{bmatrix} + R_1^T(\phi)\begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} = \begin{bmatrix} F_{N1} \\ F_{N2} - mg\sin\phi \\ -mg\cos\phi \end{bmatrix}.$$

Now that we have found F what remains to write down the Newton's equations $F = m(\widehat{\Omega}^2 + \dot{\widehat{\Omega}})X$ explicitly in the moving frame $\mathbf{b}$ is the computation of $\dot{\widehat{\Omega}}$ and $\widehat{\Omega}^2$. We know that $\widehat{\Omega} = R^T\dot{R}$. Since

$$\dot{R} = R_3\widehat{\Omega}_3R_1 + R_3R_1\widehat{\Omega}_1 = R(R_1^T\widehat{\Omega}_3R_1 + \widehat{\Omega}_1)$$

we have that

$$\widehat{\Omega} = R_1^T\widehat{\Omega}_3R_1 + \widehat{\Omega}_1.$$

Recall from (1.28) we have that $\widehat{R\Omega} = R\widehat{\Omega}R^T$. Thus using the linearity of the $\widehat{\ }$ isomorphism we find that the 3×1 version of $\widehat{\Omega}$ is explicitly given by

$$\Omega = R_1^T(\phi)\Omega_3 + \Omega_1 = \begin{bmatrix} \dot{\phi} \\ \dot{\theta}\sin\phi \\ \dot{\theta}\cos\phi \end{bmatrix}$$

and hence that

$$\dot{\Omega} = \begin{bmatrix} \ddot{\phi} \\ \ddot{\theta}\sin\phi + \dot{\theta}\dot{\phi}\cos\phi \\ \ddot{\theta}\cos\phi - \dot{\theta}\dot{\phi}\sin\phi \end{bmatrix}.$$

From these we have that the corresponding skew-symmetric matrices are given by

$$\widehat{\Omega} = \begin{bmatrix} 0 & -\dot{\theta}\cos\phi & \dot{\theta}\sin\phi \\ \dot{\theta}\cos\phi & 0 & -\dot{\phi} \\ -\dot{\theta}\sin\phi & \dot{\phi} & 0 \end{bmatrix}$$

$$\dot{\widehat{\Omega}} = \begin{bmatrix} 0 & -(\ddot{\theta}\cos\phi - \dot{\theta}\dot{\phi}\sin\phi) & (\ddot{\theta}\sin\phi + \dot{\theta}\dot{\phi}\cos\phi) \\ (\ddot{\theta}\cos\phi - \dot{\theta}\dot{\phi}\sin\phi) & 0 & -\ddot{\phi} \\ -(\ddot{\theta}\sin\phi + \dot{\theta}\dot{\phi}\cos\phi) & \ddot{\phi} & 0 \end{bmatrix}.$$

Note from (1.29) that $\widehat{\Omega}^2 = \Omega\Omega^T - ||\Omega||^2 I$ and hence that

$$\widehat{\Omega}^2 = \begin{bmatrix} -\dot{\theta}^2 & \dot{\phi}\,\dot{\theta}\,\sin\phi & \dot{\phi}\,\dot{\theta}\,\cos\phi \\ \dot{\phi}\,\dot{\theta}\,\sin\phi & -\dot{\phi}^2 - \dot{\theta}^2\cos\phi^2 & \dot{\theta}^2\cos\phi\,\sin\phi \\ \dot{\phi}\,\dot{\theta}\,\cos\phi & \dot{\theta}^2\cos\phi\,\sin\phi & -\dot{\phi}^2 - \dot{\theta}^2\sin\phi^2 \end{bmatrix}$$

Substituting these in the Newton's equations for the center of mass in the **b**-frame (1.37) we have

$$\begin{bmatrix} F_{N1} \\ F_{N2} \\ 0 \end{bmatrix} = m \begin{bmatrix} -r\ddot{\theta}\cos\phi + 2r\dot{\phi}\dot{\theta}\sin\phi \\ -r\dot{\phi}^2 - r\dot{\theta}^2\cos\phi^2 + g\sin\phi \\ r\dot{\theta}^2\sin\phi\cos\phi + r\ddot{\phi} + g\cos\phi \end{bmatrix}.$$

The first two rows can be used to find the constraint forces F_{N1} and F_{N2} that constrain the bead to stay on the hoop while the third row can be used to describe the motion of the bead as follows:

$$r\ddot{\phi} = -\cos\phi\left(g + r\dot{\theta}^2\sin\phi\right).$$

Similarly to what we have done in this problem, you are invited to try out the exercises 1.18 – 1.22 at this point in order to get an idea of how convenient it is to write down equations of motion using appropriately chosen moving frames and equation-1.37. The idea is that writing down position, velocity and the fundamental constrain force interactions are mostly convenient in a frame where the particle or the object appears fixed. Then what remains is the computation that relates this frame to an inertial frame and then use equation to write down Newton's equations in the moving frame in which the particle or the object appears to be fixed.

The following notebook contains several other similar examples worked out using the symbolic package simpy in Python: Examples_of_Describing_Motion_in_Moving_Frames.ipynb

1.2.5.3 Effects of Earth's Rotation About its Axis

In the following we also use the equations (1.36) to show the effects of Earths rotation on gravity as well as on the formation of hurricanes and the motion of a long pendulum known as the Foucault's pendulum.

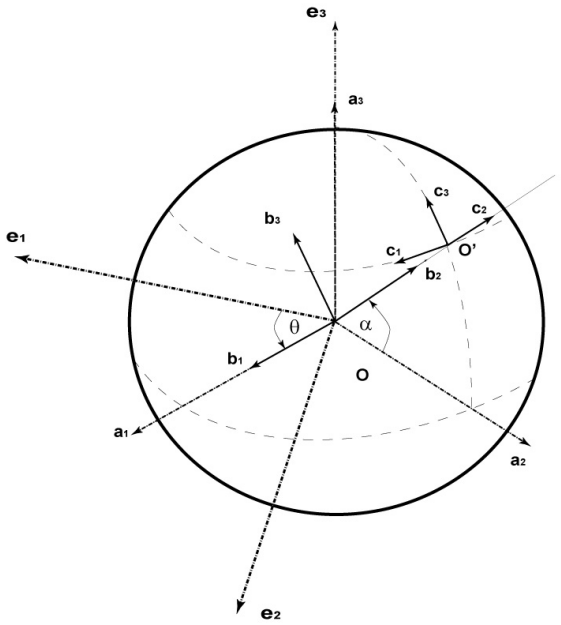


Fig. 1.14 Effects of Earth's Rotation about its Axis: Frame **c** is fixed on the surface of the Earth with origin coinciding at O'. Frame **b** is parallel to **c** with center O coinciding with the center of the earth. Frame **a** is fixed on the earth with origin at O and $\mathbf{a}_3$ aligned along the axis of rotation of the earth and $\mathbf{a}_2$ in the $\mathbf{b}_2, \mathbf{b}_3$ plane. The origin of frame **e** coincides with O and $\mathbf{e}_3$ is always aligned along $\mathbf{a}_3$.

Let us consider the effect that the Earth's rotation about its axis has on the motion of a particle as observed in an Earth fixed frame at a point O' (with latitude α) on the surface of the Earth. Consider figure 1.14. Let $\mathbf{c}(t)$ be an ortho-normal frame fixed on Earth with center at O'. The frame is oriented such that $\mathbf{c}_2$ is aligned perpendicular to the Earth (vertical direction), $\mathbf{c}_3$ points north along the local meridian (a line of constant longitude), and $\mathbf{c}_1$ points west along the local parallel of latitude (a line of constant latitude). Let **b** be a frame that is parallel to **c** and fixed on Earth with origin at the center of the Earth O. Let a be a frame with origin at O and $\mathbf{a}_3$ aligned along the axis of rotation of the Earth, $\mathbf{a}_1 \equiv \mathbf{b}_1$ and O' lies in the $\mathbf{a}_2 - \mathbf{a}_3$ plane as shown in figure-1.14. Thus if $\mathbf{b}(t) = \mathbf{a}(t)R_1(\alpha)$ where α is the latitude angle and is a constant and $R_1(\alpha)$ is one of the three basic rotations given in (1.23). Let **e** be an Earth-centered frame whose axes do not rotate relative to distant stars, with $\mathbf{e}_3$ along Earth's rotation axis. We neglect Earth's orbital acceleration and the gravitational influences of other celestial bodies, and treat **e** as inertial. We also approximate Earth as a sphere rotating at a constant angular speed. The frame is oriented such that the $\mathbf{e}_3$ direction coincides with the axis of rotation of the Earth, ie $\mathbf{a}_3 \equiv \mathbf{e}_3$ and $\mathbf{a}(t) = \mathbf{e}R_3(\theta)$ where θ is the angle of rotation of the Earth about its axis.

We are interested in analyzing the motion of a particle P (or equivalently the center of mass motion of an object) as observed in the frame $\mathbf{c}(t)$. Let the representation of the point P in the $\mathbf{c}(t)$ frame be $X_p(t)$, ie. $O'P = \mathbf{c}(t)X_p(t)$. Since $OO' = \mathbf{c}(t)o$ and **c** is parallel to **b** we have that

$OP = OO' + O'P = \mathbf{b}(t)(o + X_p)$. Since by construction O' is a distance r (r is the radius of the Earth) away on the $\mathbf{b}_2$ axis, $o = [0 \;\; r \;\; 0]^T$ and is a constant.

Thus if $x(t)$ is the representation of P in $\mathbf{e}$ we have

$$\mathbf{e}x(t) = \mathbf{b}(t)\,(o + X_p(t)) = \mathbf{e}R_3R_1 \underbrace{(o + X_p(t))}_{X} = \mathbf{e}RX.$$

Thus $x = RX(t)$ where $R = R_3R_1$ and $X = o + X_p$. From Newton's equations in the rotating frame, (1.35), we have

$$m\ddot{X}(t) = -m\,\widehat{\Omega}^2(t)X(t) - 2m\,\widehat{\Omega}(t)\dot{X}(t) - m\,\dot{\widehat{\Omega}}(t)X(t) + R^T f(t). \tag{1.38}$$

Let us decompose the total force in to two components $f = f^g + f^e$ where f^g is the gravitational force and f^e is the additional external forces. Gravity acts in the OP direction. Thus

$$f^g = -\frac{mg}{||x||}x = -\frac{mg}{||X||}RX,$$

and hence

$$R^T f^g = -\frac{mg}{||X||}X.$$

Substituting this in (1.38) and noting that the Earth is rotating at a constant rate (hence $\dot{\widehat{\Omega}} = 0$) we have

$$\ddot{X} = -\widehat{\Omega}^2 X - 2\widehat{\Omega}\dot{X} - g\frac{X}{||X||} + \frac{1}{m}R^T f^e, \tag{1.39}$$

$$= -2\widehat{\Omega}\dot{X} - \left(gI_{3\times 3} + ||X||\widehat{\Omega}^2\right)\frac{X}{||X||} + \frac{1}{m}R^T f^e. \tag{1.40}$$

Observe that these equations describe the motion of a particle as observed in a frame fixed on Earth with origin coinciding with the center of the Earth. From (1.39) it can be seen that the motion of the particle in the Earth fixed frame, in addition to the gravity and external forces, is also influenced by the Centrifugal and Coriolis terms that arise due to the ignorance of the rotation of the Earth. In equation (1.40) the Centrifugal term has been combined with the gravity term. This allows one to see that the effective gravity felt by an observer will change with the latitude of the location of the observer. We will explain this in a bit more detail at the end of this section.

Recall $X = o + X_p$ where o is a constant and X_p is the representation of the point P in the Earth fixed frame c fixed on the surface of the Earth at O'. Then $\dot{X} = \dot{X}_p$ and $\ddot{X} = \ddot{X}_p$ thus from (1.40) we have that

$$\begin{aligned}\ddot{X}_p &= -2\widehat{\Omega}\dot{X}_p - \left(gI_{3\times 3} + ||X_p + o||\widehat{\Omega}^2\right)\frac{X_p + o}{||X_p + o||} + \frac{1}{m}R^T f^e \\ &= -2\widehat{\Omega}\dot{X}_p - \widehat{\Omega}^2 X_p - \frac{g}{||X_p + o||}(X_p + o) - \widehat{\Omega}^2 o + \frac{1}{m}R^T f^e \end{aligned} \tag{1.41}$$

describes the motion of a point particle *m as observed in the Earth fixed frame with origin O' on the surface of the Earth. This is the case that applies to us when we observe particle motion.* For motion confined to a small region near O', where $\|X_p\| \ll r$, we approximate the gravitational acceleration as uniform:

$$-g\frac{X_p+o}{\|X_p+o\|} \approx -g\chi,$$

where $\chi = [0 \;\; 1 \;\; 0]^T$. Here g denotes the local gravitational acceleration before including the centrifugal contribution. Then (1.41) approximates to

$$\ddot{X}_p = -2\widehat{\Omega}\dot{X}_p - \widehat{\Omega}^2 X_p - \left(gI_{3\times 3} + r\widehat{\Omega}^2\right)\chi + \frac{1}{m}R^T f^e. \tag{1.42}$$

These equations can be used to describe many natural phenomena. For instance it explains why a Hurricane formed in the Nothern hemisphere rotates in a counter–clockwise direction and a Hurricane formed in the Southern hemisphere rotates in a clockwise direction (see figure 1.15). You are asked to show this in exercise 1.24. In Exercise-1.23 you are asked to demonstrate the precession of the oscillating plane in the Foucault's pendulum (see figure-1.16). This .ipynb interactive notebook provides python simulations and animations that describe the motion of the Foucault pendulum and the motion of an idealized atmospheric particle.



(a) ©Wikipedia. Subtropical Storm Raoni. (b) ©Wikipedia. A Northern Polar Hurricane.

Fig. 1.15 Figures show the clockwise and anti-clockwise rotation of respectively a southern hemisphere and northern hemisphere formed hurricane.

The rotational velocity of the Earth, $\Omega_e = \frac{2\pi}{23h\,56m\,4s} = 7.292\times 10^{-5} rad/s$, is very small and thus $\widehat{\Omega}$ and $\widehat{\Omega}^2$ are very small and for most applications these effects can be neglected and then the equations (1.42) reduce to the usual equations of projectile motion given by

$$\ddot{X}_p = -g\chi + \frac{1}{m}R^T f^e. \tag{1.43}$$

Explicitly written down they are:

$$\ddot{X}_{p_1} = \frac{1}{m}F_1, \tag{1.44}$$

$$\ddot{X}_{p_2} = -g + \frac{1}{m}F_2, \tag{1.45}$$

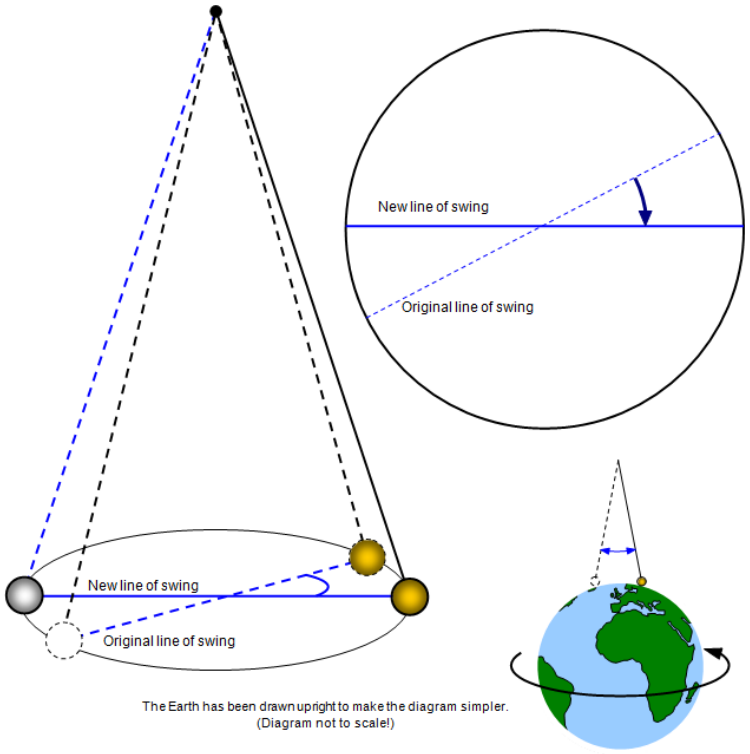


Fig. 1.16 The shift in the plane of oscillation of the Foucault Pendulum. Figure taken from https://www.schoolphysics.co.uk/age14-16/Astronomy/text/Foucaults_pendulum/index.html

$$\ddot{X}_{p_3} = \frac{1}{m} F_3. \tag{1.46}$$

Let us now explicitly consider the effects of the rotation of Earth. Let

$$\gamma(t) \triangleq -\left(g I_{3\times 3} + r\widehat{\Omega}^2\right)\chi + \frac{1}{m} R^T f^e,$$

and then (1.42) can be expressed as

$$\ddot{X}_p = -\widehat{\Omega}^2 X_p - 2\widehat{\Omega}\dot{X}_p + \gamma(t), \tag{1.47}$$

If the additional force expressed in the Earth-fixed frame is a prescribed function of time, the resulting equations form a linear system with known forcing by defining $Z_p = [X_p^T \;\; \dot{X}_p^T]^T$ and arranging this equation as

$$\dot{Z}_p = AZ_p + B(t),$$

where

$$A = \begin{bmatrix} 0 & I_{3\times 3} \\ -\widehat{\Omega}^2 & -2\widehat{\Omega} \end{bmatrix}, \quad B(t) = \begin{bmatrix} 0 \\ \gamma(t) \end{bmatrix}.$$

From linear systems theory we find that the solution to this differential equation is explicitly given by

$$Z_p(t) = e^{At} Z_p(0) + \int_0^t e^{A(t-\tau)} B(\tau)\, d\tau. \tag{1.48}$$

To compute e^{At} we need $\widehat{\Omega}$ that is given by $R^T \dot{R} = \widehat{\Omega}$. Since $R = R_3 R_1$ differentiating we have that

$$\dot{R} = \dot{R}_3 R_1 = R_3 \widehat{\Omega}_e R_1 = R_3 R_1 \left(R_1^T \widehat{\Omega}_e R_1\right) = R_3 R_1 \left(\widehat{R_1^T \Omega}_e\right) = R\widehat{\Omega},$$

Now from $\dot{R}_3 = R_3 \widehat{\Omega}_e$ we have that

$$\widehat{\Omega}_e = \begin{bmatrix} 0 & -\dot{\theta}_e & 0 \\ \dot{\theta}_e & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

where $\dot{\theta} = \dot{\theta}_e$ is the angular velocity of Earth about its axis of rotation. Hence we have that

$$\widehat{\Omega} = R_1^T \widehat{\Omega}_e R_1.$$

Thus we have

$$\Omega = \begin{bmatrix} 0 \\ \dot{\theta}_e \sin\alpha \\ \dot{\theta}_e \cos\alpha \end{bmatrix}, \qquad \widehat{\Omega} = \dot{\theta}_e \begin{bmatrix} 0 & -\cos\alpha & \sin\alpha \\ \cos\alpha & 0 & 0 \\ -\sin\alpha & 0 & 0 \end{bmatrix},$$

and

$$\widehat{\Omega}^2 = -\dot{\theta}_e^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos^2\alpha & -\cos\alpha\sin\alpha \\ 0 & -\cos\alpha\sin\alpha & \sin^2\alpha \end{bmatrix}.$$

Substituting these in (1.48) we can explicitly find $X_p(t)$ and $\dot{X}_p(t)$. Below we will do this for a special case where the observer is on the equator of the Earth.

Example 1.3. A cannonball is released from a balloon held stationary relative to the ground, directly h meters above a point O' on the equator. Where will it land? We neglect air resistance and use the local uniform-gravity approximation developed above, with $h \ll r$.

At the equator, $\alpha = 0$. There are no interaction forces other than gravity, so $f^e = 0$. The cannonball is released from rest relative to the balloon, giving

$$\dot{X}_p(0) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \qquad X_p(0) = \begin{bmatrix} 0 \\ h \\ 0 \end{bmatrix}.$$

For brevity, write $\omega_e \triangleq \dot{\theta}_e$ and define $g_{\text{eff}} = g - r\omega_e^2$, where g denotes gravitational acceleration before the centrifugal contribution is included. Then

$$\widehat{\Omega} = \omega_e \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \widehat{\Omega}^2 = -\omega_e^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \gamma = -g_{\text{eff}} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

The matrix exponential can be written compactly as

$$e^{At} = \begin{bmatrix} E(t)(I + t\widehat{\Omega}) & tE(t) \\ -tE(t)\widehat{\Omega}^2 & E(t)(I - t\widehat{\Omega}) \end{bmatrix}, \qquad E(t) = \begin{bmatrix} \cos(\omega_e t) & \sin(\omega_e t) & 0 \\ -\sin(\omega_e t) & \cos(\omega_e t) & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

where $I = I_{3\times 3}$. Hence (1.48) and the initial conditions give

$$X_p(t) = h \begin{bmatrix} \sin(\omega_e t) - \omega_e t\cos(\omega_e t) \\ \cos(\omega_e t) + \omega_e t\sin(\omega_e t) \\ 0 \end{bmatrix} - g_{\text{eff}} \int_0^t \begin{bmatrix} (t-\tau)\sin\big(\omega_e(t-\tau)\big) \\ (t-\tau)\cos\big(\omega_e(t-\tau)\big) \\ 0 \end{bmatrix} d\tau.$$

For a fall lasting only a few seconds, $\omega_e t \ll 1$. Expanding the trigonometric terms through second order gives

$$\begin{aligned} X_p(t) &\approx h \begin{bmatrix} 0 \\ 1 + \frac{\omega_e^2 t^2}{2} \\ 0 \end{bmatrix} - g_{\text{eff}} \int_0^t \begin{bmatrix} \omega_e(t-\tau)^2 \\ (t-\tau) - \frac{\omega_e^2(t-\tau)^3}{2} \\ 0 \end{bmatrix} d\tau \\ &= h \begin{bmatrix} 0 \\ 1 + \frac{\omega_e^2 t^2}{2} \\ 0 \end{bmatrix} - g_{\text{eff}} \begin{bmatrix} \frac{\omega_e t^3}{3} \\ \frac{t^2}{2} - \frac{\omega_e^2 t^4}{8} \\ 0 \end{bmatrix}. \end{aligned}$$

These expressions retain terms through second order in ω_e; higher-order contributions arising from the factor g_{eff} are beyond this accuracy.

Let T be the landing time. Approximating the ground locally by $X_{p_2} = 0$ and neglecting the small rotational corrections to the vertical motion gives

$$T \approx \sqrt{\frac{2h}{g_{\text{eff}}}}.$$

The leading horizontal displacement is therefore

$$X_{p_1}(T) \approx -\frac{g_{\text{eff}}\omega_e}{3} T^3 = -\frac{2h\omega_e}{3}\sqrt{\frac{2h}{g_{\text{eff}}}}.$$

Since the positive c_1 direction points west, the negative sign indicates an eastward displacement. Thus the cannonball lands approximately

$$d_{\text{east}} = \frac{2h\omega_e}{3}\sqrt{\frac{2h}{g - r\omega_e^2}}$$

east of O'.

For $h = 500\,\text{m}$, using $\omega_e = 7.292 \times 10^{-5}\,\text{s}^{-1}$, $r = 6.371 \times 10^6\,\text{m}$, and $g = 9.81\,\text{m/s}^2$, we obtain

$$T \approx 10.1\,\text{s}, \qquad d_{\text{east}} \approx 0.246\,\text{m}.$$

The predicted deflection is therefore about 25 cm to the east.

The eastward deflection may initially seem surprising: Earth continues to rotate eastward during the fall, which might suggest that the ground should move ahead of the cannonball and the ball should fall to the west. However, release does not leave the cannonball stationary in an inertial frame. Before release, it already moves eastward with the balloon at speed $(r+h)\omega_e$, while the ground below moves at speed $r\omega_e$.

The equations describe how this motion develops during the fall. In particular, the component of the Coriolis acceleration along the positive c_1 direction (west) is $2\omega_e\dot{X}_{p_2}$. Since the cannonball is descending, $\dot{X}_{p_2} < 0$, so this acceleration points eastward. Starting from zero velocity relative to the balloon, the cannonball therefore develops an eastward velocity relative to the ground and lands east of the release point's vertical projection.

1.2.6 Rate of Change of Angular Momentum in Moving Frames

Let $\mathbf{e}$ be an inertial frame and let $\mathbf{b} = \mathbf{e}R$ have origin O', whose position in $\mathbf{e}$ is $o(t)$. We first allow the particle to move relative to $\mathbf{b}$, so $X_i(t)$ need not be constant.

1.2.6.1 A Particle Moving Relative to the Frame

Previously we have seen that the angular momentum of P_i about O', defined using its velocity relative to $\mathbf{e}$, has the representation

$$\pi_i = R\left(\mathbb{I}_i\Omega + m_i X_i \times (R^T\dot{o} + \dot{X}_i)\right),$$

where

$$\mathbb{I}_i \triangleq -m_i\widehat{X}_i^2 = m_i\left(\|X_i\|^2 I_{3\times 3} - X_i X_i^T\right)$$

is its moment-of-inertia matrix about O', expressed in $\mathbf{b}$. Since $X_i(t)$ may change with time,

$$\dot{\mathbb{I}}_i = -m_i\left(\dot{\widehat{X}}_i\widehat{X}_i + \widehat{X}_i\dot{\widehat{X}}_i\right).$$

For brevity, set $V_o = R^T\dot{o}$. Using $\dot{R} = R\widehat{\Omega}$ gives

$$\dot{V}_o = -\Omega \times V_o + R^T\ddot{o}.$$

Differentiating π_i, and using the Jacobi identity

$$A\times(B\times C) + B\times(C\times A) + C\times(A\times B) = 0,$$

we obtain

$$\begin{aligned}\dot{\pi}_i = R\Big(&\mathbb{I}_i\dot{\Omega} + \dot{\mathbb{I}}_i\Omega + \Omega\times(\mathbb{I}_i\Omega) + m_i\Omega\times(X_i\times\dot{X}_i)\\ &- m_i V_o\times(\Omega\times X_i) + m_i\dot{X}_i\times V_o + m_i X_i\times(R^T\ddot{o} + \ddot{X}_i)\Big).\end{aligned}$$

On the other hand, differentiating $\pi_i = m_i(x_i - o) \times \dot{x}_i$ and applying Newton's law gives

$$\dot{\pi}_i = -m_i \dot{o} \times \dot{x}_i + \tau_i, \qquad \tau_i = (x_i - o) \times f_i.$$

Let $F_i = R^T f_i$ and $T_i = R^T \tau_i = X_i \times F_i$. Since $\dot{x}_i = R(V_o + \Omega \times X_i + \dot{X}_i)$, this becomes

$$\dot{\pi}_i = R\left(-m_i V_o \times (\Omega \times X_i) + m_i \dot{X}_i \times V_o + T_i\right).$$

Equating the two expressions cancels the terms involving V_o. Consequently,

$$\mathbb{I}_i \dot{\Omega} + \Omega \times (\mathbb{I}_i \Omega) + \dot{\mathbb{I}}_i \Omega + m_i \Omega \times (X_i \times \dot{X}_i) + m_i X_i \times \ddot{X}_i = T_i - m_i X_i \times R^T \ddot{o}.$$

Using the expression for $\dot{\mathbb{I}}_i$, one can verify that

$$\dot{\mathbb{I}}_i \Omega + m_i \Omega \times (X_i \times \dot{X}_i) = 2 m_i X_i \times (\Omega \times \dot{X}_i).$$

Thus the balance gives us the following result.

For a particle moving relative to **b**,

$$\mathbb{I}_i \dot{\Omega} + \Omega \times (\mathbb{I}_i \Omega) + m_i X_i \times \left(\ddot{X}_i + 2\Omega \times \dot{X}_i\right) = X_i \times F_i - m_i X_i \times R^T \ddot{o}. \tag{1.49}$$

The terms involving $\dot{X}_i$ and $\ddot{X}_i$ account for motion relative to the frame, while the term involving $\ddot{o}$ accounts for acceleration of its origin.

Here $F_i = R^T f_i$ is the representation of the force acting on the particle and expressed in the **b**-frame and since and $T_i = X_i \times F_i$ is the representation of the moment of the force acting on on the particle with respect to the origin of the moving frame $\mathbf{b}(t)$.

Refer to Exercise 1.20 for an example of calculation of these quantities for a complicated motion. The detailed treatment of a model problem of a payload transfer on a space station is presented below as another illustrative example.

Example: Payload Transfer by a Robotic Boom on an Earth-Pointing Space Station

Consider a space station moving in an inclined circular orbit of radius a about Earth. A robotic boom carries a small, heavy payload, modeled as a particle of mass m. The boom can change its orientation relative to the station and extend or retract. We wish to determine the payload's angular momentum and kinetic energy, and the force and moment exerted by the boom.

We assume a spherical Earth and neglect atmospheric drag. The station's reference point O' follows a prescribed circular orbit, while its orientation follows the radial, along-orbit, and orbit-normal directions. We neglect the boom's mass and flexibility and do not calculate the station's response to the payload reaction.

Orbit and reference frames:

We approximate the station's orbit by a circular orbit of radius a, with inclination λ to Earth's equatorial plane. For an ISS-like orbit, take $\lambda \approx 51.6^\circ$. Over the duration of the payload maneuver, we neglect changes in the orbital plane.

Let $\mathbf{e}$ be an Earth-centered inertial frame, with $\mathbf{e}_3$ pointing along Earth's northward rotation axis. The orbital plane and Earth's equatorial plane intersect along a line through Earth's center, called the *line of nodes*. This line meets the circular orbit at two points: the *ascending node*, where the station passes from the Southern Hemisphere into the Northern Hemisphere, and the *descending node*, where it passes from the Northern Hemisphere into the Southern Hemisphere. We choose $\mathbf{e}_1 = \mathbf{b}_1$ to point from Earth's center towards the ascending node and measure the orbital angle ψ from this direction in the direction of orbital motion, so that $\psi = 0$ at the ascending node.

Introduce an Earth-centered frame $\mathbf{b}$ by

$$\mathbf{b} = \mathbf{e}R_1(\lambda).$$

Then $\mathbf{b}_1 = \mathbf{e}_1$ lies along the ascending node, $\mathbf{b}_1$ and $\mathbf{b}_2$ span the orbital plane, and $\mathbf{b}_3$ is its normal in the direction of orbital angular momentum. Under the fixed-orbit-plane approximation, $\mathbf{b}$ is an inertial frame.

Let $\psi(t)$ be the orbital angle measured from the ascending node in the direction of motion. For the circular orbit,

$$\psi(t) = \psi_0 + \omega_o t, \qquad \omega_o^2 = \frac{\mu}{a^3},$$

where μ is Earth's gravitational parameter.

We assume that the station maintains an attitude for which a station-fixed frame $\mathbf{c}$ has $\mathbf{c}_1$ pointing radially outwards, $\mathbf{c}_2$ along the direction of orbital motion, and $\mathbf{c}_3 = \mathbf{b}_3$. Its origin O' is the station's reference point.

$$\mathbf{c} = \mathbf{b}R_3(\psi) = \mathbf{e}R_o, \qquad R_o = R_1(\lambda)R_3(\psi).$$

The angular velocity of $\mathbf{c}$ relative to $\mathbf{e}$, expressed in $\mathbf{c}$, is

$$\Omega_o = \begin{bmatrix} 0 \\ 0 \\ \omega_o \end{bmatrix}, \qquad \widehat{\Omega}_o = R_o^T \dot{R}_o.$$

The inertial position of O' is

$$o = aR_o \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = a \begin{bmatrix} \cos\psi \\ \cos\lambda\, \sin\psi \\ \sin\lambda\, \sin\psi \end{bmatrix}.$$

Let $\mathbf{d}$ be a frame attached to the boom, with origin at its pivot O', which we take to coincide with the station-frame origin. Define

$$\mathbf{d} = \mathbf{c}R_B, \qquad R_B = R_3(\theta)R_1(\alpha).$$

Here $\theta(t)$ describes rotation about $\mathbf{c}_3$, followed by a rotation $\alpha(t)$ about the resulting first axis. These angles describe the boom's orientation; α does not denote geographical latitude.

Choose $\mathbf{d}_2$ along the boom. A payload of mass m at distance $r(t) > 0$ from the pivot then has position

$$O'P = \mathbf{d}Y, \qquad Y = \begin{bmatrix} 0 \\ r(t) \\ 0 \end{bmatrix}.$$

Thus the payload moves along $\mathbf{d}_2$ as the boom extends or retracts, while $\theta(t)$ and $\alpha(t)$ allow the boom to change direction relative to the station.

Its position expressed in the station frame is

$$X = R_B Y = r \begin{bmatrix} -\sin\theta\cos\alpha \\ \cos\theta\cos\alpha \\ \sin\alpha \end{bmatrix}.$$

Since $\mathbf{c} = \mathbf{e}R_o$, the boom frame satisfies

$$\mathbf{d} = \mathbf{e}R_o R_B = \mathbf{e}R, \qquad R = R_o R_B, \qquad x = o + RY.$$

All representations below are expressed in $\mathbf{d}$ unless otherwise indicated.

Angular velocity of the boom frame:

Since rotations about the same axis add,

$$R = R_1(\lambda)R_3(\psi+\theta)R_1(\alpha).$$

The inclination λ is constant. Differentiating and multiplying by R^T therefore gives

$$\widehat{\Omega} = R^T\dot{R} = R_1(\alpha)^T\left(R_3(\psi+\theta)^T\frac{d}{dt}R_3(\psi+\theta)\right)R_1(\alpha) + R_1(\alpha)^T\dot{R}_1(\alpha).$$

For brevity, define

$$\nu = \omega_o + \dot{\theta}, \qquad \dot{\nu} = \ddot{\theta}.$$

Using $\dot{\psi} = \omega_o$, we obtain

$$\widehat{\Omega} = \begin{bmatrix} 0 & -\nu\cos\alpha & \nu\sin\alpha \\ \nu\cos\alpha & 0 & -\dot{\alpha} \\ -\nu\sin\alpha & \dot{\alpha} & 0 \end{bmatrix},$$

and hence

$$\boxed{\Omega = \begin{bmatrix} \dot{\alpha} \\ \nu\sin\alpha \\ \nu\cos\alpha \end{bmatrix}.}$$

Thus the boom's angular velocity includes both the station's orbital rotation and its own motion relative to the station. Its derivative is

$$\dot{\Omega} = \begin{bmatrix} \ddot{\alpha} \\ \ddot{\theta}\sin\alpha + \nu\dot{\alpha}\cos\alpha \\ \ddot{\theta}\cos\alpha - \nu\dot{\alpha}\sin\alpha \end{bmatrix}.$$

Motion of the pivot:

In the station frame,

$$R_o^T \dot{o} = \begin{bmatrix} 0 \\ a\omega_o \\ 0 \end{bmatrix}, \qquad R_o^T \ddot{o} = \begin{bmatrix} -a\omega_o^2 \\ 0 \\ 0 \end{bmatrix}.$$

Transforming these into the boom frame gives

$$V_o \triangleq R^T \dot{o} = a\omega_o \begin{bmatrix} \sin\theta \\ \cos\theta\cos\alpha \\ -\cos\theta\sin\alpha \end{bmatrix},$$

and

$$A_o \triangleq R^T \ddot{o} = -a\omega_o^2 \begin{bmatrix} \cos\theta \\ -\sin\theta\cos\alpha \\ \sin\theta\sin\alpha \end{bmatrix}.$$

These terms must be retained because the pivot moves along the station's orbit.

Payload velocity and angular momentum:

The payload's position and relative motion are

$$Y = \begin{bmatrix} 0 \\ r \\ 0 \end{bmatrix}, \qquad \dot{Y} = \begin{bmatrix} 0 \\ \dot{r} \\ 0 \end{bmatrix}, \qquad \ddot{Y} = \begin{bmatrix} 0 \\ \ddot{r} \\ 0 \end{bmatrix}.$$

Its inertial velocity expressed in **d** is

$$\begin{aligned} V = R^T \dot{x} &= V_o + \Omega \times Y + \dot{Y} \\ &= \begin{bmatrix} a\omega_o \sin\theta - r\nu\cos\alpha \\ a\omega_o\cos\theta\cos\alpha + \dot{r} \\ -a\omega_o\cos\theta\sin\alpha + r\dot{\alpha} \end{bmatrix}. \end{aligned}$$

The moment-of-inertia matrix about O' is

$$\mathbb{I} = -m\widehat{Y}^2 = mr^2 \begin{bmatrix} 1\,0\,0 \\ 0\,0\,0 \\ 0\,0\,1 \end{bmatrix}.$$

The angular momentum about O', defined using inertial velocity and expressed in **d**, is

$$\begin{aligned} \Pi = R^T \pi &= \mathbb{I}\Omega + mY \times (V_o + \dot{Y}) \\ &= mY \times V. \end{aligned}$$

Therefore,

$$\boxed{\Pi = m\begin{bmatrix} r^2\dot{\alpha} - ar\omega_o\cos\theta\sin\alpha \\ 0 \\ r^2\nu\cos\alpha - ar\omega_o\sin\theta \end{bmatrix}.}$$

The terms containing $a\omega_o$ arise from the inertial velocity of the pivot. Although $Y\times\dot{Y}=0$, the changing extension $r(t)$ changes the payload's angular momentum.

Rate of change of angular momentum:

Writing $\Pi = [\Pi_1\ 0\ \Pi_3]^T$, differentiation gives

$$\dot{\Pi}_1 = m\Big[2r\dot{r}\dot{\alpha} + r^2\ddot{\alpha} - a\omega_o\big(\dot{r}\cos\theta\sin\alpha - r\dot{\theta}\sin\theta\sin\alpha + r\dot{\alpha}\cos\theta\cos\alpha\big)\Big],$$
$$\dot{\Pi}_3 = m\Big[2r\dot{r}\nu\cos\alpha + r^2\ddot{\theta}\cos\alpha - r^2\nu\dot{\alpha}\sin\alpha - a\omega_o\big(\dot{r}\sin\theta + r\dot{\theta}\cos\theta\big)\Big].$$

Since $\pi = R\Pi$, the inertial rate of change expressed in **d** is

$$R^T\dot{\pi} = \dot{\Pi} + \Omega\times\Pi = \begin{bmatrix} \dot{\Pi}_1 + \nu\sin\alpha\,\Pi_3 \\ \nu\cos\alpha\,\Pi_1 - \dot{\alpha}\Pi_3 \\ \dot{\Pi}_3 - \nu\sin\alpha\,\Pi_1 \end{bmatrix}.$$

If T denotes the total interaction-force moment about O' expressed in **d**, the moving-origin balance gives

$$\boxed{T = \dot{\Pi} + \Omega\times\Pi + V_o\times(mV).}$$

Thus neither $\dot{\Pi}$ nor $R^T\dot{\pi}$ alone is the interaction-force moment about the moving pivot.

Applying the angular momentum balance:

The results of this section gives the equivalent expression

$$T = \mathbb{I}\dot{\Omega} + \Omega\times(\mathbb{I}\Omega) + mY\times\big(\ddot{Y} + 2\Omega\times\dot{Y}\big) + mY\times A_o.$$

Here $Y\times\ddot{Y}=0$, and the remaining terms are

$$\mathbb{I}\dot{\Omega} = mr^2\begin{bmatrix} \ddot{\alpha} \\ 0 \\ \ddot{\theta}\cos\alpha - \nu\dot{\alpha}\sin\alpha \end{bmatrix},$$

$$\Omega\times(\mathbb{I}\Omega) = mr^2\begin{bmatrix} \nu^2\sin\alpha\cos\alpha \\ 0 \\ -\nu\dot{\alpha}\sin\alpha \end{bmatrix},$$

and

$$2mY\times(\Omega\times\dot{Y}) = 2mr\dot{r}\begin{bmatrix} \dot{\alpha} \\ 0 \\ \nu\cos\alpha \end{bmatrix}.$$

Consequently,

$$\boxed{T = mr\begin{bmatrix} A_{o3} + r\ddot{\alpha} + 2\dot{r}\dot{\alpha} + r\nu^2 \sin\alpha\cos\alpha \\ 0 \\ -A_{o1} + r\ddot{\theta}\cos\alpha + 2\dot{r}\nu\cos\alpha - 2r\nu\dot{\alpha}\sin\alpha \end{bmatrix}}.$$

There is no moment component along the boom because $T = Y \times F$ and Y lies along $\mathbf{d}_2$. This statement concerns the particle payload; it does not account for a finite payload's rotational inertia.

Total force and the force exerted by the boom:

The payload's inertial acceleration expressed in $\mathbf{d}$ is

$$A \triangleq R^T\ddot{x} = A_o + \ddot{Y} + 2\Omega \times \dot{Y} + \dot{\Omega} \times Y + \Omega \times (\Omega \times Y).$$

Its components are

$$\begin{aligned} A_1 &= A_{o1} - 2\dot{r}\nu\cos\alpha - r\ddot{\theta}\cos\alpha + 2r\nu\dot{\alpha}\sin\alpha, \\ A_2 &= A_{o2} + \ddot{r} - r\left(\dot{\alpha}^2 + \nu^2\cos^2\alpha\right), \\ A_3 &= A_{o3} + 2\dot{r}\dot{\alpha} + r\ddot{\alpha} + r\nu^2\sin\alpha\cos\alpha. \end{aligned}$$

Newton's law gives $F = mA$, and hence

$$T = Y \times F = mr\begin{bmatrix} A_3 \\ 0 \\ -A_1 \end{bmatrix},$$

in agreement with the angular momentum calculation.

To separate gravity from the boom force, define the Earth-centered position expressed in $\mathbf{d}$:

$$Z \triangleq R^T x = R^T o + Y = \begin{bmatrix} a\cos\theta \\ r - a\sin\theta\cos\alpha \\ a\sin\theta\sin\alpha \end{bmatrix}.$$

Its length is

$$\rho = \|Z\| = \sqrt{a^2 + r^2 - 2ar\sin\theta\cos\alpha}.$$

Thus gravity has representation

$$F_g = -\frac{m\mu}{\rho^3}Z.$$

Assuming that gravity and the boom force are the only forces on the payload,

$$\boxed{F_{\text{boom}} = mA + \frac{m\mu}{\rho^3}Z.}$$

The moment exerted by the boom on the payload about O' is therefore

$$\boxed{T_{\text{boom}} = Y \times F_{\text{boom}} = mr \begin{bmatrix} A_3 + \mu Z_3/\rho^3 \\ 0 \\ -A_1 - \mu Z_1/\rho^3 \end{bmatrix}.}$$

Unlike the purely radial configuration, gravity generally has a nonzero moment about the station's pivot. Thus T_{boom} must be distinguished from the total moment T.

The force and moment exerted by the boom on the payload are F_{boom} and T_{boom}. The payload exerts the opposite force and moment on the boom. These reactions must be included when determining the loads transmitted to the station.

Kinetic energy:

The payload's kinetic energy relative to the inertial frame is

$$\text{KE} = \frac{1}{2} m \|V\|^2.$$

Using the velocity components above,

$$\boxed{\text{KE} = \frac{m}{2}\Big[\big(a\omega_o \sin\theta - r\nu\cos\alpha\big)^2 + \big(a\omega_o\cos\theta\cos\alpha + \dot{r}\big)^2 + \big(-a\omega_o\cos\theta\sin\alpha + r\dot{\alpha}\big)^2\Big].}$$

The quantity $\frac{1}{2}m\dot{r}^2$ accounts only for the payload's motion relative to the boom. The full inertial kinetic energy also includes the motion of the pivot and the rotation of the boom.

Short boom approximation:

For a boom much shorter than the orbital radius, $r \ll a$. For $r \ll a$,

$$\rho \approx a - r\sin\theta\cos\alpha.$$

Expanding ρ^{-3} to first order in r/a and substituting the expressions for Z and A gives

$$F_{\text{boom}} \approx m \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix}, \qquad T_{\text{boom}} \approx mr \begin{bmatrix} f_3 \\ 0 \\ -f_1 \end{bmatrix},$$

where, with $\nu = \omega_o + \dot{\theta}$,

$$\begin{aligned} f_1 &= -2\dot{r}\nu\cos\alpha - r\ddot{\theta}\cos\alpha + 2r\nu\dot{\alpha}\sin\alpha + 3r\omega_o^2\sin\theta\cos\theta\cos\alpha, \\ f_2 &= \ddot{r} - r\big(\dot{\alpha}^2 + \nu^2\cos^2\alpha\big) + r\omega_o^2\big(1 - 3\sin^2\theta\cos^2\alpha\big), \end{aligned}$$

$$f_3 = 2\dot{r}\dot{\alpha} + r\ddot{\alpha} + r\nu^2 \sin\alpha\cos\alpha + 3r\omega_o^2 \sin^2\theta \sin\alpha\cos\alpha.$$

The terms proportional to $a\omega_o^2$ cancel between gravity and the acceleration of the pivot. The kinetic energy requires no approximation and can be written as

$$\mathrm{KE} = \frac{m}{2}\Big[a^2\omega_o^2 + \dot{r}^2 + r^2\big(\dot{\alpha}^2 + \nu^2\cos^2\alpha\big) + 2a\omega_o\dot{r}\cos\theta\cos\alpha \\ - 2ar\omega_o\big(\nu\sin\theta\cos\alpha + \dot{\alpha}\cos\theta\sin\alpha\big)\Big].$$

For an illustrative slow maneuver, take $a = 6.77\times10^6\,\mathrm{m}$, $\omega_o = 1.1334\times10^{-3}\,\mathrm{s}^{-1}$, $m = 10{,}000\,\mathrm{kg}$, and $r = 15\,\mathrm{m}$. Since $r/a \approx 2.2\times10^{-6}$, the approximation $r \ll a$ is appropriate. At an instant when $\theta = -45^\circ$, $\alpha = 30^\circ$, $\dot{\theta} = \dot{\alpha} = 0.1^\circ/\mathrm{s}$, $\dot{r} = 0.02\,\mathrm{m/s}$, $\ddot{r} = 0.001\,\mathrm{m/s^2}$, and $\ddot{\theta} = \ddot{\alpha} = 0$, the preceding approximations give, in the boom frame,

$$F_{\mathrm{boom}} \approx \begin{bmatrix} -0.494 \\ 8.587 \\ 1.362 \end{bmatrix}\mathrm{N}, \qquad T_{\mathrm{boom}} \approx \begin{bmatrix} 20.42 \\ 0 \\ 7.41 \end{bmatrix}\mathrm{N\,m}.$$

The loads are modest because the motion is slow. Starting and stopping can require substantially larger moments: increasing $\dot{\alpha}$ from zero to $0.1^\circ/\mathrm{s}$ uniformly over $10\,\mathrm{s}$ gives $\ddot{\alpha} \approx 1.745\times10^{-4}\,\mathrm{s}^{-2}$. At $r = 15\,\mathrm{m}$, the resulting contribution $mr^2\ddot{\alpha}$ to the first moment component is approximately $393\,\mathrm{N\,m}$. Thus the acceleration phases can impose much greater loads than steady slow slewing. These values illustrate the model and do not represent a specified ISS maneuver.

Check: an outward radial boom:

The earlier radial-transfer example is recovered by holding

$$\theta = -\frac{\pi}{2}, \qquad \alpha = 0.$$

Then $\mathbf{d}_2 = \mathbf{c}_1$, $\nu = \omega_o$, and $\rho = a + r$. The boom force becomes

$$F_{\mathrm{boom}} = m\begin{bmatrix} -2\omega_o\dot{r} \\ \ddot{r} - \omega_o^2(a+r) + \dfrac{\mu}{(a+r)^2} \\ 0 \end{bmatrix},$$

while

$$T_{\mathrm{boom}} = \begin{bmatrix} 0 \\ 0 \\ 2mr\omega_o\dot{r} \end{bmatrix}.$$

The negative first component of the force points along the orbital direction, since in this configuration $\mathbf{d}_1 = -\mathbf{c}_2$. The kinetic energy reduces to

$$\mathrm{KE} = \frac{1}{2}m\left[\dot{r}^2 + \omega_o^2(a+r)^2\right],$$

as expected.

1.2.6.2 A System of Interacting Particles

Consider a system of N particles. The resultant interaction force on particle P_i, expressed in **b**, is

$$F_i = F_i^{\mathrm{e}} + \sum_{j\neq i} F_{ij},$$

where F_i^{e} arises from interactions with particles outside the system. The internal forces are assumed to be equal and opposite and directed along the lines joining the interacting particles. Thus their total force and total moment vanish:

$$\sum_i \sum_{j\neq i} F_{ij} = 0, \qquad \sum_i X_i \times \sum_{j\neq i} F_{ij} = \sum_{i<j} (X_i - X_j) \times F_{ij} = 0.$$

Define

$$M = \sum_i m_i, \qquad \bar{X} = \frac{1}{M}\sum_i m_i X_i, \qquad \bar{x} = o + R\bar{X}, \qquad \mathbb{I} = \sum_i \mathbb{I}_i.$$

Summing the individual particle momenta gives total angular momentum about O', both expressed in **e**:

$$\pi = R\left(\mathbb{I}\Omega + M\bar{X} \times (R^T \dot{o}) + \sum_i m_i X_i \times \dot{X}_i\right).$$

Let $T^{\mathrm{e}} = \sum_i X_i \times F_i^{\mathrm{e}}$ be the total external force moment about O', expressed in **b**. Summing the angular momentum equations obtained in (1.49) we have

$$\mathbb{I}\dot{\Omega} + \Omega \times (\mathbb{I}\Omega) + \sum_i m_i X_i \times \left(\ddot{X}_i + 2\Omega \times \dot{X}_i\right) = T^{\mathrm{e}} - M\bar{X} \times (R^T \ddot{o}).$$

Recall that (1.37) describes the motion of the center of mass of a collection of particles in a moving frame **b**. These expressions allow the particles to move relative to **b**, so $\mathbb{I}$ may depend on time.

In summary for a system of N particles, define the total mass, center of mass coordinates, and moment of inertia about O' by

$$M = \sum_i m_i, \qquad \bar{X} = \frac{1}{M}\sum_i m_i X_i, \qquad \bar{x} = o + R\bar{X}, \qquad \mathbb{I} = \sum_i \mathbb{I}_i.$$

The total angular momentum about O' expressed in **b**, $\Pi = R^T\pi$, is

$$\Pi = \left(\mathbb{I}\Omega + M\bar{X} \times (R^T \dot{o}) + \sum_i m_i X_i \times \dot{X}_i\right). \tag{1.50}$$

For equal-and-opposite central internal forces, the corresponding equations of motion in **b** are

$$M\ddot{\bar{X}} = F^{\mathrm{e}} - MR^T\ddot{o} - M\widehat{\Omega}^2\bar{X} - 2M\widehat{\Omega}\dot{\bar{X}} - M\dot{\widehat{\Omega}}\bar{X}, \tag{1.51}$$

$$\mathbb{I}\dot{\Omega} = (\mathbb{I}\Omega)\times\Omega - \sum_i m_i X_i \times \left(\ddot{X}_i + 2\Omega\times\dot{X}_i\right) - M\bar{X}\times(R^T\ddot{o}) + T^{\mathrm{e}}. \tag{1.52}$$

Here F^{e} and T^{e} denote the total external force and its moment about O', both expressed in **b**; the superscript e indicates external interactions.

Notice that for a collection of particles where the relative distance between any two particles remain fixed. Such an object is called a *rigid body*. In this case when the frame **b** is fixed to the body $\dot{X}_i = \ddot{X}_i = 0$ and the expression (1.50) and (1.52) take a particularly simple form:

$$\Pi = \left(\mathbb{I}\Omega + M\bar{X}\times(R^T\dot{o})\right). \tag{1.53}$$

$$\mathbb{I}\dot{\Omega} = (\mathbb{I}\Omega)\times\Omega - M\bar{X}\times(R^T\ddot{o}) + T^{\mathrm{e}}. \tag{1.54}$$

In this case since X_i is a constant so is the moment of inertia tensor $\mathbb{I}$. The last equation is what is famously known as *Euler's rigid body equations*.

1.3 Exercises

Exercise 1.1. Let $\mathbf{e}$ and $\mathbf{e}'$ be two inertial observers who use common units of measurement, the same positive temporal orientation, and right-handed orthonormal spatial frames. Let $(t,x) \in \mathbb{R} \times \mathbb{R}^3$ and $(\tau,\xi) \in \mathbb{R} \times \mathbb{R}^3$ be their respective coordinate representations of the same space-time event A. Assume that the coordinate transformation is twice continuously differentiable. Using the four Galilean space-time assumptions, establish the following results:

(i) Show that independence of time from space and agreement on simultaneity and elapsed time imply

$$\tau = t + a,$$

where a is constant.

(ii) The fourth assumption requires every trajectory $x(t) = x_0 + ut$ with constant velocity in one frame to have constant velocity in the other. Use this condition, for arbitrary $x_0, u \in \mathbb{R}^3$, to show that all second partial derivatives of $\xi(t,x)$ vanish. Hence show that

$$\xi = \alpha + \beta t + Rx,$$

where $\alpha, \beta \in \mathbb{R}^3$ and $R \in \mathbb{R}^{3\times 3}$ are constant.

(iii) Use agreement on distances between simultaneous events to show that

$$R^T R = RR^T = I.$$

Explain why the right-handed-frame convention additionally gives $\det R = 1$.

(iv) Let $o(t)$ denote the position coordinates of the origin O' of $\mathbf{e}'$, measured by $\mathbf{e}$. Show that

$$v = \dot{o}(t) = -R^T \beta$$

is constant, and explain why the two frames have no relative rotation.
Finally, if their clocks are synchronized ($a = 0$) and their spatial origins coincide at $t = 0$ ($\alpha = 0$), show that the same event is represented by

$$(\tau,\xi) = \big(t, R(x - vt)\big).$$

Exercise 1.2. Answer the following with clear justification.

(i) Show that the space of 3×3 special orthogonal matrices, $SO(3)$, is not a vector space under matrix addition and scalar multiplication by a real number.

(ii) A space-time event has representations (t,x) and $(t, R(x - vt))$ with respect to two inertial observers $\mathbf{e}$ and $\mathbf{b}$, respectively, where $R \in SO(3)$ and $v \in \mathbb{R}^3$ are constant. Let P, Q, S be three non-collinear points fixed with respect to $\mathbf{e}$. Show that the triangle PQS observed by $\mathbf{e}$ is congruent to the triangle PQS observed by $\mathbf{b}$ at every time instant t.

(iii) Let $R \in SO(3)$. Show that the one-to-one and onto map on $\mathbb{R}^3$ defined by $x \mapsto Rx$ is isometric.

(iv) Let P, Q, S be three non-collinear points fixed in an orthonormal frame $\mathbf{e}$, with representations x_P, x_Q, x_S, respectively. Let $R : [0,\infty) \to SO(3)$, and let P_R, Q_R, S_R have representations $R(t)x_P, R(t)x_Q, R(t)x_S$. Find an expression for the smaller, unoriented angle between the plane through P, Q, S and the plane through P_R, Q_R, S_R, taking this angle in $[0, \pi/2]$.

(v) Consider an orthonormal frame **e** with origin O. Let P,Q be two points such that O,P,Q are non-collinear, with representations x_P, x_Q. Let $R : [0,\infty) \to SO(3)$, and let P_R, Q_R have representations $R(t)x_P, R(t)x_Q$. When the planes through O,P,Q and O,P_R,Q_R are distinct, find an explicit expression for their line of intersection. Identify the condition under which the planes coincide.

Exercise 1.3. A bead that is constrained to move on a frictionless wire that lies on a horizontal plane with respect to an inertial frame. The wire is bent to a shape of a parabola (ie. $y = x^2$ and consider the coordinates to be non-dimnsional). Find the constraint forces that keep the bead on the wire and the equation of motion of the bead. Assume that the bead does not interact with any other objects other than the wire.

Exercise 1.4. Consider the 1-DOF electrostatic MEMS mirror model shown in figure 1.17. Find the governing differential equations of the system. Note that the capacitance between two parallel plate capacitors are given by $c(l(t)) = \frac{\varepsilon A}{l(t)}$ and the attractive coulomb force between the two plates is given by $f_e(t) = \frac{q(t)^2}{2\varepsilon A}$ where A is the cross sectional area of the plates.

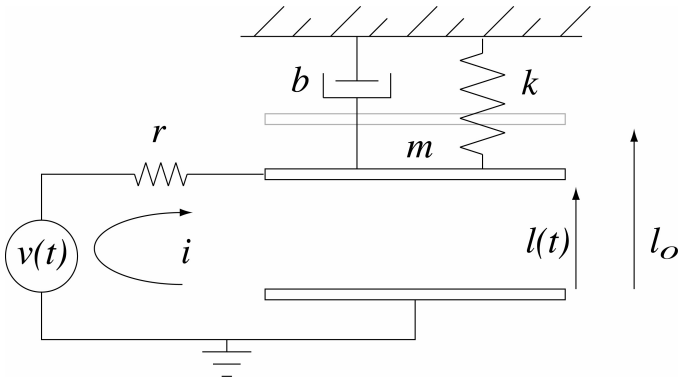


Fig. 1.17 1-DOF Electrostatic MEMS model. Let l_0 be the zero voltage gap and let $l(t)$ be the gap length, $v(t)$ be the applied source voltage, and $i(t)$ be the current in the circuit at a given time t.

Exercise 1.5. Consider the coupled spring-mass-damper system shown in Figure-1.18. Assume that the friction between the rollers and the horizontal surfaces is negligibly small. Take the displacements from the equilibrium configuration of the system.

(i) Find the equations of motion of the system.
(ii) Show that the system can be written in the form:

$$M\ddot{X} + C\dot{X} + KX = 0$$

where M, C and K are 3×3 matrices.
(iii) Show that the system can be written down in the form

$$\dot{Y} = AY$$

where $Y = [X^\top \ \ \dot{X}^\top]^\top$ and A is 6×6 matrix.

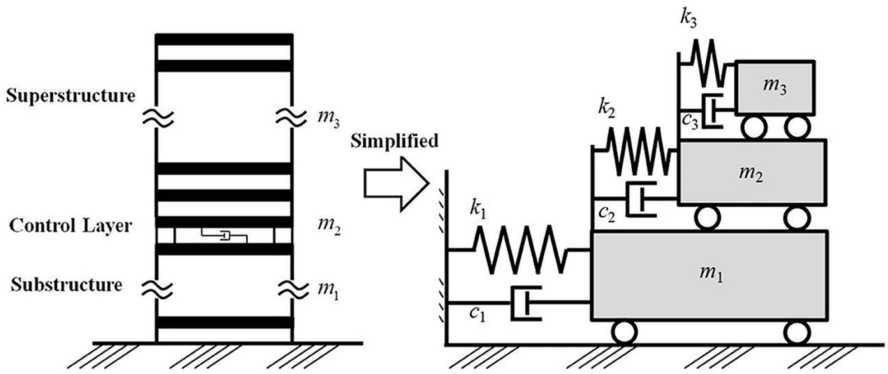


Fig. 1.18 A structural vibration model using a 3-DOF spring–mass–damper approximation.

(iv) Write down the analytical solution of the system when only the third mass (the mass denoted by m_3) is given an initial displacement of one unit and released with zero velocity.

Exercise 1.6 (The Three-Particle Universe). Consider an isolated system of three interacting particles P_1, P_2, P_3 with constant positive masses m_1, m_2, m_3. Let $x_i(t)$ denote the position of P_i in an inertial frame **e**, and let f_{ij} be the force on P_i due to P_j. Assume the pairwise forces are additive, equal and opposite, and central:

$$m_i\ddot{x}_i = \sum_{j\neq i} f_{ij}, \qquad f_{ij} = -f_{ji}, \qquad (x_i - x_j)\times f_{ij} = 0.$$

Regard P_1, P_2 as a subsystem of interest and P_3 as a simple model of its surroundings.

(i) Let

$$M_u = m_1 + m_2 + m_3, \qquad \bar{x}_u = \frac{m_1x_1 + m_2x_2 + m_3x_3}{M_u}.$$

Show that the total linear momentum is conserved and that $\ddot{\bar{x}}_u = 0$.

(ii) For the two-particle subsystem, define

$$M = m_1 + m_2, \qquad \bar{x} = \frac{m_1x_1 + m_2x_2}{M}.$$

Show that

$$M\ddot{\bar{x}} = f_{13} + f_{23}.$$

Explain why the subsystem's linear momentum need not be conserved even though the three-particle system is isolated.

(iii) Let O' be a possibly moving point with representation $o(t)$ in **e**. Define the total angular momentum of the three particles about O' by

$$\pi_u = \sum_{i=1}^{3} (x_i - o)\times m_i\dot{x}_i.$$

Show that

$$\dot{\pi}_u = -\dot{o} \times M_u \dot{\bar{x}}_u.$$

Identify where centrality of the pairwise forces is used. Deduce that the total angular momentum about the three-particle center of mass is conserved.

(iv) Define the angular momentum of the two-particle subsystem about its own center of mass by

$$\pi_{12} = \sum_{i=1}^{2} (x_i - \bar{x}) \times m_i \dot{x}_i.$$

Show that

$$\dot{\pi}_{12} = (x_1 - \bar{x}) \times f_{13} + (x_2 - \bar{x}) \times f_{23}.$$

Explain why the motion of $\bar{x}$ introduces no additional term in this balance.

(v) Suppose that $f_{13} + f_{23} = 0$ throughout an interval. Using the centrality of the interactions with P_3, show that $\dot{\pi}_{12} = 0$ throughout that interval.
Does conservation of π_{12} imply that the particles maintain a constant separation or rotate with constant angular speed? Justify your answer.

(vi) *Optional computational investigation.* Suppose the particles interact through Newtonian gravity:

$$f_{ij} = G m_i m_j \frac{x_j - x_i}{\|x_j - x_i\|^3}.$$

Derive the equations of motion and simulate them for distinct initial positions, choosing initial velocities such that the total linear momentum is zero. Use an interval over which no collision occurs.
Plot the particle paths and the centers of mass $\bar{x}_u(t)$ and $\bar{x}(t)$. Check the numerical conservation of total linear momentum, angular momentum about $\bar{x}_u$, and mechanical energy

$$E = \sum_{i=1}^{3} \frac{1}{2} m_i \|\dot{x}_i\|^2 - \sum_{i<j} \frac{G m_i m_j}{\|x_i - x_j\|}.$$

Compare the behavior of the isolated system with that of the two-particle subsystem. The special condition in part (v) is not imposed in this simulation.

Exercise 1.7. — Properties of a Rotation Matrix: Let $\mathbf{e}$ and $\mathbf{b}$ be two orthonormal, right handed oriented frames with coinciding origin O. Let $\mathbf{b} = \mathbf{e}R$ be the relationship between the two frames. Show that $R^T R = R R^T = I_{3\times 3}$ and $\det(R) = 1$.

Exercise 1.8. The three rotated frames $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are related to a fixed frame $\mathbf{e}$ as shown in figure 1.19. All frames are orthonormal. Let $\mathbf{a} = \mathbf{e}R_1(\theta_1)$, $\mathbf{b} = \mathbf{e}R_2(\theta_2)$, and $\mathbf{c} = \mathbf{e}R_3(\theta_3)$. Show using direct calculations the following

$$R_1(\theta_1) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_1 & -\sin\theta_1 \\ 0 & \sin\theta_1 & \cos\theta_1 \end{bmatrix}, \; R_2(\theta_2) = \begin{bmatrix} \cos\theta_2 & 0 & \sin\theta_2 \\ 0 & 1 & 0 \\ -\sin\theta_2 & 0 & \cos\theta_2 \end{bmatrix}, \; R_3(\theta_3) = \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 & 0 \\ \sin\theta_3 & \cos\theta_3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and

$$R_1^T\dot{R}_1 = \widehat{\Omega}_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\dot{\theta}_1 \\ 0 & \dot{\theta}_1 & 0 \end{bmatrix}, \quad R_2^T\dot{R}_2 = \widehat{\Omega}_2 = \begin{bmatrix} 0 & 0 & \dot{\theta}_2 \\ 0 & 0 & 0 \\ -\dot{\theta}_2 & 0 & 0 \end{bmatrix}, \quad R_3^T\dot{R}_3 = \widehat{\Omega}_3 = \begin{bmatrix} 0 & -\dot{\theta}_3 & 0 \\ \dot{\theta}_3 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

and

$$\widehat{\Omega}_1^2 = -\dot{\theta}_1^2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \widehat{\Omega}_2^2 = -\dot{\theta}_2^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \widehat{\Omega}_3^2 = -\dot{\theta}_3^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

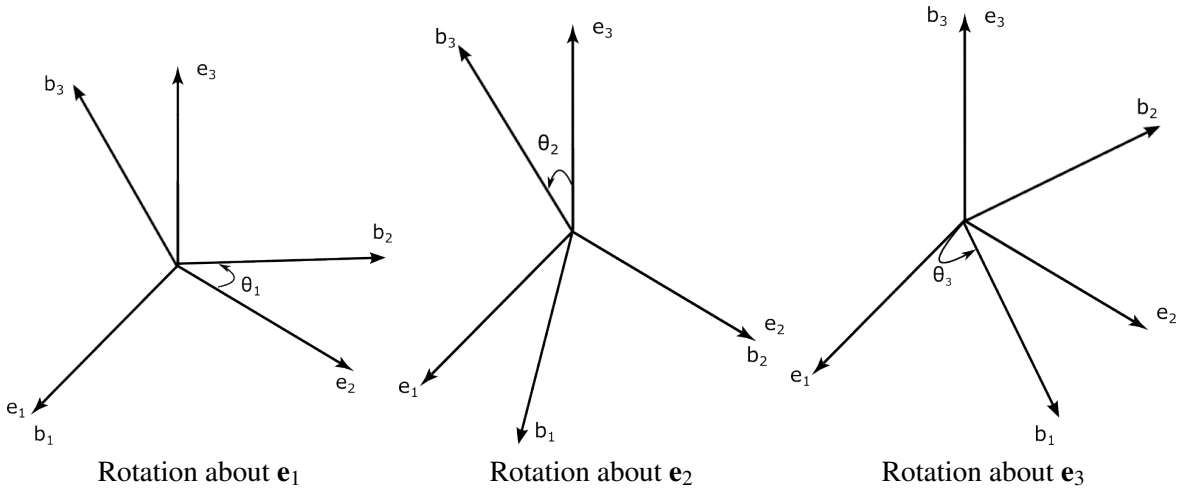


Fig. 1.19 Rotated Frames

Exercise 1.9. Show that the space of 3×3 skew-symmetric matrices is a 3-dimensional vector space and that it can be identified with $\mathbb{R}^3$ using the identification $\Omega \mapsto \widehat{\Omega}$ where

$$\widehat{\Omega} = \begin{bmatrix} 0 & -\Omega_3 & \Omega_2 \\ \Omega_3 & 0 & -\Omega_1 \\ -\Omega_2 & \Omega_1 & 0 \end{bmatrix},$$

and $\Omega = [\Omega_1 \;\; \Omega_2 \;\; \Omega_3]^T$. That is show that the map $\widehat{\;}: \mathbb{R}^3 \to so(3)$ is a one-to-one and onto map.

Exercise 1.10. Show that the cross product in $\mathbb{R}^3$ satisfies

$$\Omega \times X = \widehat{\Omega} X,$$

where $\Omega, X \in \mathbb{R}^3$ and $\widehat{\;}: \mathbb{R}^3 \to so(3)$ is the one-to-one and onto map discussed in exercise-1.9.

Exercise 1.11. Prove the following properties of the cross product in $\mathbb{R}^3$.

(i) $A \times B = -B \times A$
(ii) $A \times (B \times C) + B \times (C \times A) + C \times (A \times B) = 0$

(iii) $A\cdot(B\times C)=C\cdot(A\times B)=B\cdot(C\times A)$
(iv) $A\times(B\times C)=(A\cdot C)B-(B\cdot A)C$
(v) $(\Omega\times X)\cdot(\omega\times X)=(\Omega\cdot\omega)||X||^2-(\Omega\cdot X)(\omega\cdot X)$
(vi) $\Omega\cdot(\Omega\times X)=0=X\cdot(\Omega\times X)$

Here we use the notation $A\cdot B$ to denote the usual Euclidean inner product in $\mathbb{R}^3$.

Exercise 1.12. Using the last property proven in exercise-1.11 show that $(\Omega\times X)$ is perpendicular to both X and Ω.

Exercise 1.13. Show using exercise 1.11 part iv (or direct verification) that

$$\widehat{X}^2=XX^T-||X||^2I_{3\times 3}.$$

for any $X\in\mathbb{R}^3$.

Exercise 1.14. If $R\in SO(3)$, and X and Y are 3×1 matrices show that the following identities hold:

$$R(X\times Y)=RX\times RY$$
$$\widehat{RX}=R\widehat{X}R^T.$$

Exercise 1.15. — Einstein's box experiment: Let g be the magnitude of the gravitational acceleration. Explain why a person standing on a scale inside an elevator obtains twice the usual scale reading when the elevator accelerates upwards at g, and a zero reading when it accelerates downwards at g. Assume that the person remains at rest relative to the elevator.

Also show that, if gravity were absent but the elevator accelerated upwards at g, the scale would read mg, where m is the person's mass. This equals the usual scale reading in a stationary elevator under gravity. Explain why the scale reading alone cannot distinguish these two situations.

Exercise 1.16. A particle P of mass m is constrained to move on the surface of a sphere of radius r and origin O by attaching P to O using a tight in-elastic wire . Assume that O is fixed in an inertial frame $\mathbf{e}$ (refer to figure 1.20), and that the wire is massless and remains taut. Except for the tension of the wire no other external forces act on the particle. Let frame e be a frame with origin coinciding with O and fixed with respect to the sphere. The representation of the point P with respect to e is x. Answer the following:

(i) Show that the velocity of the particle is always tangential to the sphere (orthogonal to the position x, ie. $x^T\dot{x}=0$).
(ii) Differentiating the constraint $x^T\dot{x}=0$ show that the motion of the particle in the e frame is described by
$$\ddot{x}=\frac{1}{m}f_c(t)=-\frac{||\dot{x}||^2}{r^2}x,$$
where $f_c(t)$ is the representation of the tension force in the wire.
(iii) Write down the equations of motion as observed in a frame $\mathbf{b}(t)$ with origin coinciding with that of $\mathbf{e}$ and is moving with respect to the fixed frame $\mathbf{e}$ such that the axis $\mathbf{b}_3(t)$ always aligns along the direction of the particle.

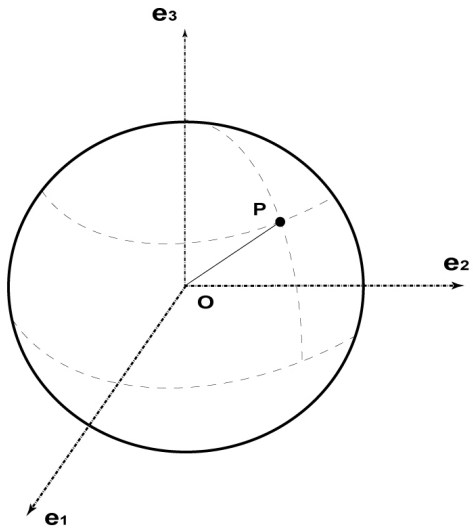


Fig. 1.20 A particle constrained to move on a sphere of radius r by a taut, massless, inextensible wire attached to the fixed center O.

Exercise 1.17. Particle motion in a circle: Consider a particle of mass m constrained to move on a fixed, frictionless circular wire of radius r. Assume that the only force acting on the particle is the constraint force exerted by the wire. Let $\mathbf{e} = [\mathbf{e}_1\ \mathbf{e}_2\ \mathbf{e}_3]$ be an inertial orthonormal frame with origin at the center of the circle and $\mathbf{e}_3$ perpendicular to its plane. Let x be the representation of the particle in $\mathbf{e}$.

(i) Using the results of Exercise 1.16, show that the velocity $\dot{x}$ is tangential to the circle.
(ii) Choose a frame $\mathbf{b}(t)$ with the same origin, with $\mathbf{b}_1$ pointing towards the particle and $\mathbf{b}_3 = \mathbf{e}_3$. Let ω be its signed angular velocity about $\mathbf{e}_3$ relative to $\mathbf{e}$. Show that

$$\|\dot{x}\| = r|\omega|.$$

(iii) Show that the constraint force is radial and is given by $f_c(t) = -m\omega^2 x$, and hence that

$$\ddot{x} = -\omega^2 x.$$

(iv) Show, using Newton's second law expressed in the $\mathbf{b}(t)$ frame, that ω is constant.
(v) Show that the position components can be written as

$$x_1(t) = r\cos(\omega t + \phi_0), \qquad x_2(t) = r\sin(\omega t + \phi_0),$$

where ϕ_0 is determined by the initial position.

Exercise 1.18. Consider a ball of mass m, modeled as a particle, constrained to move in the frictionless radial slot shown in Figure 1.21. Let $\mathbf{e}$ be an Earth-fixed orthonormal frame, treated as inertial, with origin at the fixed center of the disk and $\mathbf{e}_3$ pointing vertically upwards. The disk-fixed frame $\mathbf{b}$ satisfies

$$\mathbf{b} = \mathbf{e}R_3(\theta(t)),$$

where $\theta(t)$ is prescribed.

The disk and ball are supported on a smooth horizontal surface. Gravity acts in the negative $\mathbf{e}_3$ direction. A massless spring of stiffness k and unstretched length y_0 connects the ball to the disk's center. Let $y(t) > 0$ denote the ball's distance from the center along $\mathbf{b}_1$, so that its position in $\mathbf{b}$ is

$$X = \begin{bmatrix} y(t) \\ 0 \\ 0 \end{bmatrix}.$$

Write down the equation of motion for $y(t)$.

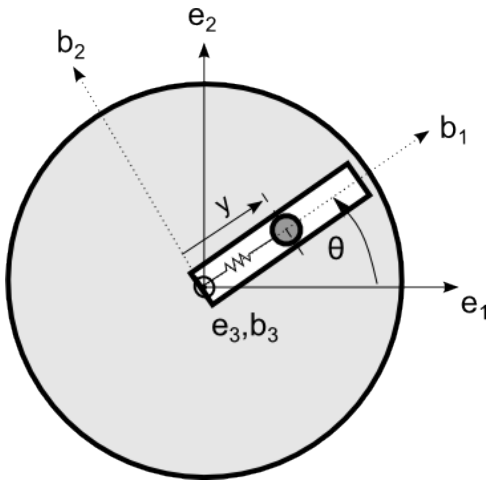


Fig. 1.21 A ball attached to a spring and constrained to a radial slot in a disk rotating about a vertical axis.

Exercise 1.19. Consider a ball of mass m, modeled as a particle, constrained to move in the frictionless straight slot shown in Figure 1.22. Let $\mathbf{e}$ be an Earth-fixed orthonormal frame, treated as inertial, with origin at the fixed center O of the disk and $\mathbf{e}_3$ pointing vertically upwards. Define the intermediate frame $\mathbf{b}$ and disk-fixed frame $\mathbf{c}$ by

$$\mathbf{b} = \mathbf{e}R_3(\theta(t)), \qquad \mathbf{c} = \mathbf{b}R_1(\phi(t)),$$

where $\theta(t)$ and $\phi(t)$ are prescribed. Thus the disk rotates about the vertical axis $\mathbf{e}_3$ and spins about its normal $\mathbf{c}_1 = \mathbf{b}_1$.

The slot is parallel to $\mathbf{c}_2$ and has a constant offset d along $\mathbf{c}_3$. Let the ball's position in $\mathbf{c}$ be

$$X = \begin{bmatrix} 0 \\ y(t) \\ d \end{bmatrix}.$$

A massless spring of stiffness k. When the spring is unstretched $y = 0$. Gravity acts vertically downwards with acceleration of magnitude g.

Write down the equation of motion for $y(t)$.

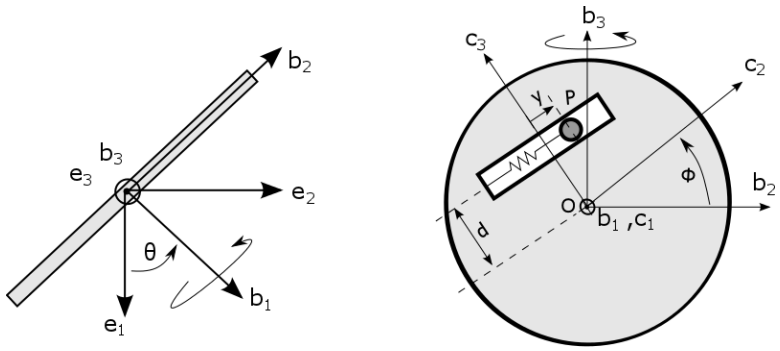


Fig. 1.22 A ball attached to a spring in an offset slot of a disk rotating about a vertical axis and spinning about its normal.

Exercise 1.20. Consider the bead of mass m constrained to move in the straight slot shown in Figure 1.22. Let $\mathbf{e}$ be an inertial frame, $\mathbf{b}$ the intermediate rotating frame, and $\mathbf{c}$ the frame fixed to the disk, with

$$\mathbf{b} = \mathbf{e}R_3(\theta), \qquad \mathbf{c} = \mathbf{b}R_1(\phi).$$

All three frames have their origin at the fixed center O of the disk. The slot is parallel to $\mathbf{c}_2$ and has a constant offset d along $\mathbf{c}_3$. Let $y(t)$ denote the bead's position along the slot, so that $OP = \mathbf{c}X$ where

$$X = \begin{bmatrix} 0 \\ y(t) \\ d \end{bmatrix}.$$

Allow $\theta(t)$, $\phi(t)$, and $y(t)$ to vary with time.

Using the results of Section 1.2.6.1:

(i) Calculate the bead's angular momentum about O, defined using its inertial velocity, and express it in $\mathbf{c}$:

$$\Pi = R^T\pi, \qquad R = R_3(\theta)R_1(\phi).$$

Express your answer in terms of m, d, y, θ, ϕ and their time derivatives.

(ii) Calculate $\dot{\Pi}$ and the representation $R^T\dot{\pi}$ of the inertial rate of change of angular momentum. Explain their relationship and identify which equals the resultant interaction-force moment about O expressed in $\mathbf{c}$.

(iii) Calculate the bead's kinetic energy relative to $\mathbf{e}$ using quantities expressed in $\mathbf{c}$. Explain why it is generally different from $\frac{1}{2}m\|\dot{X}\|^2$.

Exercise 1.21. Consider the mechanical system shown in figure 1.23. The horizontal arm has length d and rotates about the fixed vertical shaft at a prescribed constant angular velocity Ω. An ideal frictionless guide constrains the particle to the rotating vertical plane through the arm's end and perpendicular to the arm. Let $l(t) > 0$ be the spring length and let $\theta(t)$ be its angle from the downward vertical, positive towards the horizontal direction of increasing arm rotation. The mass of the point P is m. The un-stretched length of the spring is l_0. Neglecting friction and the moment of inertia of the spring derive the equations of motion of the system

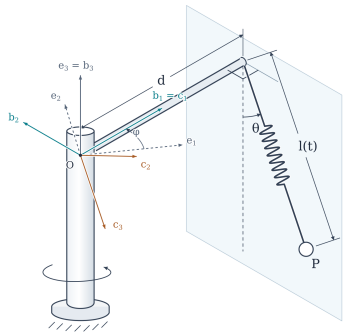


Fig. 1.23 Rotating Spring Pendulum.

Exercise 1.22. Consider the idealized gyroscope model shown in Figure 1.24. The disk rotates in a horizontal plane about its fixed center O. Measure θ positively counterclockwise as viewed from above, and take $\Omega > 0$ for disk rotation in the same sense. Keep this convention fixed when reversing the disk's rotation. Neglect Earth's rotation. A particle P of mass m moves on the disk without friction. Gravity and the vertical supporting force balance.

Let $l(t) = OP$, and let $\theta(t)$ be the angle of OP relative to a reference direction fixed on the disk, measured positively in the same sense as Ω. The outer end O' of a massless spring of stiffness k_l moves freely in the circular slot of radius r. Assume that O, P, O' remain collinear and that $0 < l(t) < r$. The spring is unstretched when $l = l_0$, and its force on P in the outward radial direction is

$$F_l = -k_l(l - l_0).$$

For small $|\theta|$, model the transverse restoring mechanism shown by an idealized force

$$F_\theta = -k_\theta l\theta,$$

acting perpendicular to OP in the plane of the disk. Here k_θ has units of $\mathrm{N\,m^{-1}}$. Assume that this mechanism does not obstruct radial or angular motion.

For this exercise, replace the capacitor shown in the figure by an ideal actuator that exerts the prescribed signed radial force

$$f_{\mathrm{act}}(t) = ma_0 \sin(\omega t),$$

where positive force is directed outwards and a_0 has units of acceleration. This idealization allows the force to act in either radial direction.

(i) Derive the equations of motion for $l(t)$ and $\theta(t)$ using a frame fixed to the disk.
(ii) With the actuator switched off, show that the equilibrium with $\theta = 0$ occurs at

$$l = L \triangleq \frac{k_l l_0}{k_l - m\Omega^2},$$

provided $k_l > m\Omega^2$ and $L < r$.

(iii) Simulate the motion with

$$r = 2\,\mathrm{mm}, \qquad l_0 = 1\,\mathrm{mm}, \qquad \frac{k_l}{m} = (2\pi \times 100)^2\,\mathrm{s}^{-2}, \qquad \frac{k_\theta}{m} = (2\pi \times 10)^2\,\mathrm{s}^{-2},$$

$$\omega = 2\pi \times 11\,\mathrm{s}^{-1}, \qquad a_0 = 0.01\,\mathrm{m\,s}^{-2}, \qquad \Omega = 300\,\mathrm{r.p.m.},$$

and initial conditions

$$l(0) = L, \qquad \dot{l}(0) = 0, \qquad \theta(0) = 0, \qquad \dot{\theta}(0) = 0.$$

Plot the radial displacement $l(t) - L$ and the transverse displacement $l(t)\sin\theta(t)$. Check that $0 < l(t) < r$ and that the small-angle assumption remains valid.

(iv) Keeping the other parameters unchanged, repeat the simulation for three values of Ω between 100 r.p.m. and 1000 r.p.m.. Use the corresponding equilibrium value L for each initial condition. Also investigate the effect of reversing the direction of rotation.

(v) Linearize the equations about $l = L$, $\theta = 0$. Use the resulting equations and your simulations to explain how the amplitude and phase of the transverse response relative to the radial vibration can be used to estimate Ω.
Since the model has no damping, free oscillations persist. Distinguish these from the response at the actuator frequency ω when interpreting your results.

(vi) Discuss how changing the stiffnesses, drive frequency, drive amplitude, or introducing damping would affect the sensitivity and operating range.

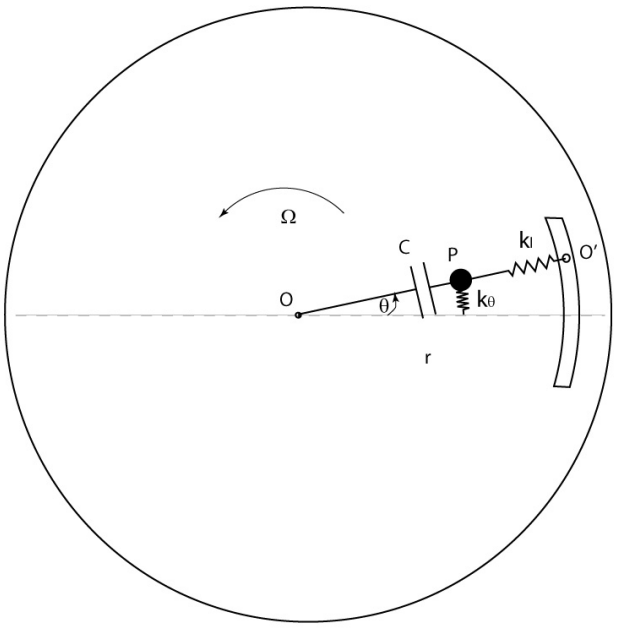


Fig. 1.24 An idealized mechanical model illustrating Coriolis-based angular-rate sensing. The capacitor C is replaced by a prescribed radial actuator in this exercise.

Exercise 1.23 (A Pendulum on the Rotating Earth). A particle of mass m is suspended by a massless, inextensible string of length l from a fixed support on Earth at latitude ϕ. Neglect air resistance and assume that the string remains taut.

Let $\mathbf{c} = [\mathbf{c}_1\ \mathbf{c}_2\ \mathbf{c}_3]$ be an Earth-fixed orthonormal frame with origin at the support, with axes pointing west, radially upwards, and north, respectively (see Figure 1.14). Approximate Earth as a sphere of radius r_e, rotating at constant angular speed ω_e. Neglect orbital effects and the height of the support compared with r_e.

Let $X(t)$ denote the position of the particle in $\mathbf{c}$. Use the local uniform-gravity model, where g is the gravitational acceleration before including centrifugal effects. The angular velocity of the frame has the representation

$$\Omega = \omega_e \begin{bmatrix} 0 \\ \sin\phi \\ \cos\phi \end{bmatrix}, \qquad \widehat{\Omega} X = \Omega \times X.$$

Write $e_2 = [0\ 1\ 0]^T$.

(i) The fixed string length requires $X^T X = l^2$. Show that

$$X^T \dot{X} = 0, \qquad X^T \ddot{X} = -\|\dot{X}\|^2.$$

Explain the geometric meaning of the first identity.

(ii) Represent the tension force by $-\lambda X$, where $\lambda(t) \geq 0$, and show that its magnitude is λl. Derive the equation of motion

$$m\ddot{X} = -m\widehat{\Omega}^2 X - 2m\widehat{\Omega}\dot{X} - m(gI_{3\times 3} + r_e\widehat{\Omega}^2)e_2 - \lambda X.$$

Identify the terms due to the rotation of Earth.

(iii) Use the constraint identities in part (i) to determine λ in terms of X and $\dot{X}$. Hence obtain a closed differential equation for $X(t)$.

What would a negative value of the required λ mean physically?

(iv) Define

$$g_b = (gI_{3\times 3} + r_e\widehat{\Omega}^2)e_2.$$

If the variation of centrifugal acceleration over the pendulum's length is neglected, show that the hanging equilibrium is

$$X_* = -l\frac{g_b}{\|g_b\|}.$$

Explain why the equilibrium string need not align exactly with the radial direction.

(v) To study the leading effect of Earth's rotation on small oscillations, neglect terms of order ω_e^2, so that

$$\ddot{X} = -2\Omega \times \dot{X} - ge_2 - \frac{\lambda}{m}X.$$

Define

$$H = X \times \dot{X}.$$

Here mH is formed using the particle's velocity relative to Earth; it is not the angular momentum defined using its inertial velocity.

Show that taking the cross product with X eliminates the tension and gives

$$\dot{H} = 2(X \cdot \Omega)\dot{X} + g\, e_2 \times X.$$

Let

$$q = (I_{3\times 3} - e_2 e_2^T)X$$

be the horizontal displacement. Near the hanging configuration, use the length constraint to show that, to first order in oscillation amplitude,

$$X = -l e_2 + q, \qquad H = -l\, e_2 \times \dot{q}, \qquad \dot{q} = \frac{1}{l} e_2 \times H.$$

Hence, defining

$$\Omega_v = (\Omega \cdot e_2)e_2 = \omega_e \sin\phi\, e_2,$$

derive

$$\ddot{H} + 2\Omega_v \times \dot{H} + \frac{g}{l} H = 0.$$

Explain why only the vertical component of Earth's angular velocity enters this linearized equation.

(vi) Introduce

$$H(t) = S(t)K(t), \qquad \dot{S} = -\widehat{\Omega_v} S, \qquad S(0) = I_{3\times 3}.$$

Show that $S(t)$ is a rotation about the vertical direction and that

$$\ddot{K} + \left(\frac{g}{l} + \omega_e^2 \sin^2\phi\right) K = 0.$$

Consistently neglecting terms of order ω_e^2, deduce that

$$H(t) \approx S(t)\left(A\cos(\omega_0 t) + B\sin(\omega_0 t)\right), \qquad \omega_0 = \sqrt{\frac{g}{l}},$$

where A and B are constant horizontal columns determined by the initial conditions.
For release from a small displacement at rest relative to Earth, let $q_0 = q(0) \neq 0$. Show that

$$H(t) \approx \frac{g}{\omega_0} \sin(\omega_0 t)\, S(t)(e_2 \times q_0).$$

Interpret this as a rapid oscillation along a slowly rotating horizontal direction. Explain why H itself does not undergo uniform rotation with constant magnitude, and why its direction is undefined when $H = 0$.
Deduce the leading precession rate of this slowly rotating direction. Show that it is clockwise in the Northern Hemisphere and counterclockwise in the Southern Hemisphere, as viewed from above. What happens at the equator, and what is the precession period away from the equator?

(vii) *Optional computational investigation.* Simulate the full constrained equations from parts (ii)–(iii), releasing the particle at rest from a small displacement from the hanging equilibrium. Choose initial data satisfying

$$X(0)^T X(0) = l^2, \qquad X(0)^T \dot{X}(0) = 0.$$

Compute $H(t) = X(t) \times \dot{X}(t)$ and compare it with the approximation in part (vi). To separate the predicted slow rotation from the rapid oscillation, also examine

$$K_{\text{num}}(t) = S(t)^T H(t).$$

Plot its components along the fixed directions

$$\frac{e_2 \times q_0}{\|q_0\|}, \qquad \frac{q_0}{\|q_0\|}, \qquad e_2,$$

and interpret the results using part (vi). Avoid normalizing H near its zeros.
Monitor the string-length and tangency errors, check that the required tension remains nonnegative, and verify that the conclusions persist when the numerical tolerances are tightened. Discuss the limitations of comparing the full equations with a small-amplitude approximation that neglects terms of order ω_e^2.

Exercise 1.24. Explain how the Coriolis term deflects air moving towards a low-pressure region, producing counterclockwise cyclonic circulation in the Northern Hemisphere and clockwise circulation in the Southern Hemisphere, as viewed from above (see figure 1.15).

Exercise 1.25. At a point on the equator, a cannonball is fired towards the west at 45° above the local horizontal, with an initial speed of $100\,\mathrm{m\,s^{-1}}$ relative to Earth. Neglect air resistance and use the local flat-ground, uniform-gravity approximation. Assume that the cannonball lands at the same elevation from which it was fired.

Taking Earth's rotation into account, find the horizontal distance from the launch point to the landing point. Compare this with the distance obtained when Earth's rotation is neglected. Use

$$r_e = 6.37 \times 10^6\,\mathrm{m}, \qquad \omega_e = 7.29 \times 10^{-5}\,\mathrm{s^{-1}}, \qquad g = 9.81\,\mathrm{m\,s^{-2}},$$

where g denotes gravitational acceleration before including centrifugal effects.

Exercise 1.26. Model Earth and the Sun as an isolated system of two point particles with masses M_E and M_S, interacting through the gravitational attraction

$$\frac{GM_S M_E}{r^2},$$

where G is the universal gravitational constant and r is their separation.

Assume that the relative motion has negative mechanical energy and nonzero angular momentum. Show that Earth's orbit relative to the Sun is an ellipse, with the Sun at one focus (Kepler's First Law). Include the circle as a special case.

Recall that an ellipse can be represented in polar coordinates about one focus by

$$r = r_0 \frac{1+e}{1-e\cos\theta},$$

where r_0 and r_1 are the minimum and maximum distances from that focus, respectively,

$$e = \frac{r_1 - r_0}{r_1 + r_0}, \qquad 0 \le e < 1, \qquad \frac{b}{a} = \sqrt{1-e^2}.$$

Here a and b are the semimajor and semiminor axes, and θ is measured from the direction towards the farthest point, as shown in Figure 1.25.

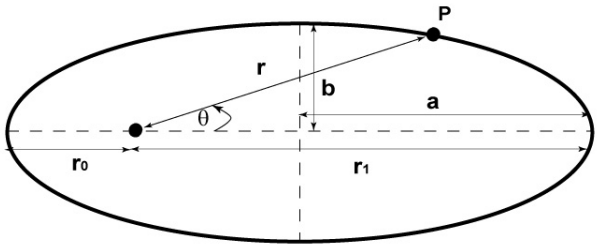


Fig. 1.25 An ellipse in polar coordinates about a focus. The minimum and maximum distances from the focus are r_0 and r_1, and the semimajor and semiminor axes have lengths a and b.

Chapter 2
Rigid Body Motion

In this chapter we consider a collection of non-collinear interacting particles $P_1, P_2, \ldots, P_n$ whose interactions keep the distance between every pair constant. Such a collection is called a *rigid body*. We specialize the equations for a system of particles developed in Section 1.2.6.2 to this case. Choosing a frame fixed to the body makes all particle coordinates constant and considerably simplifies those equations.

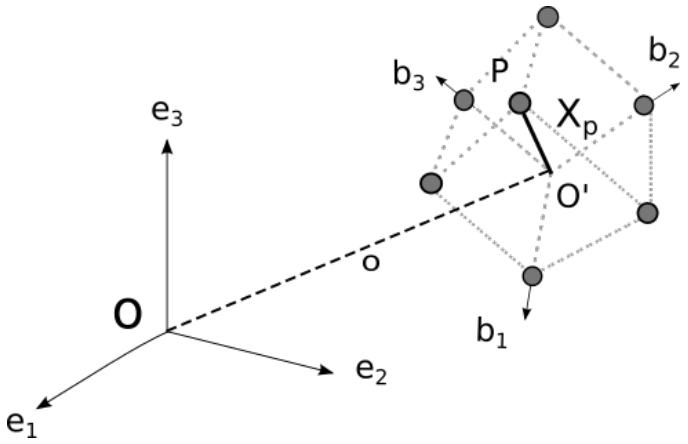


Fig. 2.1 The motion of a set of particles that appear to be fixed with respect to the moving frame **b**.

Let **e** be an inertial frame with origin O and let $\mathbf{b}(t)$ be an orthonormal frame fixed to the body, with origin O', as illustrated in Figure 2.1. We call $\mathbf{b}(t)$ the *body frame*. Writing

$$\mathbf{b}(t) = \mathbf{e}R(t), \qquad OO' = \mathbf{e}o(t),$$

the particle positions satisfy

$$x_i(t) = o(t) + R(t)X_i,$$

where each X_i is constant. Once the body frame and these coordinates have been chosen, the pair $(o(t), R(t))$ uniquely specifies the position and orientation of the body.

Thus the configuration space of an unconstrained rigid body is $\mathbb{R}^3 \times \mathrm{SO}(3)$: three coordinates describe translation and three describe orientation. A rigid body therefore has six degrees of freedom.

The pair (o,R) can also be represented by the matrix

$$E = \begin{bmatrix} R & o \\ \mathbf{0} & 1 \end{bmatrix},$$

where $\mathbf{0} = 0_{1\times 3}$. The set of these matrices forms the *special Euclidean group* $\mathrm{SE}(3)$ under matrix multiplication. We may therefore also identify the configuration space of a rigid body with $\mathrm{SE}(3)$.

2.1 Rigid Body Kinematics

Recall that Ω and $\omega = R\Omega$ represent the angular velocity in the body and inertial frames, respectively. The rotational kinematics are

$$\dot{R} = R\widehat{\Omega} = \widehat{\omega}R. \tag{2.1}$$

For a rigid body, the body coordinates X_i are constant. Thus the momentum expressions in section 1.2.6.2 reduce to

$$p = M\left(\dot{o} + R\widehat{\Omega}\bar{X}\right) = M\dot{\bar{x}},$$
$$\pi = R\Pi, \qquad \Pi = \mathbb{I}\Omega + M\bar{X}\times(R^T\dot{o}),$$

where M is the total mass, $\bar{X}$ is the center of mass represented in the body frame, and $\mathbb{I} = -\sum_i m_i\widehat{X}_i^2$ is the moment of inertia about O' expressed in that frame. Both $\bar{X}$ and $\mathbb{I}$ are constant. For positive particle masses and a non-collinear configuration, $\mathbb{I}$ is symmetric positive definite.

The linear momentum expression gives the translational kinematics:

$$\dot{o} = \frac{p}{M} - R(\Omega\times\bar{X}). \tag{2.2}$$

Substituting into the expression for Π gives

$$\Pi = \mathbb{I}\Omega - M\bar{X}\times(\Omega\times\bar{X}) + \bar{X}\times(R^Tp).$$

Since $M\bar{X}\times(\Omega\times\bar{X}) = -M\bar{X}\times(\bar{X}\times\Omega) = -M\widehat{\bar{X}}^2\Omega$ we can write the above as

$$\Pi = \mathbb{I}_c\Omega + \bar{X}\times(R^Tp), \tag{2.3}$$

where

$$\mathbb{I}_c = \mathbb{I} + M\widehat{\bar{X}}^2 = -\sum_i m_i\widehat{(X_i - \bar{X})}^2 \tag{2.4}$$

is the moment of inertia about the center of mass, expressed in axes parallel to **b**. Equivalently,

$$\mathbb{I} = \mathbb{I}_c - M\widehat{\bar{X}}^2,$$

which is the *parallel axis theorem*.

The same inertia tensor expressed in axes parallel to **e** is

$$\mathbb{I}_c^R \triangleq R\mathbb{I}_cR^T. \tag{2.5}$$

Using $\omega = R\Omega$ and $\pi = R\Pi$ in (2.3) gives the following equivalent representations of angular momentum

$$\pi = \mathbb{I}_c^R\omega + (R\bar{X})\times p. \tag{2.6}$$

If the body-frame origin is the center of mass, then $\bar{X} = 0$ and the above derived expressions simplify to

$$p = M\dot{o}, \qquad \Pi = \mathbb{I}_c\Omega, \qquad \pi = R\mathbb{I}_c\Omega = \mathbb{I}_c^R\omega.$$

2.2 Rigid Body Equations

As shown in section 1.1.6, the total momenta satisfy

$$\begin{aligned}\dot{p} &= f^e,\\ \dot{\pi} &= -\dot{o}\times p + \tau^e,\end{aligned}$$

where f^e is the total external force and τ^e is its total moment about O', both expressed in the inertial frame **e**. These balance laws apply whether or not the system is rigid. We now combine them with the rigid body kinematics.

2.2.1 Rigid Body Equations in the Spatial Frame

Let $\vec{x}' = \vec{x} - o = R\bar{X}$. Since $\bar{X}$ is constant,

$$\dot{\vec{x}}' = \omega \times \vec{x}'.$$

Substituting $\dot{o} = p/M - \omega \times \vec{x}'$ into the angular momentum balance gives

$$\dot{\pi} = (\omega \times \vec{x}') \times p + \tau^e.$$

Together with the kinematic equations, this yields the following system.

The rigid body equations in spatial variables are

$$\dot{o} = \frac{p}{M} - \omega \times \vec{x}', \tag{2.7}$$
$$\dot{R} = \widehat{\omega} R, \tag{2.8}$$
$$\dot{p} = f^e, \tag{2.9}$$
$$\dot{\pi} = (\omega \times \vec{x}') \times p + \tau^e, \tag{2.10}$$

where

$$\vec{x}' = R\bar{X}, \qquad \mathbb{I}_c^R = R\mathbb{I}_c R^T, \qquad \omega = (\mathbb{I}_c^R)^{-1}\big(\pi - \vec{x}' \times p\big).$$

Once the external forces and moments are specified as functions of the state and time, these equations form a closed system for (o, R, p, π).

If the body-frame origin is chosen at the center of mass, then $\vec{x}' = 0$ and the equations simplify to

$$\dot{o} = \frac{p}{M}, \tag{2.11}$$
$$\dot{R} = \widehat{\omega} R, \tag{2.12}$$

$$\dot{p} = f^e, \tag{2.13}$$
$$\dot{\pi} = \tau_c^e, \tag{2.14}$$

with $\omega = (\mathbb{I}_c^R)^{-1}\pi$. Here τ_c^e denotes the external moment about the center of mass.

2.2.2 Rigid Body Equations in the Body Frame

Let $F^e = R^T f^e$ and $T^e = R^T \tau^e$ denote the external force and moment expressed in **b**. Setting $\dot{X}_i = \ddot{X}_i = 0$ in the equations of section 1.2.6.2 gives

$$MR^T\ddot{o} - M\hat{\bar{X}}\dot{\Omega} = -M\hat{\Omega}^2\bar{X} + F^e,$$
$$\mathbb{I}\dot{\Omega} + \Omega \times (\mathbb{I}\Omega) = -M\bar{X} \times (R^T\ddot{o}) + T^e.$$

Eliminating $R^T\ddot{o}$ yields

$$(\mathbb{I} + M\hat{\bar{X}}^2)\dot{\Omega} = (\mathbb{I}\Omega) \times \Omega + M\bar{X} \times (\hat{\Omega}^2\bar{X}) - \bar{X} \times F^e + T^e.$$

Using

$$\bar{X} \times (\hat{\Omega}^2\bar{X}) = -\Omega \times (\hat{\bar{X}}^2\Omega), \qquad \mathbb{I}_c = \mathbb{I} + M\hat{\bar{X}}^2,$$

we obtain

$$\mathbb{I}_c\dot{\Omega} + \Omega \times (\mathbb{I}_c\Omega) = T^e - \bar{X} \times F^e.$$

The right-hand side is precisely the external moment about the center of mass, expressed in **b**:

$$T_c^e = \sum_i (X_i - \bar{X}) \times F_i^e = T^e - \bar{X} \times F^e.$$

The rigid body equations expressed using body-frame quantities are

$$\dot{R} = R\hat{\Omega}, \tag{2.15}$$
$$MR^T\ddot{o} - M\hat{\bar{X}}\dot{\Omega} = -M\hat{\Omega}^2\bar{X} + F^e, \tag{2.16}$$
$$\mathbb{I}_c\dot{\Omega} + \Omega \times (\mathbb{I}_c\Omega) = T_c^e. \tag{2.17}$$

If the body-frame origin is the center of mass, then $\bar{X} = 0$ and these reduce to

$$\dot{R} = R\hat{\Omega}, \tag{2.18}$$
$$M\ddot{o} = f^e, \tag{2.19}$$
$$\mathbb{I}_c\dot{\Omega} + \Omega \times (\mathbb{I}_c\Omega) = T_c^e. \tag{2.20}$$

The last equation is *Euler's rotational equation*; together with the translational balance it forms the *Newton–Euler equations* for a rigid body.

Since the moment of inertia tensor is always symmetric and positive definite it is always possible to find a body frame **b** such that the Inertia tensor $\mathbb{I}_c$ is diagonalized. For instance in an axi-symmetric rigid body if the body frame **b** is aligned along the axes of symmetry then it can be shown that the inertia matrix $\mathbb{I}_c$ is diagonal. The diagonal elements are called the principle moments of inertia. Let the body frame **b** be chosen such that the inertia tensor is diagonalized

$$\mathbb{I}_c = \begin{bmatrix} \mathbb{I}_1 & 0 & 0 \\ 0 & \mathbb{I}_2 & 0 \\ 0 & 0 & \mathbb{I}_3 \end{bmatrix}, \tag{2.21}$$

where the principle moments of inertia are given by $\mathbb{I}_1, \mathbb{I}_2, \mathbb{I}_3$. Based on the values of the principle moments of inertia one can classify rigid bodies into three distinct categories:

(a) Asymmetric Rigid Body: $\mathbb{I}_1 > \mathbb{I}_2 > \mathbb{I}_3$
(b) Axi-Symmetric Rigid Body: $\mathbb{I}_1 = \mathbb{I}_2 > \mathbb{I}_3$ or $\mathbb{I}_1 > \mathbb{I}_2 = \mathbb{I}_3$
(c) Symmetric Rigid Body: $\mathbb{I}_1 = \mathbb{I}_2 = \mathbb{I}_3$

When the body frame **b** coincides with the directions of the principle moments of inertia of the object, the rotation dynamics (2.20) take the form

$$\mathbb{I}_1 \dot{\Omega}_1 = (\mathbb{I}_2 - \mathbb{I}_3)\Omega_2\Omega_3 + T_1^e, \tag{2.22}$$
$$\mathbb{I}_2 \dot{\Omega}_2 = (\mathbb{I}_3 - \mathbb{I}_1)\Omega_3\Omega_1 + T_2^e, \tag{2.23}$$
$$\mathbb{I}_3 \dot{\Omega}_3 = (\mathbb{I}_1 - \mathbb{I}_2)\Omega_1\Omega_2 + T_3^e, \tag{2.24}$$

2.2.3 Purely Rotating Rigid Bodies

Suppose a point O' of the body is fixed in the inertial frame **e**. Choose the body-frame origin at this pivot, so that $\dot{o} = \ddot{o} = 0$. Let $\mathbb{I}$ be the moment of inertia about the pivot expressed in **b**, and let $\mathbb{I}^R = R\mathbb{I}R^T$.

Recall that, in general, the body-frame representation of angular momentum about O' is

$$\Pi = \mathbb{I}_c\Omega + \bar{X} \times (R^T p).$$

For a fixed pivot, the linear momentum satisfies $p = MR(\Omega \times \bar{X})$. Consequently,

$$\begin{aligned} \Pi &= \mathbb{I}_c\Omega + M\bar{X} \times (\Omega \times \bar{X}) \\ &= (\mathbb{I}_c - M\hat{\bar{X}}^2)\Omega = \mathbb{I}\Omega, \end{aligned}$$

where the last equality follows from the parallel axis theorem. Thus, even when the pivot is away from the center of mass,

$$\pi = R\Pi = R\mathbb{I}\Omega = \mathbb{I}^R\omega.$$

The rotational equations about the fixed pivot O' are, in the inertial frame $\mathbf{e}$,

$$\begin{aligned}\dot{R} &= \widehat{\omega} R,\\ \dot{\pi} &= \tau^e, \qquad \omega = (\mathbb{I}^R)^{-1}\pi.\end{aligned}$$

Equivalently, using body-frame quantities,

$$\begin{aligned}\dot{R} &= R\widehat{\Omega},\\ \mathbb{I}\dot{\Omega} &= (\mathbb{I}\Omega)\times\Omega + T^e,\end{aligned}$$

where τ^e and $T^e = R^T\tau^e$ represent the total external moment about the pivot in the two frames. Here $\mathbb{I}$ is the inertia about the pivot, which need not coincide with the center of mass.

To express the rotational equations directly in terms of Π, recall that the angular momentum about the pivot is

$$\Pi = \mathbb{I}_c\Omega + \bar{X}\times(R^T p).$$

Fixing the pivot means $\dot{o} = 0$, but does not imply $p = 0$: the center of mass may still move around the pivot, giving

$$p = M\dot{\bar{x}} = MR(\Omega\times\bar{X}).$$

Consequently,

$$\Pi = \mathbb{I}_c\Omega + M\bar{X}\times(\Omega\times\bar{X}) = (\mathbb{I}_c - M\widehat{\bar{X}}^2)\Omega = \mathbb{I}\Omega.$$

Thus the inertia $\mathbb{I}$ about the pivot incorporates the contribution from the motion of the center of mass.

Since $\mathbb{I}$ is constant in the body frame, we have $\Omega = \mathbb{I}^{-1}\Pi$ and $\dot{\Pi} = \mathbb{I}\dot{\Omega}$. Substituting these into the rotational equations gives the following result.

The motion of a rigid body about a fixed pivot O' is described in terms of its orientation R and body angular momentum Π by

$$\begin{aligned}\dot{R} &= R\widehat{\mathbb{I}^{-1}\Pi},\\ \dot{\Pi} &= \Pi\times(\mathbb{I}^{-1}\Pi) + T^e.\end{aligned}$$

Here $\mathbb{I}$, Π, and T^e are the inertia, angular momentum, and total external moment about the pivot, all expressed in the body frame.

Since $\Pi = \mathbb{I}\Omega$, the angular momentum and angular velocity need not have the same direction. For nonzero Ω, they are parallel precisely when Ω lies along a principal axis of inertia.

Consider either an isolated rigid body with moments taken about its center of mass, or a body rotating about a fixed pivot with zero resultant external moment about that pivot. In

either case, $\dot{\pi} = 0$, so the spatial angular momentum is constant. However, its body-frame representation $\Pi = R^T \pi$ generally changes as the body rotates. Its magnitude remains constant because

$$\|\Pi\| = \|R^T \pi\| = \|\pi\|.$$

More generally, when an external moment T^e acts about the chosen point, the body angular momentum equation gives

$$\frac{1}{2}\frac{d}{dt}\|\Pi\|^2 = \Pi \cdot \dot{\Pi} = \Pi \cdot \left(\Pi \times (\mathbb{I}^{-1}\Pi) + T^e\right) = \Pi \cdot T^e, \tag{2.25}$$

since $A \cdot (A \times B) = 0$. Thus only the component of the external moment along Π changes its magnitude. Whenever $\Pi \neq 0$, this can also be written as

$$\frac{d}{dt}\|\Pi\| = \frac{\Pi \cdot T^e}{\|\Pi\|}.$$

For a freely moving isolated rigid body, the spatial angular momentum π about the center of mass and the magnitude $\|\Pi\|$ are conserved. The body-frame representation Π generally varies as the body rotates.

For a heavy top with an ideal fixed pivot, gravity generally exerts a nonzero moment about the pivot, so neither π nor $\|\Pi\|$ is generally conserved. However, the component of π along the fixed vertical direction is conserved, since gravity produces no moment in that direction.

These conclusions follow from the angular momentum balance, using the assumption of equal-and-opposite central internal forces.

2.3 Kinetic Energy of a Rigid Body

From section 1.2.4, the kinetic energy of a rigid body with respect to the inertial frame **e** is

$$\mathrm{KE} = \frac{1}{2}M\|\dot{o}\|^2 + M\dot{o}^T R(\widehat{\Omega}\bar{X}) + \frac{1}{2}\Omega^T \mathbb{I}\Omega. \tag{2.26}$$

Here $\mathbb{I}$ is the inertia about the body-frame origin O'.

Choosing O' at the center of mass gives $\bar{X} = 0$ and $\mathbb{I} = \mathbb{I}_c$. Thus

$$\mathrm{KE} = \frac{1}{2}M\|\dot{o}\|^2 + \frac{1}{2}\Omega^T \mathbb{I}_c\Omega. \tag{2.27}$$

The total kinetic energy is the sum of the energy of translation of the center of mass and the energy of rotation about it.

For a body rotating about a fixed pivot, $\dot{o} = 0$, and instead

$$\mathrm{KE} = \frac{1}{2}\Omega^T \mathbb{I}\Omega,$$

where $\mathbb{I}$ is the inertia about the pivot.

With the origin at the center of mass, differentiating and using the Newton–Euler equations gives

$$\begin{aligned}\frac{d}{dt}\mathrm{KE} &= M\dot{o}\cdot\ddot{o} + \Omega\cdot(\mathbb{I}_c\dot{\Omega}) \\ &= \dot{o}\cdot f^e + \Omega\cdot\left((\mathbb{I}_c\Omega)\times\Omega + T_c^e\right) \\ &= \dot{o}\cdot f^e + \Omega\cdot T_c^e.\end{aligned} \tag{2.28}$$

Thus the rate of change of kinetic energy equals the power supplied by the external interactions. In particular, an isolated rigid body has constant kinetic energy. For a fixed pivot the corresponding power balance is $\frac{d}{dt}\mathrm{KE} = \Omega\cdot T^e$, with the moment taken about the pivot.

2.3.1 Example: Kinetic Energy of a Falling and Rolling Disk

Let **b** be a frame fixed to a thin disk of radius r, with origin at its center of mass and second axis along its symmetry axis as shown in figure 2.2. Let **e** be an Earth-fixed frame, treated as inertial, with its third axis vertically upwards. Parameterize the orientation by

$$\mathbf{b} = \mathbf{e}R, \qquad R = R_3(\theta)R_1(\alpha)R_2(\varphi).$$

Using $\widehat{\Omega} = R^T\dot{R}$ gives

$$\Omega = \dot{\theta} R_2(\varphi)^T R_1(\alpha)^T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \dot{\alpha} R_2(\varphi)^T \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \dot{\varphi} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

Thus the body angular velocity components are

$$\Omega_1 = \dot{\alpha}\cos\varphi - \dot{\theta}\cos\alpha\sin\varphi, \tag{2.29}$$

$$\Omega_2 = \dot{\varphi} + \dot{\theta}\sin\alpha, \tag{2.30}$$

$$\Omega_3 = \dot{\theta}\cos\alpha\cos\varphi + \dot{\alpha}\sin\varphi. \tag{2.31}$$

Assume the disk remains in contact with a horizontal plane and $|\alpha| < \pi/2$. Its center of mass then has height $r\cos\alpha$, so

$$o = \begin{bmatrix} x \\ y \\ r\cos\alpha \end{bmatrix}, \qquad \dot{o} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ -r\dot{\alpha}\sin\alpha \end{bmatrix}.$$

Let M be the disk's mass and $\mathbb{I}_c = \mathrm{diag}(\mathbb{I}_i, \mathbb{I}_r, \mathbb{I}_i)$ its inertia about the center of mass, expressed in $\mathbf{b}$. Its kinetic energy is

$$\begin{aligned} \mathrm{KE} &= \frac{1}{2}\left(M\|\dot{o}\|^2 + \Omega^T \mathbb{I}_c \Omega\right) \\ &= \frac{1}{2}\Big(M\dot{x}^2 + M\dot{y}^2 + \mathbb{I}_r\dot{\varphi}^2 + (\mathbb{I}_i + Mr^2\sin^2\alpha)\dot{\alpha}^2 \\ &\qquad + (\mathbb{I}_i\cos^2\alpha + \mathbb{I}_r\sin^2\alpha)\dot{\theta}^2 + 2\mathbb{I}_r\dot{\varphi}\dot{\theta}\sin\alpha\Big). \end{aligned}$$

This expression allows slipping; rolling without slipping imposes additional relations between the translational and angular velocities.

2.4 Examples of Modeling Rigid Body Motion

2.4.1 Spatial Momentum Model of a Falling and Rolling Disk

Consider a thin circular disk of mass M and radius r rolling without slipping on a horizontal plane, as shown in Figure 2.2. The disk may change its inclination and direction of motion. We assume ideal point contact, neglect rolling resistance and air resistance, and consider motion for which contact is maintained.

Let $\mathbf{e}$ be an inertial frame with $\mathbf{e}_3$ pointing vertically upwards. Let $\mathbf{b}$ be fixed to the disk, with its origin at the center of mass C and $\mathbf{b}_2$ perpendicular to the disk. Write

$$\mathbf{b} = \mathbf{e}R, \qquad R \in SO(3), \qquad \dot{R} = \widehat{\omega} R.$$

Although the figure illustrates the orientation using angles, we will work directly with R.

Let o represent C in $\mathbf{e}$, and define

$$e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \qquad n = Re_2.$$

Thus n represents the disk's symmetry axis in the inertial frame.

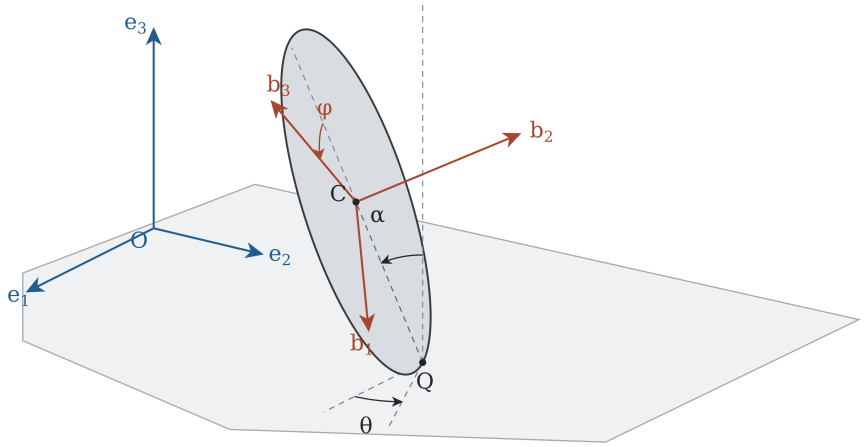


Fig. 2.2 A disk rolling on a horizontal plane. The body frame $\mathbf{b}$ has its origin at the center C, with $\mathbf{b}_2$ normal to the disk. The instantaneous contact point is Q.

2.4.1.1 Contact geometry

Let ρ be the displacement from C to the instantaneous contact point Q, expressed in $\mathbf{e}$. The downward direction within the plane of the disk is obtained by projecting $-e_3$ onto that plane. Therefore,

$$\boxed{\rho = -r\frac{e_3 - (e_3 \cdot n)n}{\sqrt{1-(e_3 \cdot n)^2}}.}$$

For convenience, define

$$\gamma = e_3 \cdot n, \qquad d = \sqrt{1-\gamma^2},$$

so that

$$\rho = -\frac{r}{d}(e_3 - \gamma n).$$

We assume $d > 0$, excluding the configuration in which the disk lies flat and the point-contact description ceases to apply.

The contact location is

$$q = o + \rho.$$

Since it lies on the horizontal plane,

$$e_3^T(o+\rho) = 0.$$

Equivalently, the center height satisfies

$$\boxed{e_3^T o = rd.}$$

2.4.1.2 Rolling constraint

The velocity of the material point of the disk instantaneously in contact with the ground is

$$\dot{o} + \omega \times \rho.$$

Rolling without slipping requires this velocity to vanish:

$$\boxed{\dot{o} + \omega \times \rho = 0.}$$

The vertical component is consistent with the contact-height constraint. The two horizontal components impose the nonholonomic no-slip conditions.

Notice that $\dot{\rho}$ is generally not equal to $\omega \times \rho$: different material points of the disk come into contact as it rolls. Consequently, the geometric contact location $q(t)$ need not have zero velocity, even though the material point instantaneously at contact has zero velocity.

2.4.1.3 Spatial momenta

Let $\mathbb{I}$ be the constant inertia tensor about C, expressed in $\mathbf{b}$. For an axisymmetric disk,

$$\mathbb{I} = \mathrm{diag}(\mathbb{I}_i, \mathbb{I}_r, \mathbb{I}_i)$$

where I_r is the moment about the symmetry axis and I_i is the moment about a diameter. For a uniform thin disk,

$$\mathbb{I}_i = \frac{1}{4}Mr^2, \qquad \mathbb{I}_r = \frac{1}{2}Mr^2.$$

Define the spatial inertia tensor

$$\mathbb{I}_R = R\mathbb{I}R^T.$$

The spatial linear momentum and angular momentum about C are

$$p = M\dot{o}, \qquad \pi = \mathbb{I}_R\omega.$$

Thus

$$\dot{o} = \frac{p}{M}, \qquad \omega = \mathbb{I}_R^{-1}\pi.$$

The rolling constraint becomes

$$\boxed{\frac{p}{M} + (\mathbb{I}_R^{-1}\pi) \times \rho = 0.}$$

2.4.1.4 Equations of motion

Let f be the total ground-contact force on the disk, expressed in **e**. It includes both the normal force and static friction.

Gravity acts through C and has zero moment about C. The contact force has moment $\rho \times f$. Therefore,

The spatial momentum equations of the rolling disk are

$$\dot{o} = \frac{p}{M}, \tag{2.32}$$

$$\dot{R} = \widehat{\omega}R, \qquad \omega = \mathbb{I}_R^{-1}\pi, \tag{2.33}$$

$$\dot{p} = f - Mge_3, \tag{2.34}$$

$$\dot{\pi} = \rho \times f. \tag{2.35}$$

They are subject to

$$e_3^T(o+\rho) = 0, \qquad \frac{p}{M} + \omega \times \rho = 0,$$

where

$$\mathbb{I}_R = R\mathbb{I}R^T, \qquad \rho = -r\frac{e_3 - (e_3 \cdot Re_2)Re_2}{\sqrt{1-(e_3 \cdot Re_2)^2}}.$$

The contact force is determined by requiring the motion to satisfy the rolling constraint. It is not a prescribed force.

2.4.1.5 Determining the contact force

Differentiate the rolling constraint:

$$0 = \frac{\dot{p}}{M} + \dot{\omega} \times \rho + \omega \times \dot{\rho}.$$

Since

$$\dot{\mathbb{I}}_R = \widehat{\omega}\mathbb{I}_R - \mathbb{I}_R\widehat{\omega}, \qquad \pi = \mathbb{I}_R\omega,$$

we have

$$\dot{\omega} = \mathbb{I}_R^{-1}\big(\dot{\pi} - \omega \times \pi\big) = \mathbb{I}_R^{-1}\big(\rho \times f - \omega \times \pi\big).$$

Substituting this and $\dot{p} = f - Mge_3$ gives

$$0 = \frac{f}{M} - ge_3 + \big[\mathbb{I}_R^{-1}\big(\rho \times f - \omega \times \pi\big)\big] \times \rho + \omega \times \dot{\rho}.$$

Hence the contact force satisfies

$$\boxed{\left(\frac{1}{M}I_{3\times 3} - \widehat{\rho}\mathbb{I}_R^{-1}\widehat{\rho}\right) f = ge_3 - \widehat{\rho}\mathbb{I}_R^{-1}(\omega \times \pi) - \omega \times \dot{\rho}.}$$

All quantities on the right can be calculated from the current configuration and momenta. In particular,

$$\dot{n} = \omega \times n, \qquad \dot{\gamma} = e_3 \cdot \dot{n},$$

and differentiation of the contact displacement gives

$$\boxed{\dot{\rho} = \frac{r}{d}\big(\dot{\gamma}n + \gamma\dot{n}\big) + \frac{\gamma\dot{\gamma}}{d^2}\rho.}$$

The coefficient matrix determining f is symmetric positive definite. Indeed, for any nonzero column z,

$$z^T\left(\frac{1}{M}I_{3\times 3} - \widehat{\rho}\mathbb{I}_R^{-1}\widehat{\rho}\right) z = \frac{\|z\|^2}{M} + (\rho \times z)^T\mathbb{I}_R^{-1}(\rho \times z) > 0.$$

Thus the contact force is uniquely determined. Substituting it into the momentum equations gives a closed system for (o, R, p, π).

2.4.1.6 Constraints on the initial conditions

The initial conditions must satisfy

$$R(0) \in SO(3), \qquad e_3^T\big(o(0) + \rho(0)\big) = 0,$$

and

$$\frac{p(0)}{M} + \big(\mathbb{I}_R(0)^{-1}\pi(0)\big) \times \rho(0) = 0.$$

The exact equations preserve these constraints.

The model applies while the disk has a unique contact point, the normal contact force is nonnegative, and sufficient static friction is available to prevent slipping. It does not describe the subsequent impact or extended contact if the disk falls flat.

The spatial momentum balances retain the simple forms $\dot{p} = f - Mge_3$ and $\dot{\pi} = \rho \times f$. The coupling between translation and rotation enters through the contact geometry and the rolling constraint.

2.4.2 Free Rigid Body Rotational Motion

In this section we analyze the dynamics (the evolution over time) of a freely rotating rigid body. Here, "free" means that the resultant external force moment about the chosen point vanishes, so $T = 0$. We take this point to be either the center of mass or a fixed pivot, and $\mathbb{I}$ denotes the inertia about that point. When moments are taken about the center of mass, the energy considered below is the rotational kinetic energy, with the translational kinetic energy treated separately. When moments are taken about a fixed pivot, it is the total kinetic energy.

We have seen that it is always possible to choose a body frame **b** in which $\mathbb{I}$ is diagonal. Thus, without loss of generality, we use such a frame.

When $T = 0$, the rotational kinetic energy KE, the spatial angular momentum π, and the magnitude of the body angular momentum $\|\Pi\|$ are conserved. We consider nonzero angular momentum, writing $h = \|\Pi\| > 0$; the case $h = 0$ corresponds to no rotation.

For free rigid body rotation,

$$\pi(t) = R(t)\begin{bmatrix}\mathbb{I}_1\Omega_1\\ \mathbb{I}_2\Omega_2\\ \mathbb{I}_3\Omega_3\end{bmatrix} = R(t)\begin{bmatrix}\Pi_1\\ \Pi_2\\ \Pi_3\end{bmatrix} = \mu, \tag{2.36}$$

$$\text{KE} = \frac{1}{2}\left(\mathbb{I}_1\Omega_1^2 + \mathbb{I}_2\Omega_2^2 + \mathbb{I}_3\Omega_3^2\right) = \frac{1}{2}\left(\frac{\Pi_1^2}{\mathbb{I}_1} + \frac{\Pi_2^2}{\mathbb{I}_2} + \frac{\Pi_3^2}{\mathbb{I}_3}\right) = E, \tag{2.37}$$

$$\|\Pi(t)\|^2 = \mathbb{I}_1^2\Omega_1^2 + \mathbb{I}_2^2\Omega_2^2 + \mathbb{I}_3^2\Omega_3^2 = \Pi_1^2 + \Pi_2^2 + \Pi_3^2 = h^2, \tag{2.38}$$

where μ, E, and h are constant.

2.4.2.1 Relative Equlibria

Notice that the energy and momentum-magnitude conservation laws are conveniently expressed using the body angular momentum Π. Thus we consider the free rotational equations in the form

$$\dot{R} = R\widehat{\mathbb{I}^{-1}\Pi}, \tag{2.39}$$

$$\dot{\Pi} = \Pi \times (\mathbb{I}^{-1}\Pi). \tag{2.40}$$

In a body frame where $\mathbb{I}$ is diagonal, the angular momentum equation becomes

$$\dot{\Pi}_1 = \frac{\mathbb{I}_2 - \mathbb{I}_3}{\mathbb{I}_2\mathbb{I}_3}\Pi_2\Pi_3, \tag{2.41}$$

$$\dot{\Pi}_2 = \frac{\mathbb{I}_3 - \mathbb{I}_1}{\mathbb{I}_3\mathbb{I}_1}\Pi_3\Pi_1, \tag{2.42}$$

$$\dot{\Pi}_3 = \frac{\mathbb{I}_1 - \mathbb{I}_2}{\mathbb{I}_1 \mathbb{I}_2} \Pi_1 \Pi_2. \tag{2.43}$$

These equations admit the equilibrium solutions

$$\Pi(t) \equiv \pm h \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \Pi(t) \equiv \pm h \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad \Pi(t) \equiv \pm h \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Since $\Omega = \mathbb{I}^{-1}\Pi$, these solutions correspond to steady rotations about the principal axes. They are equilibria of the body angular momentum equations, but the orientation R continues to change. They are therefore called *relative equilibria* of the rigid body motion. When the principal moments are distinct, these are the only equilibria on $\|\Pi\| = h$.

Equation (2.37) defines an ellipsoid, called the *energy ellipsoid*, in body angular momentum space. Equation (2.38) defines the *angular momentum sphere*. Conservation of energy and angular momentum magnitude requires every solution of (2.41)–(2.43) to lie on both surfaces.

The solutions lie in the intersection of the energy ellipsoid and angular momentum sphere. Where these surfaces intersect transversely, their intersection is a curve along which the body angular momentum evolves. Equilibria and degenerate intersections require separate consideration.

2.4.2.2 Precession

A solution being periodic in Π-space does not imply that the corresponding orientation of the rigid body has the same period. To see this, let τ be a period, so that $\Pi(\tau) = \Pi(0)$. Since $\pi(t) \equiv \mu$,

$$R^T(0)\mu = \Pi(0) = \Pi(\tau) = R^T(\tau)\mu,$$

and hence

$$\mu = R(\tau)R^T(0)\mu.$$

Thus $R(\tau)R^T(0)$ is a rotation about the μ axis, possibly the identity, and generally $R(\tau) \neq R(0)$. We may write

$$R(\tau)R^T(0) = \exp(\theta\hat{n}), \tag{2.44}$$

where $n = \mu/\|\mu\|$. The angle θ, understood modulo 2π, is called the *phase* of the rotation over one period of the body angular momentum.

It is interesting to note that

$$2\,\mathrm{KE} = \Pi(t) \cdot \Omega(t) = \pi(t) \cdot \omega(t) = \mu \cdot \omega(t) = 2E. \tag{2.45}$$

Thus the component of angular velocity along the fixed spatial angular momentum direction is constant. The angle between them need not be constant, since $\|\omega(t)\|$ may vary.

To visualize the motion of the body, let Υ be a fixed unit direction expressed in the body frame; for instance, its third principal direction. Its spatial representation is $R\Upsilon$. Since

$$\mu \cdot R\Upsilon = (R\Pi) \cdot (R\Upsilon) = \Pi \cdot \Upsilon,$$

the angle between this body-fixed direction and the constant spatial angular momentum can be computed entirely using body-frame representations.

For $\|\Upsilon\| = 1$, the angle

$$\alpha(t) = \cos^{-1}\left(\frac{\Upsilon \cdot \Pi(t)}{h}\right) \tag{2.46}$$

between the body-fixed direction $R\Upsilon$ and the constant spatial angular momentum π is called the *nutation angle* of this axis.

2.4.2.3 Stability of the Relative Equilibria

The nature of the intersections depends on the three principal moments of inertia $\{\mathbb{I}_1, \mathbb{I}_2, \mathbb{I}_3\}$. For instance, when $\mathbb{I}_1 = \mathbb{I}_2 = \mathbb{I}_3 = I_0$, as for a spherically symmetric body, the energy ellipsoid is a sphere. For compatible values $h^2 = 2I_0E$, the two surfaces coincide. Nevertheless, every solution remains at a single point on this sphere, since (2.41)–(2.43) reduce to $\dot{\Pi} = 0$. Each solution therefore corresponds to steady rotation about a fixed axis.

For an axisymmetric body with $\mathbb{I}_1 > \mathbb{I}_2 = \mathbb{I}_3$, such as a thin circular disk whose symmetry axis is the first principal axis, Π_1 is constant. The nonstationary solutions trace circles on the angular momentum sphere in planes of constant Π_1. The two poles $\Pi = \pm h[1\ 0\ 0]^T$ are isolated relative equilibria, while every point on the equatorial circle $\Pi_1 = 0$ is also a relative equilibrium.

The two polar equilibria are stable. Individual equilibria on the equatorial circle are unstable: a small nonzero Π_1 produces a slow motion around a nearby circle, eventually carrying Π away from its initial equilibrium. The equatorial circle as a set is nevertheless stable. The same conclusions apply, with the axes relabelled, when $\mathbb{I}_1 = \mathbb{I}_2 > \mathbb{I}_3$.

Figure 2.3 shows an energy ellipsoid and angular momentum spheres for the asymmetric case $\mathbb{I}_1 > \mathbb{I}_2 > \mathbb{I}_3$. The pink sphere has radius greater than $\sqrt{2\mathbb{I}_2E}$, while the light blue sphere has radius less than this value. The blue curves show the critical case $h = \sqrt{2\mathbb{I}_2E}$. At this level, four nonconstant solution curves connect the two intermediate-axis equilibria $\Pi = \pm h[0\ 1\ 0]^T$, approaching them asymptotically. These solutions are not periodic.

Using the above discussion, let us investigate the qualitative behavior of an asymmetric rigid body with $\mathbb{I}_1 > \mathbb{I}_2 > \mathbb{I}_3$ at a fixed angular momentum magnitude $\|\Pi\| = h > 0$. Figure 2.4 shows the intersections of energy ellipsoids at different energy levels with this angular momentum sphere. The relative equilibria corresponding to rotation in either direction about

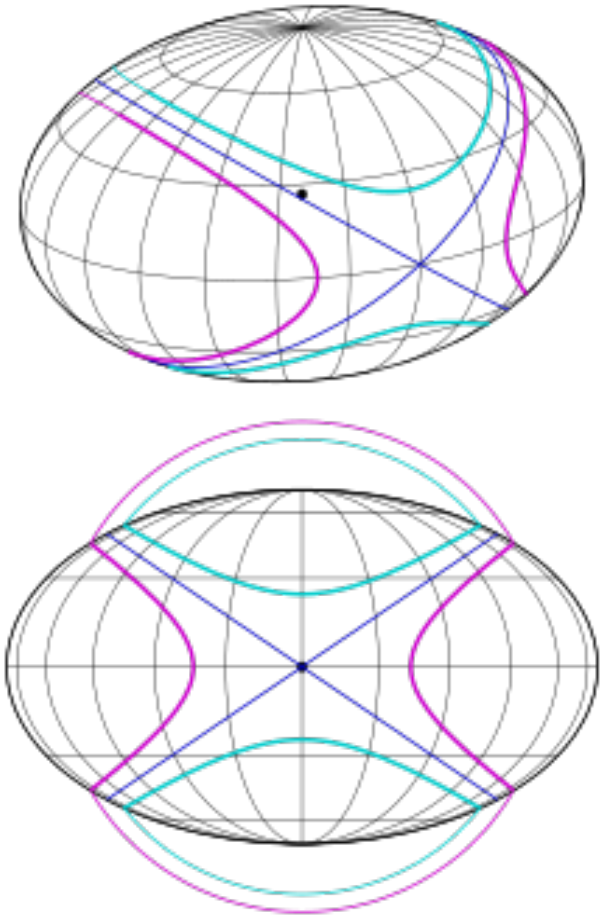


Fig. 2.3 Source: Wikipedia. Intersections of an energy ellipsoid with angular momentum spheres for $\mathbb{I}_1 > \mathbb{I}_2 > \mathbb{I}_3$. The pink and light blue spheres have radii greater than and less than $\sqrt{2\mathbb{I}_2 E}$, respectively. The blue intersection curves correspond to the critical radius $h = \sqrt{2\mathbb{I}_2 E}$ and contain the nonperiodic solutions connecting the two intermediate-axis equilibria.

the axes of largest and smallest principal moments are stable, while those corresponding to rotation about the intermediate principal axis are unstable. Every nonconstant solution in body angular momentum space is periodic except the four connecting solutions at the critical energy $E = h^2/(2\mathbb{I}_2)$.

The two conserved quantities constrain the body angular momentum to these intersections and allow the equations (2.41)–(2.43) to be reduced to a single scalar equation that can be integrated by quadrature. In this sense, the body angular momentum equations are completely integrable. In the following section we obtain their analytical solutions; the orientation is then determined by the kinematic equation (2.39).

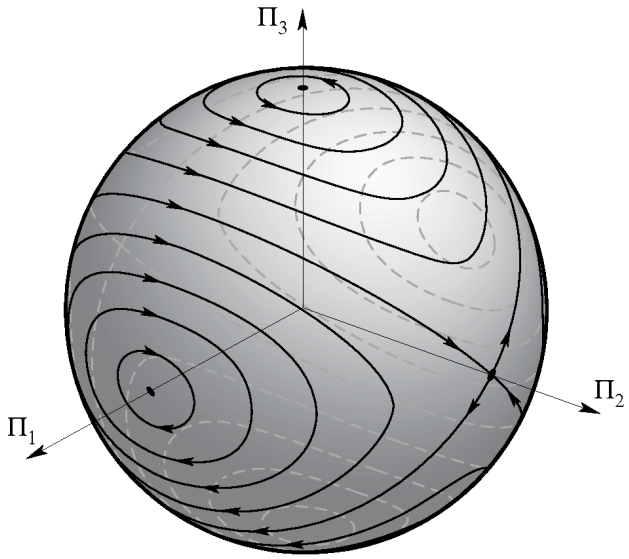


Fig. 2.4 The figure shows several intersection curves of different energy ellipsoid with a fixed constant angular momentum sphere. Figure taken from https://www.angelfire.com/rnb/pp0/poleshift.html

2.4.2.4 Analytic Solutions of Free Rigid Body Rotations

Let us first consider the asymmetric rigid body, $\mathbb{I}_1 > \mathbb{I}_2 > \mathbb{I}_3$. From the conservation laws (2.37) and (2.38), with $h = \|\Pi\|$, we have

$$\Pi_1^2 = \alpha_{21} - \beta_{21}\Pi_2^2, \tag{2.47}$$
$$\Pi_3^2 = \alpha_{23} - \beta_{23}\Pi_2^2, \tag{2.48}$$

where

$$\alpha_{21} = \frac{\mathbb{I}_1(h^2 - 2E\mathbb{I}_3)}{\mathbb{I}_1 - \mathbb{I}_3}, \qquad \alpha_{23} = \frac{\mathbb{I}_3(2E\mathbb{I}_1 - h^2)}{\mathbb{I}_1 - \mathbb{I}_3},$$

and

$$\beta_{21} = \frac{\mathbb{I}_1(\mathbb{I}_2 - \mathbb{I}_3)}{\mathbb{I}_2(\mathbb{I}_1 - \mathbb{I}_3)}, \qquad \beta_{23} = \frac{\mathbb{I}_3(\mathbb{I}_1 - \mathbb{I}_2)}{\mathbb{I}_2(\mathbb{I}_1 - \mathbb{I}_3)}.$$

For $2E\mathbb{I}_3 < h^2 < 2E\mathbb{I}_1$, both α_{21} and α_{23} are positive. The assumed ordering of the principal moments also gives $\beta_{21} > 0$ and $\beta_{23} > 0$.

Using these expressions, the square of (2.42) becomes

$$\dot{\Pi}_2^2 = \left(\frac{\mathbb{I}_1 - \mathbb{I}_3}{\mathbb{I}_1\mathbb{I}_3}\right)^2 (\alpha_{21} - \beta_{21}\Pi_2^2)(\alpha_{23} - \beta_{23}\Pi_2^2).$$

The sign of $\dot{\Pi}_2$ must still agree with the original equation

$$\dot{\Pi}_2 = -\frac{\mathbb{I}_1 - \mathbb{I}_3}{\mathbb{I}_1\mathbb{I}_3}\Pi_1\Pi_3.$$

We first treat the regime

$$2E\mathbb{I}_3 < h^2 < 2E\mathbb{I}_2.$$

Introduce

$$u \triangleq \sqrt{\frac{\beta_{21}}{\alpha_{21}}}\,\Pi_2,$$

$$\omega \triangleq \sqrt{\beta_{21}\alpha_{23}}\,\frac{\mathbb{I}_1-\mathbb{I}_3}{\mathbb{I}_1\mathbb{I}_3} = \sqrt{\frac{(\mathbb{I}_2-\mathbb{I}_3)(2E\mathbb{I}_1-h^2)}{\mathbb{I}_1\mathbb{I}_2\mathbb{I}_3}},$$

and

$$k^2 \triangleq \frac{\alpha_{21}\beta_{23}}{\alpha_{23}\beta_{21}} = \frac{(\mathbb{I}_1-\mathbb{I}_2)(h^2-2E\mathbb{I}_3)}{(\mathbb{I}_2-\mathbb{I}_3)(2E\mathbb{I}_1-h^2)}.$$

In this regime $0<k<1$, and the scalar equation reduces to

$$\dot{u}^2 = \omega^2(1-u^2)(1-k^2u^2).$$

On a branch where $\dot{u}>0$, separation of variables gives

$$\int \frac{du}{\sqrt{(1-u^2)(1-k^2u^2)}} = \omega t + c.$$

For $-1 \le u \le 1$, the integral

$$\mathrm{sn}^{-1}(u;k) = \int_0^u \frac{dv}{\sqrt{(1-v^2)(1-k^2v^2)}}$$

defines the principal inverse of the Jacobi elliptic function sn with modulus k. Inverting on this branch and continuing through the turning points gives

$$u(t) = \mathrm{sn}(\omega t + c;k),$$

where c is determined by the initial conditions. Consequently,

$$\Pi_2(t) = \sqrt{\frac{\mathbb{I}_2(h^2-2E\mathbb{I}_3)}{\mathbb{I}_2-\mathbb{I}_3}}\,\mathrm{sn}(\omega t + c;k). \tag{2.49}$$

From (2.47) and (2.48),

$$\Pi_1^2 = \alpha_{21}(1-u^2) = \alpha_{21}\mathrm{cn}^2(\omega t + c;k),$$
$$\Pi_3^2 = \alpha_{23}(1-k^2u^2) = \alpha_{23}\mathrm{dn}^2(\omega t + c;k).$$

The component signs must be chosen consistently with the original differential equations. Writing $\sigma = \mathrm{sgn}\,\Pi_3(0) \in \{-1,1\}$, the corresponding solutions are

$$\Pi_1(t) = -\sigma\sqrt{\frac{\mathbb{I}_1(h^2-2E\mathbb{I}_3)}{\mathbb{I}_1-\mathbb{I}_3}}\,\mathrm{cn}(\omega t + c;k), \tag{2.50}$$

$$\Pi_3(t) = \sigma\sqrt{\frac{\mathbb{I}_3(2E\mathbb{I}_1 - h^2)}{\mathbb{I}_1 - \mathbb{I}_3}}\,\mathrm{dn}(\omega t + c; k). \tag{2.51}$$

Here Π_3 retains its sign, while c fixes the initial values of Π_1 and Π_2.

For $0 < k < 1$, the Jacobi elliptic functions sn and cn have period $4K(k)$, while dn has period $2K(k)$, where

$$K(k) = \int_0^1 \frac{du}{\sqrt{(1-u^2)(1-k^2u^2)}} = \int_0^{\pi/2} \frac{d\psi}{\sqrt{1-k^2\sin^2\psi}}.$$

For small k,

$$K(k) = \frac{\pi}{2}\left(1 + \frac{k^2}{4} + O(k^4)\right).$$

Thus the body angular momentum $\Pi(t)$ has period

$$T = \frac{4K(k)}{\omega}.$$

For small k, this gives

$$\begin{aligned} T &= \frac{2\pi}{\omega}\left(1 + \frac{k^2}{4} + O(k^4)\right) \\ &\approx 2\pi\sqrt{\frac{\mathbb{I}_1\mathbb{I}_2\mathbb{I}_3}{(2E\mathbb{I}_1 - h^2)(\mathbb{I}_2 - \mathbb{I}_3)}}\left(1 + \frac{(\mathbb{I}_1 - \mathbb{I}_2)(h^2 - 2E\mathbb{I}_3)}{4(\mathbb{I}_2 - \mathbb{I}_3)(2E\mathbb{I}_1 - h^2)}\right). \end{aligned}$$

At the critical level $h^2 = 2E\mathbb{I}_2$, the modulus approaches $k = 1$ and $K(k)$ diverges, corresponding to the nonperiodic connecting solutions. The other regime, $2E\mathbb{I}_2 < h^2 < 2E\mathbb{I}_1$, requires a different elliptic-function parametrization and is not covered by the formulas above.

Finally, periodicity of $\Pi(t)$ does not by itself imply periodicity of the body's orientation $R(t)$, as discussed previously.

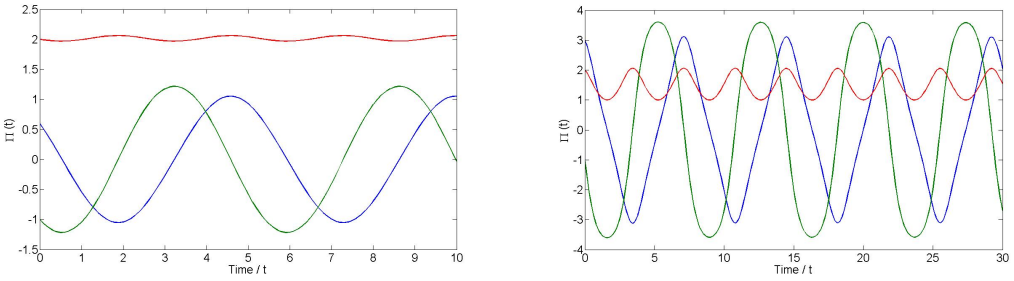
Figure 2.5 shows the simulation results of for an asymmetric rigid body of $\mathbb{I} = \mathrm{diag}[3,2,1]$.

Let us explicitly compute the nutation angle $\alpha(t)$ between the third principal axis, whose body-frame representation is $\Upsilon = [0\ 0\ 1]^T$, and the constant spatial angular momentum π. In the regime considered above, (2.46) gives

$$\begin{aligned} \alpha(t) &= \cos^{-1}\left(\frac{\Upsilon\cdot\Pi(t)}{h}\right) = \cos^{-1}\left(\frac{\Pi_3(t)}{h}\right) \\ &= \cos^{-1}\left(\sigma\sqrt{\frac{\mathbb{I}_3(2E\mathbb{I}_1 - h^2)}{h^2(\mathbb{I}_1 - \mathbb{I}_3)}}\,\mathrm{dn}(\omega t + c; k)\right), \end{aligned} \tag{2.52}$$

where $\sigma = \mathrm{sgn}\,\Pi_3(0)$.

Figure 2.6 shows the motion of the tip of this body-fixed axis in the spatial frame for an asymmetric rigid body with $\mathbb{I} = \mathrm{diag}(3,2,1)$ and initial angular momentum $\Pi(0) = [0.6\ -\ 1\ 2]^T$. For these initial conditions, $\sigma = 1$.



(a) (b)

Fig. 2.5 Components of the angular momentum vector $\Pi(t)$ for the Non-Aix-Symmetric Rigid Body. $\mathbb{I} = \text{diag}[3,2,1]$. Figure (a) corresponds to an initial condition of $\Pi(0) = [0.6,-1,2]$ thus a $k = 0.2951$ and $T = 5.4$. Figure (b) corresponds to an initial condition of $\Pi(0) = [3,-1,2]$ thus $k = 0.88$ and $T = 7.34$.

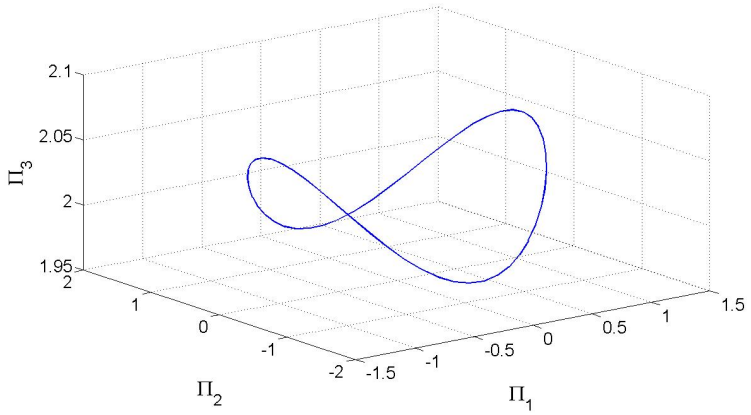


Fig. 2.6 The motion of the tip of the minor axis for the asymmetric rigid body with $\mathbb{I} = \text{diag}[3,2,1]$ and an initial conditions $\Pi(0) = [0.6,-1,2]$.

The Special Case of an Axi-symmetric Rigid Body

Consider an axi-symmetric rigid body with $\mathbb{I}_1 = \mathbb{I}_2 > \mathbb{I}_3$. For nonconstant body angular momentum, $0 < |\Pi_3| < h$, we have $k = 0$, and the elliptic functions reduce to trigonometric functions: $\text{sn}(s;0) = \sin s$, $\text{cn}(s;0) = \cos s$, and $\text{dn}(s;0) = 1$. Thus

$$\omega = \sqrt{\frac{(2E\mathbb{I}_1 - h^2)(\mathbb{I}_2 - \mathbb{I}_3)}{\mathbb{I}_1\mathbb{I}_2\mathbb{I}_3}},$$

and, with $\sigma = \text{sgn}\,\Pi_3(0)$,

$$\Pi_1(t) = -\sigma\sqrt{\frac{\mathbb{I}_1(h^2 - 2E\mathbb{I}_3)}{\mathbb{I}_1 - \mathbb{I}_3}}\cos(\omega t + c), \tag{2.53}$$

$$\Pi_2(t) = \sqrt{\frac{\mathbb{I}_2(h^2 - 2E\mathbb{I}_3)}{\mathbb{I}_2 - \mathbb{I}_3}}\sin(\omega t + c), \tag{2.54}$$

$$\Pi_3(t) = \sigma\sqrt{\frac{\mathbb{I}_3(2E\mathbb{I}_1 - h^2)}{\mathbb{I}_1 - \mathbb{I}_3}} = \text{constant}. \tag{2.55}$$

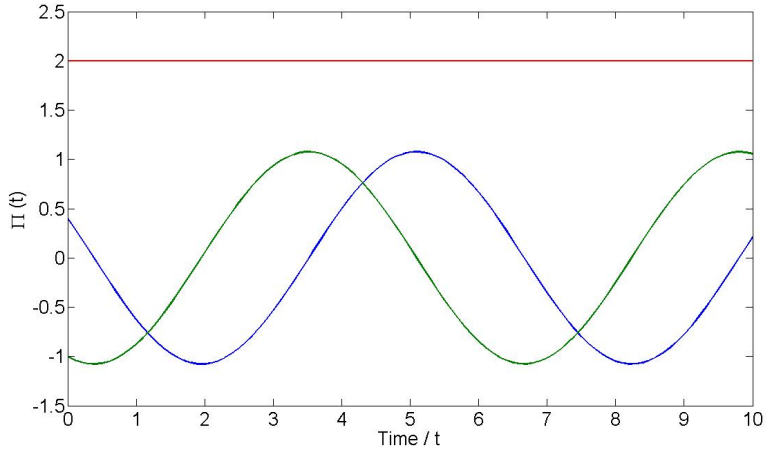


Fig. 2.7 Components of the body angular momentum $\Pi(t)$ for an axi-symmetric rigid body with $\mathbb{I} = \text{diag}(2,2,1)$ and $\Pi(0) = [0.4\ \ -1\ \ 2]^T$. The oscillation period is $T = 2\pi \approx 6.28$.

The nonconstant body angular momentum $\Pi(t)$ is periodic with period

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{\mathbb{I}_1\mathbb{I}_2\mathbb{I}_3}{(2E\mathbb{I}_1 - h^2)(\mathbb{I}_2 - \mathbb{I}_3)}}.$$

The cases $\Pi_3 = 0$ and $|\Pi_3| = h$ instead give constant Π, corresponding to steady rotations about principal axes.

Consider the symmetry axis $\Upsilon = [0\ 0\ 1]^T$, represented in the body frame. Its angle with the constant spatial angular momentum is

$$\begin{aligned}\alpha &= \cos^{-1}\left(\frac{\Upsilon\cdot\Pi}{h}\right) = \cos^{-1}\left(\frac{\Pi_3}{h}\right)\\ &= \cos^{-1}\left(\sigma\sqrt{\frac{\mathbb{I}_3(2E\mathbb{I}_1 - h^2)}{h^2(\mathbb{I}_1 - \mathbb{I}_3)}}\right) = \text{constant}.\end{aligned}$$

Thus the nutation angle of the symmetry axis remains constant.

Figure 2.7 shows the angular momentum components for $\mathbb{I} = \text{diag}(2,2,1)$ and $\Pi(0) = [0.4\ \ -1\ \ 2]^T$. Figure 2.8 shows the motion of the tip of the body angular momentum representation $\Pi(t)$ in the body frame.

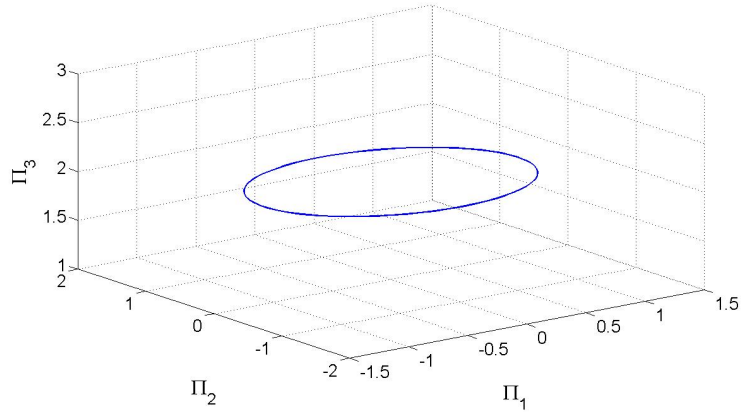


Fig. 2.8 Motion of the tip of the body angular momentum $\Pi(t)$ for an axi-symmetric rigid body with $\mathbb{I} = \text{diag}(2,2,1)$ and $\Pi(0) = [0.6 \ -1 \ 2]^T$.

2.4.3 The Axi-symmetric Heavy Top in a Gravitational Field

In this section we consider an axi-symmetric top of mass m in a uniform gravitational field, with an ideal fixed pivot on its axis of symmetry. The distance from the pivot to the center of mass is l. Let $\mathbf{a}$ be an inertial frame and $\mathbf{b} = \mathbf{a}R$ a body-fixed frame, both with origin at the pivot. Choose the third body axis to point from the pivot towards the center of mass, so that

$$\bar{X} = l\chi, \qquad \chi = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Let Υ be the upward unit direction expressed in $\mathbf{a}$. Gravity therefore has body-frame representation $-mgR^T\Upsilon$, and its moment about the pivot is

$$T = (l\chi) \times (-mgR^T\Upsilon) = mgl(R^T\Upsilon) \times \chi.$$

Neglecting friction and any other external moments, the equations of motion are

$$\dot{R} = R\widehat{\Omega}, \tag{2.56}$$

$$\mathbb{I}\dot{\Omega} = (\mathbb{I}\Omega) \times \Omega + mgl(R^T\Upsilon) \times \chi, \tag{2.57}$$

where $\mathbb{I} = \text{diag}(\mathbb{I}_1, \mathbb{I}_2, \mathbb{I}_3)$ is the inertia about the pivot, expressed in $\mathbf{b}$, and $\mathbb{I}_1 = \mathbb{I}_2$.

One way to describe $R(t)$ is to use the 3-1-3 Euler angles. Taking $\mathbf{a}_3$ vertically upwards and introducing intermediate frames $\mathbf{a}'(t)$ and $\mathbf{f}(t)$, as in Figure 2.9, we have

$$\mathbf{a}' = \mathbf{a}R_3(\varphi), \qquad \mathbf{f} = \mathbf{a}'R_1(\theta), \qquad \mathbf{b} = \mathbf{f}R_3(\psi).$$

Thus

$$R = R_3(\varphi)R_1(\theta)R_3(\psi).$$

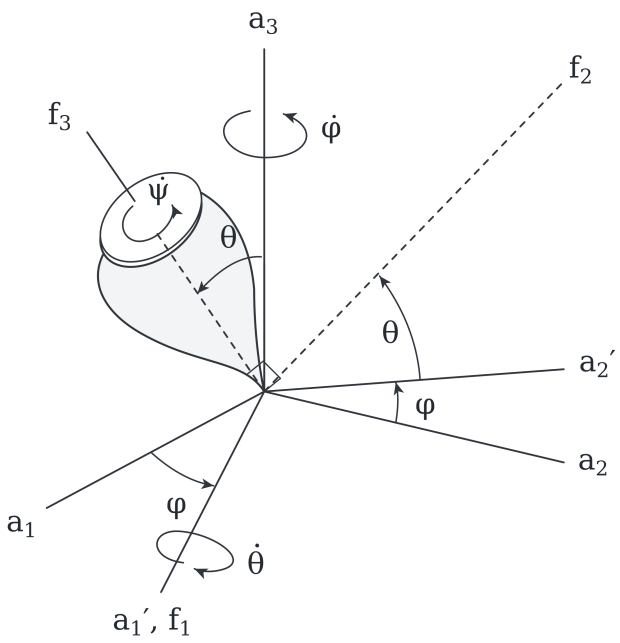


Fig. 2.9 The axi-symmetric heavy top.

Here φ is the angle of *precession*, θ the angle of *nutation*, and ψ the angle of *spin*. Below we will formulate the equations of motion without resorting to this parameterization. The parameterization can be considered for illustration purposes.

Alternatively, let $\Gamma = R^T \Upsilon$ be the upward vertical direction expressed in the body frame. Differentiating gives the reduced equations

$$\dot{\Gamma} = \Gamma \times \Omega, \tag{2.58}$$

$$\mathbb{I}\dot{\Omega} = (\mathbb{I}\Omega) \times \Omega + mgl\,\Gamma \times \chi. \tag{2.59}$$

These determine (Γ, Ω) without having to integrate the rotational kinematics $\dot{R} = R\widehat{\Omega}$.

Writing $\Pi = \mathbb{I}\Omega$, the following quantities are conserved:

$$\begin{aligned} E &= \frac{1}{2}\Omega^T \mathbb{I}\Omega + mgl\,\Gamma^T \chi \\ &= \frac{1}{2}\left(\frac{\Pi_1^2}{\mathbb{I}_1} + \frac{\Pi_2^2}{\mathbb{I}_2} + \frac{\Pi_3^2}{\mathbb{I}_3}\right) + mgl\,\Gamma_3, \end{aligned} \tag{2.60}$$

$$\Pi^T \Gamma = \Pi_1\Gamma_1 + \Pi_2\Gamma_2 + \Pi_3\Gamma_3, \tag{2.61}$$

$$\|\Gamma\|^2 = \Gamma_1^2 + \Gamma_2^2 + \Gamma_3^2 = 1, \tag{2.62}$$

$$\Pi_3 = \mathbb{I}_3\Omega_3. \tag{2.63}$$

Here E is the total mechanical energy, $\Pi^T\Gamma$ is the vertical component of spatial angular momentum, and conservation of Π_3 follows from axial symmetry.

An axi-symmetric heavy top with its pivot on the symmetry axis is called a *Lagrange top*. The upright spinning solution has $\Gamma = \chi$ and $\Omega = \Omega_3 \chi$. It is stable in the reduced variables provided

$$|\Omega_3| > \frac{2\sqrt{mgl\mathbb{I}_1}}{\mathbb{I}_3}.$$

This is stability of the spinning relative equilibrium, rather than of its accumulated spin angle (see Section 15.10 of [5]).

2.5 Exercises

Exercise 2.1. Let $\mathscr{B}_1$ and $\mathscr{B}_2$ be two rigid bodies moving in a plane. At an instant when their relative angular velocity is nonzero, show that there is a unique point in the plane at which their rigid-body velocity fields agree. This point is called the *instantaneous center of relative motion*. It need not lie within either physical body.

Discuss separately the cases of nonzero relative pure translation and zero relative motion.

Exercise 2.2. *Kennedy's Theorem:* Consider three rigid bodies moving in a plane. Assume that each pair has nonzero relative angular velocity, so that its instantaneous center is uniquely defined. Prove that the three instantaneous centers are collinear, including the degenerate case in which they coincide.

Exercise 2.3. Let $\mathbb{I}$ be the inertia tensor of a rigid body about a body-fixed point O, expressed in a body-fixed frame $\mathbf{b}$. Let $\mathbb{I}_c$ be its inertia tensor about the center of mass O_c, expressed in axes parallel to $\mathbf{b}$. Let $\bar{X}$ be the representation of OO_c in $\mathbf{b}$, and let M be the total mass. Prove the parallel-axis theorem:

$$\mathbb{I} = \mathbb{I}_c - M\widehat{\bar{X}}^2.$$

Exercise 2.4. Consider a rigid body consisting of particles with positive masses in a non-collinear configuration. Let $\mathbb{I}$ be its inertia tensor about a body-fixed point O.

(i) Prove that $\mathbb{I}$ is symmetric and positive definite.
(ii) Show that there exists a right-handed orthonormal body frame with origin O in which $\mathbb{I}$ is diagonal.

Exercise 2.5. Let

$$\mathbb{I} = -\sum_i m_i \widehat{X}_i^2, \qquad m_i > 0,$$

be the inertia tensor of a collection of particles about the origin O. Show that, for every $u \in \mathbb{R}^3$,

$$u^T \mathbb{I} u = \sum_i m_i \|X_i \times u\|^2.$$

Hence show that $\mathbb{I}$ is positive semidefinite, and determine precisely when it is positive definite. Discuss the case in which all particles lie on a line through O.

Exercise 2.6. Consider a rigid body made of three point particles. The first two particles have mass m and body-frame positions

$$X_1 = \ell \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \qquad X_2 = \ell \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix},$$

while the third has mass $2m$ and position

$$X_3 = \ell \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix},$$

where $\ell > 0$ is a fixed length. Calculate the inertia tensor $\mathbb{I}$ about the origin of the body frame **b**, expressed in that frame.

Exercise 2.7. Calculate the inertia tensors of the bodies shown in Figure 2.10. Each body has total mass M and a uniform mass distribution. Use the following idealizations and reference frames:

(i) A slender rod of length L, with origin at one end and the first axis along the rod. Neglect its transverse dimensions.
(ii) A solid cylinder of radius r and height h, with origin at its center of mass and the third axis along its symmetry axis.
(iii) A solid cone of base radius r and height h, with origin at its vertex and the third axis pointing towards the center of its base.
(iv) A thin spherical shell of radius r, with origin at its center.

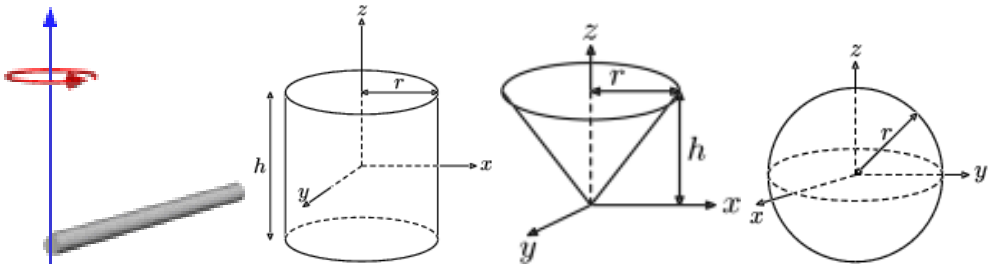


Fig. 2.10 A slender rod, solid cylinder, solid cone, and thin spherical shell. Use the reference frames specified in the exercise.

Exercise 2.8. Consider the shaft and horizontal arm shown in Figure 1.23, excluding the spring and suspended particle. Model the shaft and arm as uniform solid cylinders rigidly joined together, neglecting any correction for overlap at the joint.

The shaft has mass M_s, radius r_s, and length l_s. The arm has mass M_a, radius r_a, and length d. Let a body-fixed frame **b** have origin at the center of the lower end of the shaft, with $\mathbf{b}_3$ directed upwards along the shaft and $\mathbf{b}_1$ along the arm. The shaft center of mass has coordinates $[0\ 0\ l_s/2]^T$, and the arm center of mass has coordinates $[d/2\ 0\ l_s]^T$.

Find the inertia tensor of the assembly about the frame origin, expressed in **b**.

Exercise 2.9. Consider the system shown in Figure 2.11. A vertical circular hoop of radius r rotates freely about its fixed vertical diameter. Its moment of inertia about this axis, excluding the bead, is $\mathbb{I}_z$. A bead of mass m, modeled as a particle, slides without friction on the hoop.

Let $\theta(t)$ describe the hoop's rotation about the vertical axis, and let $\phi(t)$ describe the bead's position on the hoop, measured from the upward vertical towards a chosen horizontal direction fixed in the hoop. Gravity acts vertically downwards with magnitude g. The only applied torque about the vertical shaft is the viscous resisting torque $-C\dot{\theta}$.

Derive the coupled equations of motion. Simulate the system with

$$\mathbb{I}_z = 1\,\mathrm{kg\,m^2}, \qquad m = 1\,\mathrm{kg}, \qquad r = 1\,\mathrm{m},$$
$$C = 1\,\mathrm{N\,m\,s}, \qquad g = 1\,\mathrm{m\,s^{-2}},$$

and initial conditions

$$\theta(0) = 0, \qquad \dot{\theta}(0) = 1\,\mathrm{s^{-1}}, \qquad \phi(0) = \frac{7\pi}{6}, \qquad \dot{\phi}(0) = 0.$$

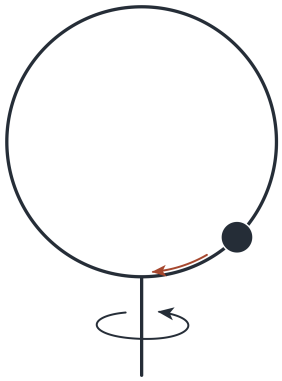


Fig. 2.11 A bead sliding on a hoop that is free to rotate about its vertical diameter.

Exercise 2.10. Figure 2.12 shows an idealized model of the free/forced vibration apparatus in the applied mechanics laboratory. A rigid beam is attached to a fixed frictionless pivot at O. A vertical viscous damper is attached at A, and a spring is attached at B. The other end of the spring, P, has a prescribed vertical displacement $z(t)$, positive upwards.

Let $\theta(t)$ be the beam's downward angular displacement from the horizontal, as shown. When $\theta = 0$ and $z = 0$, the spring is unstretched. For this idealized model, neglect gravity. The spring stiffness is k, the damping coefficient is C, and the beam's moment of inertia about the pivot axis is I. Let $OA = L_c$ and $OB = L_k$.

Neglect the beam's thickness and assume small $|\theta|$, so that the spring and damper forces may be treated as vertical. Derive the governing equation for $\theta(t)$.

Exercise 2.11. Consider the system shown in Figure 1.23. The shaft and horizontal arm are free to rotate about the fixed vertical shaft axis. Their combined moment of inertia about that axis is $\mathbb{I}_z$. An external driving torque, τ_u acts about the shaft axis.

A particle P of mass m is attached to the end of the arm by a massless spring of stiffness k and unstretched length l_0. The arm length is d. Assume that an ideal frictionless guide constrains the particle to the rotating vertical plane through the arm's end and perpendicular to the arm. Gravity acts vertically downwards with magnitude g.

Let $\varphi(t)$ be the arm's rotation about the vertical shaft, $l(t)$ the spring length, and $\theta(t)$ the spring's angle from the downward vertical, positive towards $-\mathbf{b}_2$, where $\mathbf{b}_1$ points along the arm and $\mathbf{b}_3$ points vertically upwards.

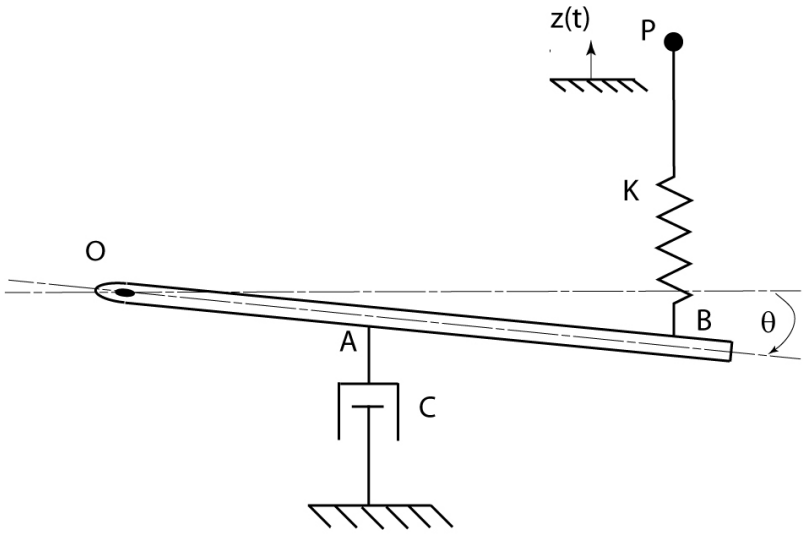


Fig. 2.12 An idealized beam–spring–damper model with prescribed vertical motion of the spring support.

Derive the coupled equations of motion and the kinetic energy of the system. Simulate the motion, choosing and reporting physically consistent parameter values and initial conditions with $l(0) > 0$ and nonzero $\dot{\varphi}(0)$. Use a time interval over which $l(t) > 0$.

Exercise 2.12. Consider the uniform slender bar pendulum shown in Figure 2.13. The bar has mass M and length L and rotates in a vertical plane about a fixed frictionless pivot at one end. Gravity acts vertically downwards with magnitude g.

Let θ be the angle from the downward vertical, and let $T(t)$ be a prescribed applied torque about the pivot, positive in the direction of increasing θ. Derive the equation of motion using Euler's rigid body equations, and find the kinetic energy. State the equation for the unforced case $T = 0$.

Exercise 2.13. Consider the planar double pendulum shown in Figure 2.14. Two point particles of masses m_1 and m_2 are connected by massless rigid links of lengths L_1 and L_2. The first link is attached to the fixed pivot O, and all joints are frictionless.

Let θ_1 and θ_2 be the angles of the first and second links, respectively, each measured from the downward vertical in the same positive sense. Find the kinetic energy of the system relative to an inertial frame.

Exercise 2.14. Two point particles P_1 and P_2 move in three-dimensional Euclidean space. The particle P_1 is constrained to a fixed sphere of radius $r > 0$ centered at O, and the distance between P_1 and P_2 is fixed at $l > 0$. There are no other constraints.

Specify the configuration space and a suitable set of local coordinates. Determine the number of degrees of freedom.

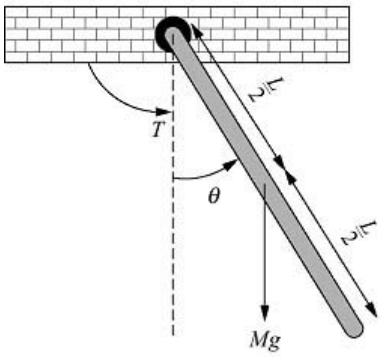


Fig. 2.13 A uniform slender bar pendulum with an applied torque about its fixed pivot.

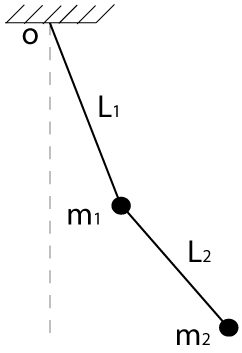


Fig. 2.14 A planar double pendulum consisting of two point masses connected by massless rigid links.

Exercise 2.15. Find the kinetic energy of the 3-DOF robot arm shown in Figure 2.15, using the following model.

Let $\mathbf{e}$ be an inertial frame with origin at the fixed base pivot O_1 and $\mathbf{e}_3$ vertical. Define

$$\mathbf{a} = \mathbf{e}R_3(\alpha), \qquad \mathbf{b} = \mathbf{a}R_1(q_1), \qquad \mathbf{c} = \mathbf{b}R_1(q_2).$$

The frames $\mathbf{a}$ and $\mathbf{b}$ have origin O_1, while $\mathbf{c}$ has origin at the elbow joint O_2. Thus α describes base rotation about the vertical axis, q_1 the first link's rotation relative to $\mathbf{a}$, and q_2 the second link's rotation relative to the first. The base rotation α is not explicitly shown in the figure.

Let G_1, G_2 be the link centers of mass and P the endpoint. Their geometry is specified by

$$O_1O_2 = \mathbf{b}\begin{bmatrix}0\\L_1\\0\end{bmatrix}, \qquad O_1G_1 = \mathbf{b}\begin{bmatrix}0\\L_2\\0\end{bmatrix},$$

$$O_2P = \mathbf{c}\begin{bmatrix}0\\L_3\\0\end{bmatrix}, \qquad O_2G_2 = \mathbf{c}\begin{bmatrix}0\\L_4\\0\end{bmatrix}.$$

The links have masses M_1, M_2 and center-of-mass inertia tensors $\mathbb{I}_{L_1}, \mathbb{I}_{L_2}$, expressed in axes parallel to $\mathbf{b}, \mathbf{c}$, respectively. Neglect the inertia of the base, motors, and joints.

Express the kinetic energy relative to $\mathbf{e}$ in terms of the joint coordinates α, q_1, q_2, their time derivatives, and the given geometric and inertial parameters.

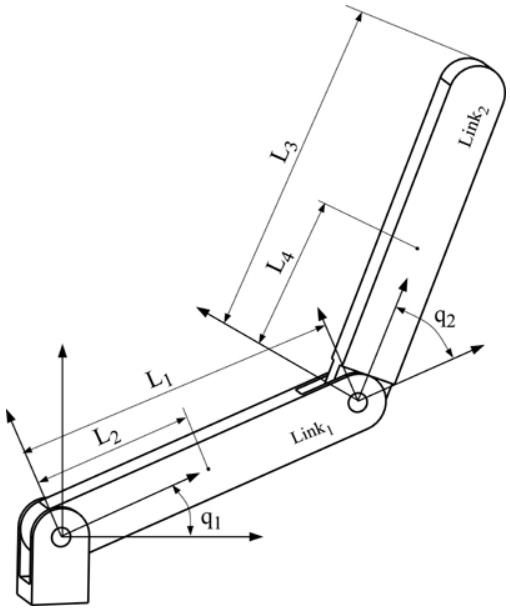


Fig. 2.15 A two-link robot arm with two illustrated joint angles and an additional rotation of the base about the vertical axis.

Exercise 2.16 (Dynamics of a Centrifugal Governor). Consider the symmetric centrifugal governor shown in Figure 2.16. Two identical upper arms are hinged at the fixed point O on a vertical rotating spindle. Each arm carries a particle of mass m at distance L from O.

Each upper arm has mass m_l, its center of mass is at distance c_l from O along the arm, and its inertia tensor about O, expressed in an arm-fixed frame, is

$$\mathbb{I} = \operatorname{diag}(\mathbb{I}_z, \mathbb{I}_l, \mathbb{I}_l).$$

Neglect the inertia of the spindle and the masses of the lower links and sliding sleeve. Assume frictionless joints, no external load on the sleeve, and symmetric motion of the two arms. The lower linkage imposes the symmetric geometry without contributing kinetic or potential energy.

Let $\mathbf{e}$ be an inertial frame with origin O and $\mathbf{e}_3$ vertically upwards. Let $\mathbf{a} = \mathbf{e}R_3(\theta)$ rotate with the spindle, with $\mathbf{a}_1$ directed horizontally towards one arm. For that arm, define

$$\mathbf{b} = \mathbf{a}R_2(\alpha), \qquad R = R_3(\theta)R_2(\alpha).$$

Thus $\mathbf{b}_1$ points along the arm from O towards the particle, and α is measured downwards from the horizontal. Consider the nondegenerate configurations $0 < \alpha < \pi/2$. The particle has body coordinates

$$X = \begin{bmatrix} L \\ 0 \\ 0 \end{bmatrix}.$$

Gravity acts vertically downwards with magnitude g. A prescribed moment $\tau_u(t)$ acts on the complete governor about the vertical spindle, positive in the direction of increasing θ.

(i) Calculate the angular velocity of the arm expressed in **b** and the inertial acceleration of the particle expressed in that frame.

(ii) Let

$$F^r = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix}$$

be the force exerted by the upper arm on its particle. Determine these components using Newton's law.

(iii) Let

$$M^r = \begin{bmatrix} M_1 \\ 0 \\ M_3 \end{bmatrix}$$

be the hinge reaction moment on the upper arm, expressed in **b**. Use Euler's rigid body equations about O to derive the equation for α and determine M_1 and M_3.

(iv) Combine the contributions of the two arms to obtain the coupled equations for θ and α. Check the spindle equation using the total vertical angular momentum.

(v) For prescribed constant spindle speed $\dot{\theta} = \Omega$, obtain the equation for α. Determine the condition for a nondegenerate steady configuration and find its inclination. State the corresponding result when the upper arms are also treated as massless.

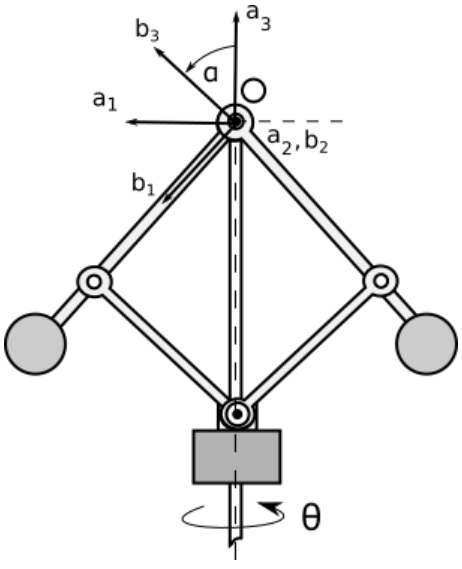


Fig. 2.16 A symmetric centrifugal governor. The angle α measures each upper arm's inclination below the horizontal. The lower links and sleeve are idealized as massless and unloaded.

Exercise 2.17 (Free Rigid Body Rotation). Consider a rigid body with no external forces or force moments. Let $\mathbf{e}$ be an inertial frame, and let $\mathbf{b} = \mathbf{e}R$ be a principal-axis body frame with origin at the center of mass. The inertia tensor in $\mathbf{b}$ is

$$\mathbb{I} = \mathrm{diag}(2,3,4)\,\mathrm{kg\,m^2}.$$

Let $\Pi = \mathbb{I}\Omega$ be the body representation of angular momentum about the center of mass.

(i) Write down the rotational equations of motion using (R,Π) as the variables. Show that

$$\pi = R\Pi, \qquad E = \frac{1}{2}\Pi^T\mathbb{I}^{-1}\Pi, \qquad h^2 = \Pi^T\Pi$$

are conserved.

(ii) Identify the steady rotations about the three principal axes. Linearize the body angular momentum equations about each nonzero steady rotation. Determine which rotations have oscillatory transverse perturbations and which have exponentially growing transverse perturbations.

(iii) Simulate the rotational motion with $R(0) = I_{3\times 3}$ and, separately,

$$\Omega(0) = \Omega_0\begin{bmatrix}1\\ \varepsilon\\ \varepsilon\end{bmatrix}, \qquad \Omega(0) = \Omega_0\begin{bmatrix}\varepsilon\\ 1\\ \varepsilon\end{bmatrix}, \qquad \Omega(0) = \Omega_0\begin{bmatrix}\varepsilon\\ \varepsilon\\ 1\end{bmatrix},$$

where $\Omega_0 = 1\,\mathrm{s}^{-1}$ and $\varepsilon = 0.01$. Use $\Pi(0) = \mathbb{I}\Omega(0)$ and simulate for at least 60 s. Plot the components of $\Omega(t)$ and $\Pi(t)$. Compare the behavior with your linearized analysis.

(iv) For each simulation, plot the path of $\Pi(t)$ on the angular momentum sphere and display the corresponding energy ellipsoid. Also plot the spatial motion of the body axis initially closest to the angular velocity direction. Explain the difference between a steady body angular momentum and a stationary body orientation.

(v) Monitor the errors in E, $\|\Pi\|$, and $\pi = R\Pi$, together with the orthogonality error $\|R^TR - I_{3\times 3}\|_F$. Check that your conclusions persist when the numerical tolerances or time step are refined.

Exercise 2.18 (Motion and Stability of an Axisymmetric Heavy Top). Consider an axisymmetric rigid body with a fixed, frictionless pivot O. Let $\mathbf{e}$ be an inertial frame with $\mathbf{e}_3$ pointing vertically upwards, and let $\mathbf{b} = \mathbf{e}R$ be a body frame with origin at O. Define

$$\chi = \begin{bmatrix}0\\0\\1\end{bmatrix}, \qquad e_3 = \begin{bmatrix}0\\0\\1\end{bmatrix}.$$

The center of mass has body coordinates $l\chi$, and the inertia tensor about the pivot is

$$\mathbb{I} = \mathrm{diag}(I_\perp, I_\perp, I_3).$$

Gravity is the only interaction producing a moment about the pivot. Define

$$\Gamma = R^Te_3, \qquad \Pi = \mathbb{I}\Omega.$$

(i) Derive the gravitational force moment in the body frame and obtain the reduced equations

$$\dot{\Gamma} = \Gamma \times \Omega, \qquad \mathbb{I}\dot{\Omega} = (\mathbb{I}\Omega) \times \Omega + mgl\,\Gamma \times \chi.$$

(ii) Show that the following quantities are conserved:

$$E = \frac{1}{2}\Omega^T \mathbb{I}\Omega + mgl\,\Gamma^T\chi, \qquad \Pi^T\Gamma, \qquad \Pi^T\chi, \qquad \|\Gamma\|^2.$$

Explain the physical meaning of the first three. Does the full spatial angular momentum $R\Pi$ remain constant?

(iii) Show that

$$\Gamma = \chi, \qquad \Omega = s\chi$$

describes an upright steady spin. Linearize the reduced equations about this motion, using perturbations consistent with $\|\Gamma\| = 1$. Determine the spin threshold separating exponentially growing transverse perturbations from oscillatory ones. You need not classify the threshold case.

(iv) Simulate the motion with

$$m = 0.5\,\mathrm{kg}, \qquad l = 0.1\,\mathrm{m}, \qquad g = 9.81\,\mathrm{m\,s^{-2}},$$

$$I_\perp = 0.02\,\mathrm{kg\,m^2}, \qquad I_3 = 0.01\,\mathrm{kg\,m^2},$$

and initial conditions

$$R(0) = R_1(0.05), \qquad \Gamma(0) = R(0)^T e_3, \qquad \Omega(0) = s\chi.$$

Compare $s = 10\,\mathrm{s^{-1}}$ and $s = 30\,\mathrm{s^{-1}}$ over a time interval of at least 10 s.
Plot the angular velocity components and the inclination angle

$$\vartheta(t) = \cos^{-1}\big(\Gamma(t)^T\chi\big).$$

Relate the observed motion to the threshold found in part (iii).

(v) Integrate $\dot{R} = R\widehat{\Omega}$ to reconstruct the orientation, and plot the path of the symmetry-axis direction $R(t)\chi$ on the unit sphere. Describe its precession and nutation.
Monitor the conserved quantities from part (ii), the consistency condition $\Gamma = R^T e_3$, and the orthogonality of R. Refine the numerical tolerances or time step to verify the results.

Exercise 2.19 (Modeling a Twin-Rotor System).

Consider the twin-rotor assembly shown in Figure 2.17, mounted on an ideal two-axis support. The support permits rotation about a fixed vertical axis and about a horizontal hinge axis carried by the rotating support. The two axes intersect at a fixed point O.

Let $\mathbf{e}$ be an inertial orthonormal frame with origin O and $\mathbf{e}_3$ pointing vertically upwards. Let $\mathbf{c}$ be attached to the rotating support, with $\mathbf{c}_3 = \mathbf{e}_3$, and let $\mathbf{b}$ be attached to the rotor assembly, with $\mathbf{b}_1 = \mathbf{c}_1$ along the horizontal hinge. Thus $\mathbf{b}_1$ always remains horizontal.

Write

$$\mathbf{b} = \mathbf{e}R, \qquad R \in SO(3), \qquad \widehat{\Omega} = R^T\dot{R}.$$

Although angles are illustrated in the figure, do not parameterize R using Euler angles or any other rotation coordinates. Define

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \qquad \Gamma = R^T e_3.$$

Model the rotor assembly as a rigid body with constant inertia tensor

$$\mathbb{I} = \mathrm{diag}(I_1, I_2, I_3), \qquad I_i > 0,$$

about O, expressed in $\mathbf{b}$. Assume that the assembly is balanced so that its center of mass is at O. Neglect the inertia of the intermediate support, bearing friction, and the additional angular momentum and reaction couples associated with rotor spin. The support joints are ideal, so their constraint moment does no work on admissible motions.

The rotor thrusts act at body-fixed positions

$$X_1 = l_1 e_2, \qquad X_2 = -l_2 e_2, \qquad l_1, l_2 > 0,$$

and have body-frame representations

$$F_1 = f_1(t) \begin{bmatrix} -\sin\alpha \\ 0 \\ \cos\alpha \end{bmatrix}, \qquad F_2 = f_2(t) \begin{bmatrix} -\sin\beta \\ 0 \\ \cos\beta \end{bmatrix}.$$

Here α, β are fixed mounting angles, measured from $\mathbf{b}_3$ towards $-\mathbf{b}_1$, and $f_1(t), f_2(t) \geq 0$ are prescribed thrust magnitudes.

(i) Show that the configuration space is

$$\mathscr{Q} = \left\{ R \in SO(3) : e_3^T R e_1 = 0 \right\},$$

and determine its number of degrees of freedom. Differentiate the configuration constraint to show that

$$n(R) \cdot \Omega = 0, \qquad n(R) \triangleq (R^T e_3) \times e_1.$$

(ii) Show that $n(R)$ is the unit representation of $\mathbf{c}_2$ in the body frame. Use the ideal-support assumption to show that the constraint moment has the form

$$T_c = \lambda n(R)$$

for some scalar $\lambda(t)$. Hence verify that $T_c \cdot \Omega = 0$.

(iii) Calculate explicitly the resultant rotor moment about O, expressed in $\mathbf{b}$:

$$T_u = X_1 \times F_1 + X_2 \times F_2.$$

Explain why gravity produces no moment about O.

(iv) Write down the constrained Euler rigid-body equations for R and Ω, including both T_u and T_c.

(v) Differentiate the velocity constraint $n(R)\cdot\Omega = 0$ and use the equations of motion to determine λ explicitly in terms of R, Ω, and the prescribed thrusts. Hence obtain a closed system of differential equations. State the compatibility conditions on $R(0)$ and $\Omega(0)$.

(vi) Find the kinetic energy and show that

$$\frac{d}{dt}\mathrm{KE} = \Omega \cdot T_u.$$

Verify this power expression directly from the thrust forces and the inertial velocities of their points of application.

When both thrusts vanish, show that the kinetic energy and the vertical component of spatial angular momentum are conserved. Is the full spatial angular momentum necessarily conserved? Explain.

Exercise 2.20 (Twin-Rotor Equations in Spatial Angular Momentum). Consider the twin-rotor system of Exercise 2.19, with the same assumptions and prescribed thrusts. Let π be the angular momentum about the fixed pivot O, expressed in $\mathbf{e}$. Define

$$\mathbb{I}^R = R\mathbb{I}R^T, \qquad K(R) = (\mathbb{I}^R)^{-1}, \qquad N(R) = e_3 \times (Re_1).$$

Without parameterizing R:

(i) Express the configuration and velocity constraints in terms of R and π.

(ii) Derive a closed system of equations for (R, π), including an explicit expression for the scalar constraint moment λ in $\tau_c = \lambda N$. State the required initial compatibility conditions.

(iii) Determine the condition on the prescribed thrusts under which the spatial rotor moment τ_u is perpendicular to N. Is this condition required for the constrained equations to be valid?

Exercise 2.21 (Spatial Momentum Model of a Constrained Two-Wheeled Vehicle). Consider the vehicle shown in Figure 2.18. We idealize the vehicle body as a rigid body and omit the wheels, replacing their effects by ideal support forces and moments at the axle midpoint O'.

Let $\mathbf{e}$ be an inertial orthonormal frame, with $\mathbf{e}_3$ pointing vertically upwards. Let $\mathbf{a}$ be an upright frame with origin O', with $\mathbf{a}_3 = \mathbf{e}_3$ and $\mathbf{a}_2$ along the axle. Let $\mathbf{b}$ be fixed to the body, with $\mathbf{b}_2 = \mathbf{a}_2$, and write

$$\mathbf{b} = \mathbf{e}R, \qquad R \in SO(3), \qquad \dot{R} = \widehat{\omega}R.$$

Do not parameterize R using rotation angles. Define the coordinate columns

$$e_2 = \begin{bmatrix}0\\1\\0\end{bmatrix}, \qquad e_3 = \begin{bmatrix}0\\0\\1\end{bmatrix}, \qquad t = Re_2, \qquad s = Re_3.$$

Let o represent O' in $\mathbf{e}$. The body has mass M, and its center of mass has representation

$$x_c = o + ls, \qquad l > 0.$$

Let $\mathbb{I}_c$ be its constant inertia tensor about the center of mass, expressed in **b**.

The ideal support keeps O' at a fixed height h and the axle horizontal:

$$e_3^T o = h, \qquad e_3^T R e_2 = 0.$$

It also prevents lateral motion along the axle:

$$t^T \dot{o} = 0.$$

Forward sliding is unrestricted; no wheel-rolling relation is imposed. Assume that the support remains engaged and that its reactions do no work on admissible motions.

Let $f_u(t)$ and $\tau_u(t)$ be the resultant applied force and applied moment about O', expressed in **e**, excluding gravity and support reactions. Gravity acts in the direction $-e_3$.

Let p be the total spatial linear momentum and π the total spatial angular momentum about the moving point O', defined using the inertial velocities of the body's particles.

(i) State the configuration space and derive the velocity constraints. With

$$N = e_3 \times t,$$

show that the admissible velocities satisfy

$$e_3^T \dot{o} = 0, \qquad t^T \dot{o} = 0, \qquad N^T \omega = 0.$$

Explain why the lateral velocity constraint is nonholonomic.

(ii) Show that the ideal support reactions can be written as a force acting at O' and a reaction couple:

$$f_c = \lambda t + \nu e_3, \qquad \tau_c = \kappa N.$$

Verify that their total power vanishes on admissible motions.

(iii) Show that

$$p = M\big(\dot{o} + l\omega \times s\big), \qquad \pi = (R\mathbb{I}_c R^T)\omega + ls \times p.$$

Hence express ω and $\dot{o}$ in terms of o, R, p, π.

(iv) Derive the spatial momentum balances for p and π, including gravity, the prescribed inputs, and the support reactions. Account explicitly for the motion of O' in the angular-momentum balance.

(v) Combine the momentum balances, reconstruction equations, and constraints to obtain a constrained model in the variables

$$(o, R, p, \pi).$$

By differentiating the velocity constraints, obtain equations determining the reaction multipliers λ, ν, κ. State the compatibility conditions on the initial configuration and momenta.

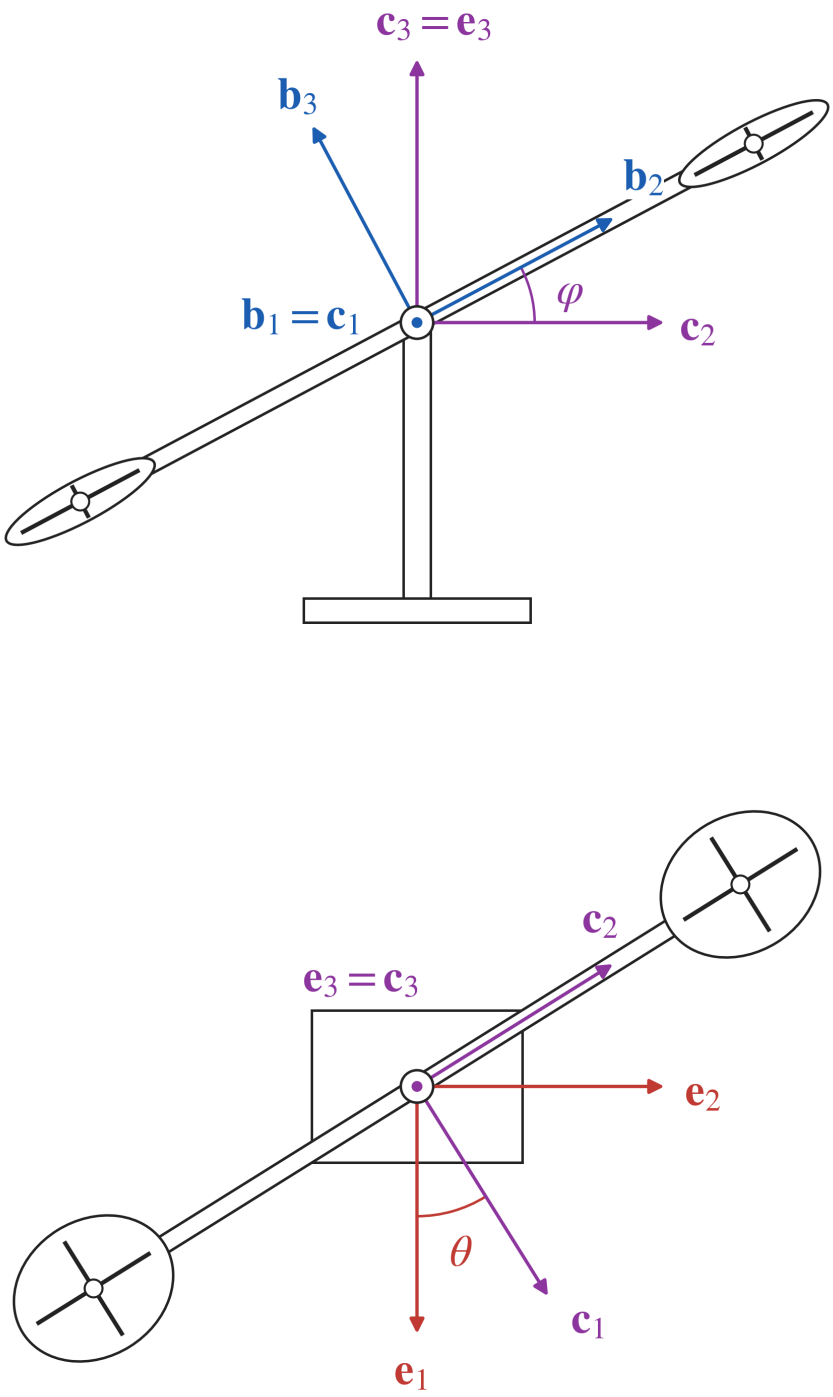


Fig. 2.17 Side view (top) and top view (bottom) of the twin-rotor assembly and its reference frames, with $\mathbf{c}_3 = \mathbf{e}_3$ and $\mathbf{b}_1 = \mathbf{c}_1$. The angles φ and θ illustrate the assembly's tilt and the support's rotation about the vertical axis, respectively. Dots indicate axes directed out of the page.

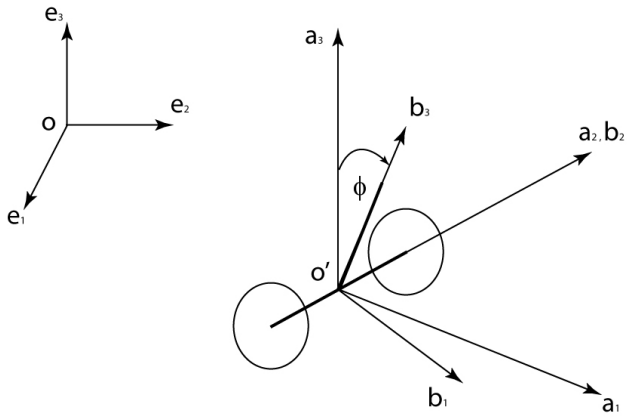


Fig. 2.18 Vehicle body and reference frames. The wheels are replaced in the model by ideal support reactions at the axle midpoint O'. The body may tilt about the horizontal axle, turn about the vertical, and move forwards without lateral sliding.

Chapter 3
The Space of Rotations $SO(3)$

For torque-free motion, the body angular momentum equations can be studied without first determining the orientation. In general, however, external moments may depend on the orientation, as in the heavy-top example. Furthermore even in these cases where the external moments do not depend on the configuration of the body we are still interested in keeping track of the configuration of the body over time. This involves the integration of $\dot{R} = R\widehat{\Omega} = \widehat{\omega}R$. In what follows we will concern ourselves with this integration that will tell us what the configuration of the rigid body is at every given time instant. If one considers element by element then $\dot{R} = R\widehat{\Omega} = \widehat{\omega}R$ represents nine time varying ODEs. However since $R^T R = I_{3\times 3}$ defines six constraints we see that all the nine differential equations defined by $\dot{R} = R\widehat{\Omega} = \widehat{\omega}R$ are not independent. Thus it is important to know how to find independent equations to describe these kinematics. This brings us to to the question of parameterizing rotations or in other words finding suitable coordinates for $SO(3)$ and thereby representing $\dot{R} = R\widehat{\Omega} = \widehat{\omega}R$ in terms of these parameterizations.

In section-1.2.2 we saw that if R can be thought of as a map that transforms points in space to other points in space in such a way that it preserves distances between points and angles between lines. That is, every $R \in \mathrm{SO}(3)$ can be viewed as a rigid rotation of space. Conversely, since we have seen that two frames with coinciding origins are related by $\mathbf{b} = \mathbf{e}R$, every rigid body rotation about a fixed point can also be identified with an $R \in \mathrm{SO}(3)$ by fixing an orthonormal frame $\mathbf{b}$ on the rigid body and finding its relationship with some inertial frame-$\mathbf{e}$. In what follows we will take a closer look at all possible rigid body rotations. Or in other words we will try to get some insight about the structure of the space $\mathrm{SO}(3)$.

Since $R^T R = I_{3\times 3}$ we see that all 9 elements of the 3×3 matrix R are not independent. The condition $R^T R = I_{3\times 3}$ gives rise to six constraints on the 9 elements of R. Thus we see that there are only 3 independent elements in R. Therefore we can conclude that in order to express R we need to know at least three parameters. This implies that the space $\mathrm{SO}(3)$ is a 3-dimensional space. However we will see below that $SO(3)$ is quite different from the the 3 dimensional vector space $\mathbb{R}^3$ and thus, in particular, there is no single smooth, nonsingular coordinate chart covering all of $\mathrm{SO}(3)$.

3.1 Non Commutativity of Rotations

In $\mathbb{R}^3$ the binary operation of addition that makes it a vector space is commutative. Below we will see that the space of rotations, $SO(3)$, even though is closed under the composition of rotations is not a vector space under composition of rotations since the composition of rotations is not commutative. This indicates a quite significant difference with $\mathbb{R}^3$.

To see this let us first consider composition of rotations by considering three frames **e**, **a** and **b**. Let $R_\alpha, R_\beta \in \mathrm{SO}(3)$. Consider the two rotations of space that take the frame **e** to **a** and and the frame **a** to **b** respectively such that $\mathbf{a} = \mathbf{e}R_\alpha$ and $\mathbf{b} = \mathbf{a}R_\beta$. Since $\mathbf{b} = \mathbf{a}R_\beta = \mathbf{e}R_\alpha R_\beta$ we see that $R = R_\alpha R_\beta$ is a rotation that takes **e** to **a** by a rotation R_α and then to **b** by a rotation R_β. Here each successive rotation is specified relative to the current frame, so it acts by right multiplication. Also consider the two rotations of space that takes the frame **e** to $\mathbf{a}'$ and and the frame $\mathbf{a}'$ to $\mathbf{b}'$ respectively such that $\mathbf{a}' = \mathbf{e}R_\beta$ and $\mathbf{b}' = \mathbf{a}'R_\alpha$. Since $\mathbf{b}' = \mathbf{a}'R_\alpha = \mathbf{e}R_\beta R_\alpha$ we see that $R = R_\beta R_\alpha$ is a rotation that takes **e** to $\mathbf{a}'$ by a rotation R_β and then to $\mathbf{b}'$ by a rotation R_α. Since in general matrix multiplication is non-commutative $R_\alpha R_\beta \neq R_\beta R_\alpha$ and therefore the final frames in the two cases above **b** and $\mathbf{b}'$ will not be the same.

Thus in general, a rotation by R_α followed up by a rotation R_β will not be the same as a rotation by R_β followed up by a rotation R_α. That is rotations do not commute.

Let us illustrate this non-commutatitivity of rotations using some specific easy to visualize examples. Now consider a counter clockwise rotation first about the third axis by an angle equal to $\pi/2$ and then about the first axis by an angle equal to $\pi/2$. The resultant frame **b** is then related to the original frame **e** by $\mathbf{b} = \mathbf{e}(R_3(\pi/2)R_1(\pi/2))$. When you reverse the oder of rotation the resultant frame $\mathbf{b}'$ is then related to the original frame **e** by $\mathbf{b}' = \mathbf{e}(R_1(\pi/2)R_3(\pi/2))$. Since

$$R_1(\pi/2)R_3(\pi/2) = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$R_3(\pi/2)R_1(\pi/2) = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

we clearly see that $\mathbf{b}' \neq \mathbf{b}$. This is illustrated in figure-3.1

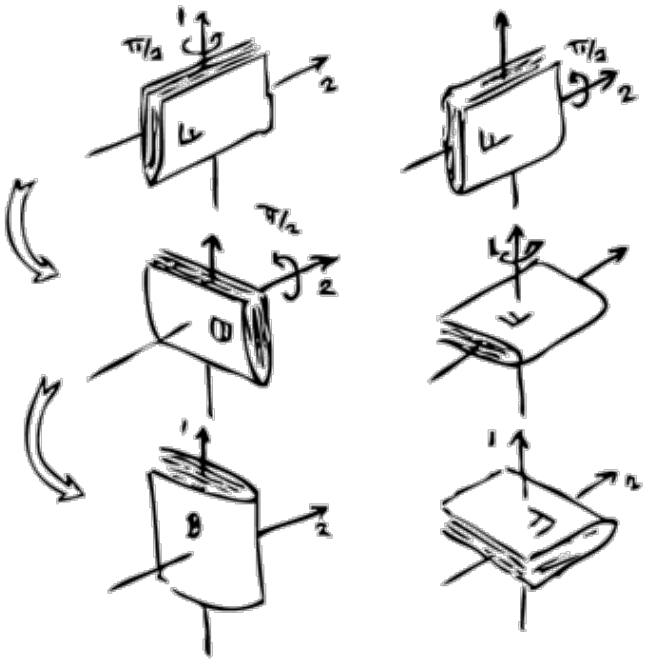


Fig. 3.1 ©ScienceBits. The non-commutativity of rigid body rotations.

3.2 Representation of Rotations

Euler's theorem shows that every rotation R can be thought of as a rotation about some axis n by some angle θ[1]. This is depicted in figure 3.2. Therefore we see that the space of rotations, $\mathrm{SO}(3)$ can be identified with the solid ball in $\mathbb{R}^3$ with the antipodal points on the boundary identified or by the space of unit tangent vectors on $\mathbb{S}^2$ that is typically denoted by $T_0\mathbb{S}^2$. This confirms our previous conclusion that the dimension of the space $\mathrm{SO}(3)$ is three. On the other hand this also confirms our observation that the space of $SO(3)$ is quite different from $\mathbb{R}^3$. In fact this shows that $SO(3)$ is not isomorphic to $\mathbb{R}^3$. No single smooth, nonsingular, one-to-one coordinate chart covers all of $\mathrm{SO}(3)$. Three-parameter representations such as Euler angles therefore require restrictions or multiple coordinate charts to avoid singularities and nonuniqueness. Below we will begin by showing how one can parameterize R using unit quaternions to avoid this problem of singularity.

3.2.1 Quaternion Representation of Rotations

Recall that in section-1.2.2 we have shown that if $\widehat{\Omega} = R^T\dot{R}$ then Ω corresponds to an instantaneous rotation about Ω at an instantaneous rate that is equal to $||\Omega||$. Thus if Ω is a constant then the solution to the following initial value problem

[1] You are asked to prove this in exercise 3.1.

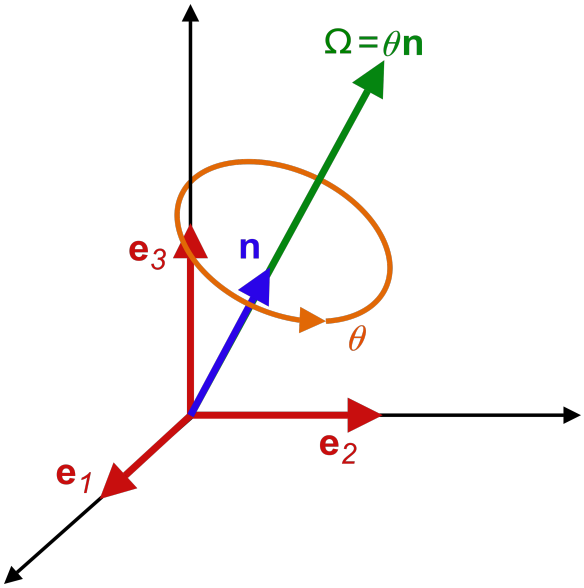


Fig. 3.2 ©Wikipedia. Euler's Theorem that says that every rotation is a rotation about some axis n by some angle θ.

$$\dot{R} = R\widehat{\Omega}, \quad R(0) = I_{3\times 3}, \tag{3.1}$$

is a pure rotation about the axis Ω at a constant angular rate of Ω. This says that if we define $\theta(t) \triangleq t||\Omega||$ then since Ω does not depend on t we see that the solution, $R(t)$, to the initial value problem (3.1) is a rotation about the axis Ω by an angle equal to $\theta = t||\Omega||$. It is easy to verify by direct computation that this solution is explicitly and uniquely given by

$$R(t) = \exp(t\widehat{\Omega}), \tag{3.2}$$

where $\exp(A)$ denotes the matrix exponential of the matrix A.

Setting $t = 1$ in (3.2) we see that $\exp(\widehat{\Omega}) \in \mathrm{SO}(3)$ for any $\widehat{\Omega} \in \mathrm{so}(3)$ and that it corresponds to a rotation about the axis Ω by an angle $\theta = ||\Omega||$. This process defines a map

$$\exp : \mathrm{so}(3) \to \mathrm{SO}(3). \tag{3.3}$$

The exponential map is locally one-to-one in a neighborhood of the zero matrix, providing local coordinates for rotations near the identity[2]. In exercise-3.3 you are asked to show[3] that every $R \in \mathrm{SO}(3)$ can be thought of as a rotation about some axis Ω and hence that for any given R there exists some $\widehat{\Omega} \in \mathrm{so}(3)$ such that $R = \exp(\widehat{\Omega})$. Thus we see that the map $\exp : \mathrm{so}(3) \to \mathrm{SO}(3)$ is onto as well.

[2] Showing this is beyond the scope of this lecture notes as it requires advanced mathematical notions involving group theory.

[3] Hint: First show that every $R \in \mathrm{SO}(3)$ has an eigenvalue that is equal to one. Let V be the corresponding eigenvector. Recall that R can be thought of as acting on space by rotations. Then since $RV = V$ we see that R is a rotation about V. Alternatively, consider an orthonormal basis for $\mathbb{R}^3$ with V as the first basis vector, we may express R in this new basis and come to the same conclusion.

In summary what we have shown is that for every Ω there is a unique corresponding R given by the matrix exponential $\exp(\widehat{\Omega})$ and that for every R there also exists some Ω such that $R = \exp(\widehat{\Omega})$.

Let us now proceed to find an explicit expression for $\exp(\widehat{\Omega})$. In exercise-3.2 you are asked to prove that

$$\widehat{\Omega}^2 = (\Omega\Omega^T - ||\Omega||^2 I).$$

From this it follows that

$$\widehat{\Omega}^3 = -||\Omega||^2\widehat{\Omega}, \quad \widehat{\Omega}^4 = -||\Omega||^2\widehat{\Omega}^2, \quad \widehat{\Omega}^5 = ||\Omega||^4\widehat{\Omega}, \quad \cdots$$

and therefore in exercise-1.11 you asked to show that

$$\exp\left(\widehat{\Omega}\right) = I + \frac{\sin||\Omega||}{||\Omega||}\widehat{\Omega} + \frac{1}{2}\left(\frac{\sin\frac{||\Omega||}{2}}{\frac{||\Omega||}{2}}\right)^2\widehat{\Omega}^2. \tag{3.4}$$

This is famously known as the *Rodrigues formula*. We stress again that this corresponds to a rotation about the axis Ω by an angle $\theta = ||\Omega||$. Thus letting $n = \Omega/||\Omega||$ be the unit length direction along Ω we can write the above equation also as

$$R = \exp(\theta\,\widehat{n}) = I + \sin\theta\,\widehat{n} + (1-\cos\theta)\,\widehat{n}^2. \tag{3.5}$$

Since we have seen that every $R \in \mathrm{SO}(3)$ can be written as $R = \exp(\widehat{\Omega})$ for some $\widehat{\Omega} \in \mathrm{so}(3)$ we can also conclude that every $R \in \mathrm{SO}(3)$ can be written down using the expression (3.5) for some angle θ and unit direction n. Observe that specifically if $E_1 = [1\ \ 0\ \ 0]^T, E_2 = [0\ \ 1\ \ 0]^T, E_3 = [0\ \ 0\ \ 1]^T$ then the Rodrigues formula (3.5) gives that $\exp(\theta_i\widehat{E}_i) = R_i(\theta_i)$ corresponds to a rotation about the axis E_i by an angle equal to θ_i.

Let $w = \sin\left(\frac{\theta}{2}\right)n$ and $q_0 = \cos\left(\frac{\theta}{2}\right)$. Observe that since $q_0^2 + ||w||^2 = 1$ the ordered quadruple of numbers $q = (q_0, w)$ represents a point on the surface of the unit sphere, $\mathbb{S}^3 = \{q \in \mathbb{R}^4 \mid ||q|| = 1\}$, in $\mathbb{R}^4$. Then re-arranging the above expression we have that every $R \in \mathrm{SO}(3)$ can be written down as

$$R = I + 2q_0\widehat{w} + 2\widehat{w}^2.$$

for some $(q_0, w) \in \mathbb{S}^3$. Conversely we also see that for every $q = (q_0, w) \in \mathbb{S}^3$ there is a unique corresponding $R \in \mathrm{SO}(3)$ that is explicitly given by the above formula. Note that, since $-q = (-q_0, -w) \in \mathbb{S}^3$ also gives the same R that is given by $q = (q_0, w) \in \mathbb{S}^3$, this correspondence is two-to-one. In summary, we have the following:

The group of rotations $\mathrm{SO}(3)$ is isomorphic to $\mathbb{S}^3/\{1,-1\}$ where the isomorphism is explicitly given by the Rodrigues formula

$$R = I + 2q_0\widehat{w} + 2\widehat{w}^2. \tag{3.6}$$

Here R corresponds to a rotation about w by an angle equal to $\theta = 2\cos^{-1}(q_0)$. Using algebraic manipulations it can also be shown that, in terms of $(q_0, w) \in \mathbb{S}^3$, the differential equation $\dot{R} = R\widehat{\Omega}$ becomes

$$\begin{bmatrix} \dot{q}_0 \\ \dot{w} \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -\Omega \cdot w \\ q_0\Omega - \Omega \times w \end{bmatrix}. \tag{3.7}$$

The unit sphere in $\mathbb{R}^4$, denoted by $\mathbb{S}^3 = \{q \in \mathbb{R}^4 \mid ||q|| = 1\}$ is also known as the space of *unit quaternions*. We conclude this section by noting that the unit quaternion $q = (q_0, w)$ that correspond to a given rotation matrix R can be found using the following two expressions obtained from (3.6):

$$\text{trace}(R) = -1 + 4q_0^2 = 2\cos\theta + 1, \tag{3.8}$$

$$R - R^T = 4q_0\widehat{w} = 4\cos\left(\frac{\theta}{2}\right)\widehat{w}. \tag{3.9}$$

The first expression determines $q_0^2 = \cos^2(\theta/2)$, so that $q_0 = \pm\cos(\theta/2)$. Choosing either sign of q_0 determines w from (3.9) when $q_0 \neq 0$. For a rotation through π, $q_0 = 0$, and w is instead chosen as a unit direction satisfying $Rw = w$. The choices q and $-q$ represent the same rotation.

3.2.2 Euler Angle Representation of Rotations

The following matrix R obtained by composing the three consecutive counter clockwise rotations about the axis $i - j - k$ respectively

$$R = R_i(\theta_1)R_j(\theta_2)R_k(\theta_3) \tag{3.10}$$

is once again a special orthogonal matrix. This corresponds to successive rotations of the frame about its current i-, j-, and k-axes by angles θ_1, θ_2, and θ_3, respectively. When $i \neq j$ and $j \neq k$, these angles provide local coordinates wherever the parameterization is nonsingular. This is called the $i - j - k$ Euler angle parameterization of R.

It is important to note that for certain angles this correspondence is not unique, meaning that there exists certain R such that the corresponding choice for $(\theta_1, \theta_2, \theta_3)$ is not unique. For example consider the parameterization of R using the three angles $(\theta_1, \theta_2, \theta_3)$ corresponding to the 3-1-3 Euler angle parameterization of R that is explicitly given by

$$R = R_3(\theta_1)R_1(\theta_2)R_3(\theta_3) = \begin{bmatrix} c_1c_3 - s_1c_2s_3 & -c_1s_3 - s_1c_2c_3 & s_1s_2 \\ s_1c_3 + c_1c_2s_3 & -s_1s_3 + c_1c_2c_3 & -c_1s_2 \\ s_2s_3 & s_2c_3 & c_2 \end{bmatrix}, \tag{3.11}$$

where we have used the notation $c_i \triangleq \cos\theta_i$, $s_i \triangleq \sin\theta_i$. This parameterization is shown in figure 3.3. The angles provide nonsingular local coordinates when $\sin\theta_2 \neq 0$; in particular, it is singular at $(\theta_1,\theta_2,\theta_3) = (0,0,0)$.

It is easy to see that

$$\theta_2 = 0 \implies R = R_3(\theta_1+\theta_3) \quad \text{and} \quad \theta_2 = \pi \implies R = R_3(\theta_1-\theta_3)R_1(\pi).$$

Hence that there exists no unique θ_1,θ_3 that describe the orientation of the resulting two frames **e** and **b**. This situation is called *gimbal lock* in the satellite and robotics communities. This turns out to be a common problem for any type of Euler angles being used. Thus in particular we find that the Euler angles only provide a local isomorphism between $\mathbb{R}^3$ and $SO(3)$.

The 3-1-3 Euler angles turn out to be a popular choice of local coordinates for $SO(3)$ in the robotics and satellite communities. In this parameterization θ_1 is called the angle of *precession*, θ_2 is called the angle of *nutation*, and θ_3 is called the angle of *spin*. For illustration

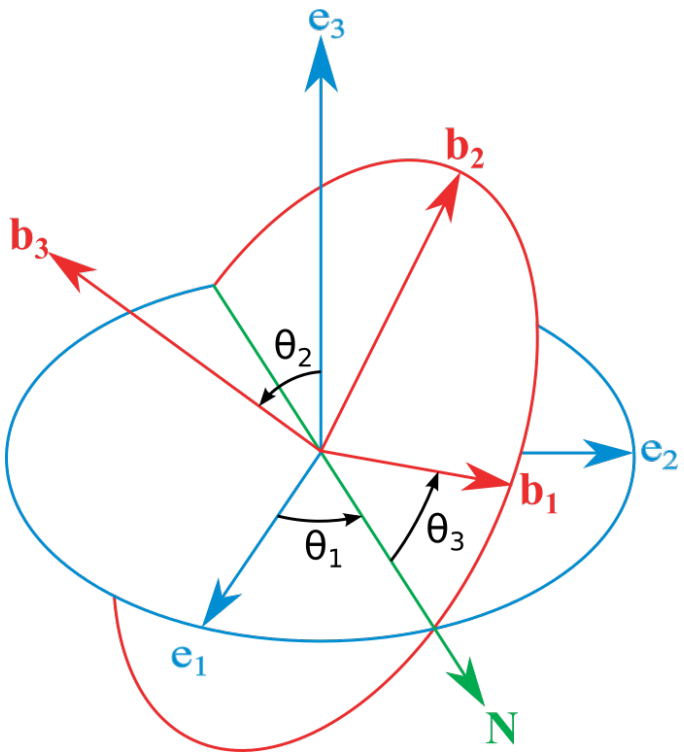


Fig. 3.3 ©Wikipedia. The 3-1-3 Euler angle parameterization on $SO(3)$.

purposes, below, we will provide explicit expressions for the representation of the angular velocity using the 3-1-3 Euler angle representation of R that is shown in figure 3.3. Consider two frames **b** and **e** where $\mathbf{b} = \mathbf{e}R$. Then if $R^T\dot{R} = \widehat{\Omega}$, we have seen earlier that Ω is the representation of the angular velocity in the **b**-frame while $\omega = R\Omega$ is the representation of the angular velocity in the **e**-frame. One can find that in terms of the 3-1-3 Euler angles they are explicitly given by:

$$\Omega = \begin{bmatrix} \dot{\theta}_1 \sin\theta_2 \sin\theta_3 + \dot{\theta}_2 \cos\theta_3 \\ \dot{\theta}_1 \sin\theta_2 \cos\theta_3 - \dot{\theta}_2 \sin\theta_3 \\ \dot{\theta}_1 \cos\theta_2 + \dot{\theta}_3 \end{bmatrix}, \tag{3.12}$$

$$\omega = \begin{bmatrix} \dot{\theta}_3 \sin\theta_2 \sin\theta_1 + \dot{\theta}_2 \cos\theta_1 \\ -\dot{\theta}_3 \sin\theta_2 \cos\theta_1 + \dot{\theta}_2 \sin\theta_1 \\ \dot{\theta}_3 \cos\theta_2 + \dot{\theta}_1 \end{bmatrix}. \tag{3.13}$$

Differentiating (3.12) we obtain the body angular acceleration

$$\dot{\Omega} = \begin{bmatrix} \ddot{\theta}_1 \sin\theta_2 \sin\theta_3 + \dot{\theta}_1\dot{\theta}_2 \cos\theta_2 \sin\theta_3 + \dot{\theta}_1\dot{\theta}_3 \sin\theta_2 \cos\theta_3 + \ddot{\theta}_2 \cos\theta_3 - \dot{\theta}_2\dot{\theta}_3 \sin\theta_3 \\ \ddot{\theta}_1 \sin\theta_2 \cos\theta_3 + \dot{\theta}_1\dot{\theta}_2 \cos\theta_2 \cos\theta_3 - \dot{\theta}_1\dot{\theta}_3 \sin\theta_2 \sin\theta_3 - \ddot{\theta}_2 \sin\theta_3 - \dot{\theta}_2\dot{\theta}_3 \cos\theta_3 \\ \ddot{\theta}_1 \cos\theta_2 - \dot{\theta}_1\dot{\theta}_2 \sin\theta_2 + \ddot{\theta}_3 \end{bmatrix}. \tag{3.14}$$

From (3.12) we have

$$\Pi = \mathbb{I}\Omega = \begin{bmatrix} \mathbb{I}_1 \left(\dot{\theta}_1 \sin\theta_2 \sin\theta_3 + \dot{\theta}_2 \cos\theta_3\right) \\ \mathbb{I}_2 \left(\dot{\theta}_1 \sin\theta_2 \cos\theta_3 - \dot{\theta}_2 \sin\theta_3\right) \\ \mathbb{I}_3 \left(\dot{\theta}_1 \cos\theta_2 + \dot{\theta}_3\right) \end{bmatrix},$$

and from (3.14) we have

$$\mathbb{I}\dot{\Omega} = \begin{bmatrix} \mathbb{I}_1 \left(\ddot{\theta}_1 \sin\theta_2 \sin\theta_3 + \dot{\theta}_1\dot{\theta}_2 \cos\theta_2 \sin\theta_3 + \dot{\theta}_1\dot{\theta}_3 \sin\theta_2 \cos\theta_3 + \ddot{\theta}_2 \cos\theta_3 - \dot{\theta}_2\dot{\theta}_3 \sin\theta_3\right) \\ \mathbb{I}_2 \left(\ddot{\theta}_1 \sin\theta_2 \cos\theta_3 + \dot{\theta}_1\dot{\theta}_2 \cos\theta_2 \cos\theta_3 - \dot{\theta}_1\dot{\theta}_3 \sin\theta_2 \sin\theta_3 - \ddot{\theta}_2 \sin\theta_3 - \dot{\theta}_2\dot{\theta}_3 \cos\theta_3\right) \\ \mathbb{I}_3 \left(\ddot{\theta}_1 \cos\theta_2 - \dot{\theta}_1\dot{\theta}_2 \sin\theta_2 + \ddot{\theta}_3\right) \end{bmatrix}.$$

Thus we see that Euler's Rigid body equations (2.20) in 3-1-3 Euler angles $(\theta_1, \theta_2, \theta_3)$ take the form

$$\begin{aligned}
\mathbb{I}_1 \sin\theta_2 \sin\theta_3\, \ddot{\theta}_1 + \mathbb{I}_1 \cos\theta_3\, \ddot{\theta}_2 &= -\mathbb{I}_1 \left(\dot{\theta}_1\dot{\theta}_2 \cos\theta_2 \sin\theta_3 + \dot{\theta}_1\dot{\theta}_3 \sin\theta_2 \cos\theta_3 - \dot{\theta}_2\dot{\theta}_3 \sin\theta_3\right) + (\mathbb{I}_2 - \mathbb{I}_3)\left(\dot{\theta}_1 \sin\theta_2 \cos\theta_3 - \dot{\theta}_2 \sin\theta_3\right)\left(\dot{\theta}_1 \cos\theta_2 + \dot{\theta}_3\right) + T_1^e \\
\mathbb{I}_2 \sin\theta_2 \cos\theta_3\, \ddot{\theta}_1 - \mathbb{I}_2 \sin\theta_3\, \ddot{\theta}_2 &= -\mathbb{I}_2 \left(\dot{\theta}_1\dot{\theta}_2 \cos\theta_2 \cos\theta_3 - \dot{\theta}_1\dot{\theta}_3 \sin\theta_2 \sin\theta_3 - \dot{\theta}_2\dot{\theta}_3 \cos\theta_3\right) + (\mathbb{I}_3 - \mathbb{I}_1)\left(\dot{\theta}_1 \cos\theta_2 + \dot{\theta}_3\right)\left(\dot{\theta}_1 \sin\theta_2 \sin\theta_3 + \dot{\theta}_2 \cos\theta_3\right) + T_2^e \\
\mathbb{I}_3 \cos\theta_2\, \ddot{\theta}_1 + \mathbb{I}_3\, \ddot{\theta}_3 &= \mathbb{I}_3\, \dot{\theta}_1\dot{\theta}_2 \sin\theta_2 + (\mathbb{I}_1 - \mathbb{I}_2)\left(\dot{\theta}_1 \sin\theta_2 \sin\theta_3 + \dot{\theta}_2 \cos\theta_3\right)\left(\dot{\theta}_1 \sin\theta_2 \cos\theta_3 - \dot{\theta}_2 \sin\theta_3\right) + T_3^e.
\end{aligned}$$

Since the 3-1-3 parameterization is singular when $\sin\theta_2 = 0$, these equations cannot uniquely determine the Euler-angle accelerations at those configurations. In section-3.2.1 we have seen how to overcome the complications involved in parameterizing R using Euler angles by instead using unit quaternions $(q_0, w) \in \mathbb{S}^3$ as given by expression (3.6). We also pointed out that in this parameterization $\dot{R} = R\widehat{\Omega}$ is given by the differential equation (3.7) and hence one can avoid any singularities by resorting to unit quaternions instead.

Example of a Forced Gyroscope

For the axi-symmetric gyroscope, $\mathbb{I}_1 = \mathbb{I}_2 = \mathbb{I}_l, \mathbb{I}_z$ and the rigid body equations take the form

$$\mathbb{I}_l \dot{\Omega}_1 = (\mathbb{I}_l - \mathbb{I}_z)\Omega_2\Omega_3 + T_1, \tag{3.15}$$



Fig. 3.4

$$\mathbb{I}_l \dot{\Omega}_2 = (\mathbb{I}_z - \mathbb{I}_l)\Omega_3\Omega_1 + T_2, \tag{3.16}$$
$$\mathbb{I}_z \dot{\Omega}_3 = T_3. \tag{3.17}$$

Let us parameterize the rotation matrix from the earth fixed frame $\mathbf{e}$ to a disk fixed frame $\mathbf{d}$ using the 3-1-3 Euler angles. We will need this to express the force moment

$$T = \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix}.$$

Let $\mathbf{e}$ be an Earth fixed frame with origin, O. A frame $\mathbf{b}$ is fixed to the outer gimbal of the Gyroscope, a frame $\mathbf{c}$ is fixed to the inner gimbal and a frame $\mathbf{d}$ is fixed on the disc. Let $\mathbf{d}(t) = \mathbf{e}R(t)$. The $\mathbf{b}(t)$ frame is fixed on the outer gimbal so that $\mathbf{b}_3(t) \equiv \mathbf{e}_3(t)$. Then $\mathbf{b}(t) = \mathbf{e}R_3(\theta_1(t))$ where θ_1 is the angle of rotation of $\mathbf{b}$ about the fixed $\mathbf{e}_3, \mathbf{b}_3$ axis. The frame $\mathbf{c}(t)$ is fixed on the inner gimbal so that $\mathbf{c}_1(t) \equiv \mathbf{b}_1(t)$. Then $\mathbf{c}(t) = \mathbf{b}(t)R_1(\theta_2(t))$ where θ_2 is the angle of rotation of $\mathbf{c}$ about the $\mathbf{c}_1, \mathbf{b}_1$ axis. The frame $\mathbf{d}(t)$ is fixed on the disc so that $\mathbf{d}_3(t) \equiv \mathbf{c}_3(t)$. Then $\mathbf{d}(t) = \mathbf{c}(t)R_3(\theta_3(t))$ where θ_3 is the angle of rotation of $\mathbf{c}$ about the $\mathbf{c}_3, \mathbf{d}_3$ axis.

Now since $\mathbf{d}(t) = \mathbf{e}R(t)$ we have that

$$R = R_3(\theta_1)R_1(\theta_2)R_3(\theta_3) = \begin{bmatrix} c_1c_3 - s_1c_2s_3 & -c_1s_3 - s_1c_2c_3 & s_1s_2 \\ s_1c_3 + c_1c_2s_3 & -s_1s_3 + c_1c_2c_3 & -c_1s_2 \\ s_2s_3 & s_2c_3 & c_2 \end{bmatrix},$$

and hence that R is naturally parameterized by the 3-1-3 Euler angles. Recall that these are singular when $\theta_2 = 0, \pi$.

Neglecting the gimbal inertia, the force moments acting on the disc are calculated as follows. The moments acting on the outer gimbal due to the fixed end.

$$T^f = \mathbf{e}T^1 = \mathbf{e}\begin{bmatrix} T_1^1 \\ T_2^1 \\ u_1 \end{bmatrix}$$

The moments acting on the inner Gimbal due to the outer Gimbal.

$$T^I = \mathbf{b}\,T^2 = \mathbf{b}\begin{bmatrix} u_2 \\ T_2^2 \\ T_3^2 \end{bmatrix}$$

The moments acting on the disc due to the inner Gimbal.

$$T^D = \mathbf{c}\,T^3 = \mathbf{c}\begin{bmatrix} T_1^3 \\ T_2^3 \\ u_3 \end{bmatrix} = \mathbf{d}\,T$$

Neglecting the inertia of the outer and inner gimbals, their moment balances give

$$\mathbf{e}\,T^1 = \mathbf{b}\,T^2 = \mathbf{c}\,T^3.$$

Here the reaction to the moment transmitted to the next body acts with the opposite sign on the preceding gimbal. Hence

$$T^1 = R_3(\theta_1)\,T^2, \qquad T^2 = R_1(\theta_2)\,T^3, \qquad T^3 = R_3(\theta_3)\,T, \tag{3.18}$$

and therefore

$$T^1 = R_3(\theta_1)R_1(\theta_2)R_3(\theta_3)\,T = R\,T.$$

From the first relation in (3.18), $T_3^2 = u_1$, and from the second, $T_1^3 = u_2$. Thus the second relation becomes

$$\begin{bmatrix} u_2 \\ T_2^2 \\ u_1 \end{bmatrix} = R_1(\theta_2)\begin{bmatrix} u_2 \\ T_2^3 \\ u_3 \end{bmatrix}.$$

Its last component gives

$$u_1 = \sin\theta_2\,T_2^3 + \cos\theta_2\,u_3.$$

For $\sin\theta_2 \neq 0$, this yields

$$T_2^3 = \frac{u_1 - \cos\theta_2\,u_3}{\sin\theta_2}.$$

Since $T^3 = R_3(\theta_3)\,T$, the resultant moment on the disc, expressed in $\mathbf{d}$, is

$$\begin{aligned} T &= R_3^T(\theta_3)\begin{bmatrix} u_2 \\ \dfrac{u_1 - \cos\theta_2\,u_3}{\sin\theta_2} \\ u_3 \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_3 & \sin\theta_3 & 0 \\ -\sin\theta_3 & \cos\theta_3 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 & 0 \\ \dfrac{1}{\sin\theta_2} & 0 & -\cot\theta_2 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}. \end{aligned}$$

Consequently,

$$T = \begin{bmatrix} \dfrac{\sin\theta_3}{\sin\theta_2} & \cos\theta_3 & -\sin\theta_3\cot\theta_2 \\ \dfrac{\cos\theta_3}{\sin\theta_2} & -\sin\theta_3 & -\cos\theta_3\cot\theta_2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}.$$

Here u_1, u_2, and u_3 are the applied moments about the rotation axes of the outer gimbal, inner gimbal, and disc, respectively, with positive signs corresponding to increasing θ_1, θ_2, and θ_3.

This expression is singular when $\sin\theta_2 = 0$. At $\theta_2 = 0$ or $\theta_2 = \pi$, the outer-gimbal and disc rotation axes are parallel or antiparallel, so they no longer provide independent rotation directions. This situation is commonly referred to as *gimbal lock*.

3.3 Exercises

Exercise 3.1. *Prove Euler's Theorem:*

1. The eigenvalues of $R \in SO(3)$ are always of the form $\{1, e^{i\theta}, e^{-i\theta}\}$.
2. Every $R \in SO(3)$ corresponds to a rotation about some axis by some angle.

Exercise 3.2. Answer the following:

1. Consider the initial value problem

$$\dot{R} = R\widehat{\Omega}, \quad R(0) = I_{3\times 3},$$

where $\widehat{\Omega}$ is a constant. Show by direct verification that

$$R(t) = \exp(t\widehat{\Omega})$$

is the unique solution of the above matrix ODE initial value problem.
2. Using the above result show that for any $\widehat{\Omega} \in so(3)$ the matrix $R = \exp(\widehat{\Omega})$ corresponds to a rotation about the axis Ω by an angle equal to $||\Omega||$.
3. Using the properties proven in exercise-1.11 show that

$$\widehat{\Omega}^2 w = (\Omega \cdot w)\Omega - ||\Omega||^2 w = (\Omega\Omega^T - ||\Omega||^2 I_{3\times 3})w$$

for any 3×1 matrix w and hence that

$$\widehat{\Omega}^3 = -||\Omega||^2\widehat{\Omega}, \quad \widehat{\Omega}^4 = -||\Omega||^2\widehat{\Omega}^2, \quad \widehat{\Omega}^5 = ||\Omega||^4\widehat{\Omega}, \quad \cdots$$

4. Using the results of the question above show that

$$\exp\left(\widehat{\Omega}\right) = I_{3\times 3} + \frac{\sin||\Omega||}{||\Omega||}\widehat{\Omega} + \frac{1}{2}\left(\frac{\sin\frac{||\Omega||}{2}}{\frac{||\Omega||}{2}}\right)^2 \widehat{\Omega}^2.$$

At $\Omega = 0$, interpret the scalar coefficients by continuity, giving $\exp(\widehat{\Omega}) = I_{3\times 3}$. This equation is known as the *Rodrigues formula*.
5. Let $E_1 = [1\ 0\ 0]^T, E_2 = [0\ 1\ 0]^T, E_3 = [0\ 0\ 1]^T$. Using the Rodrigues formula explicitly show that

$$\exp(\theta\widehat{E}_i) = R_i(\theta),$$

where $R_i(\theta)$ is the rotation matrix that corresponds to a rotation about the ith-axis by an angle equal to θ. Compare your results with what you did in exercise-1.8.

Exercise 3.3. Show that every $R \in \mathrm{SO}(3)$ can be expressed as $R = \exp(\widehat{\Omega})$ for some $\widehat{\Omega} \in so(3)$.

Chapter 4
Answers to Selected Exercises

Answer to Exercise 1.1

Time coordinates

Agreement on simultaneity implies that τ depends only on t, so write $\tau = h(t)$. Agreement on elapsed time, together with common units and the same positive temporal orientation, gives

$$h(t_2) - h(t_1) = t_2 - t_1.$$

Setting $t_1 = 0$ yields

$$\boxed{\tau = t + a,}$$

where $a = h(0) \in \mathbb{R}$ is the difference between the chosen time origins.

Affine coordinate transformation

Consider any constant-velocity trajectory

$$x(t) = x_0 + ut.$$

The fourth assumption requires its velocity in the other frame to be constant as well. Since $\tau = t + a$, differentiation with respect to τ is the same as differentiation with respect to t.

For each component ξ_k, the chain rule therefore gives

$$0 = \frac{d^2\xi_k}{dt^2} = \frac{\partial^2\xi_k}{\partial t^2} + 2\sum_{i=1}^{3} u_i \frac{\partial^2\xi_k}{\partial t\,\partial x_i} + \sum_{i,j=1}^{3} u_i u_j \frac{\partial^2\xi_k}{\partial x_i\,\partial x_j}.$$

At any fixed (t,x), this holds for every u, since we may choose $x_0 = x - ut$.

Setting $u = 0$ gives

$$\frac{\partial^2\xi_k}{\partial t^2} = 0.$$

Subtracting the equations for u and $-u$ gives

$$4\sum_{i=1}^{3} u_i \frac{\partial^2 \xi_k}{\partial t\, \partial x_i} = 0.$$

Choosing only one component of u to be nonzero shows that

$$\frac{\partial^2 \xi_k}{\partial t\, \partial x_i} = 0 \qquad \text{for every } i.$$

The remaining equation is

$$\sum_{i,j=1}^{3} u_i u_j \frac{\partial^2 \xi_k}{\partial x_i\, \partial x_j} = 0.$$

Choosing only u_i to be nonzero gives

$$\frac{\partial^2 \xi_k}{\partial x_i^2} = 0.$$

Then choosing only u_i and u_j to be nonzero, and using equality of mixed partial derivatives, gives

$$\frac{\partial^2 \xi_k}{\partial x_i\, \partial x_j} = 0 \qquad (i \neq j).$$

Thus all second partial derivatives vanish. Consequently, the first partial derivatives are constant, and

$$\xi_k(t,x) = \alpha_k + \beta_k t + \sum_{i=1}^{3} R_{ki} x_i.$$

In matrix notation,

$$\boxed{\xi = \alpha + \beta t + Rx,}$$

where α, β, R are constant.

Spatial distance preservation

For any two simultaneous events with spatial coordinates x and y, the third assumption requires

$$\|\xi(t,x) - \xi(t,y)\|^2 = \|x-y\|^2.$$

Substituting the affine transformation gives

$$(x-y)^T R^T R (x-y) = (x-y)^T (x-y).$$

Since $z = x - y$ can be arbitrary,

$$z^T (R^T R - I) z = 0 \qquad \text{for every } z \in \mathbb{R}^3.$$

The matrix $R^T R - I$ is symmetric, so this implies

$$R^T R = I.$$

Hence $R^{-1} = R^T$, and therefore

$$\boxed{R^T R = R R^T = I.}$$

Taking determinants gives $(\det R)^2 = 1$. Since both frames are right-handed, their relative spatial transformation preserves orientation. Thus

$$\boxed{\det R = 1.}$$

Relative motion of the frames

The origin O' has spatial coordinates zero in $\mathbf{e}'$. If its coordinates in $\mathbf{e}$ are $o(t)$, then

$$0 = \alpha + \beta t + R o(t).$$

Differentiating and using the constancy of α, β, R gives

$$0 = \beta + R\dot{o}(t).$$

Therefore,

$$\boxed{v = \dot{o}(t) = -R^T \beta,}$$

which is constant. Since R is constant, the relative orientation of the frames is also constant: there is no relative rotation.

Synchronizing the clocks gives $a = 0$. If the spatial origins coincide at $t = 0$, then $o(0) = 0$, so the origin equation gives $\alpha = 0$. Finally, $\beta = -Rv$, and hence

$$\xi = \beta t + Rx = R(x - vt).$$

Thus

$$\boxed{(\tau, \xi) = \big(t, R(x - vt)\big).}$$

Answer to Exercise 1.5

Consider the coupled spring-mass-damper system shown in Figure-1.18. Assume that the friction between the rollers and the horizontal surfaces is negligibly small. Take the displacements from the equilibrium configuration of the system.

Isolating each of the masses and applying Newton's equations for each of them separately we have:

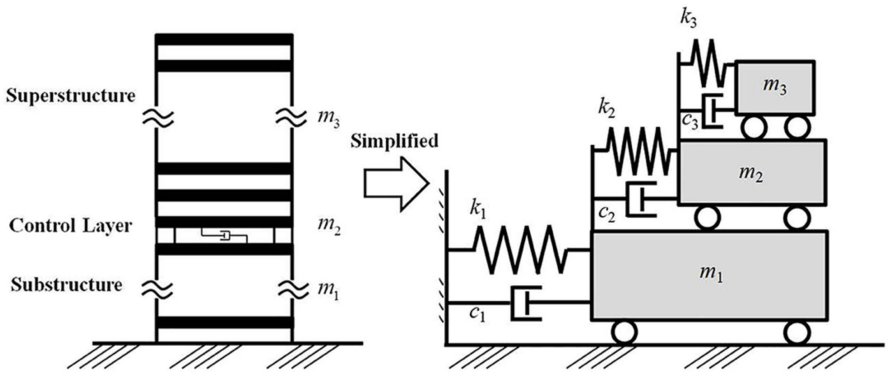


Fig. 4.1 A structural vibration model using a 3-DOF spring–mass–damper approximation.

Mass m_1

Spring k_1 and damper c_1 connect m_1 to the fixed base; spring k_2 and damper c_2 connect m_1 to m_2.

$$\begin{aligned} & m_1\ddot{x}_1 = -k_1x_1 - c_1\dot{x}_1 + k_2(x_2 - x_1) + c_2(\dot{x}_2 - \dot{x}_1) \\ \Rightarrow \quad & m_1\ddot{x}_1 + (c_1 + c_2)\dot{x}_1 - c_2\dot{x}_2 + (k_1 + k_2)x_1 - k_2x_2 = 0 \end{aligned}$$

Mass m_2

Connected to m_1 via (k_2, c_2) and to m_3 via (k_3, c_3).

$$\begin{aligned} & m_2\ddot{x}_2 = -k_2(x_2 - x_1) - c_2(\dot{x}_2 - \dot{x}_1) + k_3(x_3 - x_2) + c_3(\dot{x}_3 - \dot{x}_2) \\ \Rightarrow \quad & m_2\ddot{x}_2 - c_2\dot{x}_1 + (c_2 + c_3)\dot{x}_2 - c_3\dot{x}_3 - k_2x_1 + (k_2 + k_3)x_2 - k_3x_3 = 0 \end{aligned}$$

Mass m_3

Connected only to m_2 through (k_3, c_3).

$$\begin{aligned} & m_3\ddot{x}_3 = -k_3(x_3 - x_2) - c_3(\dot{x}_3 - \dot{x}_2) \\ \Rightarrow \quad & m_3\ddot{x}_3 - c_3\dot{x}_2 + c_3\dot{x}_3 - k_3x_2 + k_3x_3 = 0 \end{aligned}$$

In matrix form:

$$M\ddot{X} + C\dot{X} + KX = 0$$

where

$$X=\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix},\quad M=\begin{bmatrix}m_1&0&0\\0&m_2&0\\0&0&m_3\end{bmatrix},$$
$$C=\begin{bmatrix}c_1+c_2&-c_2&0\\-c_2&c_2+c_3&-c_3\\0&-c_3&c_3\end{bmatrix},\quad K=\begin{bmatrix}k_1+k_2&-k_2&0\\-k_2&k_2+k_3&-k_3\\0&-k_3&k_3\end{bmatrix}$$

$$Y=\begin{bmatrix}X\\\dot{X}\end{bmatrix}=\begin{bmatrix}x_1&x_2&x_3&\dot{x}_1&\dot{x}_2&\dot{x}_3\end{bmatrix}^\top$$

Differentiating Y gives

$$\dot{Y}=\begin{bmatrix}\dot{X}\\\ddot{X}\end{bmatrix}=\begin{bmatrix}\dot{X}\\-M^{-1}(C\dot{X}+KX)\end{bmatrix}=\begin{bmatrix}0_{3\times3}&I_{3\times3}\\-M^{-1}K&-M^{-1}C\end{bmatrix}\begin{bmatrix}X\\\dot{X}\end{bmatrix}$$

and hence

$$\dot{Y}=AY.$$

where

$$A=\begin{bmatrix}0_{3\times3}&I_{3\times3}\\-M^{-1}K&-M^{-1}C\end{bmatrix}$$

The matrix exponential solution to this first order linear ODE is.

$$Y(t)=e^{At}Y(0)$$

where

$$Y(0)=\begin{bmatrix}0&0&1&0&0&0\end{bmatrix}^\top$$

Answer to Exercise 1.6

Throughout, positions and velocities are measured relative to the inertial frame **e**. The forces satisfy

$$m_i\ddot{x}_i=\sum_{j\neq i}f_{ij},\qquad f_{ij}=-f_{ji},\qquad (x_i-x_j)\times f_{ij}=0.$$

(i) Motion of the Three-Particle Center of Mass

The total linear momentum is

$$p_u=m_1\dot{x}_1+m_2\dot{x}_2+m_3\dot{x}_3.$$

Differentiating and grouping the forces pairwise gives

$$\dot{p}_u = (f_{12} + f_{21}) + (f_{13} + f_{31}) + (f_{23} + f_{32}) = 0.$$

Thus p_u is constant. Since

$$M_u \bar{x}_u = m_1 x_1 + m_2 x_2 + m_3 x_3,$$

we have

$$M_u \dot{\bar{x}}_u = p_u, \qquad M_u \ddot{\bar{x}}_u = 0.$$

Consequently,

$$\bar{x}_u(t) = \bar{x}_u(0) + t\dot{\bar{x}}_u(0).$$

The center of mass is therefore at rest or moves with constant velocity. Only the equal-and-opposite property of the internal forces is needed for this conclusion.

(ii) Motion of the Two-Particle Center of Mass

For P_1 and P_2,

$$m_1 \ddot{x}_1 = f_{12} + f_{13}, \qquad m_2 \ddot{x}_2 = f_{21} + f_{23}.$$

Adding these equations and using $f_{12} + f_{21} = 0$ gives

$$M\ddot{\bar{x}} = m_1 \ddot{x}_1 + m_2 \ddot{x}_2 = f_{13} + f_{23}.$$

Thus the subsystem's momentum changes according to the resultant force exerted on it by P_3.

The subsystem is not isolated: it exchanges momentum with P_3. Indeed,

$$m_3 \ddot{x}_3 = f_{31} + f_{32} = -(f_{13} + f_{23}),$$

so any change in the subsystem's momentum is balanced by an opposite change in the momentum of P_3.

(iii) Angular Momentum of the Three-Particle System

About a possibly moving point O', define

$$\pi_u = \sum_{i=1}^{3} (x_i - o) \times m_i \dot{x}_i.$$

Differentiating gives

$$\dot{\pi}_u = \sum_{i=1}^{3} [(\dot{x}_i - \dot{o}) \times m_i \dot{x}_i + (x_i - o) \times m_i \ddot{x}_i]$$

$$= -\dot{o} \times p_u + \sum_{i=1}^{3} \sum_{j \neq i} (x_i - o) \times f_{ij},$$

where $\dot{x}_i \times m_i \dot{x}_i = 0$.

For each pair,

$$\begin{aligned}(x_i - o) \times f_{ij} + (x_j - o) \times f_{ji} &= (x_i - x_j) \times f_{ij} \\ &= 0.\end{aligned}$$

The first equality uses equal-and-opposite forces; the second uses centrality. Therefore,

$$\dot{\pi}_u = -\dot{o} \times p_u = -\dot{o} \times M_u \dot{\bar{x}}_u.$$

Taking $o = \bar{x}_u$ gives

$$\dot{\pi}_u = -\dot{\bar{x}}_u \times M_u \dot{\bar{x}}_u = 0.$$

Hence the angular momentum about the three-particle center of mass is conserved. It is also conserved about any point fixed in **e**, since then $\dot{o} = 0$.

(iv) Angular Momentum of the Two-Particle Subsystem

Define

$$\pi_{12} = \sum_{i=1}^{2} (x_i - \bar{x}) \times m_i \dot{x}_i.$$

Differentiating as above,

$$\dot{\pi}_{12} = -\dot{\bar{x}} \times (m_1 \dot{x}_1 + m_2 \dot{x}_2) + \sum_{i=1}^{2} (x_i - \bar{x}) \times m_i \ddot{x}_i.$$

The first term vanishes because

$$m_1 \dot{x}_1 + m_2 \dot{x}_2 = M \dot{\bar{x}}, \qquad \dot{\bar{x}} \times M \dot{\bar{x}} = 0.$$

The internal moments also cancel:

$$(x_1 - \bar{x}) \times f_{12} + (x_2 - \bar{x}) \times f_{21} = (x_1 - x_2) \times f_{12} = 0.$$

Consequently,

$$\boxed{\dot{\pi}_{12} = (x_1 - \bar{x}) \times f_{13} + (x_2 - \bar{x}) \times f_{23}.}$$

Thus its rate of change equals the resultant moment of the external interactions about the subsystem's center of mass. This remains true even when that center of mass accelerates.

(v) A Vanishing External Resultant

Suppose $f_{13} + f_{23} = 0$ throughout the interval. Part (ii) immediately gives

$$\ddot{\bar{x}} = 0.$$

Using $f_{23} = -f_{13}$ in the angular-momentum balance,

$$\dot{\pi}_{12} = (x_1 - x_2) \times f_{13}.$$

Centrality gives

$$(x_1 - x_3) \times f_{13} = 0, \qquad (x_2 - x_3) \times f_{23} = 0.$$

Hence

$$\begin{aligned} \dot{\pi}_{12} &= (x_1 - x_3) \times f_{13} - (x_2 - x_3) \times f_{13} \\ &= (x_1 - x_3) \times f_{13} + (x_2 - x_3) \times f_{23} = 0. \end{aligned}$$

Both the linear momentum and the angular momentum about the center of mass of the subsystem are therefore conserved.

This conclusion uses the geometry of the central interactions with the same particle P_3. In general, a vanishing external resultant does not imply a vanishing external moment.

To interpret the conserved angular momentum, introduce

$$q = x_1 - x_2, \qquad \mu = \frac{m_1 m_2}{m_1 + m_2}.$$

The center-of-mass definition gives

$$x_1 - \bar{x} = \frac{m_2}{M} q, \qquad x_2 - \bar{x} = -\frac{m_1}{M} q,$$

and therefore

$$\pi_{12} = \mu q \times \dot{q}.$$

If $\pi_{12} \neq 0$, the relative motion lies in the plane perpendicular to this constant quantity. Writing $q = \rho(\cos\psi\, a + \sin\psi\, b)$ in fixed orthonormal directions spanning that plane gives

$$\pi_{12} = \mu \rho^2 \dot{\psi}\, (a \times b).$$

Conservation therefore fixes $\rho^2\dot{\psi}$, not ρ and $\dot{\psi}$ separately. The separation may change, with a corresponding change in angular speed. If $\pi_{12} = 0$, purely radial relative motion is also possible.

(vi) Gravitational Interaction and Numerical Investigation

For Newtonian gravity,

$$f_{ij} = G m_i m_j \frac{x_j - x_i}{\|x_j - x_i\|^3},$$

so the equations of motion are

$$\boxed{\ddot{x}_i = G \sum_{j \neq i} m_j \frac{x_j - x_i}{\|x_j - x_i\|^3}, \qquad i = 1, 2, 3.}$$

These equations apply while all pairwise separations remain nonzero.

In normalized units, one possible set of initial data is

$$G = 1, \qquad (m_1, m_2, m_3) = (1, 2, 1.5),$$

$$x_1(0) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad x_2(0) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad x_3(0) = \begin{bmatrix} 0.5 \\ 0.5 \\ 0 \end{bmatrix},$$

and

$$\dot{x}_1(0) = \begin{bmatrix} 0 \\ 0.3 \\ 0 \end{bmatrix}, \quad \dot{x}_2(0) = \begin{bmatrix} -0.1 \\ 0 \\ 0.2 \end{bmatrix}, \quad \dot{x}_3(0) = \begin{bmatrix} 2/15 \\ -1/5 \\ -4/15 \end{bmatrix}.$$

The third velocity is chosen from

$$\dot{x}_3(0) = -\frac{m_1 \dot{x}_1(0) + m_2 \dot{x}_2(0)}{m_3},$$

so $p_u(0) = 0$. Thus the exact solution has

$$\bar{x}_u(t) = \bar{x}_u(0) = \begin{bmatrix} 11/18 \\ 1/6 \\ 0 \end{bmatrix}.$$

Writing $v_i = \dot{x}_i$ converts the equations into a first-order system:

$$\dot{x}_i = v_i, \qquad \dot{v}_i = G \sum_{j \neq i} m_j \frac{x_j - x_i}{\|x_j - x_i\|^3}.$$

This system can be integrated numerically, monitoring the minimum pairwise separation and stopping before a collision or an unresolved close encounter.

The conserved quantities used to check the computation are

$$p_u = \sum_i m_i v_i, \qquad \pi_u = \sum_i (x_i - \bar{x}_u) \times m_i v_i,$$

and

$$E = \sum_i \frac{1}{2} m_i \|v_i\|^2 - \sum_{i<j} \frac{G m_i m_j}{\|x_i - x_j\|}.$$

For the initial data above,

$$\pi_u(0) = \begin{bmatrix} -1/5 \\ -1/5 \\ 1/20 \end{bmatrix}, \qquad E(0) = \frac{31}{150} - 2 - \frac{9}{\sqrt{2}}.$$

Energy conservation follows directly from the force law. For $d_{ij} = x_i - x_j$, the rate of change of kinetic energy contributed by a pair is

$$v_i^T f_{ij} + v_j^T f_{ji} = -Gm_i m_j \frac{d_{ij}^T (v_i - v_j)}{\|d_{ij}\|^3}.$$

Meanwhile,

$$\frac{d}{dt}\left(-\frac{Gm_i m_j}{\|d_{ij}\|}\right) = Gm_i m_j \frac{d_{ij}^T (v_i - v_j)}{\|d_{ij}\|^3}.$$

Summing over pairs gives $\dot{E} = 0$.

A numerical solution should therefore keep $\bar{x}_u$ fixed and p_u, π_u, E approximately constant. Their deviations from the initial values measure numerical error and should decrease when the integration tolerances are tightened.

In contrast, the subsystem generally satisfies

$$M\ddot{\bar{x}} = f_{13} + f_{23} \neq 0, \qquad \dot{\pi}_{12} = (x_1 - \bar{x}) \times f_{13} + (x_2 - \bar{x}) \times f_{23} \neq 0.$$

Thus its center of mass may accelerate and its angular momentum may change, while the corresponding quantities for the complete isolated system remain conserved.

Answer to Exercise 1.7

Consider the representation of a point P in space. To do this we rely on the space-time assumptions of Galilean mechanics that imply space to be Euclidean. Thus we may pick another point O in space and setup an orthonormal frame $\mathbf{e}$ at O. Let x be the representation of P in $\mathbf{e}$. That is let $OP = \mathbf{e}x$. The Euclidian assumption of space implies that the distance from O to P is

$$d(O,P) = ||x|| = \sqrt{\langle\langle x, x\rangle\rangle} = \sqrt{x^T x}.$$

Let $\mathbf{b}$ be the orthonormal frame and $R_{3\times 3}$ be such that $\mathbf{b} = \mathbf{e}R$ and the origin of $\mathbf{b}$ coincides with that of $\mathbf{e}$. Now let X be the representation of P in the b frame. Then we know that $x = RX$. Then the Euclidean assumption implies that the distance from O to P can also be expressed as

$$d(O,P) = ||X|| = \sqrt{\langle\langle X, X\rangle\rangle} = \sqrt{X^T X}.$$

Note that this is true for all points P and hence for any X

Thus

$$X^T X = ||X||^2 = ||x||^2 = x^T x = (RX)^T RX = X^T (R^T R) X$$

and hence we have

$$X^T (I_{3\times 3} - R^T R) X = 0.$$

for any $X \in \mathbb{R}^3$. Thus we have that

$$R^T R = I_{3\times 3}.$$

This also says that R^T is the unique inverse of R. Hence we also have

$$RR^T = I_{3\times 3}.$$

This says that R is an orthogonal matrix.

Since

$$\det(I_{3\times 3}) = \det(R^T R) = \det(R^T)\det(R) = \det(R)\det(R) = (\det(R))^2.$$

Thus $(\det R)^2 = 1$, so $\det R = \pm 1$. To determine the sign, initially place $\mathbf{b}$ coincident with $\mathbf{e}$, so that its change-of-frame matrix is $I_{3\times 3}$. Continuously rotate the frame until its first axis aligns with the prescribed $\mathbf{b}_1$. Then rotate about that axis until its second axis aligns with the prescribed $\mathbf{b}_2$. Since both frames are right-handed, the third axis then also aligns with $\mathbf{b}_3$.

This constructs a continuous curve $C(t)$ of orthogonal matrices, with $C(0) = I_{3\times 3}$ and $C(1) = R$. Orthogonality implies $\det C(t) \in \{-1, 1\}$ throughout the motion. Since the determinant is continuous, it cannot change between these two values. Therefore,

$$\det R = \det C(1) = \det C(0) = 1.$$

Hence $R \in SO(3)$.

Answer to Exercise 1.15

Let $\mathbf{e}$ be an inertial frame and let $\mathbf{b}$ translate relative to it with corresponding axes parallel and aligned. If o represents the origin of $\mathbf{b}$ in $\mathbf{e}$, then $x = o + X$. Newton's law gives

$$m\ddot{x} = f^e,$$

where f^e is the resultant interaction force expressed in $\mathbf{e}$. Therefore,

$$m\ddot{X} = f^e - m\ddot{o}. \tag{4.1}$$

The term $-m\ddot{o}$ accounts for the acceleration of the frame; it is not an additional interaction force.

For the elevator problem, we approximate an Earth-fixed frame as inertial and take upward as positive. Let W be the magnitude of the upward supporting force exerted by the scale on the person. This is the scale reading, or apparent weight. In the presence of gravity,

$$f^e = W - mg.$$

Since the person remains at rest relative to the elevator, $\ddot{X} = 0$, and hence

$$W = m(g + \ddot{o}).$$

When the elevator is at rest or moving at constant velocity, $\ddot{o} = 0$, so

$$W = mg.$$

The scale reading then equals the gravitational weight of the person.

When the elevator accelerates upwards at g, $\ddot{o} = g$, so

$$W = 2mg.$$

The scale reading is twice the gravitational weight.

When the elevator accelerates downwards at g, $\ddot{o} = -g$, so

$$W = 0.$$

The person and elevator are in free fall, and no supporting force is required. The gravitational force remains mg, but the apparent weight is zero.

Finally, suppose gravity is absent and the elevator accelerates upwards at g, where g now denotes the same acceleration magnitude used above. The only vertical interaction force on the person is the supporting force, so $f^e = W$. Equation (4.1) then gives

$$0 = W - mg, \qquad W = mg.$$

Thus the scale gives the same reading as in a stationary elevator under gravity, even though the gravitational force is now absent.

Answer to Exercise 1.16

In a Euclidean frame **e**, with no external force fields present, a particle P is constrained to move on a sphere.

Since the particle is constrained to move on a sphere of radius r we have that

$$r^2 = x_1^2 + x_2^2 + x_3^2 = ||x||^2 = x^T x = constant,$$

where x is the representation of the particle in the e frame. Differentiating the constraint $x^T x = constant$ we have that

$$2x^T \dot{x} = 0,$$

thus the velocity in the e frame $\dot{x}$ is orthogonal to x and hence lies tangent to the sphere. Differentiating the constraint $x^T \dot{x} = 0$ we have that

$$x^T \ddot{x} + \dot{x}^T \dot{x} = 0.$$

The constraint requires the velocity to be tangential to the sphere, but the acceleration generally has a nonzero normal component. In fact,

$$x^T \ddot{x} = -\|\dot{x}\|^2.$$

Since the only force acting on the particle is the tension in the wire, directed towards the fixed center, its representation is

$$f_c = \alpha x, \qquad \alpha \leq 0.$$

Thus from Newton's equations in the **e** frame

$$\ddot{x} = \frac{1}{m} f_c = \frac{\alpha}{m} x.$$

Hence we have that

$$\frac{\alpha}{m} x^T x + \dot{x}^T \dot{x} = 0,$$

and that

$$\alpha = -m \frac{||\dot{x}||^2}{r^2},$$

and the constraint force is $f_c = -m\frac{||\dot{x}||^2}{r^2} x$. Now the motion of the particle is described by

$$\ddot{x} = \frac{1}{m} f_c = -\frac{||\dot{x}||^2}{r^2} x.$$

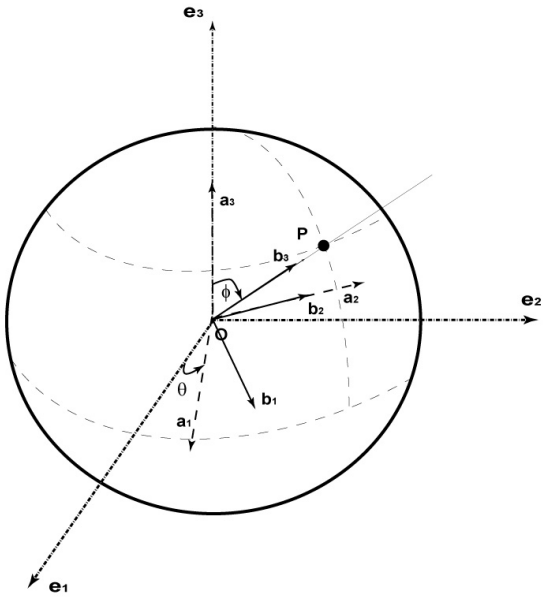


Fig. 4.2

Consider a moving frame $\mathbf{b}(t)$ such that the position of the particle appears fixed in $\mathbf{b}(t)$ and $\mathbf{b}(t)$ is related to **e** as shown in figure-4.2. At $\phi = 0$ or $\phi = \pi$, the particle lies on the vertical axis, so its position does not uniquely determine θ. The frame defined by $R = R_3(\theta)R_2(\phi)$ remains well-defined for specified angles, but these angles are singular coordinates for the particle's position at the poles.

If $\mathbf{b}(t) = \mathbf{e}R(t)$ we have that $\mathbf{a} = \mathbf{e}R_3(\theta)$ and $\mathbf{b} = \mathbf{a}R_2(\phi)$. Thus $\mathbf{b} = \mathbf{e}R = \mathbf{e}R_3(\theta)R_2(\phi)$ and $R = R_3(\theta)R_2(\phi)$.

Recall that $\dot{R} = R\widehat{\Omega}$. Differentiating $x = RX$ we have that

$$\dot{x} = \dot{R}X + R\dot{X} = R\widehat{\Omega}X.$$

where we have used the fact that $\dot{X} = 0$ since P appears to be fixed in $\mathbf{b}(t)$. Thus

$$||\dot{x}|| = ||R\widehat{\Omega}X|| = ||\widehat{\Omega}X|| = ||\Omega \times X||.$$

Newton's Equations in the $\mathbf{b}(t)$ frame are

$$m\ddot{X} = -m\widehat{\Omega}^2X - 2m\widehat{\Omega}\dot{X} - m\dot{\widehat{\Omega}}X + R^T f_c,$$

where $f_c = -m\frac{||\dot{x}||^2}{r^2}x$. Since $x = RX$ and X is a constant we have

$$0 = -m\widehat{\Omega}^2X - m\dot{\widehat{\Omega}}X - m\frac{||\widehat{\Omega}X||^2}{r^2}X,$$

and that

$$\dot{\widehat{\Omega}}X = -\widehat{\Omega}^2X - \frac{||\widehat{\Omega}X||^2}{r^2}X.$$

Now let us explicitly write down these equations. By construction of the frames $X = [0\ 0\ r]^T$. Differentiating $R = R_3(\theta)R_2(\phi)$ we have

$$\dot{R} = \dot{R}_3(\theta)R_2(\phi) + R_3(\theta)\dot{R}_2(\phi).$$

Now

$$\widehat{\Omega} = R^T\dot{R} = R_2^T(\phi)R_3^T(\theta)\dot{R}_3(\theta)R_2(\phi) + R_2^T(\phi)R_3^T(\theta)R_3(\theta)\dot{R}_2(\phi) = R_2^T(\phi)\widehat{\Omega}_3R_2(\phi) + \widehat{\Omega}_2.$$

Evaluating this we have

$$\widehat{\Omega} = \begin{bmatrix} 0 & -\dot{\theta}\cos\phi & \dot{\phi} \\ \dot{\theta}\cos\phi & 0 & \dot{\theta}\sin\phi \\ -\dot{\phi} & -\dot{\theta}\sin\phi & 0 \end{bmatrix}.$$

Then

$$\dot{\widehat{\Omega}} = \begin{bmatrix} 0 & -\ddot{\theta}\cos\phi + \dot{\theta}\dot{\phi}\sin\phi & \ddot{\phi} \\ \ddot{\theta}\cos\phi - \dot{\theta}\dot{\phi}\sin\phi & 0 & \ddot{\theta}\sin\phi + \dot{\theta}\dot{\phi}\cos\phi \\ -\ddot{\phi} & -\ddot{\theta}\sin\phi - \dot{\theta}\dot{\phi}\cos\phi & 0 \end{bmatrix}.$$

and

$$\widehat{\Omega}^2 = \begin{bmatrix} -(\dot{\phi}^2 + \dot{\theta}^2\cos^2\phi) & -\dot{\theta}\dot{\phi}\sin\phi & -\dot{\theta}^2\cos\phi\sin\phi \\ -\dot{\theta}\dot{\phi}\sin\phi & -\dot{\theta}^2 & \dot{\theta}\dot{\phi}\cos\phi \\ -\dot{\theta}^2\cos\phi\sin\phi & \dot{\theta}\dot{\phi}\cos\phi & -(\dot{\phi}^2 + \dot{\theta}^2\sin^2\phi) \end{bmatrix}.$$

Substituting these in the Newton's equations

$$\dot{\widehat{\Omega}}X = -\widehat{\Omega}^2X - \frac{||\widehat{\Omega}X||^2}{r^2}X.$$

we have

$$r\begin{bmatrix}\ddot{\phi}\\ \ddot{\theta}\sin\phi+\dot{\theta}\dot{\phi}\cos\phi\\ 0\end{bmatrix}=r\begin{bmatrix}\dot{\theta}^2\cos\phi\sin\phi\\ -\dot{\theta}\dot{\phi}\cos\phi\\ (\dot{\phi}^2+\dot{\theta}^2\sin^2\phi)\end{bmatrix}-(\dot{\phi}^2+\dot{\theta}^2\sin^2\phi)\begin{bmatrix}0\\0\\r\end{bmatrix}.$$

Which gives us

$$\ddot{\phi}=\dot{\theta}^2\cos\phi\sin\phi,$$
$$\sin\phi\ddot{\theta}=-2\dot{\theta}\dot{\phi}\cos\phi.$$

Note that these equations are not defined when $\phi=0,\pi$ since for these values the frame **b** is ill defined.

Answer to Exercise 1.19

The following typed up solution is courtesy of Kanishke Gamagedara E/09/078

Assume that the earth fixed frame **e** is inertial.

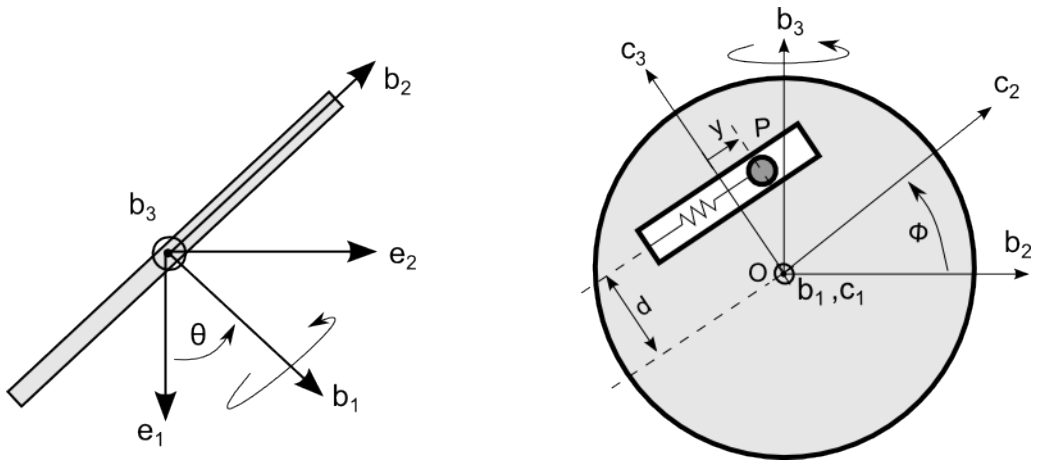


Fig. 4.3 Used frames

Considering the figure,

$$\mathbf{c}=\mathbf{b}R_1(\phi) \tag{4.2}$$
$$\mathbf{b}=\mathbf{e}R_3(\theta) \tag{4.3}$$

where,

$$R_1(\phi)=\begin{bmatrix}1&0&0\\0&\cos\phi&-\sin\phi\\0&\sin\phi&\cos\phi\end{bmatrix},\ R_3(\theta)=\begin{bmatrix}\cos\theta&-\sin\theta&0\\\sin\theta&\cos\theta&0\\0&0&1\end{bmatrix}.$$

Then we also have

$$\widehat{\Omega}_1 = R_1^T \dot{R}_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\dot{\phi} \\ 0 & \dot{\phi} & 0 \end{bmatrix}, \quad \widehat{\Omega}_3 = R_3^T \dot{R}_3 = \begin{bmatrix} 0 & -\dot{\theta} & 0 \\ \dot{\theta} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

From (4.2) and (4.3), we get $\mathbf{c} = \mathbf{e}\underbrace{R_3 R_1}_{R} = \mathbf{e}R$.

The position of the small ball x can be written as,

$$\mathbf{e}x = \mathbf{c}X = \mathbf{e}RX$$

Distance to the ball from the center of the disk (OP) can be written using the c frame as follows,

$$OP = \underbrace{\begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix}}_{\mathbf{c}} \underbrace{\begin{bmatrix} 0 \\ y \\ d \end{bmatrix}}_{X} = \mathbf{c}X.$$

The same position of the ball P can be written using the $\mathbf{e}$ frame as

$$OP = \underbrace{\begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \end{bmatrix}}_{\mathbf{e}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{x} = \mathbf{e}x = \mathbf{c}X = \mathbf{e}RX$$

The position representation of the point P with respect to the e frame given by x and the representation with respect to the frame c that is given by X are related by,

$$x = RX \tag{4.4}$$

Differentiating (4.4) twice with respect to time,

$$\ddot{x} = R\left(\widehat{\Omega}^2 X + 2\widehat{\Omega}\dot{X} + \dot{\widehat{\Omega}}X + \ddot{X}\right) \tag{4.5}$$

where,

$$\dot{X} = \begin{bmatrix} 0 \\ \dot{y} \\ 0 \end{bmatrix}, \quad \ddot{X} = \begin{bmatrix} 0 \\ \ddot{y} \\ 0 \end{bmatrix},$$

$$\widehat{\Omega} = R^T \dot{R} = (R_1^T \widehat{\Omega}_3 R_1 + \widehat{\Omega}_1) = \begin{bmatrix} 0 & -\dot{\theta}\cos\phi & \dot{\theta}\sin\phi \\ \dot{\theta}\cos\phi & 0 & -\dot{\phi} \\ -\dot{\theta}\sin\phi & \dot{\phi} & 0 \end{bmatrix},$$

$$\dot{\widehat{\Omega}} = \begin{bmatrix} 0 & -\ddot{\theta}\cos\phi + \dot{\theta}\dot{\phi}\sin\phi & \ddot{\theta}\sin\phi + \dot{\theta}\dot{\phi}\cos\phi \\ +\ddot{\theta}\cos\phi - \dot{\theta}\dot{\phi}\sin\phi & 0 & -\ddot{\phi} \\ -\ddot{\theta}\sin\phi - \dot{\theta}\dot{\phi}\cos\phi & \ddot{\phi} & 0 \end{bmatrix}$$

$$\widehat{\Omega}^2 = \begin{bmatrix} -\dot{\theta}^2 & \dot{\phi}\dot{\theta}\sin\phi & \dot{\phi}\dot{\theta}\cos\phi \\ \dot{\phi}\dot{\theta}\sin\phi & -(\dot{\theta}^2\cos^2\phi + \dot{\phi}^2) & \dot{\theta}^2\sin 2\phi/2 \\ \dot{\phi}\dot{\theta}\cos\phi & \dot{\theta}^2\sin 2\phi/2 & -(\dot{\theta}^2\sin^2\phi + \dot{\phi}^2) \end{bmatrix}$$

Assuming the Earth fixed **e**-frame is inertial we have $f = m\ddot{x}$ in the **e**-frame,

$$f = mR\left(\widehat{\Omega}^2 X + 2\widehat{\Omega}\dot{X} + \dot{\widehat{\Omega}}X + \ddot{X}\right) \tag{4.6}$$

The fundamental forces on the mass are ,

$$\mathbf{e}f = \mathbf{c}F + \mathbf{e}f_g = \mathbf{e}R\underbrace{\begin{bmatrix} F_{N1} \\ -ky \\ F_{N3} \end{bmatrix}}_{F} + \mathbf{e}\underbrace{\begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix}}_{f_g}$$

$f = RF + f_g$ and hence $R^T f = F + R^T f_g$. Substituting values in (4.6) and simplifying, we get,

$$\begin{bmatrix} F_{N1} \\ -ky \\ F_{N3} \end{bmatrix} - \begin{bmatrix} 0 \\ mg\sin\phi \\ mg\cos\phi \end{bmatrix} = \begin{bmatrix} m\left(\ddot{\theta}(d\sin\phi - y\cos\phi) - 2\dot{\theta}\dot{y}\cos\phi + 2\dot{\phi}\dot{\theta}(d\cos\phi + y\sin\phi)\right) \\ m\left(\ddot{y} - d\ddot{\phi} - y(\dot{\phi}^2 + \dot{\theta}^2\cos^2\phi) + d\dot{\theta}^2\cos\phi\sin\phi\right) \\ m\left(y\ddot{\phi} + 2\dot{\phi}\dot{y} - d\left(\dot{\phi}^2 + \dot{\theta}^2\sin^2\phi\right) + y\dot{\theta}^2\sin\phi\cos\phi\right) \end{bmatrix}$$

From the second row in the set of expressions above, we can solve this system. First and third rows can be used to find F_{N1} and F_{N3}. Simplifying the second row,

$$\ddot{y} + y\left(\frac{k}{m} - \dot{\phi}^2 - \dot{\theta}^2\cos^2\phi\right) + d\dot{\theta}^2\sin\phi\cos\phi - d\ddot{\phi} + g\sin\phi = 0 \tag{4.7}$$

Answer to Exercise 1.20

The rotation matrix relating the disk frame to the inertial frame is

$$R = R_3(\theta)R_1(\phi), \qquad \mathbf{c} = \mathbf{e}R.$$

Since the common origin O is fixed,

$$x = RX, \qquad X = \begin{bmatrix} 0 \\ y \\ d \end{bmatrix}, \qquad \dot{X} = \begin{bmatrix} 0 \\ \dot{y} \\ 0 \end{bmatrix}.$$

The angular velocity of **c** relative to **e**, expressed in **c**, satisfies $\widehat{\Omega} = R^T\dot{R}$ and is

$$\Omega = \begin{bmatrix} \dot{\phi} \\ \dot{\theta}\sin\phi \\ \dot{\theta}\cos\phi \end{bmatrix}.$$

For brevity, define

$$q = y\cos\phi - d\sin\phi, \qquad s = y\sin\phi + d\cos\phi.$$

Then

$$\dot{q} = \dot{y}\cos\phi - \dot{\phi}s.$$

The inertial velocity expressed in **c** is

$$V = R^T\dot{x} = \dot{X} + \Omega \times X = \begin{bmatrix} -\dot{\theta}q \\ \dot{y} - d\dot{\phi} \\ y\dot{\phi} \end{bmatrix}.$$

(i) Angular momentum.

The angular momentum about O, expressed in **c**, is

$$\Pi = R^T\pi = mX \times V.$$

Consequently,

$$\boxed{\Pi = m\begin{bmatrix} (y^2+d^2)\dot{\phi} - d\dot{y} \\ -d\dot{\theta}q \\ y\dot{\theta}q \end{bmatrix}.}$$

Equivalently, the formula from Section 1.2.6.1 gives

$$\Pi = \mathbb{I}\Omega + mX \times \dot{X}, \qquad \mathbb{I} = m\begin{bmatrix} y^2+d^2 & 0 & 0 \\ 0 & d^2 & -yd \\ 0 & -yd & y^2 \end{bmatrix}.$$

Notice that $\mathbb{I}$ generally varies with time because the bead moves along the slot.

(ii) Rate of change of angular momentum.

Differentiating the components of Π gives

$$\dot{\Pi} = m\begin{bmatrix} (y^2+d^2)\ddot{\phi} + 2y\dot{y}\dot{\phi} - d\ddot{y} \\ -d(\ddot{\theta}q + \dot{\theta}\dot{q}) \\ \dot{y}\dot{\theta}q + y\ddot{\theta}q + y\dot{\theta}\dot{q} \end{bmatrix}.$$

However, since $\pi = R\Pi$,

$$\dot{\pi} = R\big(\dot{\Pi} + \Omega \times \Pi\big).$$

Thus the inertial rate of change of angular momentum, expressed in **c**, is

$$\boxed{R^T\dot{\pi} = \dot{\Pi} + \Omega \times \Pi.}$$

Substitution gives explicitly

$$R^T\dot{\pi} = m\begin{bmatrix} (y^2+d^2)\ddot{\phi} + 2y\dot{y}\dot{\phi} - d\ddot{y} + \dot{\theta}^2 qs \\ -d\ddot{\theta}q - 2d\dot{\theta}\dot{y}\cos\phi + 2d\dot{\theta}\dot{\phi}s \\ y\ddot{\theta}q + 2y\dot{\theta}\dot{y}\cos\phi - 2y\dot{\theta}\dot{\phi}s \end{bmatrix}.$$

Because O is fixed in the inertial frame, $\dot{\pi} = \tau$, where τ is the resultant moment of the interaction forces about O. Hence, writing $T = R^T\tau$,

$$T = R^T\dot{\pi} = \dot{\Pi} + \Omega \times \Pi.$$

The derivative $\dot{\Pi}$ alone measures the change of the components in the rotating frame; the term $\Omega \times \Pi$ accounts for the rotation of that frame.

(iii) Kinetic energy.

The kinetic energy relative to **e** can be calculated using the inertial velocity expressed in **c**, since rotations preserve length:

$$\begin{aligned} \text{KE} &= \frac{1}{2}m\|\dot{x}\|^2 = \frac{1}{2}m\|V\|^2 \\ &= \frac{m}{2}\left[\dot{\theta}^2q^2 + (\dot{y} - d\dot{\phi})^2 + y^2\dot{\phi}^2\right]. \end{aligned}$$

Therefore,

$$\boxed{\text{KE} = \frac{m}{2}\left[\dot{y}^2 - 2d\dot{y}\dot{\phi} + (y^2+d^2)\dot{\phi}^2 + \dot{\theta}^2(y\cos\phi - d\sin\phi)^2\right].}$$

In contrast,

$$\frac{1}{2}m\|\dot{X}\|^2 = \frac{1}{2}m\dot{y}^2$$

accounts only for motion relative to the disk. The inertial velocity also contains $\Omega \times X$, so these energies are generally different.

Answer to Exercise 1.23

All components below are expressed in the Earth-fixed frame **c** (see figure 1.14), whose axes point west, radially upwards, and north. Write

$$\Omega = \omega_e\begin{bmatrix} 0 \\ \sin\phi \\ \cos\phi \end{bmatrix}, \qquad g_b = (gI_{3\times3} + r_e\widehat{\Omega}^2)e_2.$$

(i) The String Constraint

Since the string remains taut and has constant length,

$$X^T X = l^2.$$

Differentiating gives

$$2X^T \dot{X} = 0,$$

so

$$X^T \dot{X} = 0.$$

Thus the velocity relative to the support is perpendicular to the string: the particle moves tangentially to the sphere of radius l centered at the support.

Differentiating once more,

$$\dot{X}^T \dot{X} + X^T \ddot{X} = 0,$$

and hence

$$\boxed{X^T \ddot{X} = -\|\dot{X}\|^2.}$$

(ii) Equation of Motion

Let $e_2 = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T$. The tension acts towards the support, so it can be written as $-\lambda X$, with $\lambda \geq 0$. Its magnitude is

$$\| -\lambda X \| = \lambda l.$$

The interaction forces on the particle are gravity and tension. In the local uniform-gravity approximation, their resultant has the representation

$$F = -mge_2 - \lambda X.$$

The support rotates with Earth. Neglecting orbital effects, its inertial acceleration, expressed in **c**, is

$$R^T \ddot{o} = r_e \widehat{\Omega}^2 e_2.$$

Since Ω is constant, the moving-frame expression of Newton's law gives

$$F = mR^T \ddot{o} + m\widehat{\Omega}^2 X + 2m\widehat{\Omega}\dot{X} + m\ddot{X}.$$

Therefore,

$$\boxed{m\ddot{X} = -m\widehat{\Omega}^2 X - 2m\widehat{\Omega}\dot{X} - mg_b - \lambda X.}$$

The term $-2m\widehat{\Omega}\dot{X}$ is the Coriolis apparent force. The combined centrifugal contribution is

$$-m\widehat{\Omega}^2 (r_e e_2 + X).$$

Its constant part is included in $-mg_b$, while $-m\widehat{\Omega}^2 X$ describes its variation over the pendulum's displacement from the support. There is no Euler term because Earth's angular velocity is constant.

(iii) Determination of the Tension

Taking the scalar product of the equation of motion with X gives

$$mX^T\ddot{X} = -mX^T\widehat{\Omega}^2X - 2mX^T\widehat{\Omega}\dot{X} - mX^Tg_b - \lambda l^2.$$

Using part (i), we obtain

$$\boxed{\lambda = \frac{m}{l^2}\left(\|\dot{X}\|^2 - X^T\widehat{\Omega}^2X - 2X^T\widehat{\Omega}\dot{X} - X^Tg_b\right).}$$

Thus the closed equation is

$$\ddot{X} = -\widehat{\Omega}^2X - 2\widehat{\Omega}\dot{X} - g_b - \frac{X}{l^2}\left(\|\dot{X}\|^2 - X^T\widehat{\Omega}^2X - 2X^T\widehat{\Omega}\dot{X} - X^Tg_b\right).$$

Together with initial data satisfying the two constraints, this determines the motion while the string remains taut.

A negative required value of λ would mean that the constraint requires the string to push the particle. A string cannot do this: it becomes slack, and the fixed-length model must then be replaced by motion with zero tension until the string becomes taut again.

(iv) Hanging Equilibrium

Neglect the variation of centrifugal acceleration over the pendulum's length, namely $-\widehat{\Omega}^2X$. At equilibrium, $\dot{X} = \ddot{X} = 0$, so

$$0 = -mg_b - \lambda_*X_*.$$

For the hanging equilibrium, $\lambda_* > 0$. Since $\|X_*\| = l$, this gives

$$\boxed{X_* = -l\frac{g_b}{\|g_b\|}, \qquad \lambda_* = \frac{m\|g_b\|}{l}.}$$

The equilibrium tension is therefore $m\|g_b\|$.

Using $\widehat{\Omega}^2 = \Omega\Omega^T - \|\Omega\|^2I$, we find

$$g_b = \begin{bmatrix} 0 \\ g - r_e\omega_e^2\cos^2\phi \\ r_e\omega_e^2\sin\phi\cos\phi \end{bmatrix}.$$

Except at the equator and poles, this has a horizontal component. Consequently, the string aligns with the effective downward acceleration $-g_b$, which is not exactly radial. In this spherical-Earth model, the horizontal component of $-g_b$ points towards the equator.

(v) Small Oscillations

We now investigate small oscillations about the hanging equilibrium, retaining terms to first order in Earth's angular speed ω_e. Neglecting terms of order ω_e^2, the equation of motion is

$$\ddot{X} = -2\Omega \times \dot{X} - ge_2 - \frac{\lambda}{m}X, \qquad \Omega = \omega_e \begin{bmatrix} 0 \\ \sin\phi \\ \cos\phi \end{bmatrix}, \qquad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

Here ϕ is the latitude, and e_2 represents the upward vertical direction in the local frame.

Rather than calculating the tension, we eliminate it by taking the cross product with X. Define

$$H = X \times \dot{X}.$$

The quantity mH is the angular momentum formed using the particle's velocity relative to the Earth-fixed support. It is generally different from the angular momentum defined using the inertial velocity.

Since $\dot{X} \times \dot{X} = 0$, we have

$$\dot{H} = X \times \ddot{X}.$$

The tension is parallel to X and therefore contributes no cross product. Consequently,

$$\dot{H} = -2X \times (\Omega \times \dot{X}) - gX \times e_2.$$

Using the triple-product identity and the constraint $X \cdot \dot{X} = 0$,

$$X \times (\Omega \times \dot{X}) = \Omega(X \cdot \dot{X}) - \dot{X}(X \cdot \Omega) = -(X \cdot \Omega)\dot{X}.$$

Thus

$$\boxed{\dot{H} = 2(X \cdot \Omega)\dot{X} + g\, e_2 \times X.}$$

The tension has disappeared without requiring a small-oscillation approximation.

Linearization about the hanging equilibrium.

Within the approximation that neglects ω_e^2, the hanging equilibrium is

$$X_* = -le_2.$$

The position X itself is not small, since $\|X\| = l$. Instead, its departure from X_* is small.

Define the horizontal displacement

$$q = (I_{3\times 3} - e_2 e_2^T)X, \qquad e_2^T q = 0.$$

On the branch near the hanging equilibrium, the length constraint gives

$$X = q - \sqrt{l^2 - \|q\|^2}\, e_2.$$

The change in vertical position is therefore second order in the horizontal displacement. To first order in oscillation amplitude,

$$X = -le_2 + q, \qquad \dot{X} = \dot{q}.$$

All equalities below involving the linearized motion are understood to this order.

It follows that

$$H = -l\, e_2 \times \dot{q}.$$

In particular, H is horizontal to first order. Taking the cross product with e_2 gives

$$e_2 \times H = l\dot{q}, \qquad \dot{q} = \frac{1}{l} e_2 \times H.$$

Let

$$a = \Omega \cdot e_2 = \omega_e \sin\phi, \qquad \Omega_v = a e_2.$$

Thus Ω_v is the vertical component of Earth's angular velocity. Since

$$X \cdot \Omega = -la + q \cdot \Omega,$$

the product $(q \cdot \Omega)\dot{q}$ is second order in oscillation amplitude and is discarded. The equation for H consequently becomes

$$\begin{aligned} \dot{H} &= -2la\,\dot{q} + g\, e_2 \times q \\ &= -2\Omega_v \times H + g\, e_2 \times q. \end{aligned}$$

Differentiating and using $e_2 \times \dot{q} = -H/l$ yields

$$\boxed{\ddot{H} + 2\Omega_v \times \dot{H} + \frac{g}{l} H = 0.}$$

Only the vertical component of Earth's angular velocity enters this linearized equation.

Separating the oscillation from the slow rotation.

To interpret this equation, introduce a rotation $S(t) \in SO(3)$ satisfying

$$\dot{S} = -\widehat{\Omega_v} S, \qquad S(0) = I.$$

Since Ω_v is constant,

$$S(t) = \exp\left(-t\widehat{\Omega_v}\right).$$

This is a rotation about the local vertical, with signed angular velocity $-a$.

Write

$$H = SK.$$

The column K describes H after removing this rotation. Since S rotates about Ω_v, it commutes with $\widehat{\Omega_v}$. Hence

$$\dot{H} = S\left(\dot{K} - \Omega_v \times K\right),$$

and

$$\ddot{H} = S\left(\ddot{K} - 2\Omega_v \times \dot{K} + \Omega_v \times (\Omega_v \times K)\right).$$

Substituting into the equation for H gives

$$\ddot{K} + \frac{g}{l}K - \Omega_v \times (\Omega_v \times K) = 0.$$

Because K is horizontal,

$$\Omega_v \times (\Omega_v \times K) = -a^2 K.$$

Therefore,

$$\ddot{K} + \left(\frac{g}{l} + a^2\right)K = 0.$$

This last equation follows exactly from the linearized equation for H. However, the original approximation discarded terms of order ω_e^2. We therefore consistently neglect a^2 and write

$$\ddot{K} + \omega_0^2 K = 0, \qquad \omega_0 = \sqrt{\frac{g}{l}}.$$

Its solution is

$$K(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t),$$

where A and B are constant horizontal columns. They are determined by

$$A = H(0), \qquad B = \frac{\dot{H}(0) + \Omega_v \times H(0)}{\omega_0}.$$

To first order in oscillation amplitude and Earth's angular speed,

$$X \times \dot{X} \approx S(t)\left[A\cos(\omega_0 t) + B\sin(\omega_0 t)\right], \qquad S(t) = \exp\left(-t\widehat{\Omega_v}\right).$$

Thus $X \times \dot{X}$ undergoes a rapid harmonic oscillation carried by a slow rotation about the vertical. The angular velocity of this slow rotation is

$$-\Omega_v = -\omega_e \sin\phi \, e_2.$$

(vi) Interpretation of the Precession

The quantity $X \times \dot{X}$ is perpendicular to both the string direction and the instantaneous relative velocity. Its evolution therefore provides information about the orientation of the oscillatory motion.

The expression above does not assert that $X \times \dot{X}$ has constant magnitude or that its instantaneous direction rotates uniformly. It separates its rapid oscillation from the slow rotation $S(t)$.

For example, suppose the particle is released from rest relative to Earth with a small horizontal displacement q_0. Then

$$H(0) = 0, \qquad \dot{H}(0) = g\, e_2 \times q_0.$$

Consequently,

$$H(t) \approx \frac{g}{\omega_0} \sin(\omega_0 t)\, S(t)(e_2 \times q_0).$$

Away from its zeros, H lies along the slowly rotating direction $S(t)(e_2 \times q_0)$, reversing sign during each oscillation. This exhibits the precession without requiring an exact oscillation plane to be defined at every instant.

Since e_2 points upwards, the rotation $-\omega_e \sin\phi\, e_2$ is clockwise in the Northern Hemisphere and counterclockwise in the Southern Hemisphere, as viewed from above. The leading precession rate vanishes at the equator. Away from the equator, the time for a full 2π rotation is

$$T_{\text{prec}} = \frac{2\pi}{|\omega_e \sin\phi|}.$$

At either pole, this equals one sidereal day.

(vii) Numerical Investigation

To simulate the constrained motion, introduce $V = \dot{X}$ and integrate

$$\dot{X} = V, \qquad \dot{V} = -\widehat{\Omega}^2 X - 2\widehat{\Omega} V - g_b - \frac{\lambda(X,V)}{m} X,$$

with $\lambda(X,V)$ given in part (iii). These equations retain the centrifugal terms that were neglected in the first-order precession calculation.

For example, define

$$d = \frac{g_b}{\|g_b\|}, \qquad e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Since $e_1^T d = 0$, the initial data

$$X(0) = -l\cos\varepsilon\, d + l\sin\varepsilon\, e_1, \qquad V(0) = 0$$

satisfy both constraints exactly. Here ε is a small initial angular displacement from the approximate hanging equilibrium in part (iv).

Observing the precession.

From the numerical solution, calculate

$$H(t) = X(t) \times V(t).$$

To compare with the first-order calculation, define

$$q_0 = (I_{3\times 3} - e_2 e_2^T)X(0), \qquad \omega_0 = \sqrt{\frac{g}{l}}, \qquad \Omega_v = \omega_e \sin\phi\, e_2,$$

and

$$S(t) = \exp\left(-t\widehat{\Omega_v}\right).$$

For release from rest, part (vi) predicts

$$H_{\text{app}}(t) = \frac{g}{\omega_0}\sin(\omega_0 t)\, S(t)(e_2 \times q_0).$$

Compare the components of $H(t)$ with those of $H_{\text{app}}(t)$. Agreement is expected within the small-amplitude and first-order Earth-rotation approximations.

The slow rotation can also be examined by removing it from the numerical result:

$$K_{\text{num}}(t) = S(t)^T H(t).$$

The approximation predicts

$$K_{\text{num}}(t) \approx \frac{g}{\omega_0}\sin(\omega_0 t)\,(e_2 \times q_0).$$

Thus, after removing the rotation $S(t)$, the calculated quantity should approximately oscillate along a fixed horizontal direction.

For $q_0 \neq 0$, define

$$d_0 = \frac{q_0}{\|q_0\|}, \qquad n_0 = e_2 \times d_0.$$

Plot the projections

$$n_0^T K_{\text{num}}(t), \qquad d_0^T K_{\text{num}}(t), \qquad e_2^T K_{\text{num}}(t).$$

To the retained order, the first should be

$$\frac{g\|q_0\|}{\omega_0}\sin(\omega_0 t),$$

while the other two vanish. Small nonzero values reflect higher-order effects and numerical error.

Avoid normalizing H near its zeros, where its direction is undefined or numerically sensitive. Its components and the projections above remain suitable for comparison.

A simulation covering only a few oscillations may show little precession, since

$$|\omega_e \sin\phi| \ll \omega_0.$$

Use a sufficiently long interval to observe the slow rotation. Differences between the full constrained model and the first-order approximation may accumulate over long times; they should not automatically be attributed to numerical error.

Constraint and energy checks.

During integration, monitor

$$\varepsilon_l(t) = \frac{\big|\|X(t)\| - l\big|}{l}, \qquad X(t)^T V(t), \qquad \lambda(t).$$

The first two quantities should remain close to zero. The required tension must remain non-negative for the taut-string model to apply. Check numerical convergence by refining the integration tolerances or time step.

An additional check is the conserved quantity of the autonomous constrained model,

$$\mathscr{E} = \frac{1}{2} m \|V\|^2 + m g_b^T X + \frac{1}{2} m X^T \widehat{\Omega}^2 X.$$

Indeed, $\widehat{\Omega}^2$ is symmetric, and

$$\begin{aligned} \dot{\mathscr{E}} &= V^T \left(m\dot{V} + m g_b + m\widehat{\Omega}^2 X \right) \\ &= -2m V^T \widehat{\Omega} V - \lambda V^T X = 0. \end{aligned}$$

The first term vanishes by skew symmetry and the second by the string constraint. This is an effective energy for the rotating-frame model, rather than the inertial-frame kinetic energy alone.

This .ipynb interactive notebook provides a python simulation and animation of this pendulum motion.

Answer to Exercise 1.24

Let n denote the local upward unit direction, expressed in an Earth-fixed frame, and let V be the horizontal velocity of an air parcel relative to Earth. The Coriolis contribution to its acceleration is

$$a_{\mathrm{C}} = -2\Omega \times V,$$

where Ω represents Earth's angular velocity in the same frame.

To examine the horizontal deflection, decompose Ω into its vertical and horizontal parts:

$$\Omega = (\Omega \cdot n)n + \Omega_h.$$

Here $\Omega_h = \Omega - (\Omega \cdot n)n$ is the horizontal component of Earth's angular velocity, expressed in the Earth-fixed frame.

Since both Ω_h and V are horizontal, $\Omega_h \times V$ is vertical. Thus the horizontal Coriolis acceleration is

$$a_{\mathrm{C},h} = -2(\Omega \cdot n)\, n \times V.$$

At latitude ϕ, $\Omega \cdot n = \omega_e \sin\phi$.

The pressure-gradient force drives air towards a low-pressure region. Let ρ be the horizontal unit direction pointing outwards from its center. For air moving directly inwards at speed $v > 0$,

$$V = -v\rho.$$

Consequently,

$$a_{C,h} = 2v\omega_e \sin\phi \, n \times \rho.$$

Viewed from above, $n \times \rho$ points in the counterclockwise tangential direction around the low-pressure region.

In the Northern Hemisphere, $\sin\phi > 0$, so inward-moving air is deflected counterclockwise around the low-pressure region. In the Southern Hemisphere, $\sin\phi < 0$, so the deflection is clockwise. These are the respective directions of cyclonic circulation.

Equivalently, the horizontal Coriolis acceleration deflects moving air to its right in the Northern Hemisphere and to its left in the Southern Hemisphere. As circulation develops, the pressure-gradient force, Coriolis term, and friction together determine the subsequent motion. Near the surface, friction allows air to continue spiraling inwards.

The Coriolis term does not supply kinetic energy, since

$$V \cdot (-2\Omega \times V) = 0.$$

It changes the direction of motion; the pressure-gradient force drives the inflow. At the equator, $\Omega \cdot n = 0$, so this horizontal deflection mechanism vanishes.

This .ipynb interactive notebook provides a python simulation and animation of this idealized situation.

Answer to Exercise 2.1

Consider the frames $\mathbf{e}$ and $\mathbf{b}$ fixed on $\mathscr{B}_1$ and $\mathscr{B}_2$ respectively. O_1 is the origin of $\mathbf{e}$ and O_2 is the origin of $\mathbf{b}$. Now let us consider points on $\mathscr{B}_2$ as viewed from $\mathscr{B}_1$. A point P on $\mathscr{B}_2$ will have representations x and X with respect to $\mathbf{e}$ and $\mathbf{b}$ respectively. Let $e_3 = [0\,0\,1]^T$, $O_1O_2 = \mathbf{e}\,y$ and $\mathbf{b} = \mathbf{e}\,R_3(\phi)$ then

$$x = y + R_3(\phi)X, \tag{4.8}$$

$$\dot{x} = \dot{y} + \dot{\phi}R_3(\phi)\widehat{e}_3 X. \tag{4.9}$$

If there exists a point $O_{21}P$ on $\mathscr{B}_2$ that appears to be fixed in $\mathscr{B}_1$ then there exists X_I such that $\dot{x}_I = 0$. If we can show the existence of such a X_I then we have shown the existence of an instantaneous center. If $\dot{x}_I = 0$, from (4.9) it follows that

$$\dot{\phi}\widehat{e}_3 X_I = -R_3^T(\phi)\,\dot{y}, \tag{4.10}$$

$$X_I = \frac{1}{\dot{\phi}} \widehat{e}_3 R_3(\phi)^T \, \dot{y}. \tag{4.11}$$

Here we have used the fact that $\widehat{e}_3^2 X_I = -X_I$ since $e_3^T X_I = 0$.

This expression applies when $\dot{\phi} \neq 0$ and determines a unique instantaneous center, which need not lie within either physical body. If $\dot{\phi} = 0$, then every point has the same relative velocity $\dot{y}$. If $\dot{y} \neq 0$, the relative motion is pure translation and there is no finite instantaneous center. If $\dot{y} = 0$, the relative velocity vanishes everywhere, so every point is instantaneously fixed relative to the other body and there is no unique instantaneous center.

The point in $\mathscr{B}_2$ with representation X_I with respect to **b** and given by (4.11) has zero velocity with respect to $\mathscr{B}_1$. The point O_{21} is said to be the instantaneous center of $\mathscr{B}_2$ with respect to $\mathscr{B}_1$. With respect to $\mathscr{B}_1$, every point on $\mathscr{B}_2$ appears to instantaneously rotate about O_{21} with angular velocity $\dot{\phi}$. You are asked to show this in the exercises. In addition if the bodies $\mathscr{B}_1$ and $\mathscr{B}_2$ are two convex shaped rigid bodies that are moving relative to each other such that at each time instance their surfaces are in contact only at one point O_{21} then from the assumption that the bodies are rigid it follows that the relative velocity of the contact point should lie along the surface (that is perpendicular to the common surface normal through the contact point) . Thus the instantaneous center of rotation of $\mathscr{B}_2$ with respect to $\mathscr{B}_1$ (or visa versa) lies along the common normal to the surfaces at the contact point.

Answer to Exercise 2.2

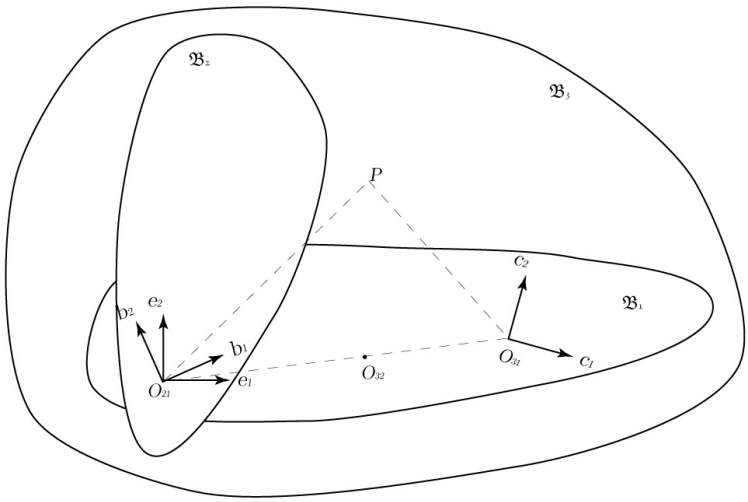


Fig. 4.4 Kennedy's Theorem

To prove Kennedy's theorem consider three rigid bodies $\mathscr{B}_1$, $\mathscr{B}_2$ and $\mathscr{B}_3$ moving in 2-dimensional Euclidean space. Let **e**, **b** and **c** be frames fixed on $\mathscr{B}_1$, $\mathscr{B}_2$ and $\mathscr{B}_3$ respectively. At a given particular time instant, without loss of generality, we pick the origins of **e** and **b**

to coincide with the instantaneous center O_{21} of $\mathscr{B}_2$ with respect to $\mathscr{B}_1$ and the origin c to coincide with the instantaneous center O_{31} of $\mathscr{B}_3$ with respect to $\mathscr{B}_1$. What we need to show is that the instantaneous center O_{32} of $\mathscr{B}_3$ with respect to $\mathscr{B}_2$ lies on the line joining O_{21} to O_{31}. To do this we observe points on $\mathscr{B}_3$ with reference to $\mathscr{B}_2$. Let P be a point on $\mathscr{B}_3$. The representation of P with respect to $\mathbf{e}$ is x, with respect to $\mathbf{b}$ is X_b and with respect to $\mathbf{c}$ is X_c and let $\mathbf{b} = \mathbf{e}R_3(\theta)$, $\mathbf{c} = \mathbf{e}R_3(\phi)$. Let Let $e_3 = [0\ 0\ 1]^T$ and $O_{21}O_{31} = \mathbf{e}y$. Then

$$x = R_3(\theta)X_b = y + R_3(\phi)X_c. \tag{4.12}$$

Now differentiating (4.12) we have

$$\dot{x} = R_3(\theta)\dot{X}_b + \dot{\theta}R_3(\theta)\widehat{e}_3X_b = \dot{y} + R_3(\phi)\dot{X}_c + \dot{\phi}R_3(\phi)\widehat{e}_3X_c. \tag{4.13}$$

Let $P = O_{23}$. The instantaneous velocity of O_{32} with respect to $\mathscr{B}_2$ (and hence the $\mathbf{b}$ frame) and $\mathscr{B}_3$ (and hence the c frame) is zero. Thus $\dot{X}_b = 0 = \dot{X}_c$. Since O_{31} is the instantaneous center of $\mathscr{B}_3$ with respect to $\mathscr{B}_1$ the point O_{31} has zero instantaneous velocity with respect to $\mathscr{B}_1$. Thus $\dot{y} = 0$. Substituting these in (4.13) we have

$$\dot{x} = \dot{\theta}R_3(\theta)\widehat{e}_3X_b = \dot{\phi}R_3(\phi)\widehat{e}_3X_c. \tag{4.14}$$

Using (4.12) and $\widehat{R_3^T e_3} = \widehat{e_3}$ we then have

$$\dot{\theta}\widehat{e}_3x = \dot{\phi}\widehat{e}_3(x-y). \tag{4.15}$$

and finally that

$$x = \frac{\dot{\phi}}{(\dot{\phi}-\dot{\theta})}y. \tag{4.16}$$

From (4.16) we see that O_{32} lies on the line joining the instantaneous centers O_{21} and O_{31} and we have proved Kennedy's theorem that the instantaneous centers of three rigid bodies moving in 2-dimensional Euclidean space lie on the same line. Furthermore (4.16) gives the ratio of the distances to the instantaneous centers as a function of the relative angular rotations of the two bodies. Explicitly

$$O_{21}O_{23} = \frac{\dot{\phi}}{(\dot{\phi}-\dot{\theta})}O_{21}O_{31}. \tag{4.17}$$

By substituting $O_{21}O_{31} = O_{21}O_{23} + O_{23}O_{31}$ in this we also have

$$O_{21}O_{23}\dot{\theta} + O_{23}O_{31}\dot{\phi} = 0. \tag{4.18}$$

Answer to Exercise 2.10

Consider figure 4.5. Assume that when $z \equiv 0$ and $\theta \equiv 0$ the system is in equilibrium. By applying Euler's 3D rigid body equations,

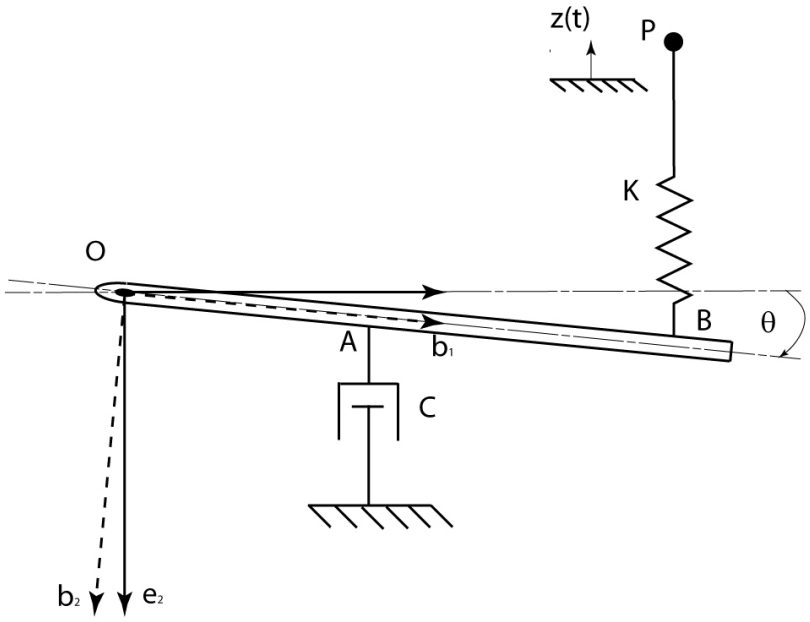


Fig. 4.5 A schematic representation of a free vibration apparatus.

[1]

$$\dot{R} = R\widehat{\Omega} \tag{4.19}$$

$$\mathbb{I}\dot{\Omega} = \mathbb{I}\Omega \times \Omega + T. \tag{4.20}$$

Here

$$R = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

and hence

$$\widehat{\Omega} = \begin{bmatrix} 0 & -\dot{\theta} & 0 \\ \dot{\theta} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \Omega = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix}.$$

The inertia tensor of the beam is

$$\mathbb{I} = \begin{bmatrix} \mathbb{I}_1 & 0 & 0 \\ 0 & \mathbb{I}_2 & 0 \\ 0 & 0 & \mathbb{I}_3 \end{bmatrix},$$

where $\mathbb{I}_1, \mathbb{I}_2, \mathbb{I}_3$ are the inertia components of the body about the b_1, b_2, b_3 axis respectively. Thus it is easily seen that $\mathbb{I}\Omega \times \Omega = 0$ and hence (4.20) reduces to:

$$\mathbb{I}\dot{\Omega} = T, \tag{4.21}$$

The external Torque$T = T_c + T_k$, where T_c and T_k are the force moments (torques) expressed in the b frame due to the damper force and spring force respectively and T_g is the force moment

due to the gravitational forces.

For small angles $\sin\theta \approx \theta \approx 0$ and $\cos\theta \approx 1$. For small deflections the above small angle approximations are valid and in the b frame the spring force has the representation,

$$F_k = \begin{bmatrix} 0 \\ -K(L_k\theta + z) \\ 0 \end{bmatrix} \tag{4.22}$$

The point at which this force acts has the representation in the b frame given by

$$X_k = \begin{bmatrix} L_k \\ 0 \\ 0 \end{bmatrix}$$

Thus the torque in the b frame due to the spring force is

$$T_k = X_k \times F_k = \begin{bmatrix} 0 \\ 0 \\ -K(L_k\theta + z)L_k \end{bmatrix} \tag{4.23}$$

In the b frame the damper force has the representation,

$$F_c = \begin{bmatrix} 0 \\ -CL_c\dot{\theta} \\ 0 \end{bmatrix} \tag{4.24}$$

The representation in the b frame of the point of action of the damper force is

$$X_c = \begin{bmatrix} L_c \\ 0 \\ 0 \end{bmatrix}.$$

The the force moment due to the damper force is

$$T_c = X_c \times F_c = \begin{bmatrix} 0 \\ 0 \\ -CL_c^2\dot{\theta} \end{bmatrix}. \tag{4.25}$$

Since gravity is neglected, the only force moments about the pivot are those exerted by the spring and damper. To first order in θ, the spring extension is $L_k\theta + z$, and the governing equation is

$$I\ddot{\theta} = -CL_c^2\dot{\theta} - kL_k(L_k\theta + z).$$

Consequently,

$$I\ddot{\theta} + CL_c^2\dot{\theta} + kL_k^2\theta = -kL_kz.$$

Answer to Exercise 2.11

Fix a frame $\mathbf{b}(t)$ on the shaft and the arm such that its origin, O, is at the point of intersection of the arm and the shaft with $\mathbf{b}_1(t)$ aligned along the arm, $\mathbf{b}_3(t)$ aligned along the shaft and pointing upwards and $\mathbf{b}_2(t)$ aligned so that the frame is right hand oriented. Let $\mathbf{e}$ be a fixed frame such that the origin coincides with O and $\mathbf{e}_3$ is aligned along $\mathbf{b}_3$. Let ϕ be the counter clockwise angle between $\mathbf{e}_1$ and $\mathbf{b}_1(t)$. Then $\mathbf{b}(t) = \mathbf{e}R_3(t)$ where

$$R_3(t) = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and $\dot{R}_3 = R_3\widehat{\Omega}_3$ where

$$\widehat{\Omega}_3 = \begin{bmatrix} 0 & -\dot{\phi} & 0 \\ \dot{\phi} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Let θ be the counter clockwise angle between the spring and the downward vertical and $l(t)$ be the length of the spring. Introduce another frame $\mathbf{c}(t)$ such that its origin also coincides with the origin of $\mathbf{e}$ and $\mathbf{b}(t)$ while $\mathbf{c}_1(t) = \mathbf{b}_1(t)$ and $\mathbf{c}_3(t)$ is along the spring. Then we see that $\mathbf{c}(t) = \mathbf{b}(t)R_1(t)$ where

$$R_1(t) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\pi+\theta) & -\sin(\pi+\theta) \\ 0 & \sin(\pi+\theta) & \cos(\pi+\theta) \end{bmatrix}.$$

Then we also have that $\dot{R}_1 = R_1\widehat{\Omega}_1$ where

$$\widehat{\Omega}_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\dot{\theta} \\ 0 & \dot{\theta} & 0 \end{bmatrix}.$$

Then we have $\mathbf{c}(t) = \mathbf{b}(t)R_1(t) = \mathbf{e}R_3(t)R_1(t) = \mathbf{e}R(t)$ and hence that $R(t) = R_3(t)R_1(t)$ and $\dot{R} = \dot{R}_3R_1 + R_3\dot{R}_1 = R_3\widehat{\Omega}_3R_1 + R_3R_1\widehat{\Omega}_1 = R_3R_1(R_1^T\widehat{\Omega}_3R_1 + \widehat{\Omega}_1) = R\widehat{\Omega}$. Thus

$$\widehat{\Omega} = R_1^T\widehat{\Omega}_3R_1 + \widehat{\Omega}_1 = \begin{bmatrix} 0 & \dot{\phi}\cos\theta & -\dot{\phi}\sin\theta \\ -\dot{\phi}\cos\theta & 0 & -\dot{\theta} \\ \dot{\phi}\sin\theta & \dot{\theta} & 0 \end{bmatrix}$$

$$\widehat{\Omega}^2 = \begin{bmatrix} -\dot{\phi}^2 & -\dot{\phi}\dot{\theta}\sin\theta & -\dot{\phi}\dot{\theta}\cos\theta \\ -\dot{\phi}\dot{\theta}\sin\theta & -(\dot{\phi}^2\cos^2\theta + \dot{\theta}^2) & \dot{\phi}^2\sin 2\theta/2 \\ -\dot{\phi}\dot{\theta}\cos\theta & \dot{\phi}^2\sin 2\theta/2 & -(\dot{\phi}^2\sin^2\theta + \dot{\theta}^2) \end{bmatrix}$$

$$\dot{\widehat{\Omega}} = \begin{bmatrix} 0 & \ddot{\phi}\cos\theta - \dot{\phi}\dot{\theta}\sin\theta & -\ddot{\phi}\sin\theta - \dot{\phi}\dot{\theta}\cos\theta \\ -\ddot{\phi}\cos\theta + \dot{\phi}\dot{\theta}\sin\theta & 0 & -\ddot{\theta} \\ \ddot{\phi}\sin\theta + \dot{\phi}\dot{\theta}\cos\theta & \ddot{\theta} & 0 \end{bmatrix}$$

Let us consider the point mass. The representation of the point mass P in e is $x(t)$ and in $c(t)$ is $X(t)$. Then $x(t) = R(t)X(t)$ and it easily seen that

$$X(t) = \begin{bmatrix} d \\ 0 \\ l \end{bmatrix}.$$

Differentiating we have

$$\dot{X}(t) = \begin{bmatrix} 0 \\ 0 \\ \dot{l} \end{bmatrix} \quad \ddot{X}(t) = \begin{bmatrix} 0 \\ 0 \\ \ddot{l} \end{bmatrix}.$$

Recall that Newton's equations for the particle expressed using the c-frame quantities are

$$F = R^T f = mR^T \ddot{x} = m(\widehat{\Omega}^2 X + 2\widehat{\Omega}\dot{X} + \dot{\widehat{\Omega}}X + \ddot{X}), \tag{4.26}$$

where the matrix F is the $\mathbf{c}(t)$-frame representation of the resultant of the fundamental forces acting on the particle. We find that it is

$$F = \begin{bmatrix} N \\ mg\sin\theta \\ -k(l-l_0) + mg\cos\theta \end{bmatrix}.$$

where N is the constraint force on the particle in the $\mathbf{c}_1$ direction and constraints the particle to only move in the vertical plane perpendicular to the arm. Now simplifying the Newton's equations (4.26) we have

$$m\left(-d\dot{\phi}^2 - 2\dot{\phi}\left(\dot{l}\sin\theta + l\dot{\theta}\cos\theta\right) - l\ddot{\phi}\sin\theta\right) = N, \tag{4.27}$$

$$l\ddot{\theta} + 2\dot{l}\dot{\theta} - \dot{\phi}^2\sin\theta\cos\theta\, l + d\ddot{\phi}\cos\theta = -g\sin\theta. \tag{4.28}$$

$$\ddot{l} + \frac{k}{m}(l-l_0) - (\dot{\phi}^2\sin^2\theta + \dot{\theta}^2)l + d\ddot{\phi}\sin\theta = g\cos\theta, \tag{4.29}$$

Now consider the shaft–arm assembly with its attached guide. We neglect only the guide's mass and inertia. The assembly rotates about the fixed axis $\mathbf{b}_3 = \mathbf{e}_3$. Let τ_u be the applied drive moment about this axis. Since the ideal shaft bearings exert no moment about the permitted rotation axis, the corresponding rigid-body equation is

$$\mathbb{I}_z\ddot{\phi} = \tau_u + T_{\text{load},3}, \tag{4.30}$$

where $T_{\text{load},3}$ is the axial component of the moment transmitted by the spring and guide.

To determine the guide reaction, consider the guide separately. The particle exerts the force $-N\mathbf{b}_1$ on the guide at P. Let H^b and C^b be the force and moment exerted by the arm on the guide at A, expressed in $\mathbf{b}$. Since the guide has negligible mass and inertia, its force and moment balances give

$$H^b - N\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 0, \qquad C^b + \begin{bmatrix} 0 \\ l\sin\theta \\ -l\cos\theta \end{bmatrix} \times \begin{bmatrix} -N \\ 0 \\ 0 \end{bmatrix} = 0.$$

Consequently, the force and moment exerted by the guide on the arm at A are

$$-H^b = \begin{bmatrix} -N \\ 0 \\ 0 \end{bmatrix}, \qquad M_A^b = -C^b = \begin{bmatrix} 0 \\ Nl\cos\theta \\ Nl\sin\theta \end{bmatrix}.$$

Thus the transmitted moment follows from the guide's moment balance.

Including the spring force, the total force acting on the arm at A, expressed in $\mathbf{b}$, is

$$F_A^b = \begin{bmatrix} -N \\ k(l-l_0)\sin\theta \\ -k(l-l_0)\cos\theta \end{bmatrix}.$$

Since

$$OA = \mathbf{b}X_A, \qquad X_A = \begin{bmatrix} d \\ 0 \\ 0 \end{bmatrix},$$

the resultant load moment about O is

$$T_{\text{load}}^b = X_A \times F_A^b + M_A^b = \begin{bmatrix} 0 \\ \left[dk(l-l_0)+Nl\right]\cos\theta \\ \left[dk(l-l_0)+Nl\right]\sin\theta \end{bmatrix}.$$

The axial equation therefore becomes

$$\mathbb{I}_z\ddot{\phi} = \tau_u + dk(l-l_0)\sin\theta + Nl\sin\theta. \tag{4.31}$$

The motion is therefore governed by (4.31), (4.29) and (4.28) with N eliminated using (4.27). These are three coupled second-order equations for $\phi(t), l(t)$ and $\theta(t)$.

The kinetic energy of the system is

$$KE = KE_{rigid\ body} + KE_{particle}.$$

where the kinetic energy of the rigid body is given by

$$KE_{rigid\ body} = \frac{\mathbb{I}_z}{2}\dot{\phi}^2.$$

The particle's inertial velocity, expressed in the shaft-fixed frame $\mathbf{b}$, is

$$R_3^T\dot{x} = \begin{bmatrix} -l\dot{\phi}\sin\theta \\ d\dot{\phi} + \dot{l}\sin\theta + l\dot{\theta}\cos\theta \\ -\dot{l}\cos\theta + l\dot{\theta}\sin\theta \end{bmatrix}.$$

Adding the kinetic energies of the shaft–arm assembly and the particle gives

$$\text{KE} = \frac{1}{2}\left(\mathbb{I}_z + md^2 + ml^2\sin^2\theta\right)\dot{\phi}^2 + \frac{1}{2}m\dot{l}^2 + \frac{1}{2}ml^2\dot{\theta}^2 + md\dot{\phi}\left(\dot{l}\sin\theta + l\dot{\theta}\cos\theta\right).$$

Answer to Exercise 2.16

Consider the Governor shown in Figure 2.16. We first consider the arm for which

$$\mathbf{b} = \mathbf{e}R, \qquad R = R_3(\theta)R_2(\alpha).$$

All moments of inertia and force moments for this arm are taken about the fixed hinge O.

Angular velocity and particle acceleration.

Using $\widehat{\Omega} = R^T\dot{R}$, we obtain

$$\Omega = \begin{bmatrix} -\dot{\theta}\sin\alpha \\ \dot{\alpha} \\ \dot{\theta}\cos\alpha \end{bmatrix}, \qquad \widehat{\Omega} = \begin{bmatrix} 0 & -\dot{\theta}\cos\alpha & \dot{\alpha} \\ \dot{\theta}\cos\alpha & 0 & \dot{\theta}\sin\alpha \\ -\dot{\alpha} & -\dot{\theta}\sin\alpha & 0 \end{bmatrix}.$$

In particular,

$$\widehat{\Omega}^2 = \begin{bmatrix} -(\dot{\alpha}^2+\dot{\theta}^2\cos^2\alpha) & -\dot{\alpha}\dot{\theta}\sin\alpha & -\dot{\theta}^2\sin\alpha\cos\alpha \\ -\dot{\alpha}\dot{\theta}\sin\alpha & -\dot{\theta}^2 & \dot{\alpha}\dot{\theta}\cos\alpha \\ -\dot{\theta}^2\sin\alpha\cos\alpha & \dot{\alpha}\dot{\theta}\cos\alpha & -(\dot{\alpha}^2+\dot{\theta}^2\sin^2\alpha) \end{bmatrix}.$$

The particle is fixed in $\mathbf{b}$, with

$$X = \begin{bmatrix} L \\ 0 \\ 0 \end{bmatrix}, \qquad \dot{X} = \ddot{X} = 0.$$

Since O is fixed in the inertial frame,

$$R^T\ddot{x} = (\dot{\widehat{\Omega}} + \widehat{\Omega}^2)X.$$

Consequently,

$$\boxed{R^T\ddot{x} = L\begin{bmatrix} -(\dot{\alpha}^2+\dot{\theta}^2\cos^2\alpha) \\ \ddot{\theta}\cos\alpha - 2\dot{\theta}\dot{\alpha}\sin\alpha \\ -\ddot{\alpha} - \dot{\theta}^2\sin\alpha\cos\alpha \end{bmatrix}}.$$

Force exerted on the particle.

Gravity expressed in $\mathbf{b}$ is

$$R^T f^g = -mg\begin{bmatrix} -\sin\alpha \\ 0 \\ \cos\alpha \end{bmatrix}.$$

For the unloaded, massless lower linkage in the symmetric nondegenerate configuration, the lower links transmit no force. Thus the forces on the particle are gravity and the upper-arm force F^r.

Newton's law gives

$$R^T f^g + F^r = mR^T \ddot{x},$$

and hence

$$F_1 = -mL\left(\dot{\alpha}^2 + \dot{\theta}^2 \cos^2\alpha\right) - mg\sin\alpha, \tag{4.32}$$

$$F_2 = mL\left(\ddot{\theta}\cos\alpha - 2\dot{\theta}\dot{\alpha}\sin\alpha\right), \tag{4.33}$$

$$F_3 = mg\cos\alpha - mL\left(\ddot{\alpha} + \dot{\theta}^2 \sin\alpha\cos\alpha\right). \tag{4.34}$$

Euler's equations for one upper arm.

The particle exerts the opposite force $-F^r$ on the arm. The hinge permits rotation about $\mathbf{b}_2$, so its ideal reaction moment is

$$M^r = \begin{bmatrix} M_1 \\ 0 \\ M_3 \end{bmatrix}.$$

The arm's own gravitational force acts at $c_l e_1$ and contributes the moment

$$\begin{bmatrix} 0 \\ m_l g c_l \cos\alpha \\ 0 \end{bmatrix}.$$

Therefore, the total moment on the arm is

$$T = \begin{bmatrix} M_1 \\ LF_3 + m_l g c_l \cos\alpha \\ M_3 - LF_2 \end{bmatrix}.$$

Euler's equation,

$$\mathbb{I}\dot{\Omega} = (\mathbb{I}\Omega)\times\Omega + T,$$

gives

$$\begin{aligned} M_1 &= -\mathbb{I}_z\left(\ddot{\theta}\sin\alpha + \dot{\theta}\dot{\alpha}\cos\alpha\right), \\ LF_3 + m_l g c_l \cos\alpha &= \mathbb{I}_l\ddot{\alpha} + (\mathbb{I}_l - \mathbb{I}_z)\dot{\theta}^2\sin\alpha\cos\alpha, \\ M_3 - LF_2 &= \mathbb{I}_l\ddot{\theta}\cos\alpha - (2\mathbb{I}_l - \mathbb{I}_z)\dot{\theta}\dot{\alpha}\sin\alpha. \end{aligned}$$

Define

$$A = \mathbb{I}_l + mL^2, \qquad B = \mathbb{I}_z, \qquad G = g(mL + m_l c_l).$$

Substitution of F_3 into the second equation gives

$$\boxed{A\ddot{\alpha} + (A-B)\dot{\theta}^2\sin\alpha\cos\alpha - G\cos\alpha = 0.}$$

Similarly, substitution of F_2 gives

$$M_3 = A\ddot{\theta}\cos\alpha - (2A - B)\dot{\theta}\dot{\alpha}\sin\alpha.$$

Equation for the spindle rotation.

The vertical direction expressed in **b** is

$$R^T e_3 = \begin{bmatrix} -\sin\alpha \\ 0 \\ \cos\alpha \end{bmatrix}.$$

Thus the vertical component of the hinge moment on one arm is

$$-\sin\alpha M_1 + \cos\alpha M_3.$$

Both arms contribute equally. Since spindle inertia is neglected,

$$\tau_u = 2\left(-M_1\sin\alpha + M_3\cos\alpha\right).$$

Therefore,

$$\boxed{2\left(A\cos^2\alpha + B\sin^2\alpha\right)\ddot{\theta} - 4(A-B)\dot{\theta}\dot{\alpha}\sin\alpha\cos\alpha = \tau_u.}$$

This result can also be checked using the total vertical angular momentum. The inertia of one arm together with its particle, about O and expressed in **b**, is

$$\mathrm{diag}(B,A,A).$$

Its vertical angular momentum is therefore

$$e_3^T R \begin{bmatrix} -B\dot{\theta}\sin\alpha \\ A\dot{\alpha} \\ A\dot{\theta}\cos\alpha \end{bmatrix} = \left(A\cos^2\alpha + B\sin^2\alpha\right)\dot{\theta}.$$

For both arms,

$$H_3 = 2\left(A\cos^2\alpha + B\sin^2\alpha\right)\dot{\theta}.$$

The balance $\dot{H}_3 = \tau_u$ reproduces the spindle equation.

The symmetric governor equations are

$$\begin{aligned} 2\left(A\cos^2\alpha + B\sin^2\alpha\right)\ddot{\theta} - 4(A-B)\dot{\theta}\dot{\alpha}\sin\alpha\cos\alpha &= \tau_u, \\ A\ddot{\alpha} + \left((A-B)\dot{\theta}^2\sin\alpha - G\right)\cos\alpha &= 0, \end{aligned}$$

where

$$A = \mathbb{I}_l + mL^2, \qquad B = \mathbb{I}_z, \qquad G = g(mL + m_l c_l).$$

Prescribed constant spindle speed.

If $\dot{\theta} = \Omega$ is maintained by a drive, the inclination satisfies

$$\boxed{A\ddot{\alpha} + \big((A-B)\Omega^2 \sin\alpha - G\big)\cos\alpha = 0.}$$

The drive moment required during a change of inclination is

$$\tau_u = -4(A-B)\Omega\,\dot{\alpha}\sin\alpha\cos\alpha.$$

Thus constant spindle speed does not generally mean zero applied moment during transient motion.

For a steady configuration $0 < \bar{\alpha} < \pi/2$, $\cos\bar{\alpha} \neq 0$, so

$$\boxed{\sin\bar{\alpha} = \frac{G}{(A-B)\Omega^2}.}$$

Such a configuration exists precisely when

$$(A-B)\Omega^2 > G.$$

For $A > B$, this gives the threshold

$$|\Omega| > \sqrt{\frac{G}{A-B}}.$$

Increasing the speed decreases $\bar{\alpha}$, so the arms rise and the masses move outwards.

The equation also formally admits $\alpha = \pm\pi/2$. These are collapsed vertical configurations outside the nondegenerate operating range assumed here; the upper configuration is also outside the illustrated linkage arrangement.

Massless upper-arm limit.

If the upper arms are also treated as massless, set

$$m_l = 0, \qquad \mathbb{I}_l = \mathbb{I}_z = 0.$$

Then

$$A = mL^2, \qquad B = 0, \qquad G = mgL,$$

and the constant-speed equation reduces to

$$\ddot{\alpha} + \left(\Omega^2 \sin\alpha - \frac{g}{L}\right)\cos\alpha = 0.$$

The nondegenerate steady configuration satisfies

$$\sin\bar{\alpha} = \frac{g}{L\Omega^2}, \qquad \Omega^2 > \frac{g}{L}.$$

Answer to Exercise 2.15

Let $\mathbf{e},\mathbf{a},\mathbf{b},\mathbf{c}$ be four orthonormal frames. The Earth-fixed frame $\mathbf{e}$, treated as inertial, has origin at the fixed base point O_1. The frames $\mathbf{a}$ and $\mathbf{b}$ also have origin at O_1, while $\mathbf{c}$ has origin at the joint O_2 between the two links. Their orientations satisfy

$$\mathbf{a} = \mathbf{e}R_a, \qquad \mathbf{b} = \mathbf{a}R_b, \qquad \mathbf{c} = \mathbf{b}R_c,$$

where

$$R_a = R_3(\alpha), \qquad R_b = R_1(\theta), \qquad R_c = R_1(\phi).$$

Here we have set $\theta = q_1$ and $\phi = q_2$. The frames $\mathbf{b}$ and $\mathbf{c}$ are fixed to links 1 and 2, respectively. Their orientations relative to $\mathbf{e}$ are

$$R_{L_1} = R_3(\alpha)R_1(\theta), \qquad R_{L_2} = R_3(\alpha)R_1(\theta+\phi).$$

Let G_1, G_2 be the link centers of mass, and define

$$O_1G_1 = \mathbf{b}X_{g_1}, \qquad O_2G_2 = \mathbf{c}X_{g_2}, \qquad O_1O_2 = \mathbf{b}X_{o_2}, \qquad O_2P = \mathbf{c}X_{cp},$$

with

$$X_{g_1} = \begin{bmatrix} 0 \\ L_2 \\ 0 \end{bmatrix}, \quad X_{g_2} = \begin{bmatrix} 0 \\ L_4 \\ 0 \end{bmatrix}, \quad X_{o_2} = \begin{bmatrix} 0 \\ L_1 \\ 0 \end{bmatrix}, \quad X_{cp} = \begin{bmatrix} 0 \\ L_3 \\ 0 \end{bmatrix}.$$

The end-point and center-of-mass positions are therefore

$$\begin{aligned} x_p &= R_{L_1}X_{o_2} + R_{L_2}X_{cp}, \\ x_{g_1} &= R_{L_1}X_{g_1}, \\ x_{g_2} &= R_{L_1}X_{o_2} + R_{L_2}X_{g_2}. \end{aligned}$$

Calculating $\widehat{\Omega}_{L_j} = R_{L_j}^T \dot{R}_{L_j}$ gives

$$\Omega_{L_1} = \begin{bmatrix} \dot{\theta} \\ \dot{\alpha}\sin\theta \\ \dot{\alpha}\cos\theta \end{bmatrix}, \qquad \Omega_{L_2} = \begin{bmatrix} \dot{\theta}+\dot{\phi} \\ \dot{\alpha}\sin(\theta+\phi) \\ \dot{\alpha}\cos(\theta+\phi) \end{bmatrix}.$$

The center-of-mass velocities satisfy

$$\begin{aligned} \dot{x}_{g_1} &= R_{L_1}\widehat{\Omega}_{L_1}X_{g_1}, \\ \dot{x}_{g_2} &= R_{L_1}\widehat{\Omega}_{L_1}X_{o_2} + R_{L_2}\widehat{\Omega}_{L_2}X_{g_2}. \end{aligned}$$

Explicitly,

$$\dot{x}_{g_1} = L_2 \begin{bmatrix} \dot{\theta}\sin\alpha\sin\theta - \dot{\alpha}\cos\alpha\cos\theta \\ -\dot{\alpha}\sin\alpha\cos\theta - \dot{\theta}\cos\alpha\sin\theta \\ \dot{\theta}\cos\theta \end{bmatrix}.$$

To write the second velocity compactly, define

$$\psi = \theta + \phi, \qquad A = L_1 \cos\theta + L_4 \cos\psi, \qquad B = L_1 \dot{\theta} \sin\theta + L_4 \dot{\psi} \sin\psi.$$

Then

$$\dot{x}_{g_2} = \begin{bmatrix} B\sin\alpha - \dot{\alpha} A \cos\alpha \\ -B\cos\alpha - \dot{\alpha} A \sin\alpha \\ L_1 \dot{\theta} \cos\theta + L_4 \dot{\psi} \cos\psi \end{bmatrix}.$$

The total kinetic energy of the two links is

$$\text{KE} = \frac{1}{2} M_1 \|\dot{x}_{g_1}\|^2 + \frac{1}{2} M_2 \|\dot{x}_{g_2}\|^2 + \frac{1}{2} \Omega_{L_1}^T \mathbb{I}_{L_1} \Omega_{L_1} + \frac{1}{2} \Omega_{L_2}^T \mathbb{I}_{L_2} \Omega_{L_2},$$

where $\mathbb{I}_{L_1}$ and $\mathbb{I}_{L_2}$ are the inertia tensors about the respective link centers of mass, expressed in axes parallel to **b** and **c**.

Answer to Exercise 2.19

All angular momenta and force moments below are taken about the fixed pivot O. We work directly with $R \in SO(3)$ and

$$\widehat{\Omega} = R^T \dot{R}, \qquad \Gamma = R^T e_3,$$

without introducing rotation coordinates. Here Γ represents the fixed vertical direction $\mathbf{e}_3$ in the body frame **b**.

Configuration and velocity constraints:

Since $\mathbf{b}_1$ remains horizontal, it is perpendicular to $\mathbf{e}_3$. Therefore

$$e_3^T R e_1 = 0.$$

Conversely, this condition makes $\mathbf{b}_1$ a horizontal hinge direction, with the remaining orientation determined by rotation about that hinge. Thus the configuration space is

$$\mathcal{Q} = \left\{ R \in SO(3) : e_3^T R e_1 = 0 \right\}.$$

This imposes one independent constraint on the three-dimensional space $SO(3)$, leaving two degrees of freedom.

Recall the scalar triple-product identity

$$A \cdot (B \times C) = B \cdot (C \times A) = C \cdot (A \times B).$$

Differentiating the configuration constraint gives

$$\begin{aligned} 0 &= e_3^T R \widehat{\Omega} e_1 \\ &= \Gamma \cdot (\Omega \times e_1) \\ &= e_1 \cdot (\Gamma \times \Omega) \end{aligned}$$

$$= -\Omega \cdot (\Gamma \times e_1).$$

This establishes both

$$e_1 \cdot (\Gamma \times \Omega) = 0$$

and the velocity constraint

$$\boxed{(\Gamma \times e_1) \cdot \Omega = 0.}$$

Define

$$n(R) \triangleq \Gamma \times e_1 = (R^T e_3) \times e_1.$$

Since $\|\Gamma\| = 1$ and $\Gamma \cdot e_1 = 0$, we have $\|n\| = 1$ on $\mathcal{Q}$.

Constraint moment:

The arm is free to rotate about $\mathbf{c}_1 = \mathbf{b}_1$ and $\mathbf{c}_3 = \mathbf{e}_3$. For an ideal support, the reaction moment is perpendicular to both permitted rotation axes, and hence lies along

$$\mathbf{c}_2 = \mathbf{c}_3 \times \mathbf{c}_1 = \mathbf{e}_3 \times \mathbf{b}_1.$$

Consequently,

$$\mathbf{b}T_c = \lambda (\mathbf{e}_3 \times \mathbf{b}_1) = \lambda \mathbf{b}(\Gamma \times e_1),$$

so that

$$\boxed{T_c = \lambda n = \lambda (\Gamma \times e_1).}$$

The velocity constraint then gives

$$T_c \cdot \Omega = \lambda n \cdot \Omega = 0.$$

Thus the constraint moment supplies no power and does no work along admissible motions.

Rotor moments:

The moment of the first thrust about O is

$$X_1 \times F_1 = l_1 f_1 \begin{bmatrix} \cos\alpha \\ 0 \\ \sin\alpha \end{bmatrix},$$

while that of the second is

$$X_2 \times F_2 = -l_2 f_2 \begin{bmatrix} \cos\beta \\ 0 \\ \sin\beta \end{bmatrix}.$$

Hence

$$\boxed{T_u = \begin{bmatrix} l_1 f_1 \cos\alpha - l_2 f_2 \cos\beta \\ 0 \\ l_1 f_1 \sin\alpha - l_2 f_2 \sin\beta \end{bmatrix}.}$$

Gravity produces no moment about O because the center of mass coincides with the pivot.

Euler's rigid body equations:

Since O is fixed, the equations are

$$\dot{R} = R\widehat{\Omega}, \qquad \mathbb{I}\dot{\Omega} + \Omega \times (\mathbb{I}\Omega) = T_u + \lambda n.$$

They must satisfy

$$e_3^T R e_1 = 0, \qquad n^T \Omega = 0.$$

Determining the constraint moment:

Differentiating $\Gamma = R^T e_3$ gives

$$\dot{\Gamma} = -\widehat{\Omega}\Gamma.$$

Thus differentiation of the velocity constraint yields

$$0 = \frac{d}{dt}(n \cdot \Omega) = n \cdot \dot{\Omega} - \Omega \cdot \left((\widehat{\Omega}\Gamma) \times e_1\right).$$

To simplify the second term, use the vector triple-product identity:

$$\begin{aligned}
\Omega \cdot \left((\widehat{\Omega}\Gamma) \times e_1\right) &= \Omega \cdot \left((\Omega \times \Gamma) \times e_1\right) \\
&= (\Omega \cdot e_1)(\Omega \cdot \Gamma) - \|\Omega\|^2 (e_1 \cdot \Gamma) \\
&= e_1^T \left(\Omega\Omega^T - \|\Omega\|^2 I_{3\times 3}\right)\Gamma \\
&= e_1^T \widehat{\Omega}^2 \Gamma.
\end{aligned}$$

Since $e_1^T \Gamma = 0$, this also equals $\Omega_1 \Gamma^T \Omega$, where $\Omega_1 = e_1^T \Omega$. The acceleration constraint is therefore

$$n^T \dot{\Omega} = e_1^T \widehat{\Omega}^2 \Gamma = \Omega_1 \Gamma^T \Omega.$$

Euler's equations give

$$\dot{\Omega} = \mathbb{I}^{-1}\left(T_u - \Omega \times (\mathbb{I}\Omega) + \lambda n\right).$$

Substitution into the acceleration constraint gives

$$n^T \mathbb{I}^{-1}\left(T_u - \Omega \times (\mathbb{I}\Omega)\right) + \lambda n^T \mathbb{I}^{-1} n = e_1^T \widehat{\Omega}^2 \Gamma.$$

Hence

$$\boxed{\lambda = \frac{e_1^T \widehat{\Omega}^2 \Gamma - n^T \mathbb{I}^{-1}\left(T_u - \Omega \times (\mathbb{I}\Omega)\right)}{n^T \mathbb{I}^{-1} n}.}$$

The denominator is strictly positive because $\mathbb{I}$ is positive definite and $\|n\| = 1$. Thus the constraint moment is uniquely determined.

The closed model is

$$\begin{aligned}\dot{R} &= R\widehat{\Omega},\\ \dot{\Omega} &= \mathbb{I}^{-1}\left(T_u - \Omega\times(\mathbb{I}\Omega) + \lambda n\right),\end{aligned}$$

where

$$\Gamma = R^T e_3, \qquad n = \Gamma \times e_1,$$

and

$$\lambda = \frac{\Omega_1 \Gamma^T \Omega - n^T \mathbb{I}^{-1}\left(T_u - \Omega\times(\mathbb{I}\Omega)\right)}{n^T\mathbb{I}^{-1}n}.$$

The prescribed thrusts $f_1(t)$ and $f_2(t)$ determine T_u.

Compatible initial conditions satisfy

$$R(0)\in SO(3), \qquad e_3^T R(0) e_1 = 0, \qquad n(R(0))^T\Omega(0) = 0.$$

For compatible initial data, the exact equations preserve the configuration and velocity constraints.

Kinetic energy and power:

Because the pivot is fixed, the kinetic energy is

$$\mathrm{KE} = \frac{1}{2}\Omega^T\mathbb{I}\Omega.$$

Differentiating and using the equations of motion,

$$\begin{aligned}\frac{d}{dt}\mathrm{KE} &= \Omega^T\mathbb{I}\dot{\Omega}\\ &= \Omega\cdot\left(T_u + \lambda n - \Omega\times(\mathbb{I}\Omega)\right)\\ &= \Omega\cdot T_u.\end{aligned}$$

Both the constraint-moment term and the cross-product term vanish. Explicitly,

$$\boxed{\frac{d}{dt}\mathrm{KE} = \Omega_1\left(l_1 f_1\cos\alpha - l_2 f_2\cos\beta\right) + \Omega_3\left(l_1 f_1\sin\alpha - l_2 f_2\sin\beta\right).}$$

This can also be checked directly from the thrust forces. Since the inertial frame has origin O and the application points are fixed in the body,

$$x_i = RX_i, \qquad \dot{x}_i = R(\Omega\times X_i).$$

The inertial representation of each thrust is RF_i. Therefore the total thrust power is

$$\begin{aligned}\sum_{i=1}^{2}(RF_i)\cdot\dot{x}_i &= \sum_{i=1}^{2}F_i\cdot(\Omega\times X_i)\\ &= \Omega\cdot\sum_{i=1}^{2}(X_i\times F_i)\\ &= \Omega\cdot T_u.\end{aligned}$$

Motion with the rotors switched off:

When $f_1 = f_2 = 0$, the kinetic energy is conserved. There is also conservation of the vertical component of spatial angular momentum.

Indeed, define

$$\Pi = \mathbb{I}\Omega, \qquad \pi = R\Pi, \qquad J = e_3^T\pi = \Gamma^T\Pi.$$

Since O is fixed,

$$\dot{\pi} = R(T_u + \lambda n).$$

Consequently,

$$\dot{J} = e_3^T R(T_u + \lambda n) = \Gamma^T T_u,$$

because $\Gamma^T n = 0$. Thus $\dot{J} = 0$ when $T_u = 0$.

The full spatial angular momentum need not be constant, since the support may still exert the nonzero moment $R\lambda n$. Its magnitude is also not generally conserved.

Answer to Exercise 2.20

The spatial angular velocity is determined by

$$\omega = K\pi.$$

The horizontal hinge condition and its time derivative give

$$e_3^T Re_1 = 0, \qquad N^T K\pi = 0.$$

On the configuration constraint, $\|N\| = 1$. The ideal support moment is $\tau_c = \lambda N$.

The spatial rotor moment is

$$\tau_u = RT_u = ARe_1 + BRe_3,$$

where

$$A = l_1 f_1\cos\alpha - l_2 f_2\cos\beta, \qquad B = l_1 f_1\sin\alpha - l_2 f_2\sin\beta.$$

Since O is fixed, the equations of motion are

$$\dot{R} = \widehat{K\pi}R, \qquad \dot{\pi} = \tau_u + \lambda N.$$

To determine λ, differentiate the momentum constraint $N^T K\pi = 0$. Using $\omega = K\pi$, we have

$$\dot{K} = \widehat{\omega} K - K \widehat{\omega},$$

and hence

$$\frac{d}{dt}(K\pi) = K\left(\tau_u + \lambda N - \omega \times \pi\right).$$

Also,

$$\dot{N} = e_3 \times (\omega \times Re_1) = -(Re_1)(e_3^T \omega),$$

where the configuration constraint was used. Therefore

$$0 = -\left((Re_1)^T K\pi\right)\left(e_3^T K\pi\right) + N^T K\left(\tau_u + \lambda N - (K\pi) \times \pi\right).$$

The closed equations in (R, π) are

$$\boxed{\dot{R} = \widehat{K\pi} R, \qquad \dot{\pi} = \tau_u + \lambda N,}$$

where

$$\boxed{\lambda = \frac{\left((Re_1)^T K\pi\right)\left(e_3^T K\pi\right) - N^T K\left(\tau_u - (K\pi) \times \pi\right)}{N^T K N}.}$$

Here

$$K = (R\mathbb{I}R^T)^{-1}, \qquad N = e_3 \times (Re_1).$$

The denominator is positive because K is positive definite and $\|N\| = 1$.

The initial conditions must satisfy

$$R(0) \in SO(3), \qquad e_3^T R(0) e_1 = 0, \qquad N(R(0))^T K(R(0))\pi(0) = 0.$$

Finally,

$$N^T \tau_u = B(e_3 \times Re_1) \cdot Re_3 = -B\, e_3^T Re_2.$$

Thus the required orthogonality condition is

$$\boxed{\left(e_3^T Re_2\right)\left(l_1 f_1 \sin\alpha - l_2 f_2 \sin\beta\right) = 0.}$$

When $e_3^T Re_2 \neq 0$, this requires $B = 0$, leaving the rotor moment along the hinge axis Re_1.

This orthogonality is not required for the constrained equations to be valid. Any component of τ_u along N is accounted for in the support reaction determined by λ.

Answer to Exercise 2.21

All quantities below are represented in the inertial frame **e**, unless otherwise stated. The angular momentum π and the applied moment τ_u are taken about the moving axle midpoint O'.

For convenience, define

$$c = ls, \qquad s = Re_3, \qquad t = Re_2, \qquad N = e_3 \times t, \qquad J = R\mathbb{I}_c R^T.$$

Thus c is the displacement from O' to the center of mass, expressed in $\mathbf{e}$.

Configuration and velocity constraints.

The configuration space is

$$\mathscr{Q} = \left\{(o,R) \in \mathbb{R}^3 \times SO(3) : e_3^T o = h, \quad e_3^T R e_2 = 0\right\}.$$

These are two independent configuration constraints, so $\mathscr{Q}$ is four-dimensional.

Using

$$\dot{R} = \widehat{\omega} R, \qquad \dot{t} = \omega \times t,$$

differentiation gives

$$e_3^T \dot{o} = 0,$$

and

$$0 = e_3 \cdot (\omega \times t) = -(e_3 \times t) \cdot \omega.$$

Including the imposed lateral constraint, we obtain

$$\boxed{e_3^T \dot{o} = 0, \qquad t^T \dot{o} = 0, \qquad N^T \omega = 0.}$$

Let

$$h_f = t \times e_3$$

be the horizontal forward direction. Since t and e_3 are orthogonal unit directions, the admissible velocities can be written as

$$\dot{o} = v h_f, \qquad \omega = \chi t + \eta e_3.$$

The three instantaneous motions are forward translation, body tilt, and turning.

The lateral constraint is nonholonomic. Indeed, consider the admissible motions consisting of forward translation and turning about the vertical. Under an infinitesimal turn,

$$\frac{dh_f}{d\theta} = e_3 \times h_f = t.$$

Consequently, the Lie bracket of the turning and forward-translation fields produces lateral translation, which is not admissible instantaneously. The admissible velocity distribution is therefore not integrable.

Ideal support reactions.

The support prevents vertical and lateral translation, so its reaction force at O' has the form

$$f_c = \lambda t + \nu e_3.$$

It also prevents angular velocity in the direction N, giving a reaction couple

$$\tau_c = \kappa N.$$

Their combined power is

$$f_c \cdot \dot{o} + \tau_c \cdot \omega = \lambda\, t \cdot \dot{o} + \nu\, e_3 \cdot \dot{o} + \kappa N \cdot \omega = 0.$$

Thus these reactions do no work on admissible motions.

Momentum and reconstruction.

The center of mass has position

$$x_c = o + c.$$

Since c is fixed in the body,

$$\dot{c} = \omega \times c,$$

and hence

$$p = M\dot{x}_c = M(\dot{o} + \omega \times c).$$

The spatial angular momentum about the center of mass is $J\omega$. Shifting the reference point to O' gives

$$\pi = J\omega + c \times p.$$

Therefore,

$$\boxed{\omega = J^{-1}(\pi - c \times p), \qquad \dot{o} = \frac{p}{M} - \omega \times c, \qquad \dot{R} = \widehat{\omega} R.}$$

Notice that π is generally not the inertia about O' multiplied by ω, because O' is translating.

Spatial momentum balances.

The total interaction force consists of the prescribed force, gravity, and the support force:

$$\boxed{\dot{p} = f_u - Mge_3 + \lambda t + \nu e_3.}$$

For angular momentum about a moving point,

$$\dot{\pi} = -\dot{o} \times p + \tau^{O'},$$

where $\tau^{O'}$ is the total interaction-force moment about O'.

The support force acts at O' and has zero moment about that point. Gravity acts at the center of mass, giving the moment

$$c \times (-Mge_3) = -Mgc \times e_3.$$

Thus

$$\boxed{\dot{\pi} = -\dot{o} \times p + \tau_u - Mgc \times e_3 + \kappa N.}$$

The term $-\dot{o} \times p$ must be retained because O' is moving.

Complete constrained model.

Combining the preceding results gives

$$\begin{aligned}
\dot{o} &= \frac{p}{M} - \omega \times c, \\
\dot{R} &= \widehat{\omega} R, \\
\dot{p} &= f_u - Mge_3 + \lambda t + \nu e_3, \\
\dot{\pi} &= -\dot{o} \times p + \tau_u - Mgc \times e_3 + \kappa N,
\end{aligned}$$

where

$$c = lRe_3, \qquad t = Re_2, \qquad N = e_3 \times t, \qquad \omega = (R\mathbb{I}_c R^T)^{-1}(\pi - c \times p).$$

The configuration constraints are

$$e_3^T o = h, \qquad e_3^T t = 0,$$

and the momentum-dependent velocity constraints are

$$e_3^T \left(\frac{p}{M} - \omega \times c\right) = 0, \qquad t^T \left(\frac{p}{M} - \omega \times c\right) = 0, \qquad N^T \omega = 0.$$

Determining the reaction multipliers.

To express the differentiated constraints compactly, introduce the auxiliary quantities

$$L = \pi - c \times p = J\omega, \qquad z = \begin{bmatrix} p/M \\ \omega \end{bmatrix}.$$

Here L is the angular momentum about the center of mass; it is calculated from the chosen state variables and is not an additional state.

The velocity constraints can be written as

$$Az = 0, \qquad A = \begin{bmatrix} t^T & -(c\times t)^T \\ e_3^T & -(c\times e_3)^T \\ 0_{1\times 3} & N^T \end{bmatrix}.$$

Define

$$q_c = \begin{bmatrix} \lambda \\ \nu \\ \kappa \end{bmatrix}, \qquad D = \begin{bmatrix} M^{-1}I_{3\times 3} & 0 \\ 0 & J^{-1} \end{bmatrix}.$$

The support force and its moment about the center of mass are then

$$A^T q_c = \begin{bmatrix} \lambda t + \nu e_3 \\ -c\times(\lambda t + \nu e_3) + \kappa N \end{bmatrix}.$$

The prescribed moment about the center of mass is $\tau_u - c\times f_u$. Therefore,

$$\dot{L} = \tau_u - c\times f_u - c\times(\lambda t + \nu e_3) + \kappa N.$$

Since

$$\dot{J} = \widehat{\omega}J - J\widehat{\omega},$$

we have

$$J\dot{\omega} = \dot{L} - \omega\times L.$$

It follows that

$$\dot{z} = a_0 + DA^T q_c,$$

where

$$a_0 = \begin{bmatrix} M^{-1}f_u - ge_3 \\ J^{-1}\big(\tau_u - c\times f_u - \omega\times L\big) \end{bmatrix}.$$

Differentiating $Az = 0$ gives

$$0 = \dot{A}z + A\dot{z} = \dot{A}z + Aa_0 + ADA^T q_c.$$

Hence

$$\boxed{q_c = -(ADA^T)^{-1}\left(\dot{A}z + Aa_0\right).}$$

All quantities on the right are determined by (o, R, p, π) and the prescribed inputs.

In particular,

$$\dot{t} = \omega\times t, \qquad \dot{c} = \omega\times c, \qquad \dot{N} = e_3\times(\omega\times t),$$

so

$$\dot{A} = \begin{bmatrix} \dot{t}^T & -(\dot{c}\times t + c\times\dot{t})^T \\ 0_{1\times 3} & -(\dot{c}\times e_3)^T \\ 0_{1\times 3} & \dot{N}^T \end{bmatrix}.$$

For $M > 0$ and positive-definite $\mathbb{I}_c$, D is positive definite. On $\mathscr{Q}$, the three rows of A are independent, so ADA^T is positive definite and the multipliers are uniquely determined.

Compatible initial conditions.

The initial configuration must satisfy

$$R(0) \in SO(3), \qquad e_3^T o(0) = h, \qquad e_3^T R(0) e_2 = 0.$$

Compute

$$\omega(0) = J(0)^{-1}\big(\pi(0) - c(0) \times p(0)\big),$$

and

$$v_o(0) = \frac{p(0)}{M} - \omega(0) \times c(0).$$

The initial momenta must satisfy

$$e_3^T v_o(0) = 0, \qquad t(0)^T v_o(0) = 0, \qquad N(0)^T \omega(0) = 0.$$

For compatible initial data, the exact equations with the multipliers derived above preserve both the configuration and velocity constraints.

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